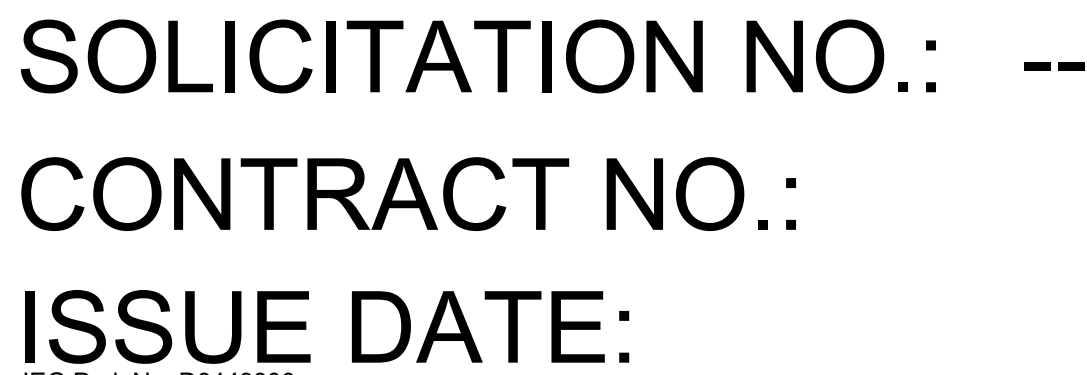




Jacobs
JACOBS ENGINEERING GROUP INC.
777 MAIN STREET, FORT WORTH, TX 76102
USA

ATIONS ANNEX,



PLOT DATE: 8/29/2025 10:51:31 AM PLOTTED BY: PLOT SCALE: 1:1

A map of Japan with labels for 'SEA OF JAPAN', 'PACIFIC OCEAN', and 'JAPAN'. An arrow points from the Okinawa region to a circular inset. The inset shows a detailed map of the Okinawa archipelago with a label 'PROJECT LOCATION OKINAWA, JAPAN' pointing to a specific site.

PROJECT LOCATION

Adaniya

81

330

29

Chus-Dori St.

Suzuran-Dori Ave.

Futemma2

Nodake4

Homachi-Dori St.

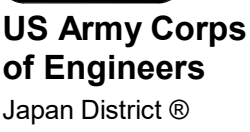
Homachi-Dori St.

Nodake3

Aragusuku1

Aragusuku2

wa Expressway



Christopher M. Surles	
TX 23638	NAME
09/04/2023	DATE
REGISTRATION #	
TEXAS REGISTERED	
ENGINEERING FIRM F-2966	

[illegible]

09/04/2025	CHRIS SURLS	DRAWN BY:	FILE NAME:
SOLICITATION NO.:	LAMONT FALLON	CHECKED BY:	ISO-A1
CONTRACT NO.:	KELVIN CARLSON	SUBMITTED BY:	G-001
DRAWING CODE:	A. GUERRA		

OKINAWA, JAPAN

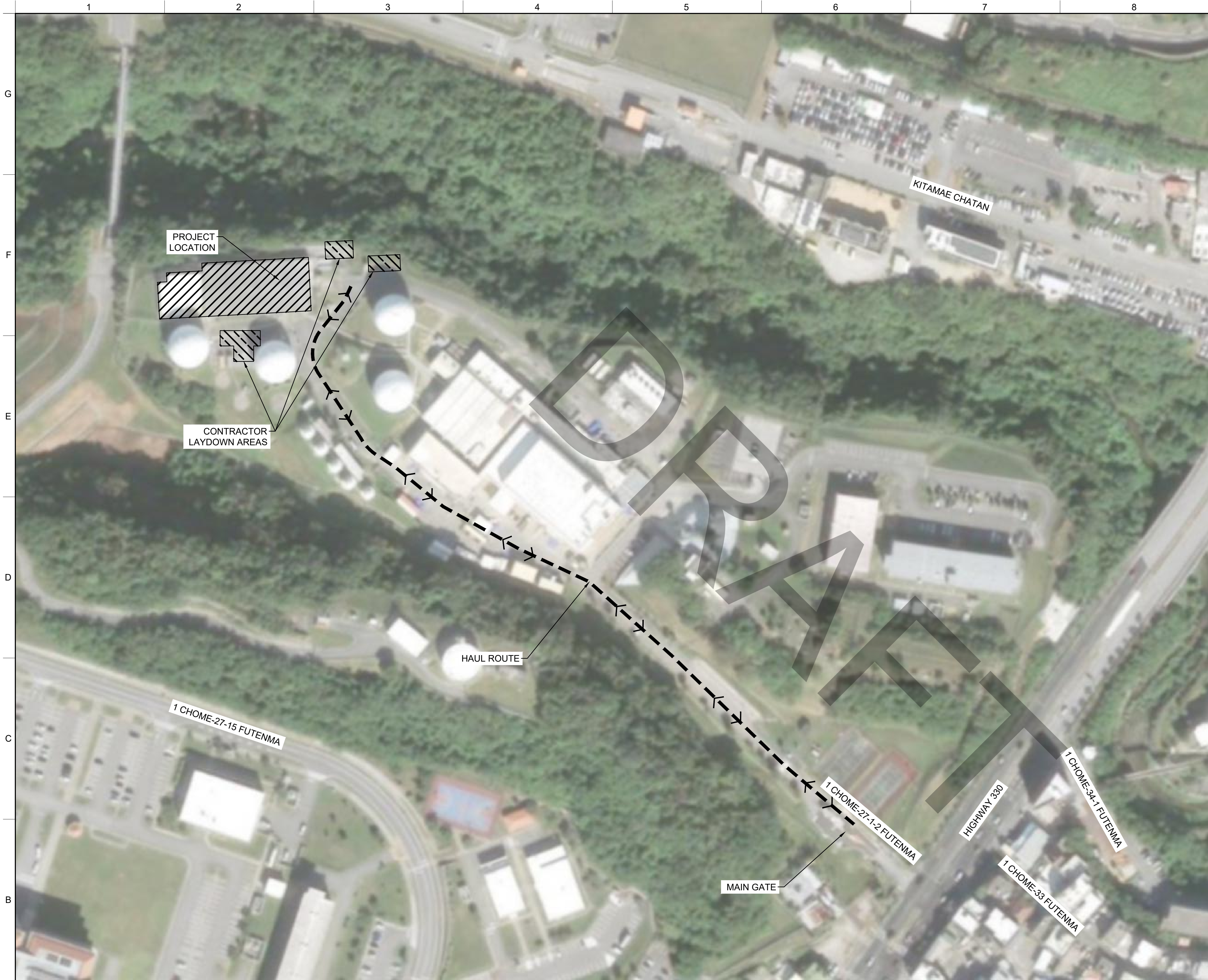


**CAMP BUCKNER COMMUNICATIONS ANNEX
OKINAWA, JAPAN**

SHEET ID

G-001

BACKCHECK SUBMITTAL DESIGN



A1 HAUL ROUTE & CONTRACTOR LAYDOWN
1 : 1000

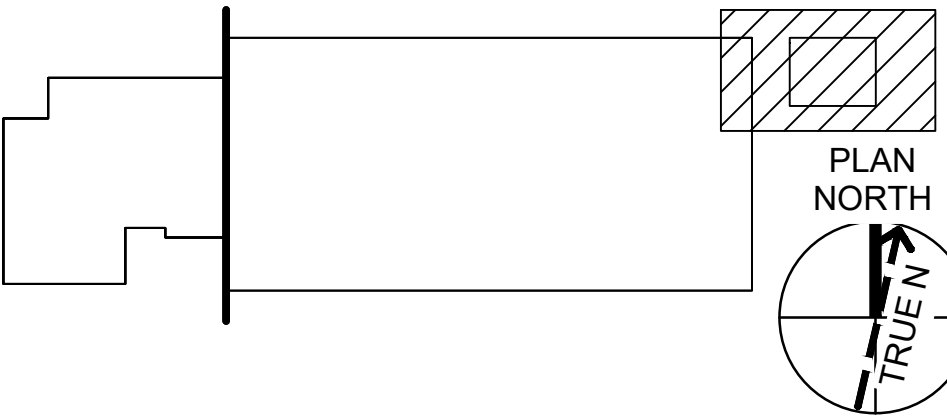
GENERAL NOTES

- A CONTRACTOR IS RESPONSIBLE FOR SECURITY OF ALL EQUIPMENT AND STORED MATERIALS. NO COVERED STORAGE IS AVAILABLE.
- B CONTRACTOR IS RESPONSIBLE FOR ANY DAMAGE CAUSED BY CONSTRUCTION ACTIVITY.
- C ALL GRUBBING AND CLEARING RESIDUE, DEMOLISHED MATERIAL RUBBISH AND DEBRIS GENERATED BY THIS PROJECT SHALL BE HAULED OFFSITE TO AN APPROVED LANDFILL.
- D MICROSOFT PRODUCT SCREEN SHOT REPRESENTED WITH PERMISSION FROM MICROSOFT CORPORATION.

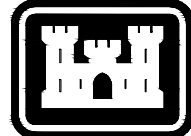
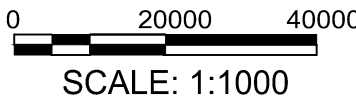
LEGEND

- TWO WAY HAUL ROUTE
- PROJECT LOCATION
- CONTRACTOR LAYDOWN AREA

KEY PLAN



GRAPHIC SCALE(S)



**US Army Corps
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Japan District ®

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THEY WERE PREPARED BY, OR
UNDER SUPERVISION OF:
LISA M. HERLOUX
TX 156321 W/VE 09/04/2025
REGISTRATION # DATE
TEXAS REGISTERED
ENGINEERING FIRM F-2966

DESIGNED BY: L. HERLOUX	ISSUE DATE: 09/04/2025	SOLICITATION NO.: -
DRAWN BY: B. CAPPELLA		CONTRACT NO.: -
CHECKED BY: D. GIMBEL		DRAWING CODE: -
SUBMITTED BY: A. GUERRA	FILE NAME: G-100	
SIZE: ISO A1		

US ARMY CORPS OF ENGINEERS
JAPAN ENGINEER DISTRICT
OKINAWA, JAPAN



FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX
OKINAWA, JAPAN
PROJECT NUMBER: D3449305

HAUL ROUTE & CONTRACTOR LAYDOWN

SHEET ID

G-100

DRAWING ABBREVIATIONS

O	
OA	OUTDOOR AIR, OVERALL
OC	ON CENTER
OCEW	ON CENTER EACH WAY
OD	OUTSIDE DIAMETER
OFD	OVERFLOW DRAIN
OFF	OFFICE
OH	OPPOSITE HAND
OH OPT	OPPOSITE HAND OPTIONAL
OHCD	OVERHEAD COILING DOOR
OHE	OVERHEAD ELECTRIC
OHT	OVERHEAD TELEPHONE
OHW	OVERHEAD WIRE
OPP	OPOSIT
ORD	OVERFLOW ROOF DRAIN
ORIG	ORIGINAL
OS	OCCUPANCY SENSOR
OSP	OUTSIDE PLANT
OWS	OIL/WATER SEPARATOR
OWSJ	OPEN WEB STEEL JOIST

F	P	PAGE, POLE
	P/T	POST-TENSION (-ED, -ING)
	PA	PUBLIC ADDRESS
	PAT	PATTERN
	PB	PULL BOX
	PC	POINT OF CURVATURE
	PCC	POINT OF COMPOUND CURVATURE, PORTLAND CEMENT CONCRETE, PRECAST CONCRETE
	PCF	POUNDS PER CUBIC FOOT
	PCT	PERCENT
	E	PDS
PE		POLYETHYLENE
PEJ		PREMOLDED EXPANSION JOINT
PERF		PERFORATED
PERIM		PERIMETER
PERP		PERPENDICULAR
PF		POWER FACTOR
PG		PLATE GIRDER
PH		PHASE
PI		POINT OF INTERSECTION
D	PICV	PRESSURE INDEPENDENT CONTROL VALVE
	PIL	PILASTER
	PIR	PASSIVE INFRARED
	PIV	POST INDICATOR VALVE
	PL	PLATE, PROPERTY LINE
	PL GL	PLATE GLASS
	PLAM	PLASTIC LAMINATED
	PLAS	PLASTER
	PLBG	PLUMBING
	PLG	PILING
	PLY WD	PLYWOOD
	PNEU	PNEUMATIC
	PNL	PANEL, PANELBOARD
	PORT	PORTABLE
	POV	PRIVATELY OWNED VEHICLE
	PP	POWER POLE
	PP PL	PUSH/PULL PLATE
	PR	PAIR, PIPE RAIL
	PRCST	PRECAST
	PREFAB	PREFABRICATED
	PREFIN	PREFINISHED
	PRELIM	PRELIMINARY
	PRKG	PRKG
	PROJ	PROJECT, PROJECTION PRESTRESSED GIRDER
	PROT	PROTECTION

P	
PS	PARKING STRUCTURE
PS CONC	PRESTRESSED CONCRETE
PSF	POUNDS PER SQUARE FOOT
PSI	POUNDS PER SQUARE INCH
PT	POINT OF TANGENCY, PORCELAIN TILE, PRESSURE TREATED
PTD	PAPER TOWEL DISPENSER
PTDR	PAPER TOWEL DISPENSER AND RECEPTACLE
PTN	PARTITION
PTR	PAPER TOWEL RECEPTACLE
PVC	POINT OF VERTICAL CURVATURE, POLYVINYL CHLORIDE PIPE, POLYVINYL CHLORIDE
PVI	POINT OF VERTICAL TANGENCY
PVMT	PAVEMENT
PWR	POWER

Q	
QT	QUARRY TILE
QTY	QUANTITY

R	
RA	RETURN AIR
RAD	RADIUS
RB	RESILIENT BASE
RBM	REINFORCED BRICK MASONRY
RBR	RUBBER
RC	REINFORCED CONCRETE
RCP	RECEPTACLE CONTROL PANEL, REFLECTED CEILING PLAN
RD	ROOF DRAIN
RDC	REDUCER
RE, REF	REFER, REFERENCE, REFRIGERATOR
REBAR	REINFORCING STEEL BARS
REC	RECEPTACLE, RECESSED
REC RM	RECREATION ROOM
RECD	RECEIVED
RECT	RECTIFIER
RED	REDUCER
REINF	REINFORCING (-ED, -MENT)
REM	REMOVABLE
REP	REPAIR
REPL	REPLACE
REQ	REQUIRE
REQD	REQUIRED
RESIL	RESILIENT
RESIST	RESISTANT
REST	RESTROOM
RET	RETAINING
REV	REVISION
RF	RESILIENT FLOORING
RFG	ROOFING
RGS	RIGID GALVANIZED STEEL
RH	ROOF HATCH, ROOF HOOD, RIGHT HAND
RHR	RIGHT HAND REVERSE
RL	ROOF LADDER
RLG	RAILING
RM	ROOM
RMC	RIGID METAL CONDUIT
RND	ROUND
RO	ROUGH OPENING
ROW	RIGHT OF WAY
RP	RADIUS POINT
RPM	REVOLUTIONS PER MINUTE
RPZ	REDUCED PRESSURE ZONE
RSD	ROLLING STEEL DOOR
RT	RIGHT
RV	ROOF VENT
RVL	REVEAL

R	
RVTR	RADON VENT THROUGH ROOF
RW	RETAINING WALL

S	
SA	SINK, SOUTH (FOR COORDINATES), STANDARD SHAPE
SAN	SUPPLY AIR
SAT	SANITARY
SB	SUSPENDED ACOUSTICAL TILE
SC	SPLASH BLOCK
SCD	SEALED CONCRETE, SHEAR CONNECTORS, SLIP CRITICAL
SCHD	SEAT COVER DISPENSER
SCWD	SCHEDULE
SD	SOLID CORE WOOD DOOR
SDI	SMOKE DETECTOR, STORM DRAIN
SDJ	STAIR DECK INSTITUTE
SE	SFB/CFE-EXTRACTOR

SEC	SECTION
SECT	SECTION
SEER	SEASONAL ENERGY EFFICIENT RATIO
SEG	SEGMENT
SEP	SEPARATE
SF	SQUARE FOOT (FEET)
SG	STATIC GROUND
SGL	SINGLE
SH	SHOWER
SHT	SHEET
SHTHG	SHEATHING
SHV	SHELVING
SIM	SIMILAR
SJ	SAW JOINT, SCORED JOINT
SJI	STEEL JOIST INSTITUTE
SKLT	SKYLIGHT
SLNT	SEALANT
SMK	SMOKE
SMLS	SEAMLESS
SND	SANITARY NAPKIN DISPENSER
SP	SINGLE POLE, SUMP PIT
SP EL	SPOT ELEVATION
SP, SPA	SURGE(S)
SPD	SPACE PROTECTION DEVICE
SPEC	SPECIFICATION
SQ	SQUARE
SQ IN	SQUARE INCH
SQ YD	SQUARE YARD
SS	STAINLESS STEEL, SANITARY SEWER
ST	SINGLE THROW, STREET, STAIRS, STORAGE TANK
STA	STATION
STC	SOUND TRANSMISSION CLASS
STIFF	STIFFENER
STL	STEEL
STL JST	STEEL JOIST
STL RF DK	STEEL ROOF DECK
STND	STANDARD
STOR	STORAGE
STR	STRINGERS
STRB/H RN	STROBE/HORN
STRUCT	STRUCTURE (-AL)
SUB FL	SUBFLOOR
SV	SHEET VINYL
SVC	SERVICE
SW	SIDEWALK
SW	SWITCH
SYM	SYMBOL
SYMM	SYMMETRICAL

S	
SYS	SYSTEM
T	

T	TANGENT, TREAD
T&B	TOP & BOTTOM
T&G	TONGUE AND GROOVE
T/G	TOP OF GRATE
T/PIPE	TOP OF PIPE
T/S	TUB/SHOWER
TA	TRANSFER AIR
TB	THROUGH BOLT, TOWEL BAR
TBM	TEMPORARY BENCHMARK
TC	TERRA COTTA, TOP OF CURB
TCE	TRICHLOROETHYLENE
TD	TRENCH DRAIN
TEL	TELEPHONE
TEMP	TEMPERATURE, TEMPORARY
TER	TELECOMMUNICATIONS EQUIPMENT ROOM
TERM	TERMINAL
TF	TOP OF FINISH FLOOR
TGB	TELECOMMUNICATIONS GROUNDING BUSBAR
THK	THICKNESS
THRES	THRESHOLD
TK BD	TACKBOARD
TMBG	TELECOMMUNICATIONS MAIN GROUNDING BUSBAR
TMPTD	TEMPERED GLASS
GL	
TMR	TIMER
TMV	THERMOSTATIC MIXING VALVE
TN	TRUE NORTH
TO	TOP OF
TOC	TABLE OF CONTENT, TOP OF CONCRETE
TOPO	TOPOGRAPHY
TOS	TOP OF SLAB, TOP OF STEEL
TP	TOP OF PAVEMENT
TPD	TOILET PAPER DISPENSER
TR	TELEPHONE ROOM
TRANS	TRANSOM
TRTD	TREATED
TS	TUBE STEEL
TSTAT	THERMOSTAT
TV	TELEVISION
TW	TILT WALL, TOP OF WALL
TYP	TYPICAL

U	
U/G	UNDERGROUND
UC	UNDERGROUND CONDUIT
UGC	UNDERGROUND CABLE
UGE	UNDERGROUND ELECTRIC
UGT	UNDERGROUND TELEPHONE
UH	UNIT HEATER
UL	UNDERWRITER'S LABORATORIES
UNK	UNKNOWN
UON	UNLESS OTHERWISE NOTED
UPS	UNINTERRUPTED POWER SUPPLY

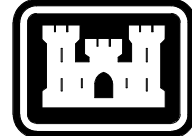
V	VENT, VOLT
V/PH/Hz	VOLT/PHASE/FREQUENCY
VA	VOLT-AMPERE
VAC	VACUUM
VAR	VARIABLE
VAV	VARIABLE AIR VOLUME
VB	VINYL COMPOSITION TILE
VC	VERTICAL CURVE
VD	VOLUME DAMPER

V	
VERT	VERTICAL
VFD	VARIABLE FREQUENCY DRIVE
VFI	VACUUM FAULT INTERRUPTER
VIF	VERIFY IN FIELD
VOL	VOLUME
VP	VACUUM PUMP FOR RADIANT HEATER
VTR	VENT THROUGH ROOF
VWF	VINYL WALL FABRIC

W	
W	WATER, WATT, WEST (FOR COORDINATES), WIDE, WIDE FLANGE, WIDTH
W/	WITH
W/O	WITHOUT
WB	WALL BOX, WET BULB
WC	WATER CLOSET
WCO	WALL CLEANOUT
WD	WOOD
WDW	WINDOW
WGL	WIRED GLASS
WH	WALL HUNG, WATER HEATER
WHM	KILOWATT-HOUR METER
WL	WATER LINE
WP	WEATHERPROOF, WORK POINT
WPJ	WEAKENED PLANE JOINT
WSCT	WAINSCOT
WT	HECTARE
WT	STRUCTURAL TEE CUT FROM WIDE FLANGE BEAM, WEIGHT
WV	WATER VALVE
WWF	WELDED WIRE FABRIC

X	
XFMR	TRANSFORMER
XFR	TRANSFER
XPS	EXTRUDED POLYSTYRENE BOARD (INSULATION)

Y	
Y	YARD
Z	
Z	IMPEDANCE



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Christopher M. Surles

NAME	
23638	09/04/2025
ISTRATION #	DATE
TEXAS REGISTERED ENGINEERING FIRM F-2966	

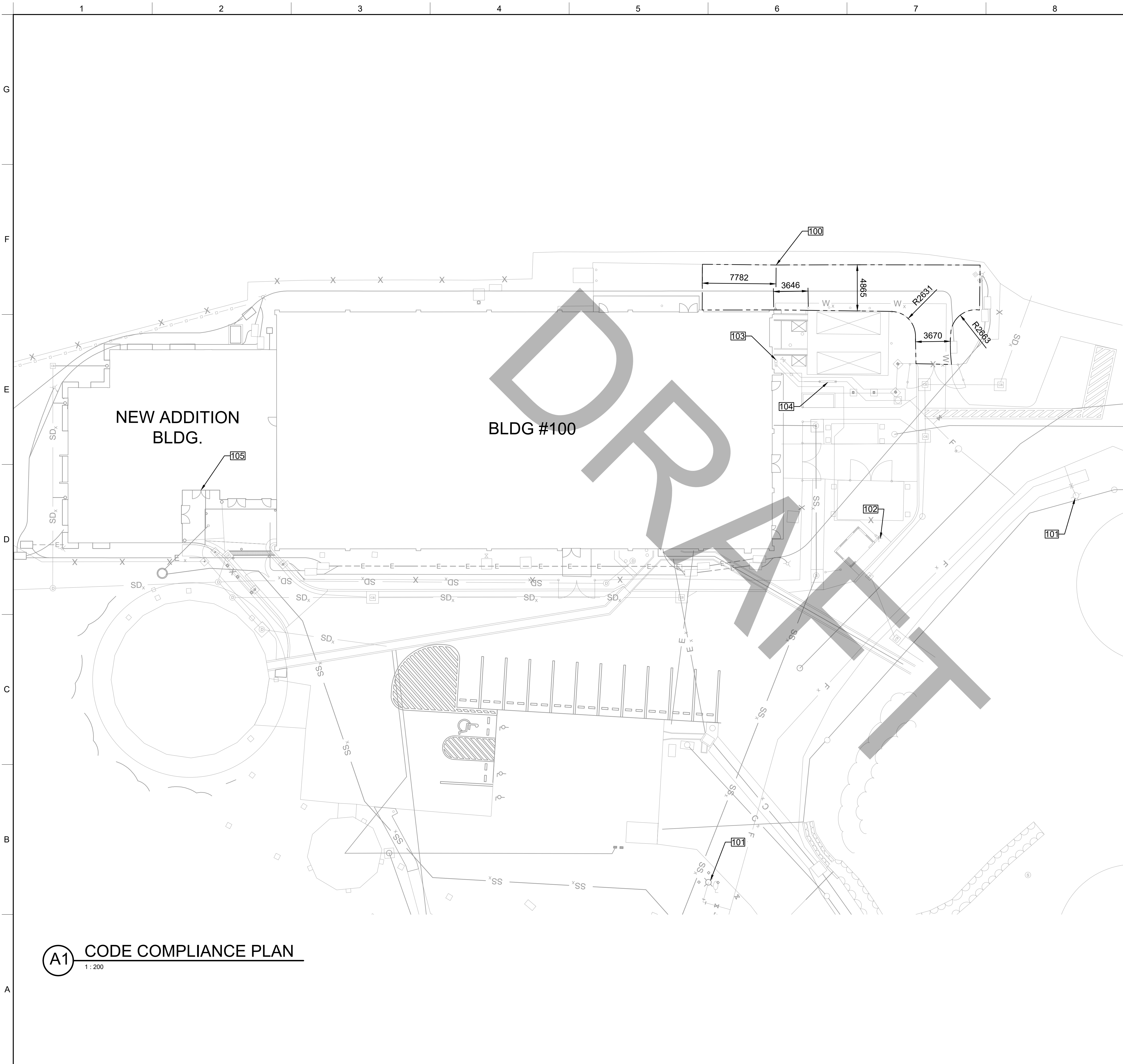
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		177 MAIN STREET, SUITE 100, WORTH, TX 76180 WORTH, TEXAS, U.S.A.	
U.S. ARMY CORPS OF ENGINEERS JAPAN ENGINEER DISTRICT OKINAWA, JAPAN		DRAWN BY: LAMONT FALLON CHECKED BY: SELVIN CARLSON DESIGNED BY: A. GUERRA	
CONTRACT NO.: -		DRAWING CODE:	
SOLICITATION NO.: -		FILE NAME:	
DATE: 09/04/2025		ISO A1	
DESIGNED BY: CHRIS SURLES		G-1003	

OKINAWA, JAPAN
PROJECT NUMBER:D3449305

SHEET ID

G-1003



GENERAL NOTES

A REFER TO ARCHITECTURAL PLANS FOR BUILDING DIMENSIONS

SHEET KEYNOTES

100 FIRE LANE

101 EXISTING FIRE HYDRANT

102 EXISTING FIRE DEPARTMENT CONNECTION

103 EXISTING FIRE RISER

104 EXISTING BACKFLOW PREVENTER

105 PRIMARY FIRE DEPARTMENT ACCESS

LEGEND

W

EXISTING WATER MAIN LINE

F

EXISTING FIRE PROTECTION WATER SUPPLY LINE

X

EXISTING GATE VALVE

SS

EXISTING FIRE RISER

FD

EXISTING FIRE DEPARTMENT CONNECTION (FDC)

HD

EXISTING FIRE HYDRANT

FIRE LANE

↔

EXISTING BACKFLOW PREVENTER

KEY PLAN

PLAN NORTH

GRAPHIC SCALE(S)

0 4000 8000

SCALE: 1:200

US Army Corps of Engineers

Japan District @

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LISA M. HEROUX

NAME

TX 156321

REGISTRATION #

09/04/2025

DATE

TEXAS REGISTERED ENGINEERING FIRM F-2966

ISSUE DATE:

09/04/2025

SOLICITATION NO.:

CONTRACT NO.:

DRAWING CODE:

DESIGNED BY:

L. HEROUX

CHECKED BY:

B. CAPPELLA

SUBMITTED BY:

D. GIMBEL

A. GUERRA

FILE NAME:

GH110

SIZE:

ISO A1

US ARMY CORPS OF ENGINEERS

JAPAN ENGINEER DISTRICT

OKINAWA, JAPAN

Jacobs

JACOBS GOVERNMENT SERVICES COMPANY

777 MAIN STREET, SUITE 1000, FORT WORTH, TX 76102

USA

FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100

CAMP BUCKNER COMMUNICATIONS ANNEX

OKINAWA, JAPAN

PROJECT NUMBER: D3449305

CODE COMPLIANCE PLAN

SHEET ID

GI1110

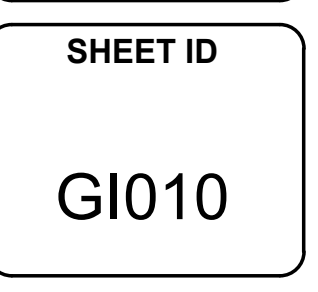
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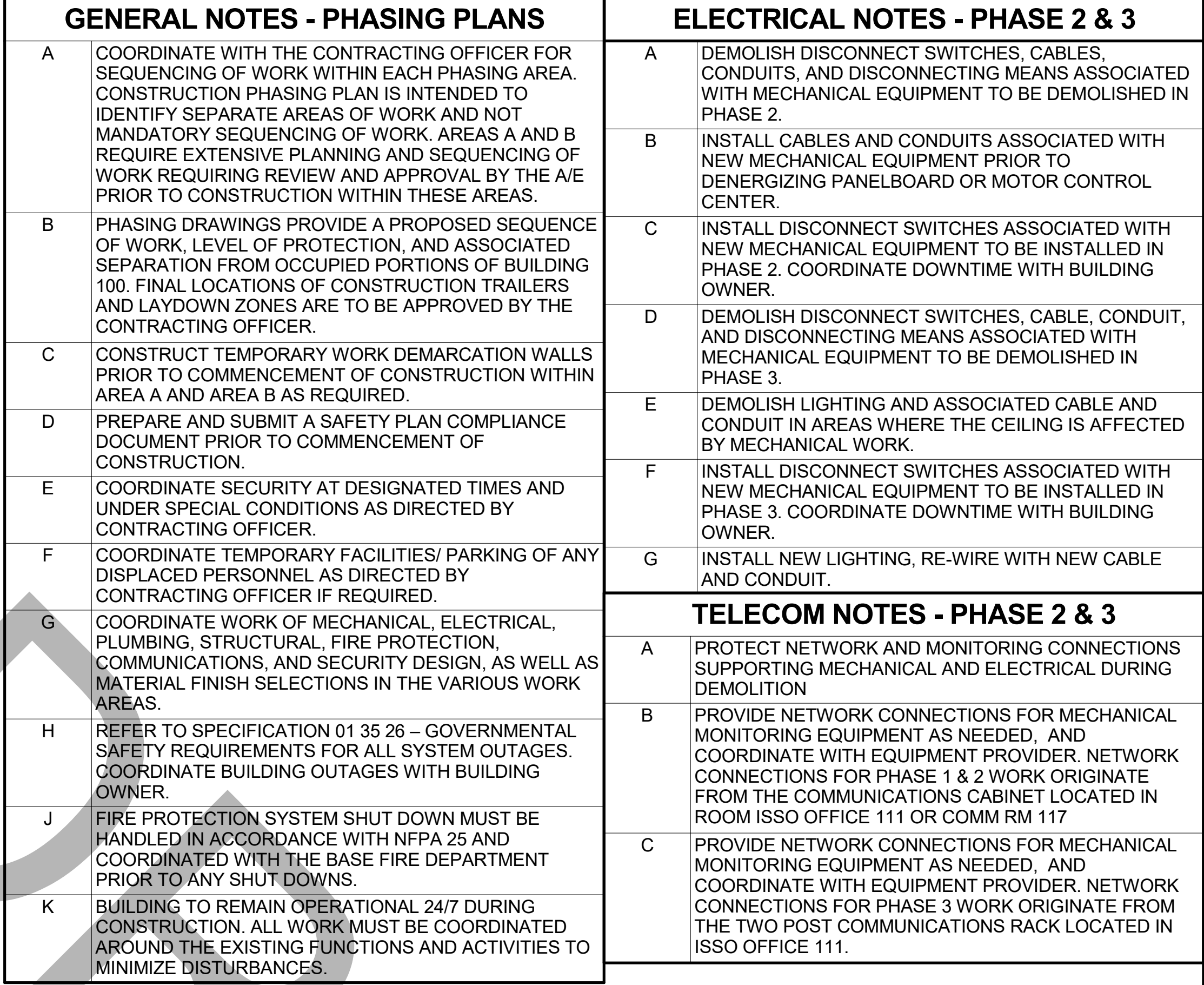
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PLOTTED BY:

PLOT SCALE:

BACKCHECK SUBMITTAL DESIGN





MECHANICAL NOTES - PHASE 2 & 3	
A	REMOVE EXISTING CHILLER 1, CHILLED PUMP 1, CONDENSING UNIT 1, AND ALL ASSOCIATED PIPING AND APPURTENANCES. CLEAN AND PREPARE AREAS FOR NEW CONSTRUCTION. INSTALL CHILLER 1, CHILLED PUMP 1, AND CONDENSING UNIT 1.
B	CONTRACTOR TO TEST AND BALANCE THE EXISTING AIRFLOWS AS A BASELINE FOR THE NEW EQUIPMENT. SUBMIT A CERTIFIED TAB REPORT IN ACCORDANCE WITH SPEC 23 05 93.
C	REMOVE EXISTING AHU AND ALL ASSOCIATED PIPING AND APPURTENANCES. CLEAN AND PREPARE AREAS FOR NEW CONSTRUCTION AND INSTALL AHU.
D	REMOVE EXISTING DOAS UNIT AND ALL ASSOCIATED PIPING AND APPURTENANCES. CLEAN AND PREPARE AREAS FOR NEW CONSTRUCTION AND INSTALL DOAS UNIT.
E	REMOVE EXISTING CONFERENCE ROOM 104 SPLIT SYSTEM AND ALL ASSOCIATED PIPING AND APPURTENANCES. CLEAN AND PREPARE AREAS FOR NEW CONSTRUCTION AND INSTALL CONFERENCE ROOM 104 SPLIT SYSTEM.
F	REMOVE EXISTING COMM ROOM MINI SPLIT AND ALL ASSOCIATED PIPING AND APPURTENANCES. CLEAN AND PREPARE AREAS FOR NEW CONSTRUCTION AND INSTALL COMM ROOM MINI SPLIT.
G	REMOVE EXISTING EXHAUST FANS AND ASSOCIATED APPURTENANCES. CLEAN AND PREPARE AREAS FOR NEW CONSTRUCTION AND INSTALL EXHAUST FANS.
H	REMOVE EXISTING REFRIGERANT PIPING. CLEAN AND PREPARE AREAS FOR NEW CONSTRUCTION AND INSTALL REFRIGERANT PIPING THROUGHOUT AREA A. REFER MECHANICAL SHEETS.

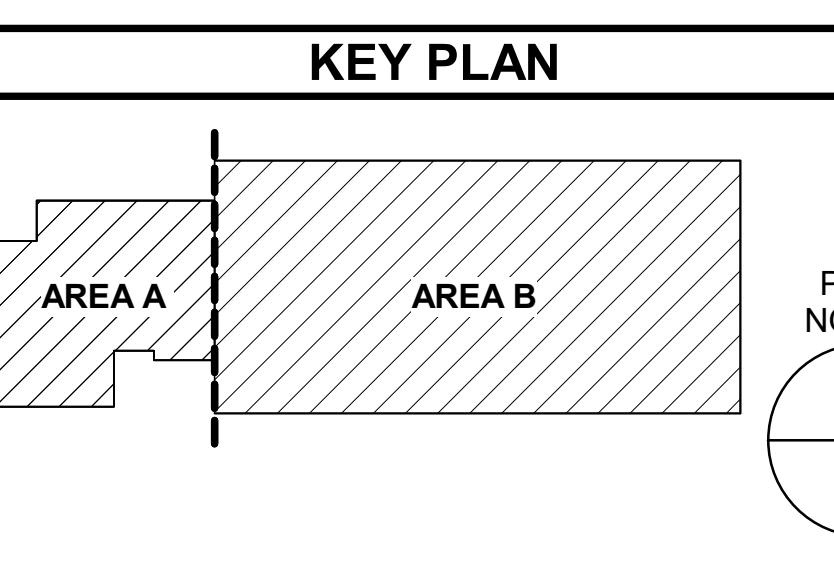
FIRE PROTECTION NOTES - PHASE 2 & 3	
A	DEMOLISH EXISTING WAVES MASS NOTIFICATION SYSTEM AND INSTALL NEW SPEAKERS FOR OCCUPANT MASS NOTIFICATION THROUGHOUT.
B	CONNECT NEW SPEAKERS TO EXISTING MONACO FACILITY TO CREATE COMBINATION FIRE ALARM/MASS NOTIFICATION SYSTEM.
C	PERFORM INTEGRATED TESTING TO CONFIRM FUNCTIONALITY WITH NEW HVAC EQUIPMENT AND FIRE ALARM EQUIPMENT.

ARCHITECTURAL NOTES - PHASE 2 & 3	
A	CONSTRUCTION OF AREAS A AND B FOR PHASE 2 & 3 MUST NOT COMMENCE UNTIL PERMANENT CONSTRUCTION MATERIALS AND EQUIPMENT ARE ON SITE AND VERIFIED BY CONTRACTING OFFICER.
B	AREA B PHASE 2 WILL INCLUDE: AREA AS INDICATED ON SHEET G1011/A1 AND ANY ASSOCIATED ITEMS REQUIRED TO COMPLETE PHASE 2 IN ITS ENTIRETY. COORDINATE FULL SCOPE AMONGST ALL DISCIPLINES.
C	AREA A PHASE 3 WILL INCLUDE: AREA AS INDICATED ON SHEET G1011/A1 AND ANY ASSOCIATED ITEMS REQUIRED TO COMPLETE PHASE 3 IN ITS ENTIRETY. COORDINATE FULL SCOPE AMONGST ALL DISCIPLINES.
D	PERFORM DEMOLITION WORK AS DESCRIBED IN THE DEMOLITION PLANS ON SHEETS AD101-AD131 WITHIN THE AREAS DEFINED UNDER PHASE 2 & 3, AREA A AND AREA B.
E	RENOVATIONS TO THE ARCHITECTURAL AND INTERIOR DESIGN ELEMENTS, ELECTRICAL, FIRE PROTECTION, AND PLUMBING SYSTEMS WILL BE COMPLETED TO THE EXTENT REQUIRED TO RETURN THE ROOMS TO A COMPLETE AND FINISHED CONDITION.
F	COORDINATE WITH THE CONTRACTING OFFICER FOR LOCATION OF TEMPORARY BARRIER PARTITION AS REQUIRED FOR RENOVATION WORK

OVERALL PHASING PLAN LEGEND

	PHASE 2
	PHASE 3

KEY PLAN



GRAPHIC SCALE(S)

0 4000 8000

SCALE: 1:200



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Jacobs JACOBS ENGINEERING GROUP INC. 777 MAIN STREET FORT WORTH, TX 76102 USA		part number:	
US ARMY CORPS OF ENGINEERS JAPAN ENGINEER DISTRICT OKINAWA, JAPAN		SIZE: A QUERRA	FILE NAME: G011
CHECKED BY: J. METZLER	SUBMITTED BY: A. GUERRA	DRAWING CODE	
DESIGNED BY: C. SHERMAN		CONTRACT NO.	
ISSUED DATE: -		SOLICITATION NO.:	-

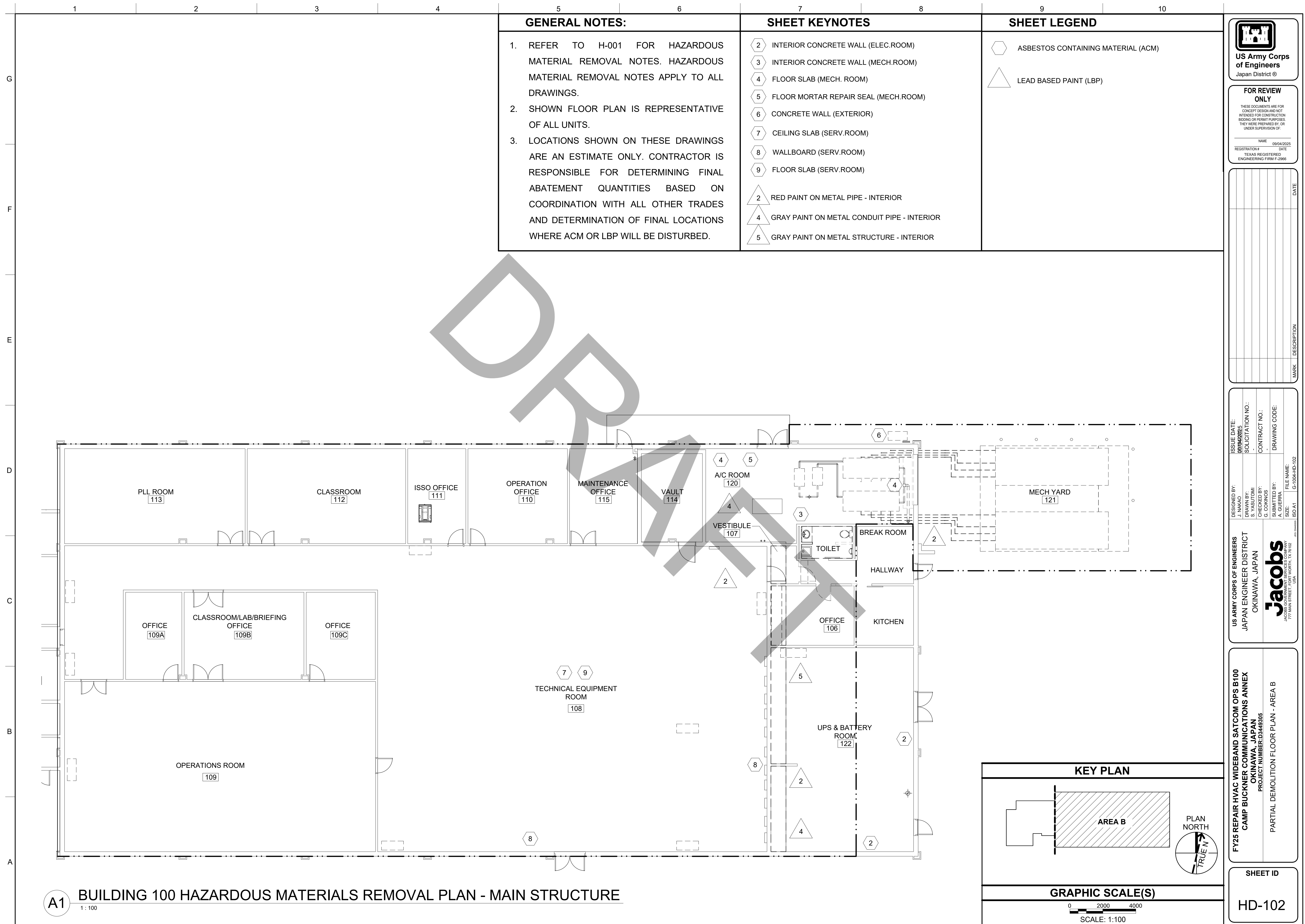
FY25 REPAIR HVAC WIDESPAN SATCOM OPS B100
 CAMP BUCKNER COMMUNICATIONS ANNEX
 OKINAWA, JAPAN
 PROJECT NUMBER:03449305

OVERALL PHASING PLAN - PHASE 2 & 3

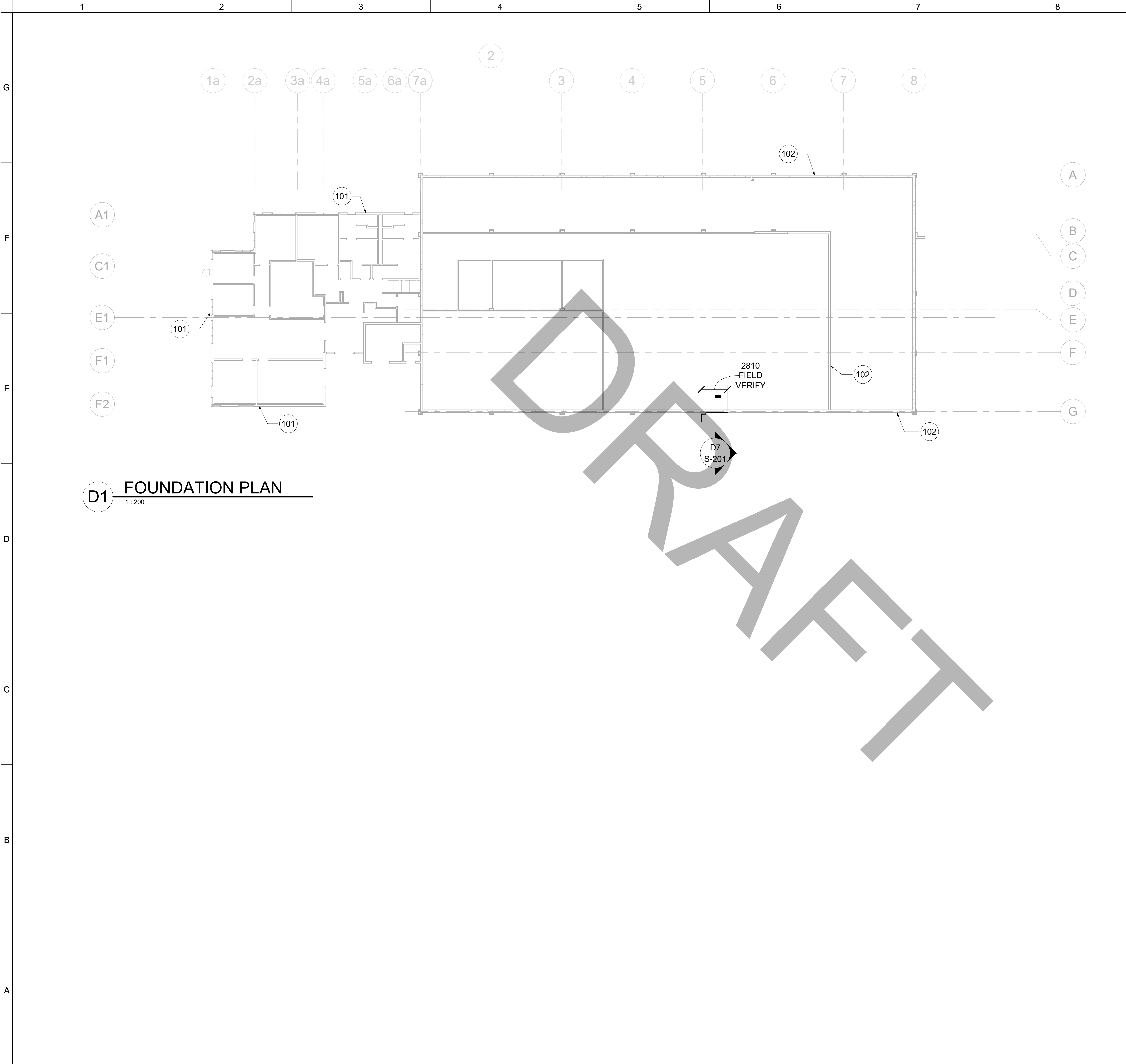
SHEET ID
GI011

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S S-100-



GENERAL SHEET NOTES

ESTIMATE OF CRACK AND SPALL QUANTITIES IN THE CONCRETE WALLS AND SLABS (FOR BIDDING PURPOSES ONLY):

TOTAL LENGTH OF WALL CRACKS=168 M
TOTAL AREA OF WALL SPALLS=17 M2
TOTAL LENGTH OF SLAB CRACKS=336 M
TOTAL AREA OF SLAB SPALLS=34 M2

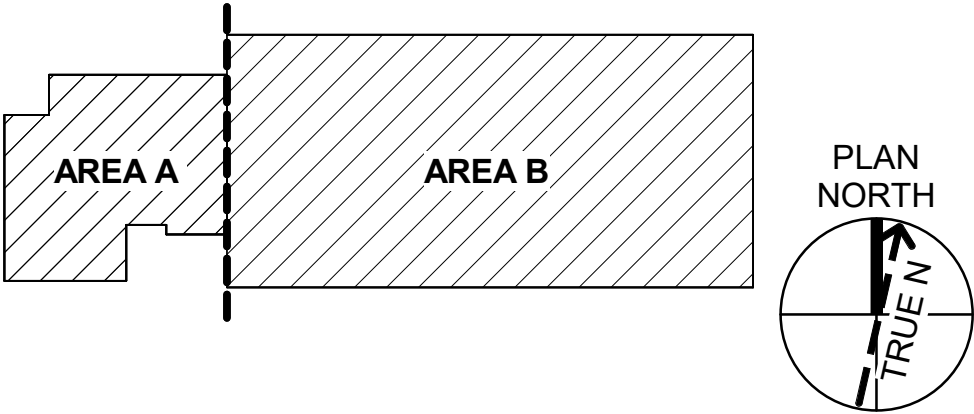
THE ABOVE ESTIMATES ARE NOT BASED ON ACTUAL OBSERVATIONS SINCE MOST OF THE WALLS AND FLOORS/ROOF ARE COVERED WITH ARCHITECTURAL FINISHES. THEY ARE BASED ON THE FOLLOWING ASSUMPTIONS.

WALL AND SLAB CRACK LENGTH (M)=0.1M/M2 OF FLOOR AREA
WALL AND SLAB SPALL AREA(M2)=0.01*FLOOR AREA
FLOOR AREA=1675 M2
SLAB CRACK AND SPALL ESTIMATES INCLUDE BOTH THE FIRST FLOOR SLAB AND THE ROOF SLAB

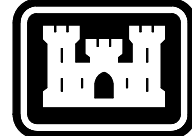
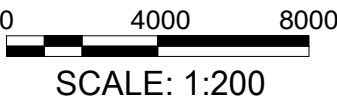
SHEET KEYNOTES

- | | |
|-----|---|
| 101 | REPAIR THE CONCRETE CRACKS AND SPALLS IN EXTERIOR CONCRETE WALLS AND INTERIOR SLABS CURRENTLY VISIBLE AND THAT ARE UNCOVERED DURING THE HVAC EQUIPMENT REPLACEMENT ACCORDING TO DETAILS D1/S-201 AND D4/S-201 |
| 102 | INFILL WALL OPENING DUE TO EXISTING LOUVER REMOVAL. REFER TO DETAIL A4/S-201 |

KEY PLAN



GRAPHIC SCALE(S)



US Army Corps
of Engineers
Japan District @

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HAKKI MURATLI
NAME: HAKKI MURATLI
REGISTRATION #: 01086
DATE: 09/04/2025
TEXAS REGISTERED ENGINEERING FIRM F-2966

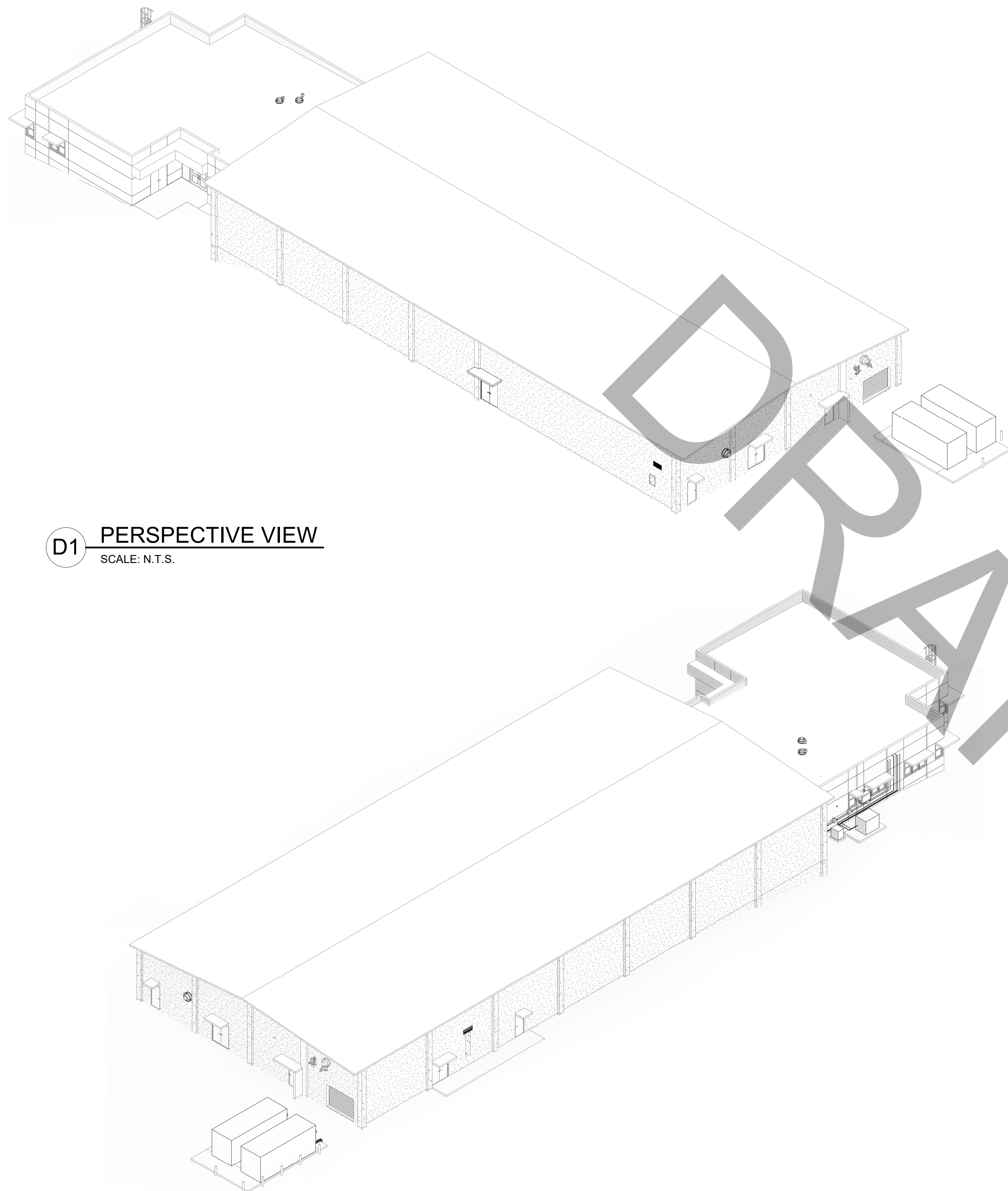
DESIGNED BY: H. MURATLI	ISSUE DATE: 09/04/2025	SOLICITATION NO.:
CHECKED BY: S. DRASKOVIC	CONTRACT NO.:	
SUBMITTED BY: G. SERNA	DRAWING CODE:	
FILE NAME: A. GUERRA	ISO A1	S-100

FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX
OKINAWA, JAPAN
PROJECT NUMBER: D3448305
FOUNDATION PLAN

SHEET ID

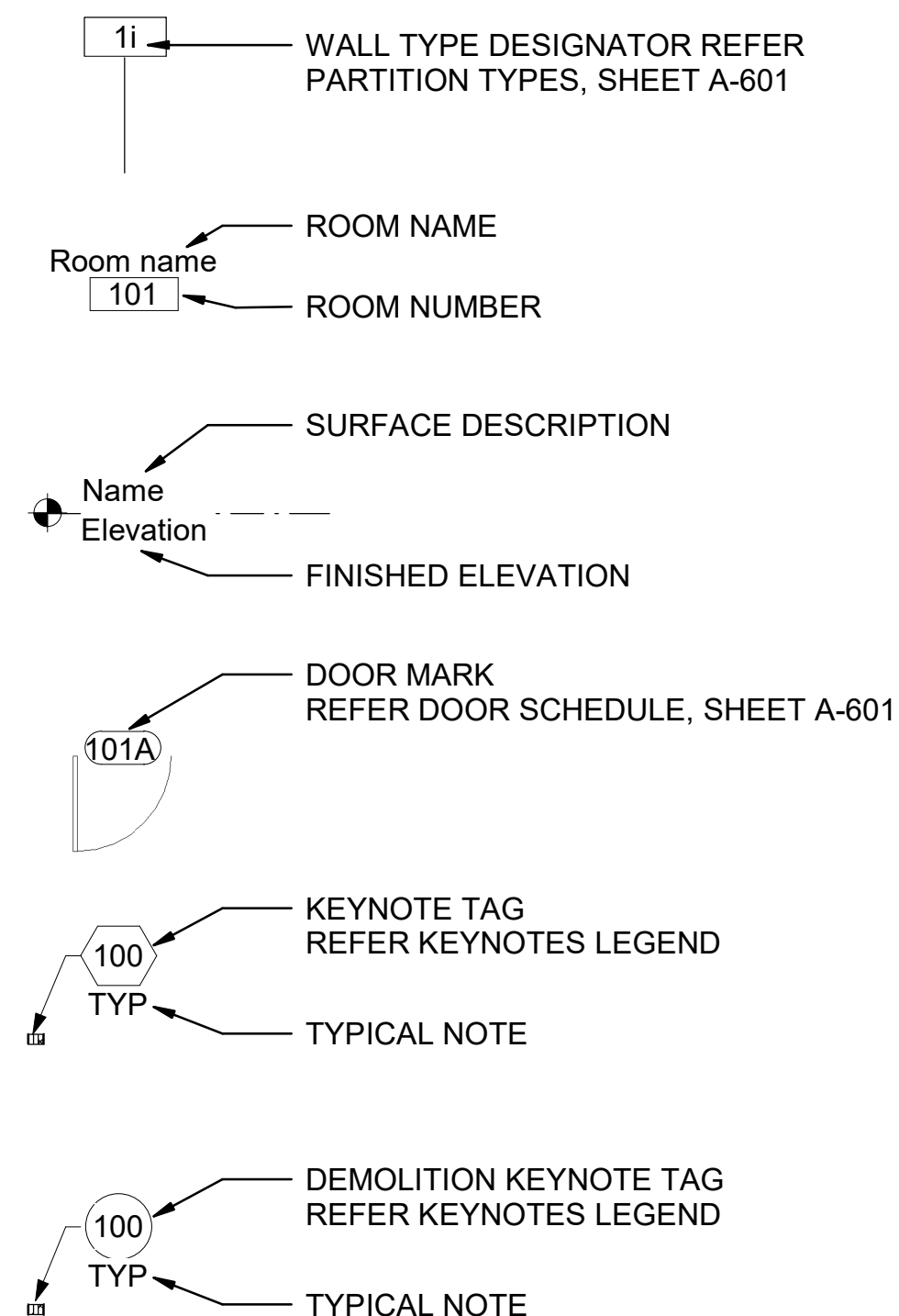
S-100

3D PERSPECTIVES

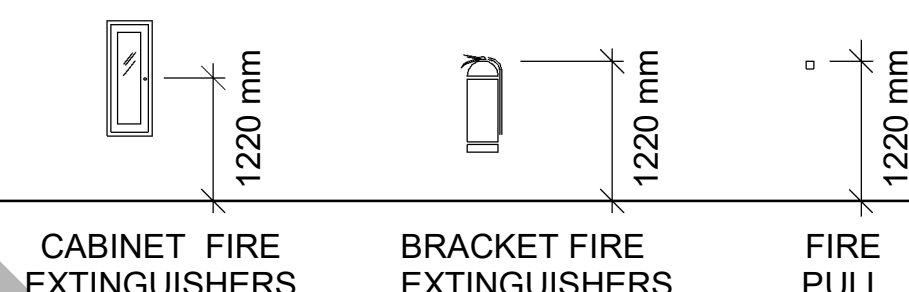


ARCHITECTURAL LEGEND

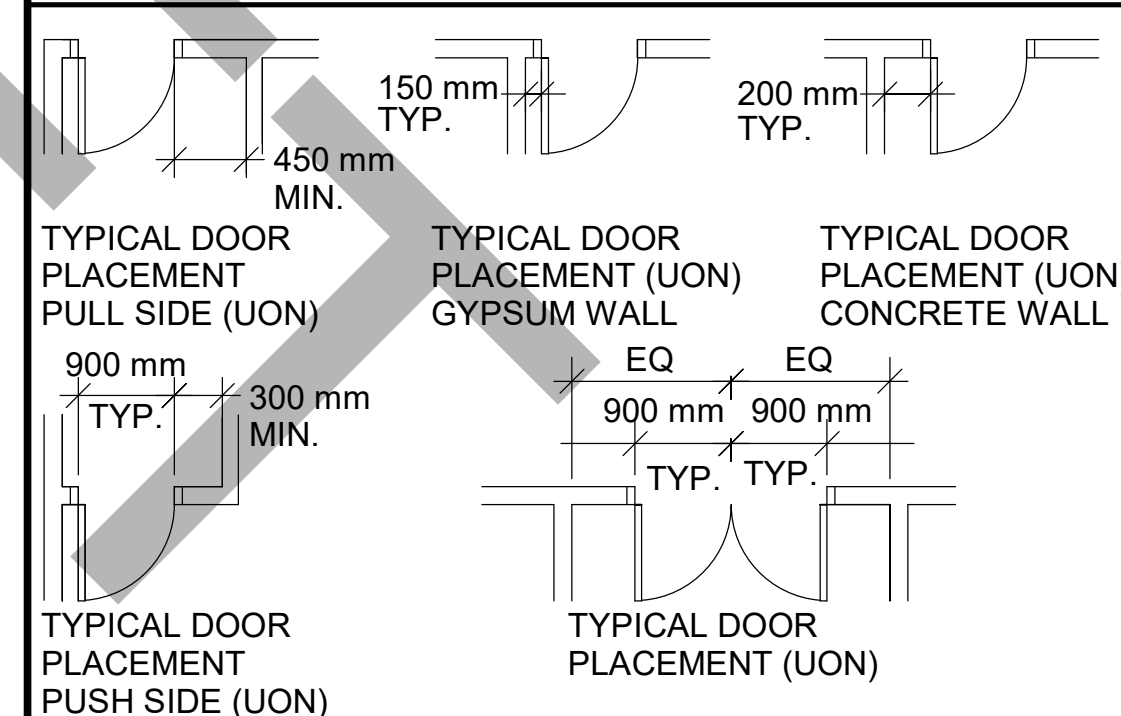
DRAWING SYMBOLS



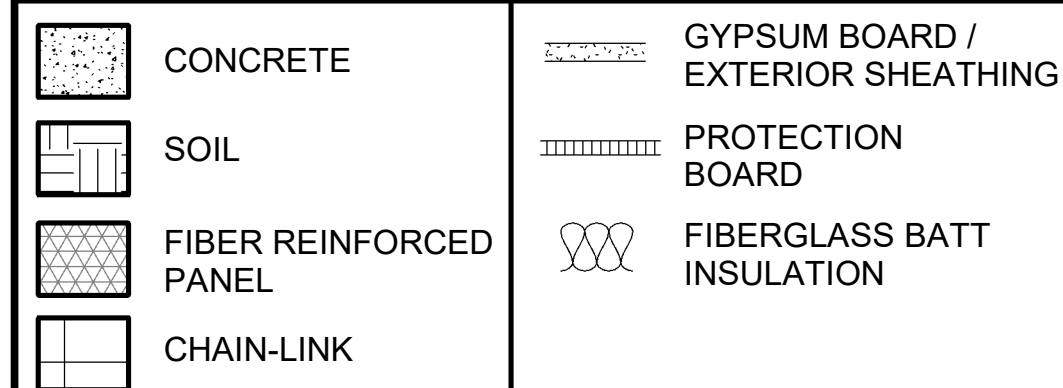
MOUNTING HEIGHTS



STANDARD DOOR LOCATIONS

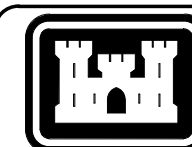


MATERIALS LEGEND



PROJECT GENERAL NOTES

- A INSTALLATION AND WORK FOR THIS PROJECT SHALL BE IN ACCORDANCE WITH THE CONTRACT DRAWINGS AND SPECIFICATIONS.
- B FIELD VERIFY EXISTING CONDITIONS AND VERIFY CONSTRUCTION LAYOUT PRIOR TO COMMENCEMENT OF CONSTRUCTION.
- C BUILDING RENOVATION DRAWINGS INDICATE THE GENERAL DESIGN INTENT FOR THE PURPOSE OF CONSTRUCTING SYSTEMS AND ASSEMBLIES. EXISTING SERVICES HAVE BEEN IDENTIFIED BASED UPON THE EXISTING AS-BUILT DRAWINGS. ANY UNDOCUMENTED SERVICES AND ASSEMBLIES ENCOUNTERED DURING CONSTRUCTION SHALL BE IDENTIFIED AND DOCUMENTED TO THE CONTRACTING OFFICER. VERIFY SIZE, LOCATION AND CHARACTERISTIC OF ALL EQUIPMENT TO BE FURNISHED WITH MANUFACTURERS OR SUPPLIERS BEFORE BEGINNING CONSTRUCTION.
- D DAMAGE CAUSED BY REMOVAL OPERATIONS OR CONSTRUCTION ACTIVITY IN AREAS OF THE PROJECT SITE SHALL BE REPAIRED/RESTORED TO THE APPROXIMATE ORIGINAL CONDITIONS OR BETTER AND SATISFACTORY TO THE CONTRACTING OFFICER AT NO COST TO THE GOVERNMENT.
- E MECHANICAL, ELECTRICAL, FURNISHINGS AND OTHER MISCELLANEOUS ITEMS WHICH ARE REQUIRED TO BE REMOVED PRIOR TO THE INSTALLATION OF WORK SHALL BE TEMPORARILY REMOVED AND REINSTALLED AS DIRECTED BY THE CONTRACTING OFFICER.
- F COORDINATE WITH THE CONTRACTING OFFICER THE SALVAGING OF SPECIFIC ITEMS AND PROTECTION OF EXISTING EQUIPMENT TO REMAIN IN PLACE.
- G COORDINATE WITH THE CONTRACTING OFFICER FOR DEMOLITION OF EXISTING SYSTEMS LOCATED BEYOND THE PERIMETER OF THE BUILDING AND PROTECTION/SEPARATION OF ADJACENT AREAS TO REMAIN IN USE.
- H COORDINATE WITH CONTRACTING OFFICER TO IDENTIFY AND MAINTAIN SECURE AREAS WHILE THE BUILDING IS BEING RENOVATED.
- J REMOVE AND REINSTALL ANY SIGN, FIXTURES, RECEPTACLES, ETC. AS NEEDED TO ACCOMMODATE CONSTRUCTION.
- K SECURE AND WEATHER PROTECT ROUGH OPENINGS DURING NON-WORKING HOURS.
- L FOR ADDITIONAL PHASING AND DEMOLITION NOTES REFER TO SHEETS G1010 THRU G1011 AND AD101 THRU AD131
- M INDICATED SIZE AND DIMENSIONS ARE IN MILLIMETERS UNLESS NOTED OTHERWISE.
- N DO NOT SCALE THE DRAWINGS FOR CONSTRUCTION PURPOSES.
- P FINISH FLOOR ELEVATION OF 0mm AT B100 IS A RELATIVE ELEVATION USED FOR DESIGN COORDINATION.
- Q FLOOR PLAN AND REFLECTED CEILING PLAN DIMENSIONS ARE TO FACE OF EXISTING FINISH, GYPSUM BOARD. OR FACE OF CONCRETE.



**US Army Corps
of Engineers**
Japan District ®

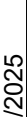
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THEY WERE PREPARED BY, OR
UNDER SUPERVISION OF:

Christopher M. Surles

NAME	
TX 23638	09/04/2025
REGISTRATION #	DATE
TEXAS REGISTERED ENGINEERING FIRM F-2966	

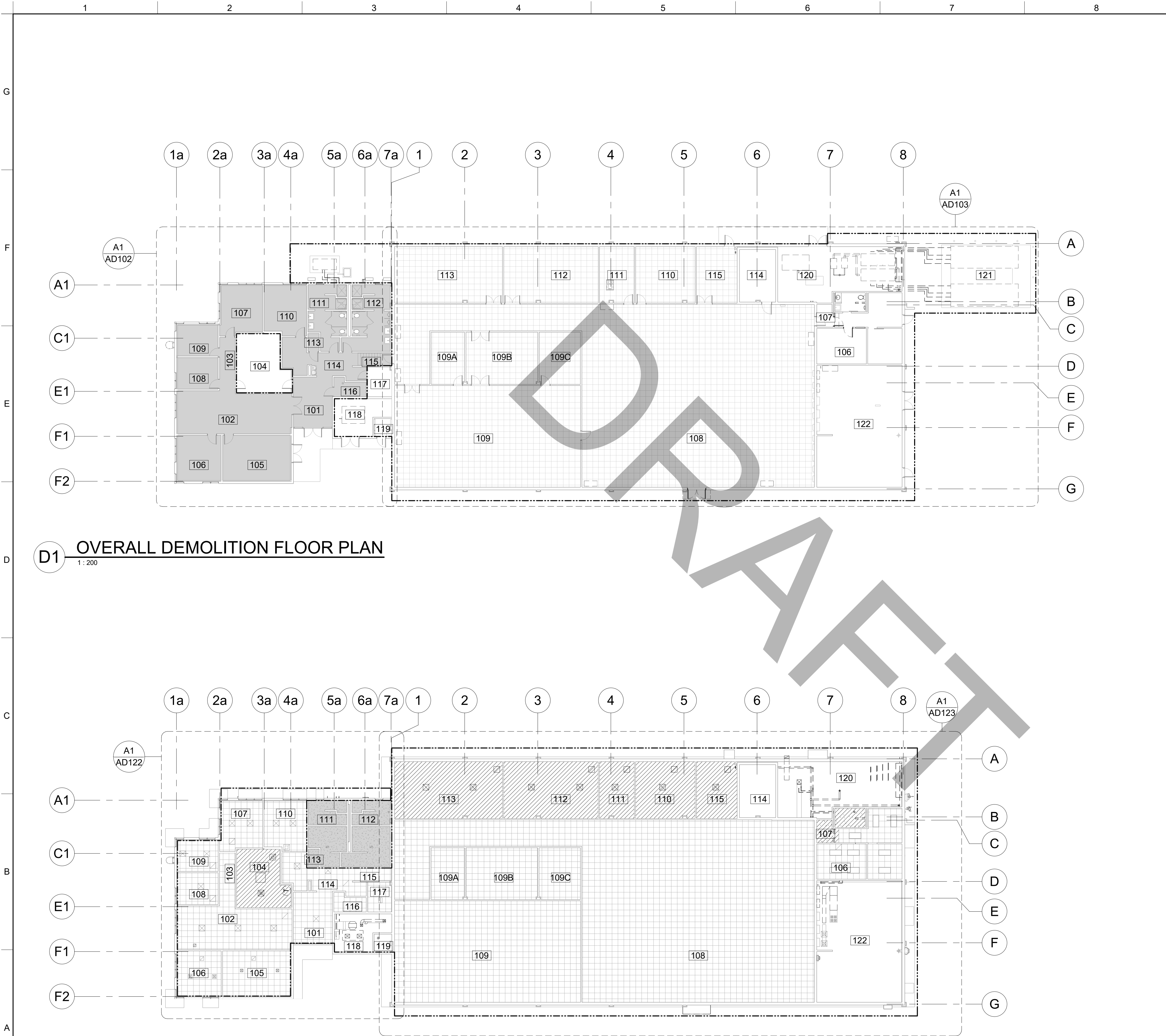
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 Jacobs Engineering Group Inc. 777 MAIN STREET, FORT WORTH, TX 76102	DESIGNED BY: C. SUMLES	ISSUE DATE: 09/04/2025
	CHECKED BY: J. METZLER	SOLICITATION NO.: -
OKINAWA, JAPAN	CONTRACT NO.: -	DRAWING CODE: -
DATE: 09/04/2025	SUBMITTED BY: K. CARLSON	FILE NAME: -

FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX
OKINAWA, JAPAN
PROJECT NUMBER:D3449305

SHEET ID

A-001



D1 OVERALL DEMOLITION FLOOR PLAN
1 : 200

A1 OVERALL DEMOLITION REFLECTED CEILING PLAN
1 : 200

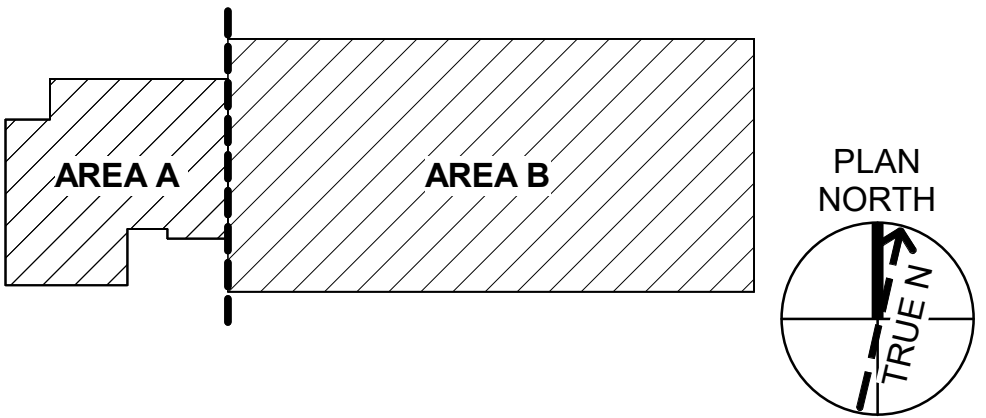
GENERAL NOTES - DEMOLITION PLANS

- A EXISTING CONSTRUCTION SHOWN BASED UPON OWNER FURNISHED PLANS AND ONSITE OBSERVATION. FIELD VERIFY EXISTING CONDITIONS AND VERIFY CONSTRUCTION LAYOUT PRIOR TO COMMENCEMENT OF CONSTRUCTION. REPORT ANY DISCREPANCIES BETWEEN DRAWINGS AND ACTUAL FIELD CONDITIONS TO THE CONTRACTING OFFICER PRIOR TO PROCEEDING WITH WORK
- B COORDINATE THE PROJECT SCHEDULE OF WORK AND THE PROJECT SAFETY DEMARCATIONS WITH THE CONTRACTING OFFICER
- C COORDINATE WITH THE CONTRACTING OFFICER FOR DEMOLITION AND PHASING SCHEDULE PRIOR TO COMMENCEMENT OF DEMOLITION
- D PROTECT AND MAINTAIN EXISTING ROOF, WALLS, FLOORING, PLUMBING, ELECTRICAL, LIGHTING, AND FIRE PROTECTION THAT IS REQUIRED TO REMAIN IN SERVICE
- E REFER AND COORDINATE WITH CIVIL, STRUCTURAL, MECHANICAL, PLUMBING, FIRE PROTECTION, AND ELECTRICAL DRAWINGS FOR ADDITIONAL DEMOLITION SCOPE AND PHASING
- F IN THE AREAS DESIGNATED FOR CONSTRUCTION, REMOVE HARDWARE, HANGERS, ANCHOR SYSTEMS, MISC. METAL ATTACHMENTS, ETC. AS REQUIRED. PATCH AND FINISH TO MATCH ADJACENT SURFACE
- G REFER STRUCTURAL, MECHANICAL, PLUMBING, FIRE PROTECTION, AND ELECTRICAL DRAWINGS FOR LOCATIONS OF ADDITIONAL PENETRATIONS IF REQUIRED
- H PATCH AND REPAIR EXISTING WALLS, FLOORS, AND ROOF DAMAGED FROM THE REMOVAL OF ASSOCIATED ACCESSORIES TO BE FREE OF DEFECTS AND PREPARE TO RECEIVE NEW CONSTRUCTION IN THE AREA OF WORK
- J COORDINATE DISPOSAL OF DEMOLISHED MATERIALS WITH CONTRACTING OFFICER
- K REMOVE EXISTING FLOOR FINISHES, WALL BASES, CEILING SYSTEMS, UTILITIES AND UTILITY SUPPORTS AND OTHER MISCELLANEOUS CONSTRUCTION IN AFFECTED AREAS AS REQUIRED TO PROVIDE A CLEAN SPACE TO INITIATE NEW CONSTRUCTION
- L FIELD VERIFY EXISTING CONDITIONS, DIMENSIONS AND VERIFY CONSTRUCTION LAYOUT PRIOR TO COMMENCEMENT OF CONSTRUCTION
- M STOP WORK IMMEDIATELY AND NOTIFY THE CONTRACTING OFFICER IF ASBESTOS CONTAINING MATERIALS ARE DISCOVERED NOT IDENTIFIED IN THE ASBESTOS SURVEY
- N COORDINATE PAINTED SURFACES ASSUMED TO CONTAIN LEAD IN ACCORDANCE WITH THE OSHA LEAD IN CONSTRUCTION STANDARD 39 CFR 1926.62
- P REMOVE AND SALVAGE OR REPLACE EXISTING DOOR THRESHOLDS AS REQUIRED FOR NEW FLOORING FINISHES
- Q COORDINATE WITH THE CONTRACTING OFFICER FOR LOCATION OF TEMPORARY BARRIER PARTITION AS REQUIRED FOR RENOVATION WORK

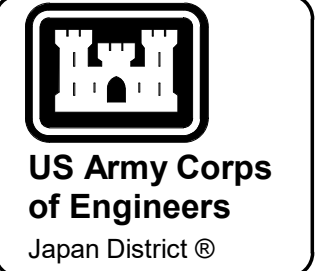
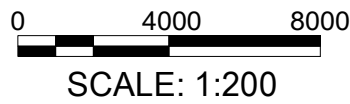
DEMOLITION OVERALL PLANS LEGEND

- APPROXIMATE AREAS OF DEMOLITION
- EXISTING ITEMS TO BE REMOVED
- EXISTING TO REMAIN
- EXISTING CEILING TO BE REMOVED
- REMOVE EXISTING DOOR AND/OR DOOR FRAME SYSTEM WHERE INDICATED.
- NOT IN SCOPE

KEY PLAN



GRAPHIC SCALE(S)



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Christopher M. Surles
TX 23638 NAME 09/04/2025
REGISTRATION # DATE
TEXAS REGISTERED ENGINEERING FIRM F-2966

ISSUE DATE: 09/04/2025
SOLICITATION NO.:
CONTRACT NO.:
DRAWING CODE:
FILE NAME: AD101
ISO A1

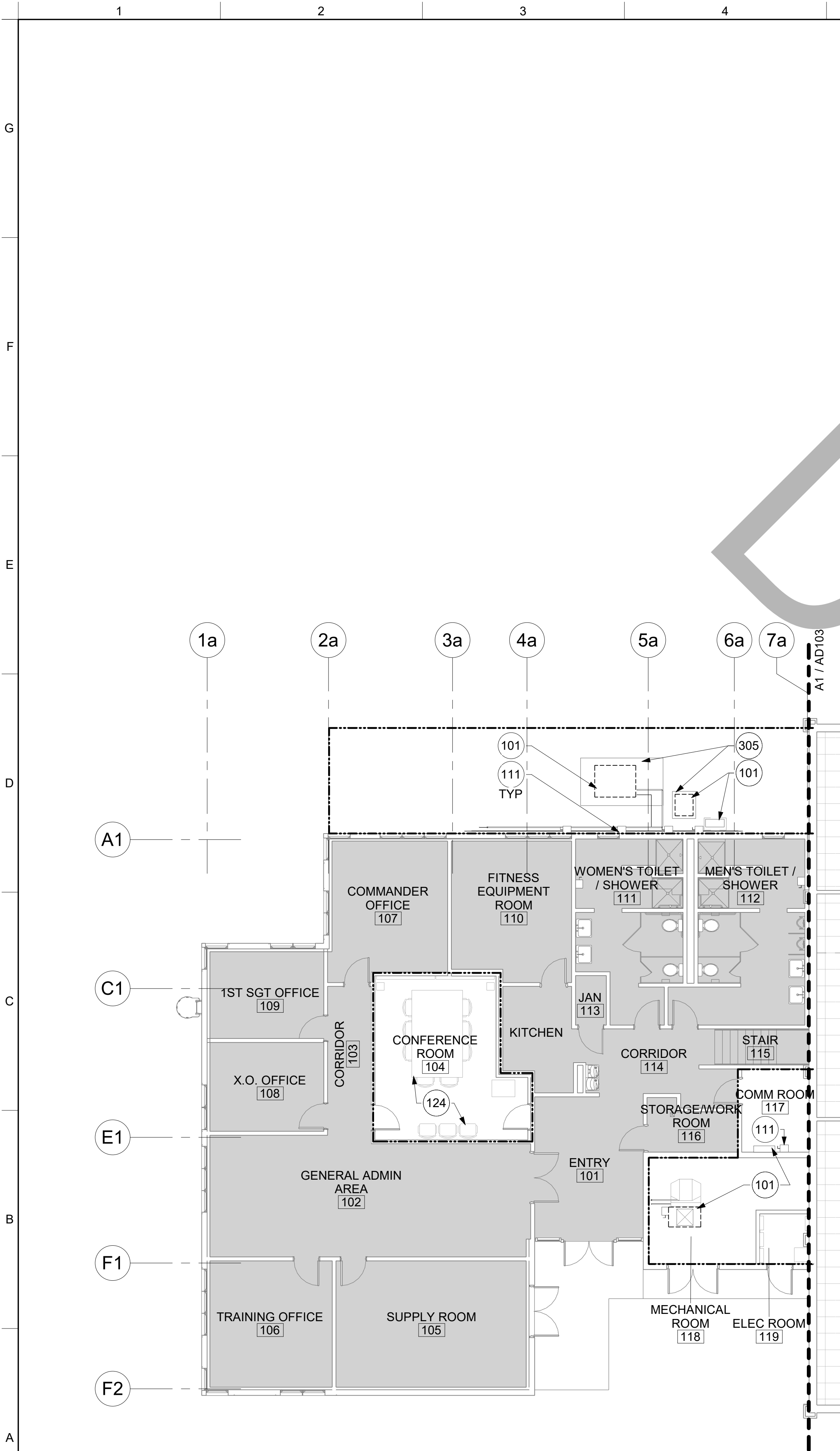
DESIGNED BY: C. SURLES
CHECKED BY: J. METZLER
SUBMITTED BY: K. CARLSON
SIZE: 11" X 17" (A1)

US ARMY CORPS OF ENGINEERS
JAPAN ENGINEER DISTRICT
OKINAWA, JAPAN
Jacobs
JACOBS ENGINEERING GROUP, INC.
77 WILSON STREET, SUITE 100
HOUSTON, TEXAS 77002
USA

FY25 REPAIR HVAC WIDEAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX
OKINAWA, JAPAN
PROJECT NUMBER: D3449305
OVERALL DEMOLITION PLANS

SHEET ID
AD101

AD AD102- AD



A1 PARTIAL DEMOLITION FLOOR PLAN - AREA A
1 : 100

DEMOLITION OVERALL PLANS LEGEND	
	APPROXIMATE AREAS OF DEMOLITION
	EXISTING ITEMS TO BE REMOVED
	EXISTING TO REMAIN
	EXISTING CEILING TO BE REMOVED
	REMOVE EXISTING DOOR AND/OR DOOR FRAME SYSTEM WHERE INDICATED.
	NOT IN SCOPE

GENERAL NOTES - DEMOLITION PLANS	
A	EXISTING CONSTRUCTION SHOWN BASED UPON OWNER FURNISHED PLANS AND ONSITE OBSERVATION. FIELD VERIFY EXISTING CONDITIONS AND VERIFY CONSTRUCTION LAYOUT PRIOR TO COMMENCEMENT OF CONSTRUCTION. REPORT ANY DISCREPANCIES BETWEEN DRAWINGS AND ACTUAL FIELD CONDITIONS TO THE CONTRACTING OFFICER PRIOR TO PROCEEDING WITH WORK
B	COORDINATE THE PROJECT SCHEDULE OF WORK AND THE PROJECT SAFETY DEMARCATIONS WITH THE CONTRACTING OFFICER
C	COORDINATE WITH THE CONTRACTING OFFICER FOR DEMOLITION AND PHASING SCHEDULE PRIOR TO COMMENCEMENT OF DEMOLITION
D	PROTECT AND MAINTAIN EXISTING ROOF, WALLS, FLOORING, PLUMBING, ELECTRICAL, LIGHTING, AND FIRE PROTECTION THAT IS REQUIRED TO REMAIN IN SERVICE
E	REFER AND COORDINATE WITH CIVIL, STRUCTURAL, MECHANICAL, PLUMBING, FIRE PROTECTION, AND ELECTRICAL DRAWINGS FOR ADDITIONAL DEMOLITION SCOPE AND PHASING
F	IN THE AREAS DESIGNATED FOR CONSTRUCTION, REMOVE HARDWARE, HANGERS, ANCHOR SYSTEMS, MISC. METAL ATTACHMENTS, ETC. AS REQUIRED. PATCH AND FINISH TO MATCH ADJACENT SURFACE
G	REFER STRUCTURAL, MECHANICAL, PLUMBING, FIRE PROTECTION, AND ELECTRICAL DRAWINGS FOR LOCATIONS OF ADDITIONAL PENETRATIONS IF REQUIRED
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L	FIELD VERIFY EXISTING CONDITIONS, DIMENSIONS AND VERIFY CONSTRUCTION LAYOUT PRIOR TO COMMENCEMENT OF CONSTRUCTION
M	STOP WORK IMMEDIATELY AND NOTIFY THE CONTRACTING OFFICER IF ASBESTOS CONTAINING MATERIALS ARE DISCOVERED NOT IDENTIFIED IN THE ASBESTOS SURVEY
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P	REMOVE AND SALVAGE OR REPLACE EXISTING DOOR THRESHOLDS AS REQUIRED FOR NEW FLOORING FINISHES
Q	COORDINATE WITH THE CONTRACTING OFFICER FOR LOCATION OF TEMPORARY BARRIER PARTITION AS REQUIRED FOR RENOVATION WORK

SHEET KEYNOTES	
101	EXISTING MECHANICAL EQUIPMENT TO BE REMOVED. CLEAN, PATCH AND PREPARE EXISTING AFFECTED SURFACES FOR INSTALL. REFER MECHANICAL.
111	EXISTING ELECTRICAL EQUIPMENT. REFER ELECTRICAL FOR REMOVAL.
124	REMOVE AND SALVAGE EXISTING FURNITURE IN THEIR ENTIRETY FOR CEILING DEMOLITION AND CONSTRUCTION; REINSTALL AFTER CEILING AND OTHER DISCIPLINE WORK IS COMPLETE.
305	EXISTING CONCRETE SLAB AT MECHANICAL UNITS TO REMAIN UNDISTURBED.

--	--

KEY PLAN	
GRAPHIC SCALE(S)	
SCALE: 1:100	

US Army Corps of Engineers
Japan District ®

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ISSUE DATE: 09/04/2025	SOLICITATION NO.:
DESIGNED BY: C. SURLES	CHECKED BY: J. METZLER
CONTRACT NO.:	DRAWING CODE:
FILE NAME: A. GUERRA	ISO A1

US ARMY CORPS OF ENGINEERS
JAPAN ENGINEER DISTRICT
OKINAWA, JAPAN

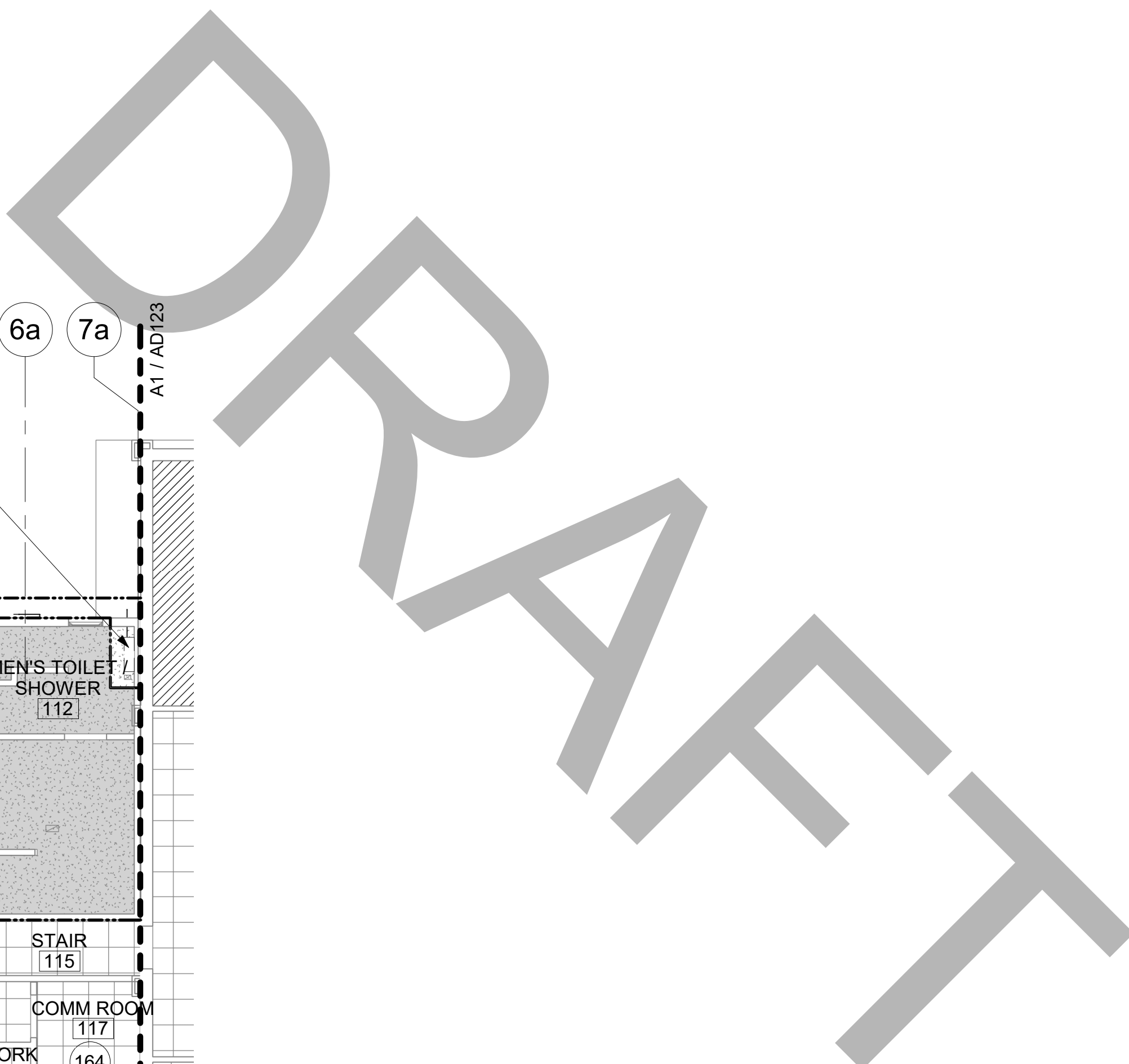
Jacobs
JACOBS ENGINEERING GROUP, INC.
77 MAVERICK STREET, SUITE 100
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USA

FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX
OKINAWA, JAPAN
PROJECT NUMBER: D3449305

PARTIAL DEMOLITION FLOOR PLAN - AREA A

SHEET ID

AD102



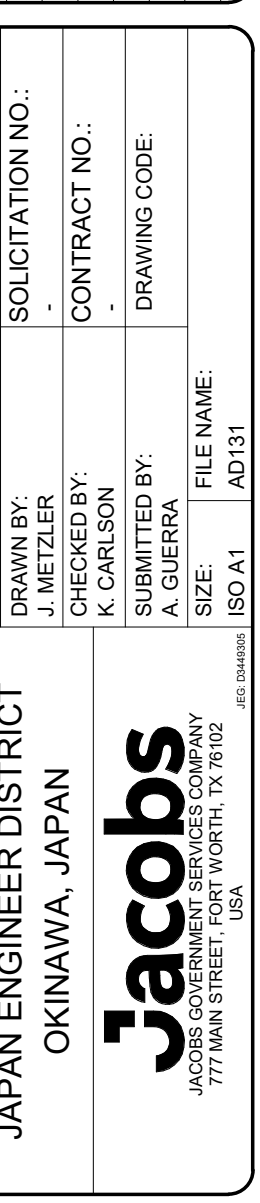
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L	FIELD VERIFY EXISTING CONDITIONS, DIMENSIONS AND VERIFY CONSTRUCTION LAYOUT PRIOR TO COMMENCEMENT OF CONSTRUCTION
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P	REMOVE AND SALVAGE OR REPLACE EXISTING DOOR THRESHOLDS AS REQUIRED FOR NEW FLOORING FINISHES
Q	COORDINATE WITH THE CONTRACTING OFFICER FOR LOCATION OF TEMPORARY BARRIER PARTITION AS REQUIRED FOR RENOVATION WORK

SHEET KEYNOTES	
139	REMOVE EXISTING ROOF EXHAUST TIE-DOWNS AT ROOF MOUNTED EQUIPMENT. ROOF ATTACHMENT TO REMAIN UNDISTURBED. COORDINATE WITH MECHANICAL FOR ROOF EXHAUST FAN REMOVAL AND REINSTALL.
142	REMOVE EXISTING ROOF EXHAUST FANS, REFER MECHANICAL. PATCH AND REPAIR ROOF TO BE FREE OF DEFECTS AND PREPARE TO RECEIVE CONSTRUCTION.
259	REMOVE EXISTING CANOPY AND PATCH AND REPAIR CONCRETE WALL WHERE CANOPY EXISTED. PREPARE SURFACE TO RECEIVE CONSTRUCTION, REFER STRUCTURAL.
322	EXISTING ROOF LADDER AND CAGE TO REMAIN.

	APPROXIMATE AREAS OF DEMOLITION
	EXISTING ITEMS TO BE REMOVED
	EXISTING TO REMAIN
	NOT IN SCOPE

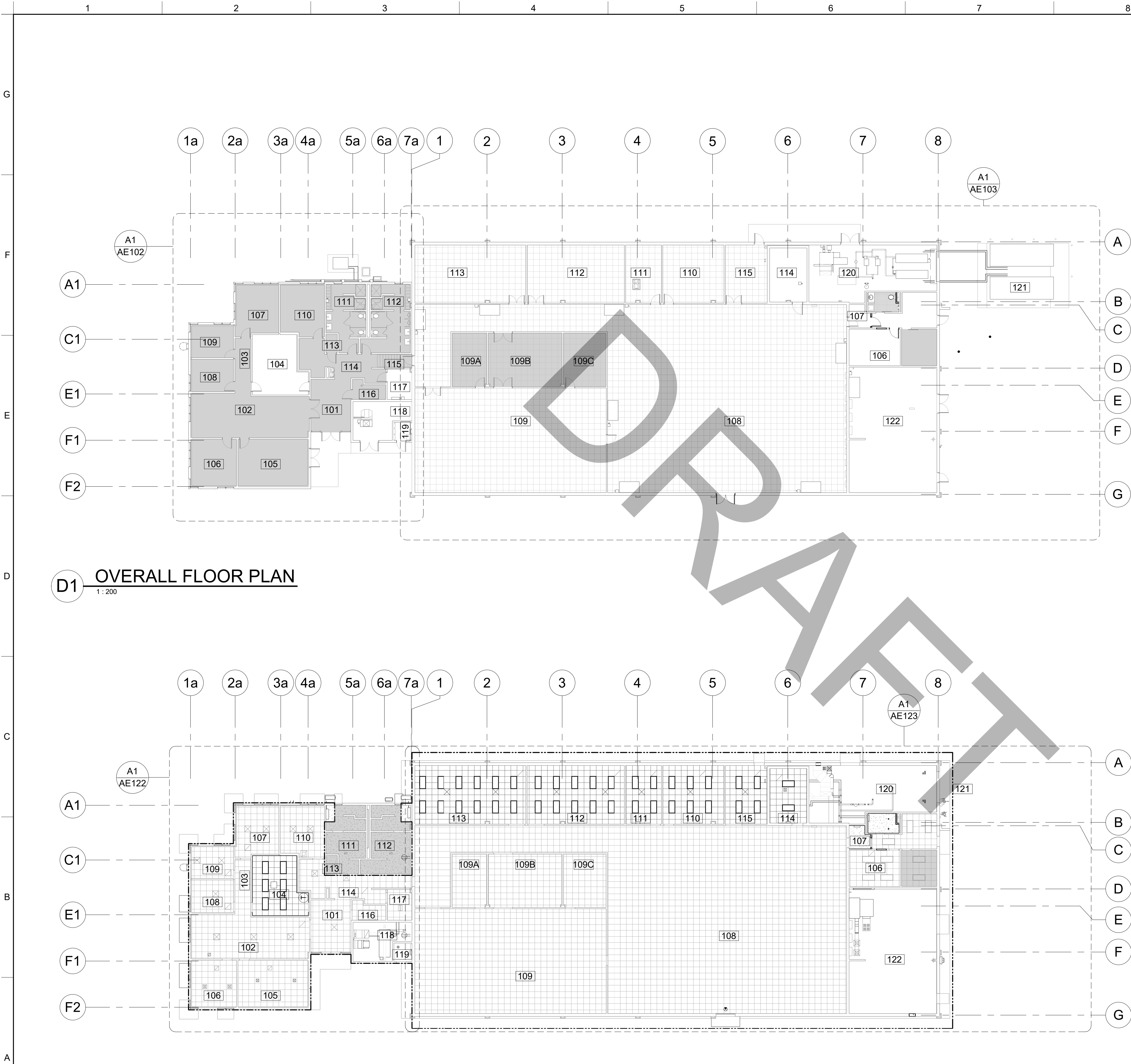
Diagram showing the plan view of a structure with two areas, AREA A and AREA B, separated by a vertical line. AREA A is on the left and AREA B is on the right. A compass rose indicates the orientation, with 'PLAN NORTH' and 'TRUE N' labels.

0 4000 8000
SCALE: 1:200



SHEET ID
AD131

AE AE101- XXX



D1 OVERALL FLOOR PLAN
1 : 200

A1 OVERALL REFLECTED CEILING PLAN
1 : 200

GENERAL NOTES - OVERALL PLAN

- A REFER TO SHEETS AE102 THRU AE123 FOR PARTIAL PLANS
- B REFER TO SHEET A-601 FOR PARTITION TYPES, DOOR AND FRAME SCHEDULE
- C COORDINATE PHASING SCHEDULE WITH CONTRACTING OFFICER
- D COORDINATE LAYOUTS OF ALL WALL MOUNTED EQUIPMENT AND PROVIDE BACKING AND WALL SUPPORT AS REQUIRED PER MANUFACTURERS RECOMMENDATIONS
- E REPLACE EXISTING WALL BASE WHERE IMPACTED BY CONSTRUCTION ACTIVITIES INCLUDED BUT NOT LIMITED TO DEMOLITION OF FLOOR FINISHES, REPAINTING, AND NEW CONSTRUCTION. MATCH EXISTING ADJACENT WALL BASE FOR MATERIAL, FINISH, AND COLOR
- F PROVIDE FLOORING TRANSITIONS AT INTERSECTIONS OF EXISTING FLOORING AT NEW FLOORING AND DOORS
- G DIMENSIONS AND ANNOTATIONS OF EXISTING AREAS PROVIDED FOR CLARITY, VERIFY ALL DIMENSIONS AS INDICATED PRIOR TO COMMENCEMENT OF WORK
- H REINSTALL RAISED ACCESS FLOORING TILES SALVAGED AND STORED DURING DEMOLITION. REFER MECHANICAL FOR PLACEMENT OF APPROXIMATELY THIRTY ADDITIONAL PERFORATED RAISED ACCESS FLOORING TILES

GENERAL NOTES - REFLECTED CEILING PLANS

- A REFER TO AE122 AND AE123 FOR PARTIAL REFLECTED CEILING PLANS.
- B PROVIDE MOISTURE RESISTANT GYPSUM BOARD IN TOILET ROOM.
- C REPLACE ACOUSTICAL CEILING TILES REMOVED DURING DEMOLITION AT ALL MECHANICAL WORK LOCATIONS AS REQUIRED. REFER MECHANICAL FOR MORE INFORMATION
- D GYPSUM BOARD AND ACOUSTICAL CEILINGS TO BE 2400mm ABOVE FINISH FLOOR UNLESS OTHERWISE NOTED.
- E ADHERE GYPSUM BOARD CEILING TILES TO VAULT ROOM 114 EXISTING CONCRETE LID SURFACE AS REQUIRED TO MATCH EXISTING DESIGN FINISH.

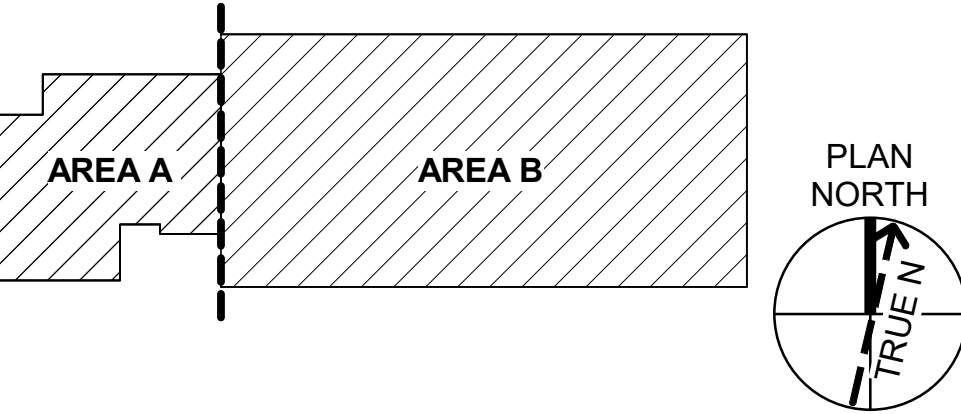
PLAN LEGEND

- APPROXIMATE AREAS OF CONSTRUCTION
- NOT IN SCOPE

REFLECTED CEILING PLAN LEGEND

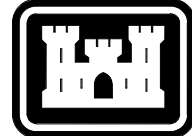
- GYPSUM BOARD CEILING AS SCHEDULED CONTROL JOINT
- 600x600mm ACOUSTICAL CEILING TILE AS SCHEDULED CENTERED IN SPACE AS INDICATED
- SPRINKLERS, SMOKE DETECTORS, AND SPEAKER STROBES; REFER TO FIRE PROTECTION DRAWINGS.
- LIGHT FIXTURE; REFER TO ELECTRICAL DRAWINGS.
- EXIT LIGHT FIXTURE; REFER TO ELECTRICAL DRAWINGS.
- RETURN AIR GRILLE; REFER TO MECHANICAL DRAWINGS.
- SUPPLY AIR DIFFUSERS; REFER TO MECHANICAL DRAWINGS.

KEY PLAN



GRAPHIC SCALE(S)

0 4000 8000
SCALE: 1:200



US Army Corps
of Engineers
Japan District @

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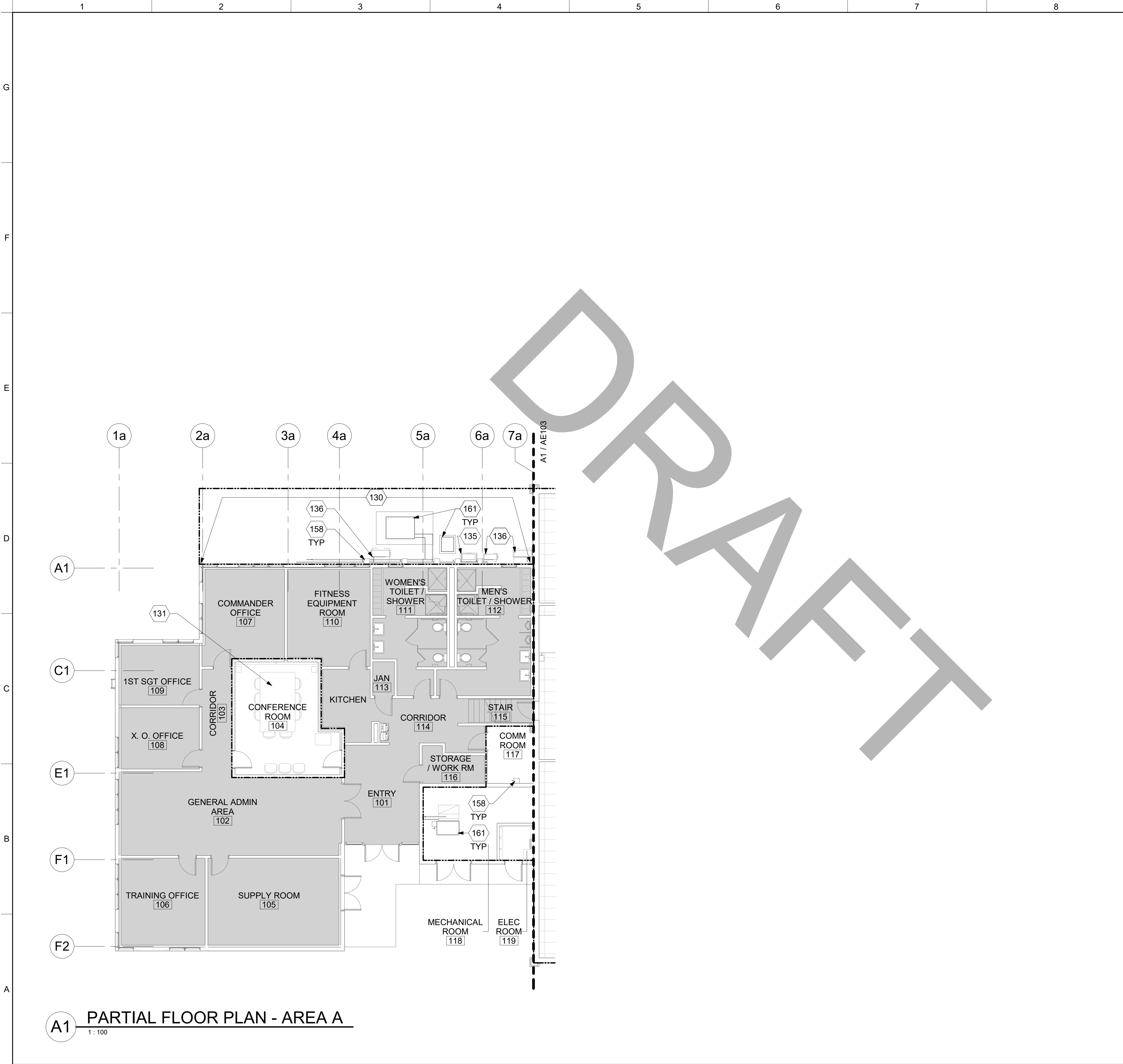
Christopher M. Surles
NAME 09/04/2025
DATE
REGISTRATION #
TEXAS REGISTERED
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ISSUE DATE: 09/04/2025	SOLICITATION NO.:	CONTRACT NO.:	DRAWING CODE:
DESIGNED BY: C. SURLES	CHECKED BY: J. METZLER	SUBMITTED BY: K. CARLSON	FILE NAME: A. GUERRA
US ARMY CORPS OF ENGINEERS JAPAN ENGINEER DISTRICT OKINAWA, JAPAN	JACOBS JACOBS ENGINEERING GROUP, INC. 171 MAIN STREET, SUITE 100 PORT WORTH, TEXAS 76150 USA	ISO A1	

FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX
OKINAWA, JAPAN
PROJECT NUMBER: D3449305
OVERALL PLANS

SHEET ID

AE101



GENERAL NOTES - PARTIAL PLANS

A

REFER TO A-601 FOR PARTITION TYPES, DOOR AND FRAME TYPES.

B

REFER ELECTRICAL DRAWINGS FOR REQUIRED FLOOR CLEARANCES OF PANELS, TRANSFORMERS, GEAR, SWITCHES IN ELECTRICAL ROOMS; ELECTRICAL EQUIPMENT DESCRIPTIONS ARE PROVIDED IN ELECTRICAL DRAWINGS.

C

REFER SHEET A-001 FOR TYPICAL DOOR LOCATION PLACEMENT.

D

DIMENSIONS PROVIDED FOR CLARITY, VERIFY ALL DIMENSIONS AS INDICATED PRIOR TO COMMENCEMENT OF WORK.

SHEET KEYNOTES

130

PATCH AND REPAIR EXISTING WALL AT LOCATIONS OF REMOVED EQUIPMENT TO PREPARE FOR PAINT. CLEAN AND PREPARE WALL AT EXTENTS SHOWN IN ITS ENTIRETY AS INDICATED TO RECEIVE PAINT FINISH, EX-1.

131

REINSTALL EXISTING SALVAGED FURNITURE AS FURNISHED.

135

PROVIDE HOUSEKEEPING PAD FOR MECHANICAL UNIT, REFER STRUCTURAL AND MECHANICAL.

136

EXISTING WALL MOUNTED BRACKETS TO BE RE-USED FOR MECHANICAL UNITS, REFER MECHANICAL AND MANUFACTURER FOR MOUNTING REQUIREMENTS.

158

ELECTRICAL EQUIPMENT, REFER ELECTRICAL.

161

MECHANICAL EQUIPMENT, REFER MECHANICAL.

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Christopher M. Surles
TX 23638 NAME 09/04/2025 DATE
REGISTRATION # TEXAS REGISTERED ENGINEERING FIRM F-2966

ISSUE DATE: 09/04/2025

SOLICITATION NO.:

CONTRACT NO.:

DRAWING CODE:

DESIGNED BY: C. SURLES

CHECKED BY: J. METZLER

SUBMITTED BY: K. CARLSON

FILE NAME: A. GUERRA

US ARMY CORPS OF ENGINEERS

JAPAN ENGINEER DISTRICT

OKINAWA, JAPAN

JACOBS

JACOBS ENGINEERING GROUP, INC.

77 MANA STREET, SUITE 100, FORT WORTH, TEXAS 76104

FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100

CAMP BUCKNER COMMUNICATIONS ANNEX

OKINAWA, JAPAN

PROJECT NUMBER: D3449305

PARTIAL FLOOR PLAN - AREA A

SHEET ID

AE102

PLAN LEGEND

APPROXIMATE AREAS OF CONSTRUCTION

NOT IN SCOPE

KEY PLAN

AREA A

PLAN NORTH

TRUE N

GRAPHIC SCALE(S)

0 2000 4000

SCALE: 1:100

FILE PATH: C:_Revit_Local_Models\2022\D3449305-A-3D_Jake.Metzler.rvt

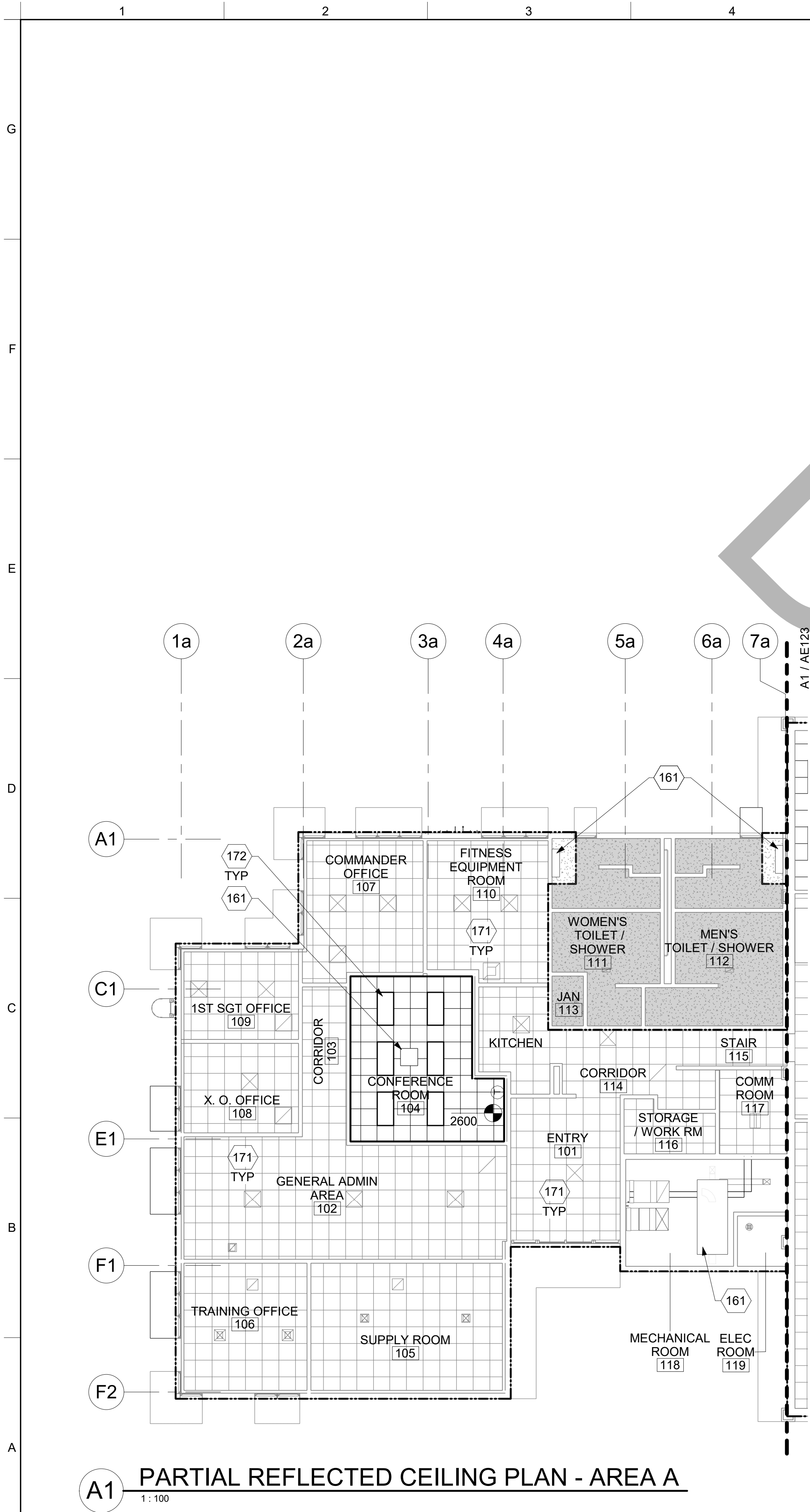
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PLOTTED BY:

PLOT SCALE: As indicated

BACKCHECK SUBMITTAL DESIGN

AE AE122-



GENERAL NOTES - REFLECTED CEILING PLANS

- A REFER TO AE122 AND AE123 FOR PARTIAL REFLECTED CEILING PLANS.
- B PROVIDE MOISTURE RESISTANT GYPSUM BOARD IN TOILET ROOM.
- C REPLACE ACOUSTICAL CEILING TILES REMOVED DURING DEMOLITION AT ALL MECHANICAL WORK LOCATIONS AS REQUIRED. REFER MECHANICAL FOR MORE INFORMATION
- D GYPSUM BOARD AND ACOUSTICAL CEILINGS TO BE 2400mm ABOVE FINISH FLOOR UNLESS OTHERWISE NOTED.
- E ADHERE GYPSUM BOARD CEILING TILES TO VAULT ROOM 114 EXISTING CONCRETE LID SURFACE AS REQUIRED TO MATCH EXISTING DESIGN FINISH.

SHEET KEYNOTES

- 161 MECHANICAL EQUIPMENT, REFER MECHANICAL.
- 171 ACOUSTICAL CEILING SYSTEM TO BE RE-INSTALLED AS REQUIRED. REPLACE IN KIND DAMAGED ACOUSTICAL CEILING TILES TO MATCH ADJACENT CEILING. COORDINATE CEILING MOUNTED EQUIPMENT WITH MECHANICAL, ELECTRICAL AND FIRE PROTECTION.
- 172 LIGHT FIXTURE, REFER ELECTRICAL.

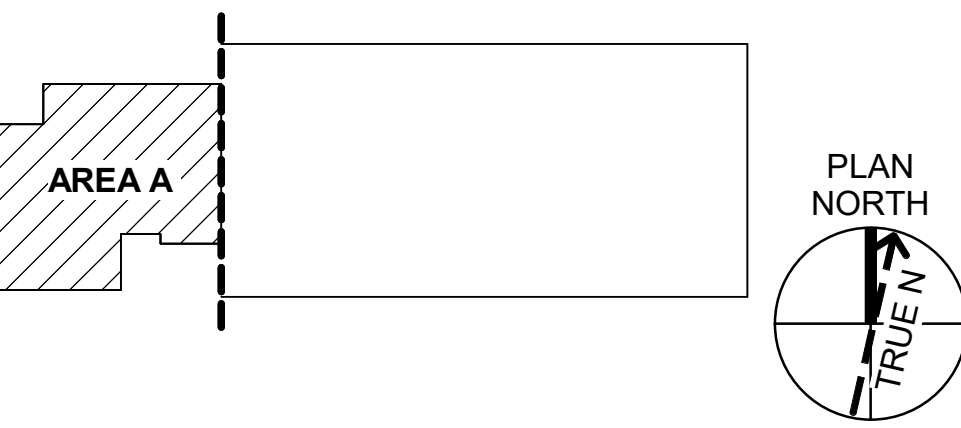
PLAN LEGEND

- APPROXIMATE AREAS OF CONSTRUCTION
- NOT IN SCOPE

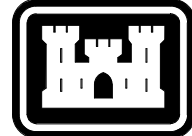
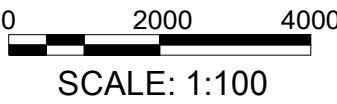
REFLECTED CEILING PLAN LEGEND

- GYPSUM BOARD CEILING AS SCHEDULED CEILING CONTROL JOINT
- 600x600mm ACOUSTICAL CEILING TILE AS SCHEDULED CENTERED IN SPACE AS INDICATED
- SPRINKLERS, SMOKE DETECTORS, AND SPEAKER STROBES; REFER TO FIRE PROTECTION DRAWINGS.
- LIGHT FIXTURE; REFER TO ELECTRICAL DRAWINGS.
- EXIT LIGHT FIXTURE; REFER TO ELECTRICAL DRAWINGS.
- RETURN AIR GRILLE; REFER TO MECHANICAL DRAWINGS.
- SUPPLY AIR DIFFUSERS; REFER TO MECHANICAL DRAWINGS.

KEY PLAN



GRAPHIC SCALE(S)



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Japan District ®

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NAME
TX 23636 09/04/2025
REGISTRATION # DATE
TEXAS REGISTERED
ENGINEERING FIRM F-2966

ISSUE DATE:
09/04/2025
SOLICITATION NO.:
CONTRACT NO.:
DRAWING CODE:
FILE NAME:
AE122

DESIGNED BY:
C. SURLES
CHECKED BY:
J. METZLER
SUBMITTED BY:
K. CARLSON
SIZE:
A4

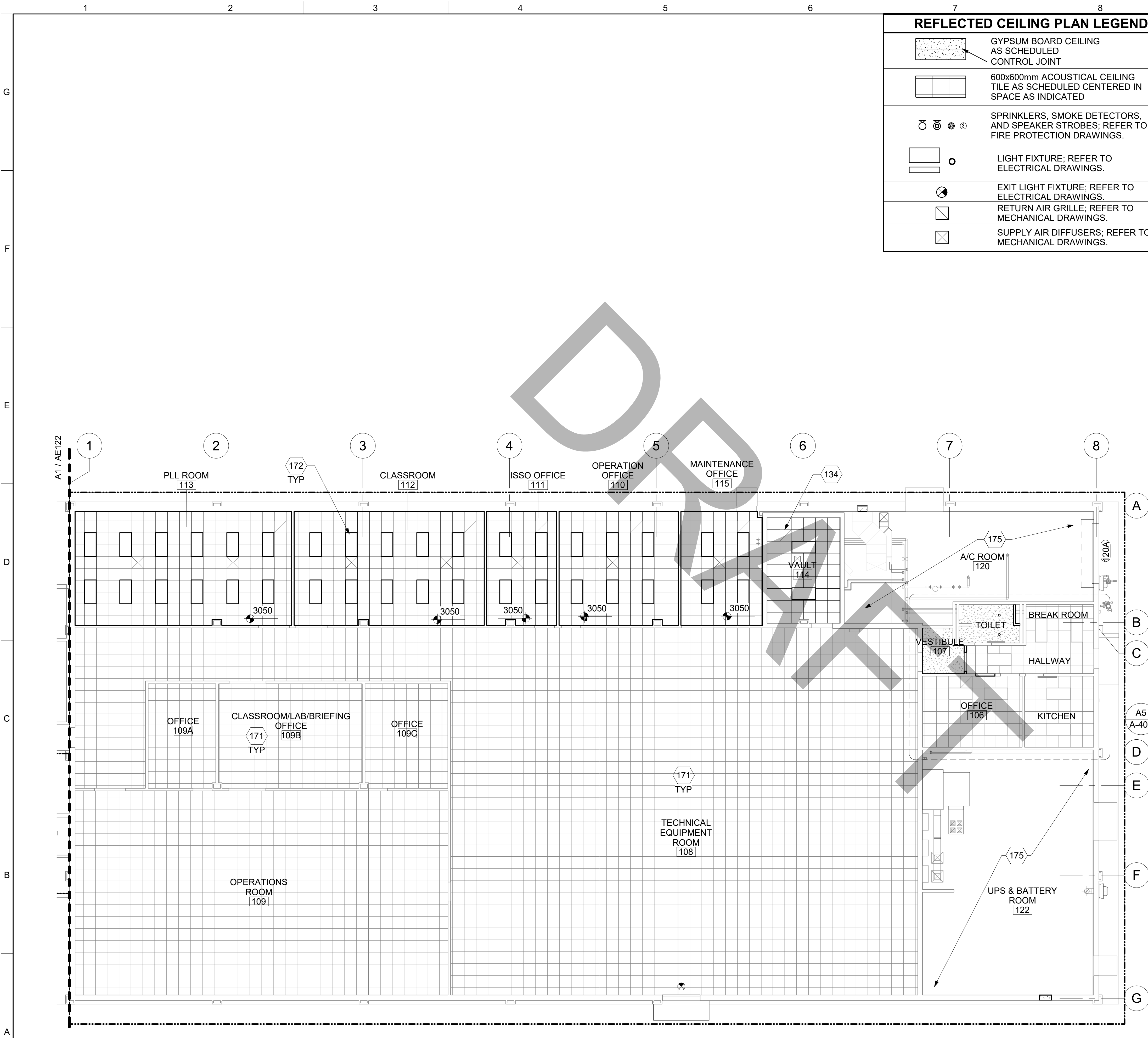
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JAPAN ENGINEER DISTRICT
OKINAWA, JAPAN

Jacobs
JACOBS ENGINEERING GROUP, INC.
771 MAIN STREET, SUITE 100
HOUSTON, TX 77002
USA

FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX
OKINAWA, JAPAN
PROJECT NUMBER: D3449305
PARTIAL REFLECTED CEILING PLAN - AREA A

SHEET ID

AE122



A1 PARTIAL REFLECTED CEILING PLAN - AREA B
1 : 100

REFLECTED CEILING PLAN LEGEND	
	GYPSUM BOARD CEILING AS SCHEDULED CONTROL JOINT
	600x600mm ACOUSTICAL CEILING TILE AS SCHEDULED CENTERED IN SPACE AS INDICATED
	SPRINKLERS, SMOKE DETECTORS, AND SPEAKER STROBES; REFER TO FIRE PROTECTION DRAWINGS.
	LIGHT FIXTURE; REFER TO ELECTRICAL DRAWINGS.
	EXIT LIGHT FIXTURE; REFER TO ELECTRICAL DRAWINGS.
	RETURN AIR GRILLE; REFER TO MECHANICAL DRAWINGS.
	SUPPLY AIR DIFFUSERS; REFER TO MECHANICAL DRAWINGS.

GENERAL NOTES - REFLECTED CEILING PLANS	
A	REFER TO AE122 AND AE123 FOR PARTIAL REFLECTED CEILING PLANS.
B	PROVIDE MOISTURE RESISTANT GYPSUM BOARD IN TOILET ROOM.
C	REPLACE ACOUSTICAL CEILING TILES REMOVED DURING DEMOLITION AT ALL MECHANICAL WORK LOCATIONS AS REQUIRED. REFER MECHANICAL FOR MORE INFORMATION
D	GYPSUM BOARD AND ACOUSTICAL CEILINGS TO BE 2400mm ABOVE FINISH FLOOR UNLESS OTHERWISE NOTED.
E	ADHERE GYPSUM BOARD CEILING TILES TO VAULT ROOM 114 EXISTING CONCRETE LID SURFACE AS REQUIRED TO MATCH EXISTING DESIGN FINISH.
SHEET KEYNOTES	
134	COORDINATE WITH MECHANICAL FOR SECURE PENETRATION AS REQUIRED.
171	ACOUSTICAL CEILING SYSTEM TO BE RE-INSTALLED AS REQUIRED. REPLACE IN KIND DAMAGED ACOUSTICAL CEILING TILES TO MATCH ADJACENT CEILING. COORDINATE CEILING MOUNTED EQUIPMENT WITH MECHANICAL, ELECTRICAL AND FIRE PROTECTION.
172	LIGHT FIXTURE, REFER ELECTRICAL.
175	OPEN TO STRUCTURE ABOVE.

PLAN LEGEND	
	APPROXIMATE AREAS OF CONSTRUCTION
	NOT IN SCOPE
KEY PLAN	
	PLAN NORTH
GRAPHIC SCALE(S)	
SCALE: 1:100	

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TX 23638 NAME 09/04/2025
REGISTRATION # DATE
TEXAS REGISTERED ENGINEERING FIRM F-2966

ISSUE DATE: 09/04/2025	SOLICITATION NO.:	CONTRACT NO.:	DRAWING CODE:
DESIGNED BY: C. SURLES	J. METZLER	K. CARLSON	A. GUERRA
CHECKED BY:			
SUBMITTED BY:			
SIZE:	FILE NAME: AE123	ISO A1	ISO A1

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JAPAN ENGINEER DISTRICT
OKINAWA, JAPAN

Jacobs
JACOBS ENGINEERING GROUP, INC.
171 MAIN STREET, SUITE 100
OKINAWA, JAPAN 904-1001

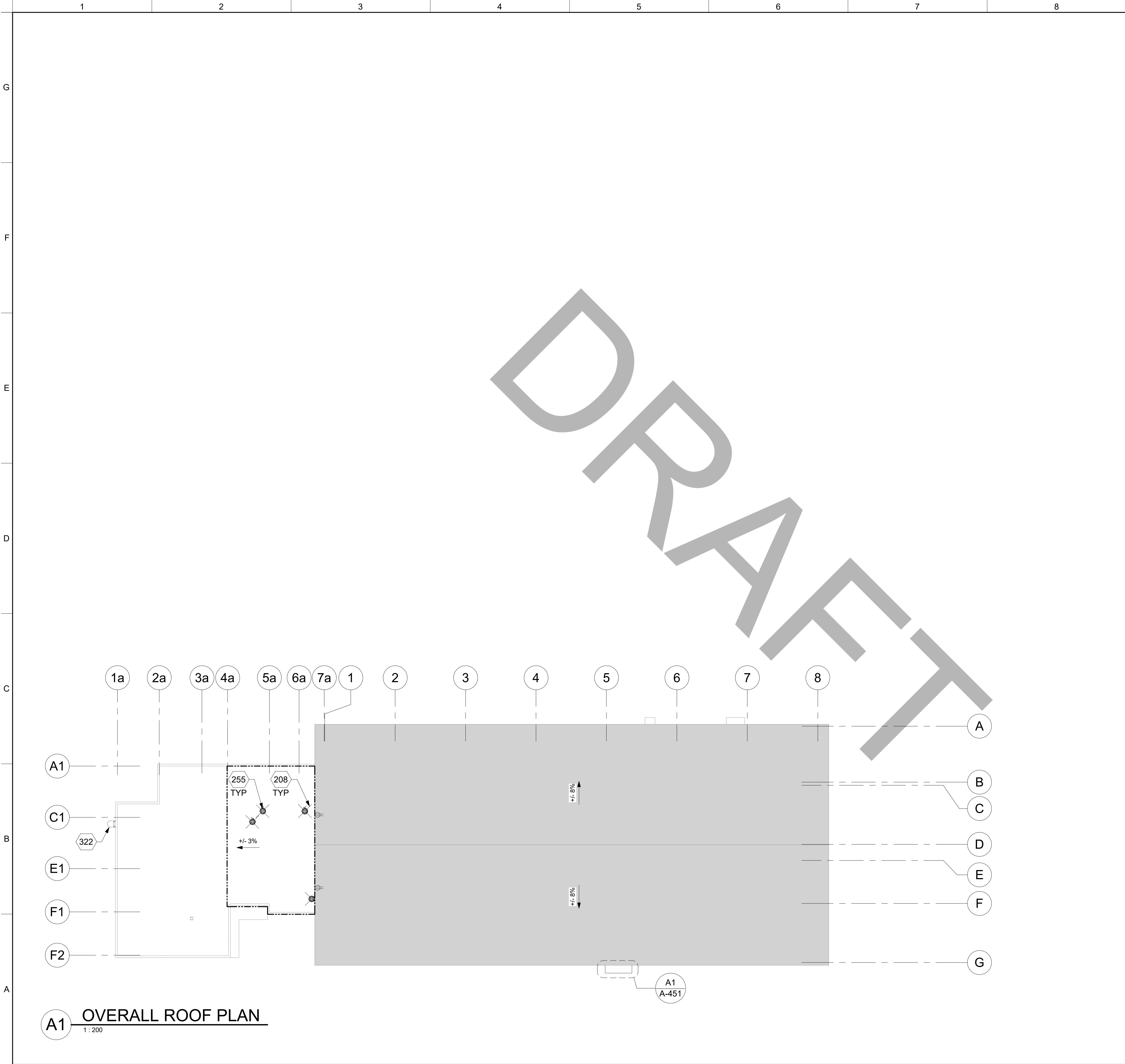
FY25 REPAIR HVAC WIDE-BAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX
OKINAWA, JAPAN
PROJECT NUMBER: D3449305

PARTIAL REFLECTED CEILING PLAN - AREA B

SHEET ID

AE123

AE AE131-



GENERAL NOTES - ROOF PLANS

A

COORDINATE THE PROPOSED ROOF RENOVATION WORK WITH CURRENT ROOF WARRANTY

B

EXISTING CONSTRUCTION SHOWN BASED UPON OWNER FURNISHED PLANS. FIELD VERIFY EXISTING CONDITIONS AND VERIFY CONSTRUCTION LAYOUT PRIOR TO COMMENCEMENT OF CONSTRUCTION. REPORT ANY DISCREPANCIES BETWEEN DRAWINGS AND ACTUAL FIELD CONDITIONS TO THE CONTRACTING OFFICER PRIOR TO PROCEEDING WITH WORK.

SHEET KEYNOTES

208

PROVIDE STAINLESS STEEL TYPHOON ROOF EQUIPMENT TIE DOWN ASSEMBLY TO INCLUDE 5MM, 7 X 7 STAINLESS STEEL WIRE ROPE, BRACKETS, TURNBUCKLES, WIRE ROPE CLIPS, FLANGE NUTS, FLAT WASHERS AND CABLE THIMBLES, AS REQUIRED FOR A COMPLETE ASSEMBLY BASED ON THE DESIGN WIND LOADS INDICATED.

255

ROOF EXHAUST FAN; REFER MECHANICAL.

322

EXISTING ROOF LADDER AND CAGE TO REMAIN.

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Christopher M. Surles
NAME
09/04/2025
DATE
TX 23638
REGISTRATION #
TEXAS REGISTERED ENGINEERING FIRM F-2966

ISSUE DATE:
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SOLICITATION NO.:
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DESIGNED BY:
C. SURLES
CHECKED BY:
J. METZLER
SUBMITTED BY:
K. CARLSON
FILE NAME:
A. GUERRA
SIZE:
ISO A1

US ARMY CORPS OF ENGINEERS
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OKINAWA, JAPAN
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USA

FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX
OKINAWA, JAPAN
PROJECT NUMBER: D3449305
OVERALL ROOF PLAN

SHEET ID

AE131

PLAN LEGEND

APPROXIMATE AREAS OF CONSTRUCTION

NOT IN SCOPE

KEY PLAN

AREA A

AREA B

PLAN NORTH

TRUE N

GRAPHIC SCALE(S)

0 4000 8000

SCALE: 1:200

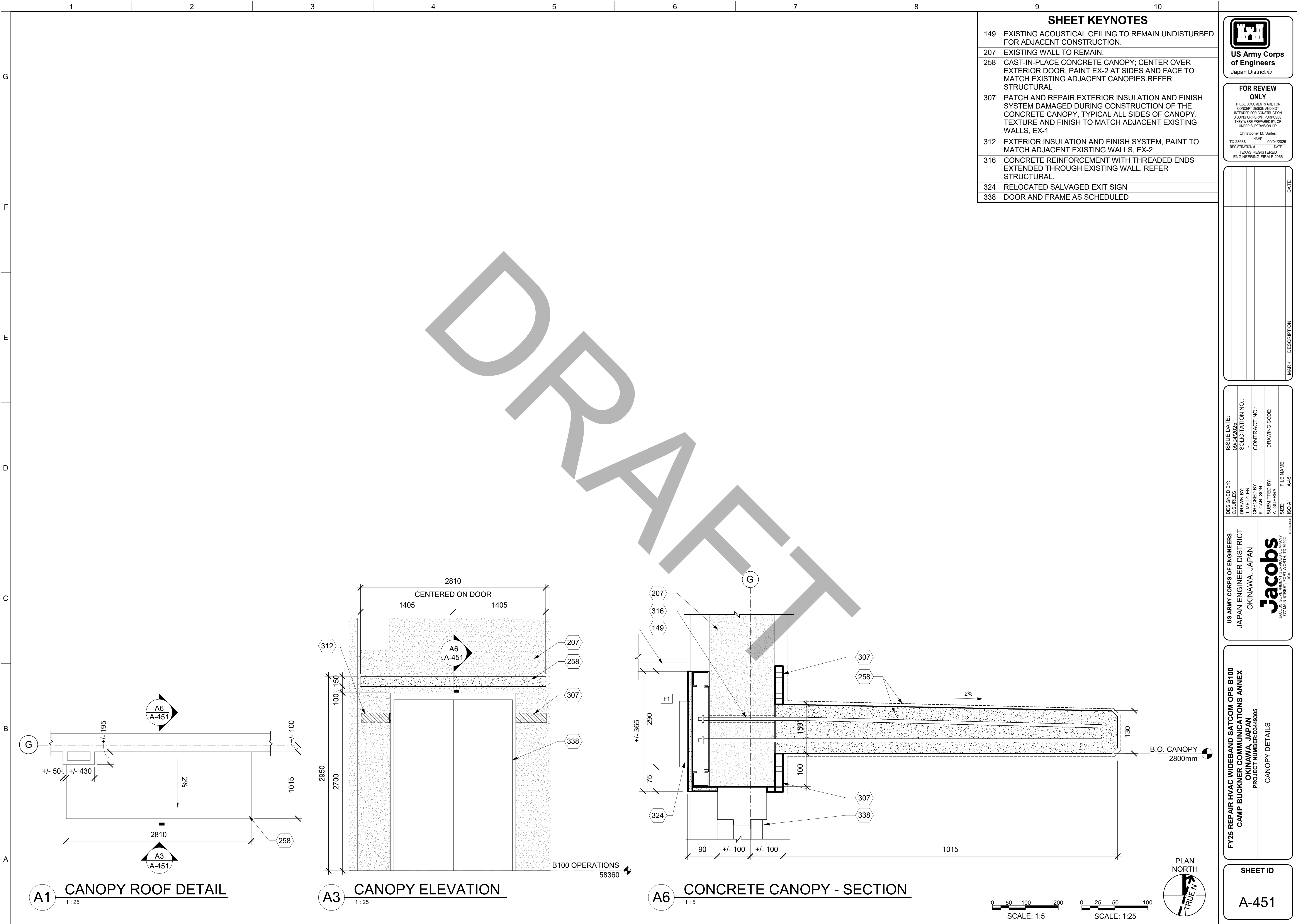
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PLOTTED BY:

PLOT SCALE: As indicated

BACKCHECK SUBMITTAL DESIGN



A A-451-



SUSPENDED ACOUSTICAL CEILING

The diagram illustrates the assembly of a suspended acoustical ceiling. A main runner is shown at the bottom, with a fastening tube attached to its top. A 3.2mm vertical hanger wire is threaded through the fastening tube and secured with a 3mm bolt and a locking nut. The hanger wire is also connected to a grid level. The fastening tube is secured to the main runner with a bolt. The hanger wire is splayed at 90 degrees from each other and 45 degrees from the ceiling plane. The fastening tube is lapped over the main runner by at least 100mm.

(4) BRACE WIRES, SPAYED 90 DEG FROM EACH OTHER AND 45 DEG MAXIMUM FROM CEILING PLANE

INSTALL BOLT TO SECURE ONE TUBE TO THE OTHER. TUBES SHALL LAP ONE ANOTHER AT LEAST 100mm IN FULLY EXTENDED POSITION

EMT TIGHT TO STRUCTURE ABOVE

3.2mm VERTICAL HANGER WIRE SET CEILING GRID LEVEL PRIOR TO FASTENING TUBE

DRILL 3mm HOLE AND INSTALL 3mm BOLT AND LOCKING NUT

SLOT END OF TUBE TO FIT OVER MAIN RUNNER



F7 FIXTURE BRACING DETAIL

SCALE: NOT TO SCALE





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Christopher M. Surles
NAME
TX 23636
REGISTRATION #
DATE
09/04/2025
TEXAS REGISTERED
ENGINEERING FIRM F-2966

MARK	DESCRIPTION	DATE

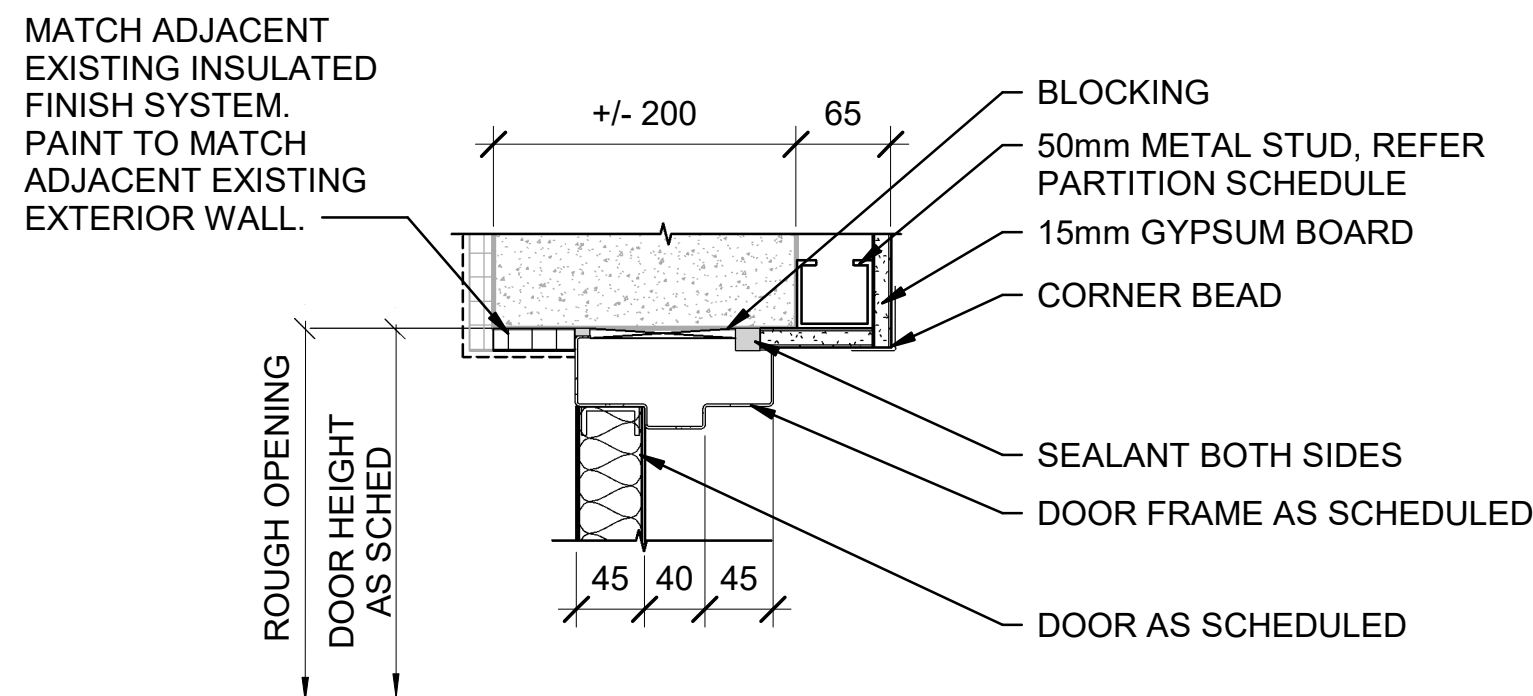
DESIGNED BY: C. SURLES	ISSUE DATE: 09/04/2025	SOLICITATION NO.:
CHECKED BY: J. METZLER	CONTRACT NO.:	
SUBMITTED BY: K. CARLSON	DRAWING CODE:	
SIZE: A. GUERRA	FILE NAME: A-602	ISO A1

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JACOBS ENGINEERING GROUP, INC.
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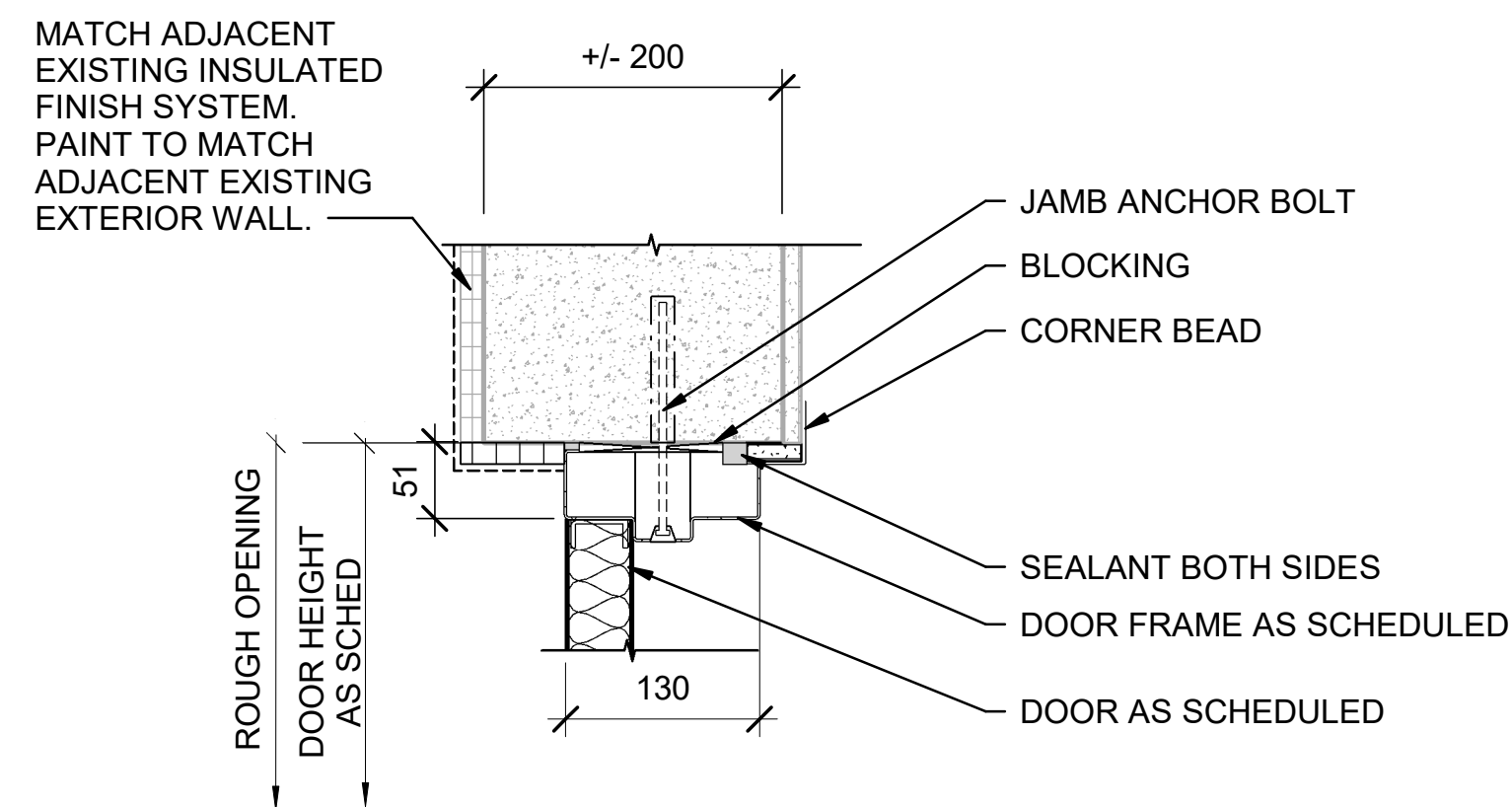
FY25 REPAIR HVAC WIDEAND SATCOM OPS B100 CAMP BUCKNER COMMUNICATIONS ANNEX OKINAWA, JAPAN PROJECT NUMBER: D3449305 DOOR DETAILS

SHEET ID

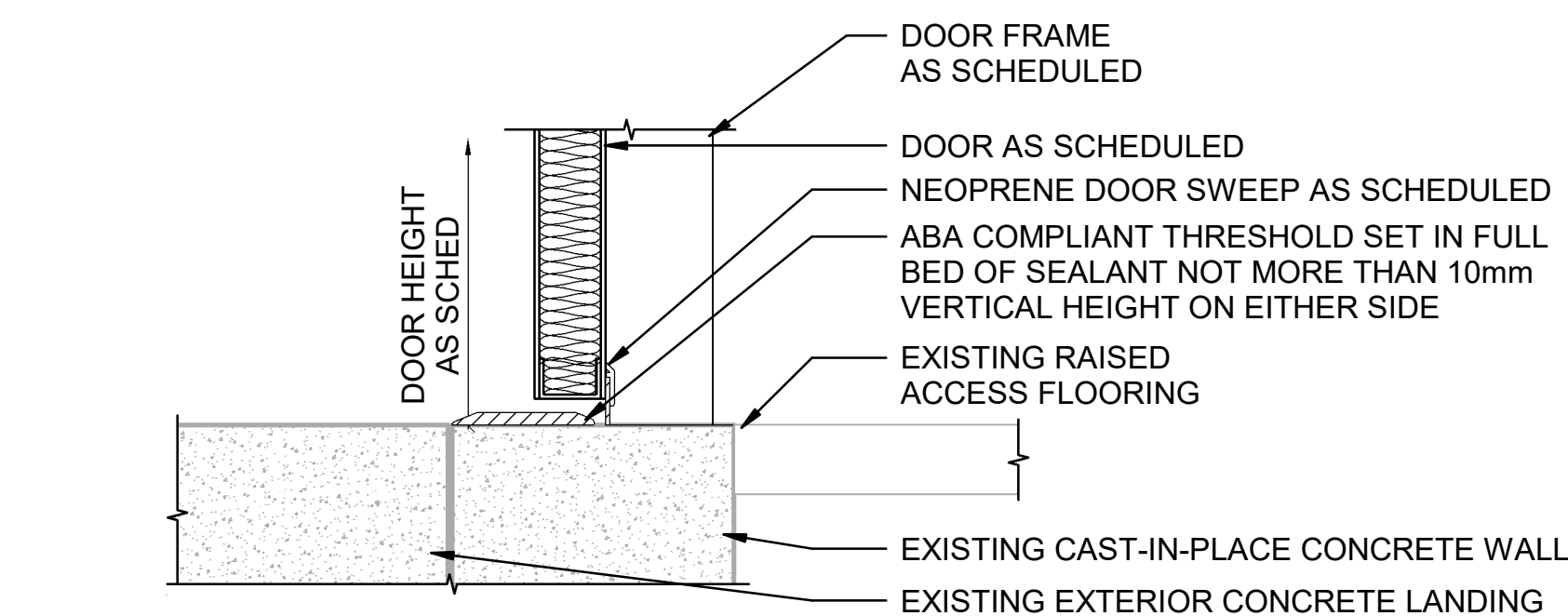
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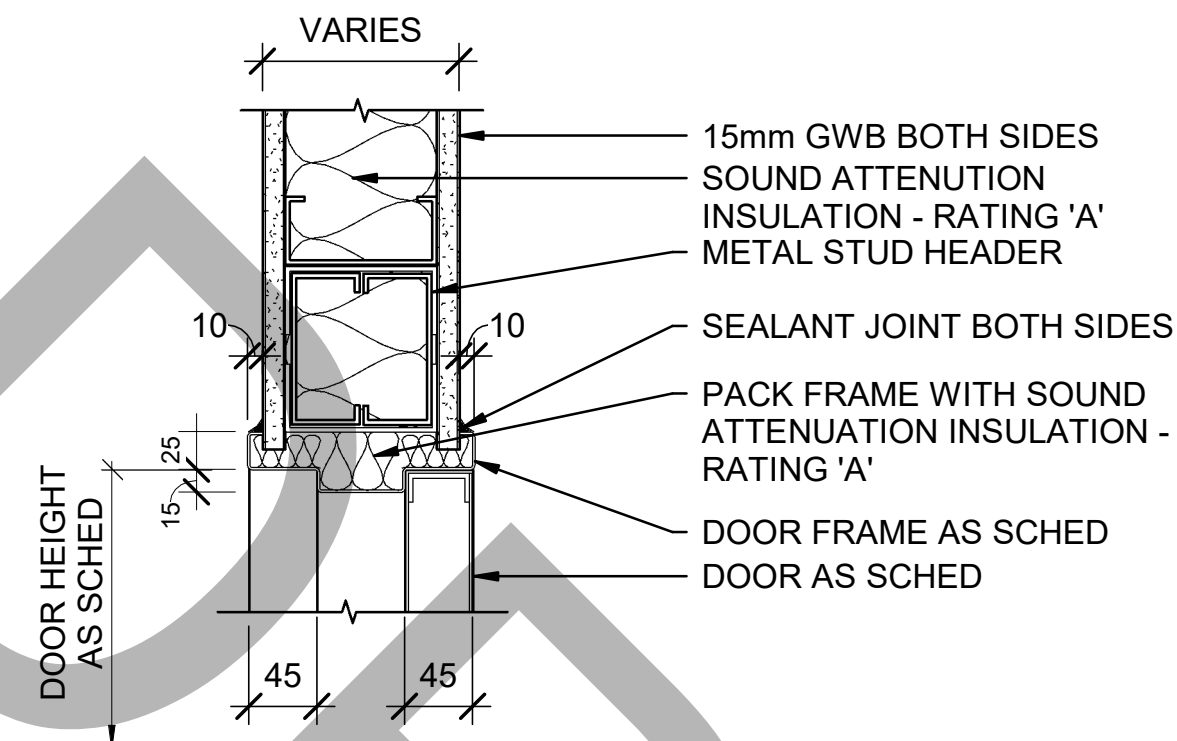
D1 HEAD DETAIL AT EXISTING CONCRETE WALL
1:5



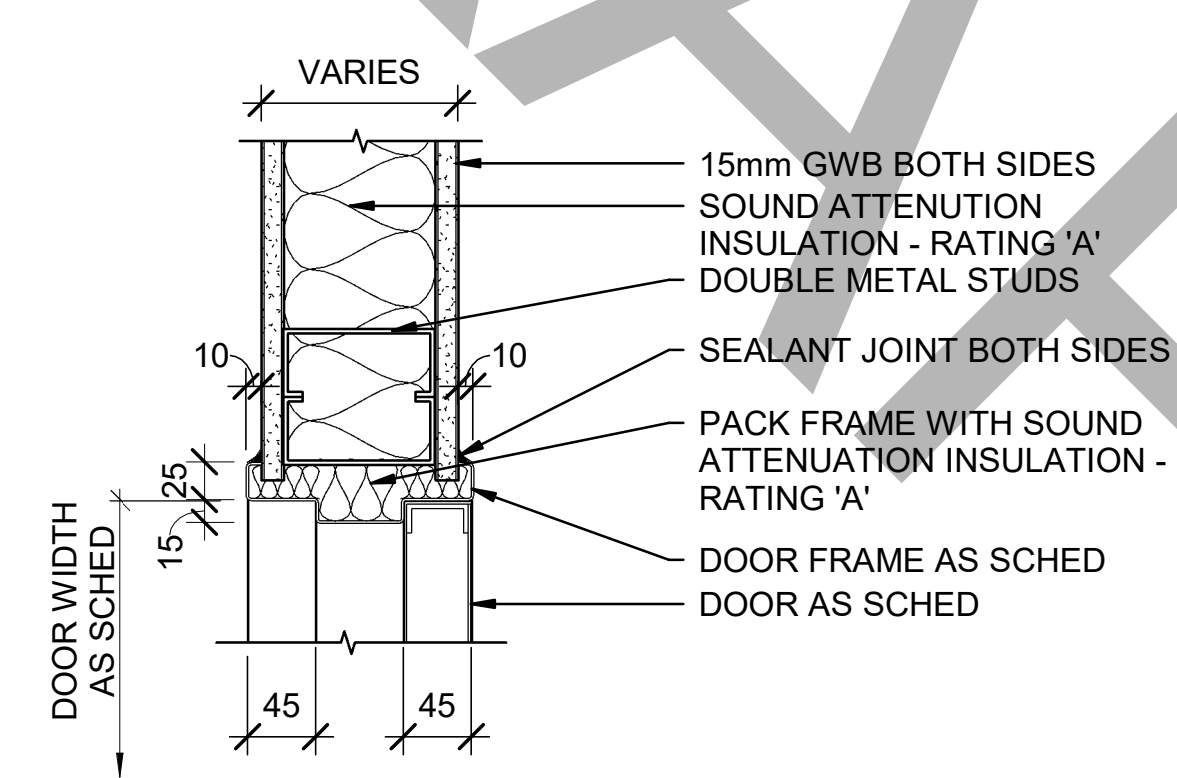
B1 JAMB DETAIL AT EXISTING CONCRETE WALL
1:5



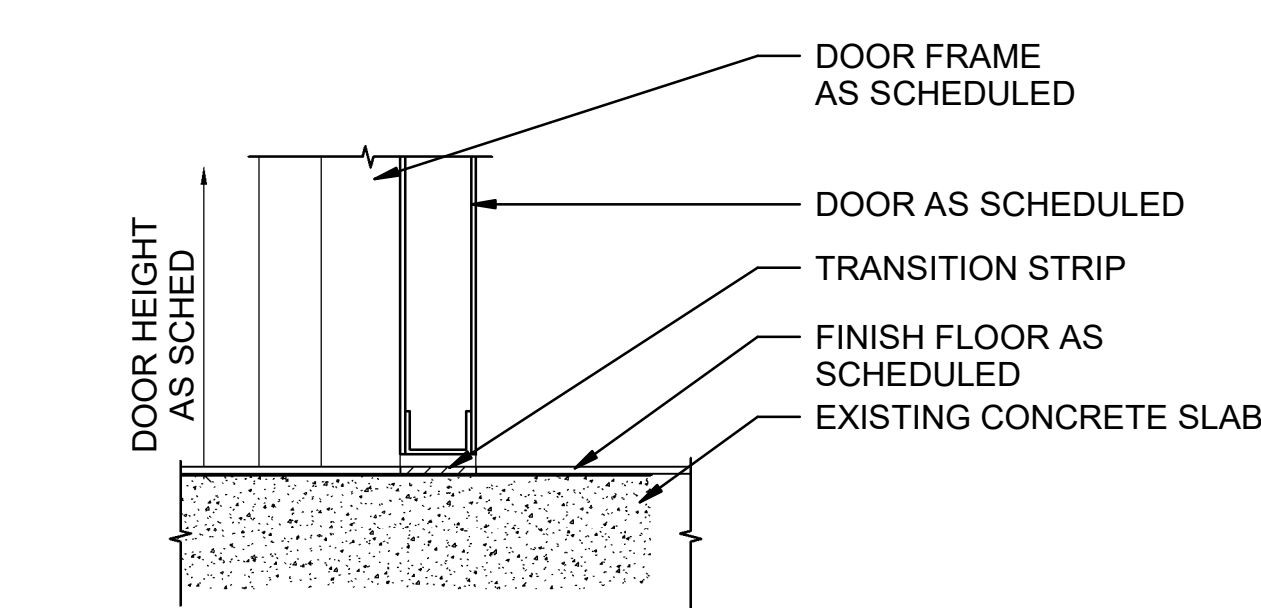
A1 SILL DETAIL AT EXISTING CONCRETE WALL
1:5



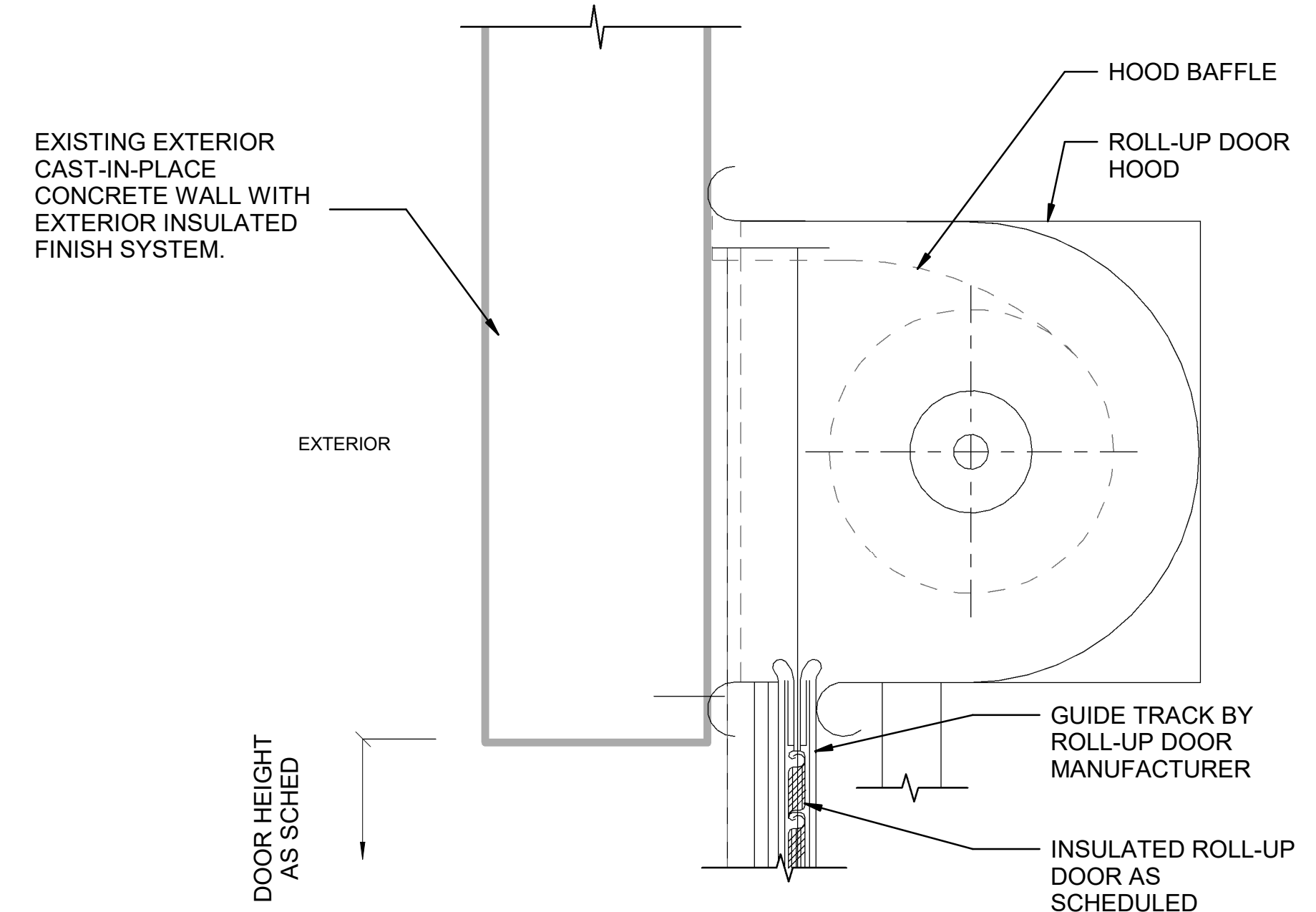
D4 HEAD DETAIL AT METAL STUD WALL
1:5



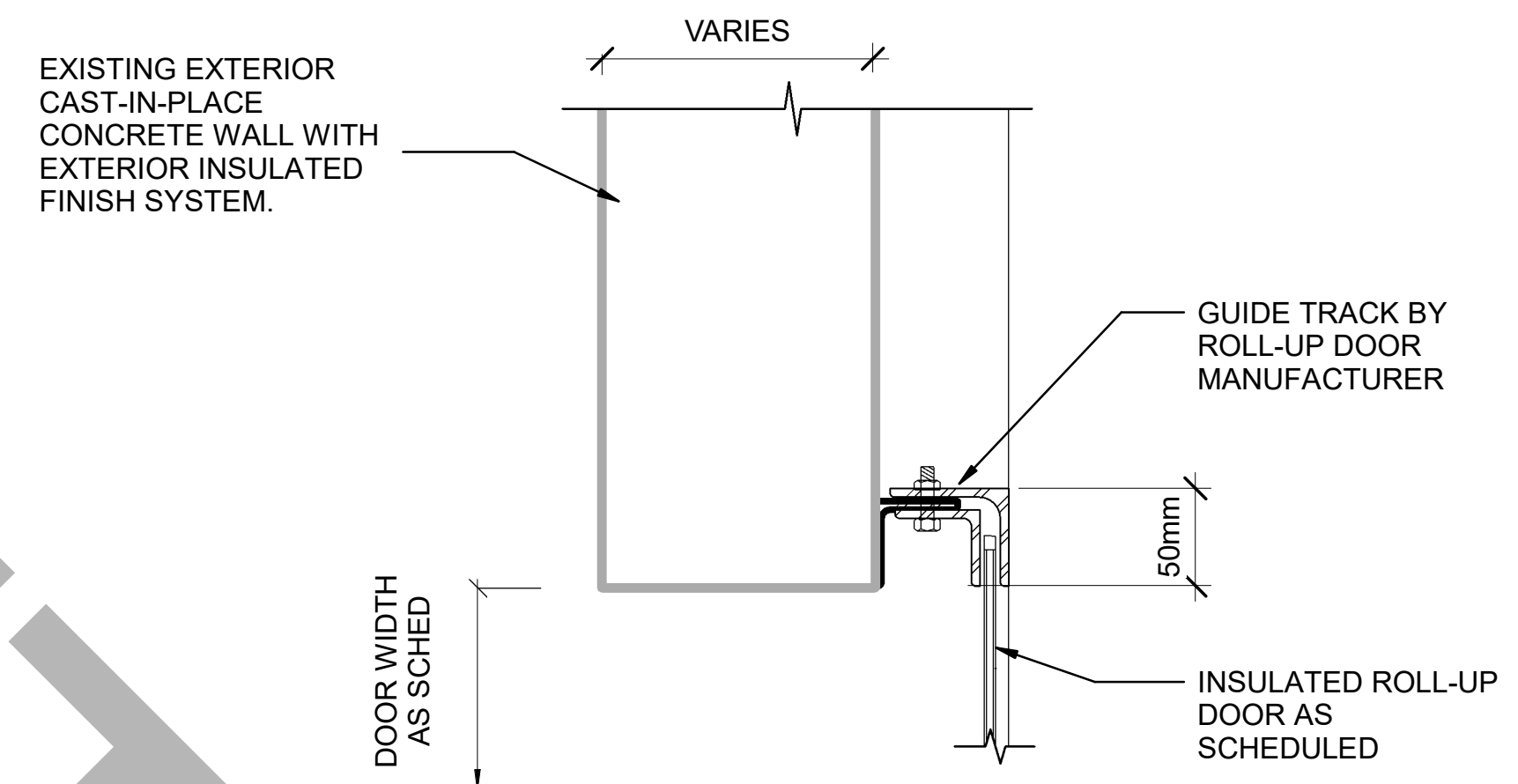
B4 JAMB DETAIL AT METAL STUD WALL
1:5



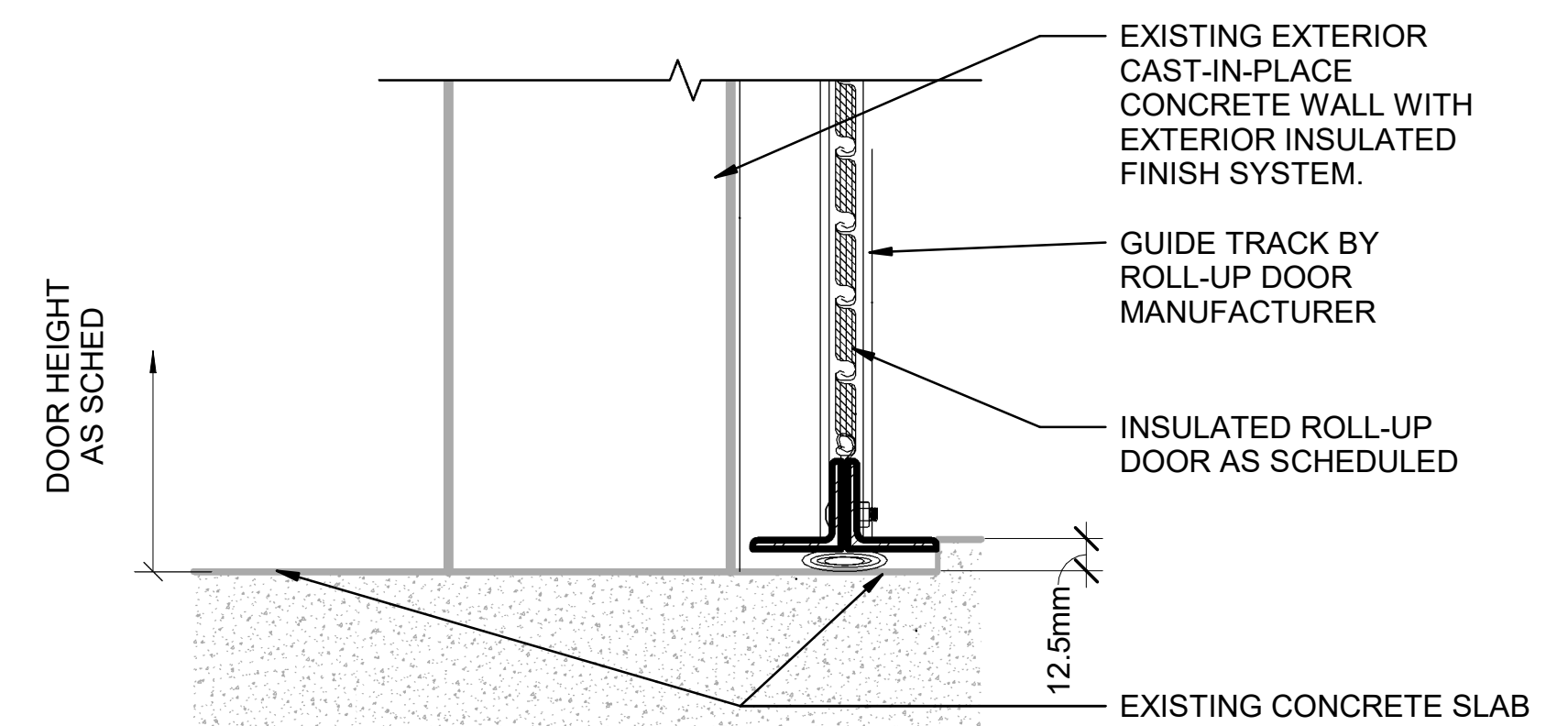
A4 SILL DETAIL AT METAL STUD WALL
1:5



E7 OVERHEAD COILING DOOR HEAD DETAIL
1:5



C7 OVERHEAD COILING DOOR JAMB DETAIL
1:5



A7 OVERHEAD COILING DOOR SILL DETAIL
1:5

0 50 100 200
SCALE: 1:5



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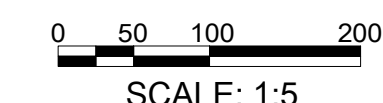
NAME	
TX 23638	09/04/2025
REGISTRATION #	DATE
TEXAS REGISTERED ENGINEERING FIRM F-2966	

[illegible]

US ARMY CORPS OF ENGINEERS JAPAN ENGINEER DISTRICT OKINAWA, JAPAN	DESIGNED BY:	ISSUE DATE:
	C. SURLS	09/04/2025
 JACOBS ENGINEERING SERVICES COMPANY 777 MAIN STREET, FORT WORTH, TX 76102	DRAWN BY:	SOLICITATION NO.:
	J. METZLER	-
	CHECKED BY:	CONTRACT NO.:
	K. CARLSON	-
	SUBMITTED BY:	DRAWING CODE:
	A. GUERRA	
	SIZE:	FILE NAME:
	A4	

FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100 CAMP BUCKNER COMMUNICATIONS ANNEX OKINAWA, JAPAN PROJECT NUMBER:D3449305	PENETRATION AND LOUVER DETAILS
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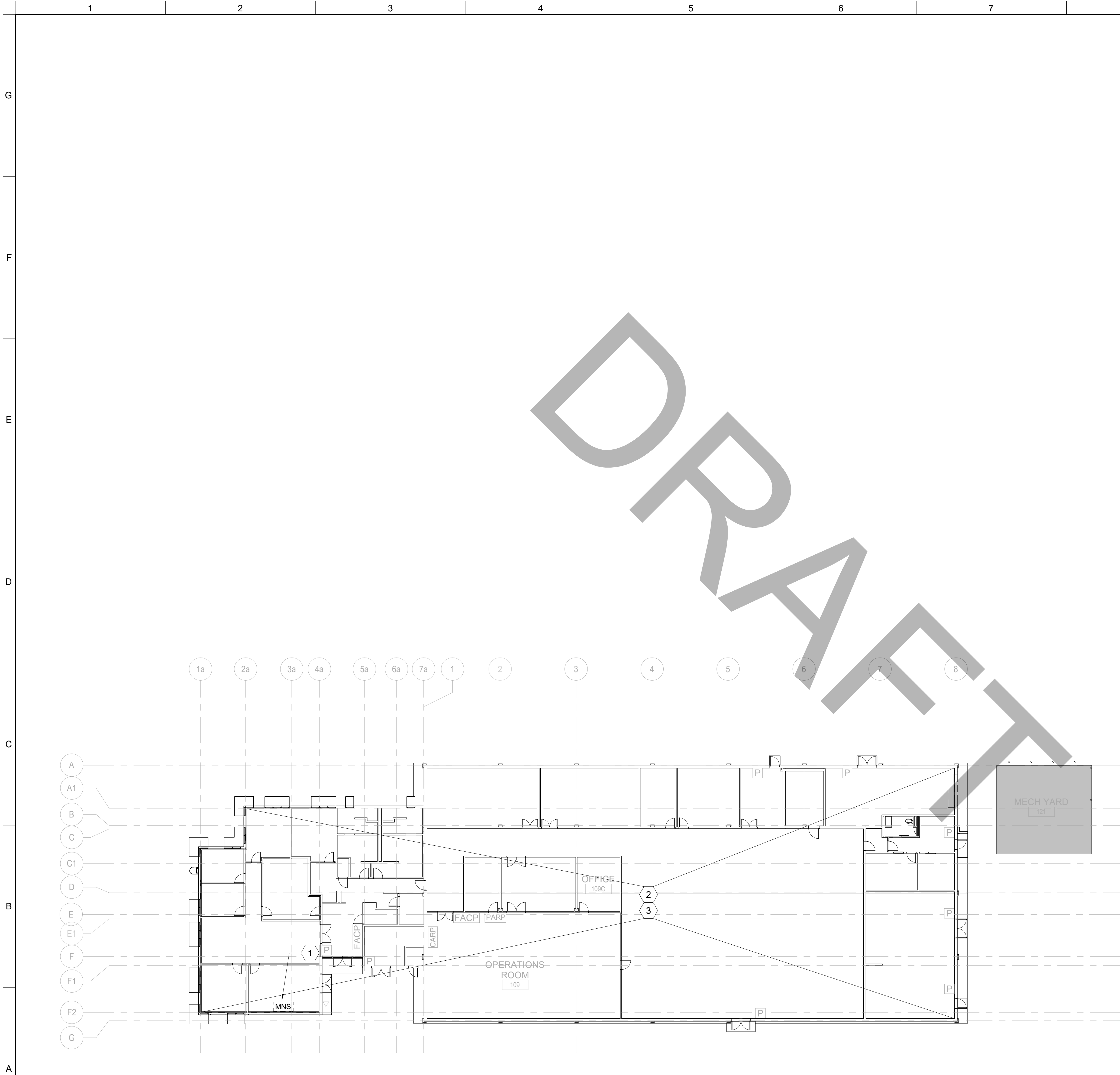
A-603



F FA001-

	ACOUSTICALLY DISTINGUISHABLE SPACE LEGEND				FIRE ALARM/MASS NOTIFICATION SYMBOLS			FIRE ALARM/MASS NOTIFICATION SYSTEM GENERAL NOTES																																																																																			
	SPACE	AMBIENT SPL (dBA)	MAXIMUM SPL (dBA)	MINIMUM INTELLIGIBILITY (CIS)	FA FCP FIRE ALARM/MASS NOTIFICATION CONTROL PANEL	A. MODIFY EXISTING FIRE ALARM COVERAGE TO ACCOMMODATE NEW HVAC EQUIPMENT AND OTHER ADJUSTMENTS AS A RESULT OF THE THIS PROJECTS SCOPE OF WORK INCLUDING SMOKE DETECTION IN THE NEW CRAC UNITS. ADD NEW MASS NOTIFICATION TO EXISTING ADDRESSABLE FIRE ALARM SYSTEM FOR COMPLETE COVERAGE THROUGHOUT THE ENTIRE FACILITY. PERFORM ALL DESIGN AND INSTALLATION WORK IN ACCORDANCE WITH NFPA 72, UFC 3-600-01, UFC 4-021-01, AND PROJECT SPECIFICATIONS. B. REFER TO FIRE ALARM FLOOR PLAN FOR MAKE AND MODEL OF EXISTING EQUIPMENT. NEW DEVICES MUST BE COMPATIBLE WITH EXISTING SYSTEMS CURRENTLY IN PLACE. C. RUN ALL WIRING IN MINIMUM 19 MM CONDUIT. EXCEPTION: 19 MM CONDUIT MAY BE USED FOR 120 VAC POWER FROM A CIRCUIT BREAKER PANEL TO THE FIRE ALARM/MASS NOTIFICATION SYSTEM. D. INSTALL ALL WALL-MOUNTED NOTIFICATION DEVICES WITH THE BOTTOM OF THE STROBE LENS AT 2.03 METERS AFF, OR 150 MM BELOW THE CEILING, WHICHEVER IS LOWER. E. PROVIDE POWER LIMITED CIRCUITS. F. DO NOT LOCATE DETECTORS IN DIRECT AIR OR CLOSER THAN 910 MM FROM AN AIR SUPPLY OR RETURN DIFFUSER. G. PROVIDE CONNECTIONS TO THE LIGHT AND POWER SERVICE IN ACCORDANCE WITH NFPA 72. H. PROVIDE 18 AWG MINIMUM SIGNALING LINE CIRCUITS. I. PROVIDE 16 AWG MINIMUM NOTIFICATION APPLIANCE CIRCUITS AND INITIATING DEVICE CIRCUITS. J. PROVIDE CLASS B SIGNALING LINE CIRCUITS (SLC), INITIATING DEVICE CIRCUITS (IDC), AND NOTIFICATION APPLIANCE CIRCUITS (NAC). K. PROVIDE STROBES AND SPEAKER/STROBES WITH WHITE FACEPLATES THROUGHOUT THE BUILDING. STROBES TO FEATURE CLEAR LENS AND BE MARKED WITH "ALERT". SPEAKERS ARE TO BE SHARED BY THE MASS NOTIFICATION SYSTEM AND FIRE ALARM SYSTEMS FOR AUDIBLE NOTIFICATION. PROVIDE ADDITIONAL SPEAKERS AND STROBES AS NECESSARY TO MEET THE AUDIBILITY AND INTELLIGIBILITY REQUIREMENTS OF NFPA 72 SECTION 18.4 AND UFC 4-021-01. L. UPON ACTIVATION OF THE MASS NOTIFICATION SYSTEM, TEMPORARILY DEACTIVATE FIRE ALARM SYSTEM NOTIFICATION DEVICES DURING EMERGENCY VOICE MESSAGING. FIRE SIGNAL TRANSMISSION TO THE BASE REPORTING STATION IS TO REMAIN FUNCTIONAL. IF NOT MANUALLY ENDED, THE MASS NOTIFICATION SYSTEM MESSAGE WILL AUTOMATICALLY STOP AFTER 10 MINUTES. REACTIVATE THE FIRE ALARM AUDIBLES WHEN THE MESSAGE IS COMPLETED. M. PROVIDE DUCT SMOKE DETECTION IN ALL SUPPLY DUCTS FOR AIR HANDLING UNITS GREATER THAN 944 L/S. PROVIDE DUCT SMOKE DETECTION ON RETURN AND SUPPLY SIDE OF AIR HANDLING UNITS OVER 7,079 L/S PER NFPA 90A AND UFC 3-600-01. N. THE SOUND POWER OF AUDIBLE NOTIFICATION DEVICES SHALL BE 15 DBA OVER AMBIENT CONDITIONS OR 5 DBA ABOVE THE MAXIMUM SOUND LEVEL HAVING A DURATION OF AT LEAST 60 SECONDS WHICHEVER IS GREATER, MEASURED AT 1.5 METERS ABOVE THE FLOOR. O. THESE DOCUMENTS DEPICT A PERFORMANCE LEVEL ENGINEERING DESIGN LAYOUT TO BE UTILIZED AS GUIDANCE FOR THE PLANNING OF THE FIRE ALARM AND MASS NOTIFICATION SYSTEM BY THE CONTRACTOR. PROVIDE COMPLETE DOCUMENTS FOR REVIEW AND APPROVAL FROM THE QFPE AND THE AUTHORITY HAVING JURISDICTION PRIOR TO INSTALLATION. INCLUDE SHOP DRAWINGS, CALCULATIONS, AND ANY ADDITIONAL EQUIPMENT NECESSARY, INCLUDING INITIATING DEVICES AND NOTIFICATION APPLIANCES, TO PROVIDE A COMPLETE INSTALLATION AND COMPLY WITH LOCAL CODES AND AMENDMENTS. PROVIDE COMPLETE VOLTAGE DROP AND BATTERY CALCULATIONS IN ACCORDANCE WITH NFPA 72. PROVIDE ADDITIONAL CIRCUITS AND/OR BATTERIES WHERE NECESSARY FOR COMPLIANCE WITH NFPA 72 AND THE SPECIFICATIONS.																																																																																					
G	FITNESS EQUIPMENT, STORAGE, SUPPLY ROOMS	30	40	0.75	FA ANN ANNUNCIATOR PANEL																																																																																						
	OFFICES, CORRIDORS, RESTROOMS, ENTRY, ADMIN AREAS, KITCHEN, BREAK ROOMS	55	65	0.75	FA AMP AMPLIFIER PANEL																																																																																						
	ELECTRICAL, COMM, UPS ROOMS	55	65	0.75	FA RPS NOTIFICATION CIRCUIT POWER BOOSTER PANEL																																																																																						
	CONFERENCE, CLASSROOMS	60	70	0.75	FA LOC LOCAL OPERATING CONSOLE																																																																																						
	MECHANICAL, TECHNICAL EQUIPMENT ROOMS	80	90	0.75	FA FAT FIRE ALARM RADIO TRANSCEIVER PANEL																																																																																						
F					FA JPC JOCKEY PUMP CONTROLLER				MARK DESCRIPTION DATE																																																																																		
					FA FPC FIRE PUMP CONTROLLER																																																																																						
					FA TXT MASS NOTIFICATION EMERGENCY TEXTUAL DISPLAY																																																																																						
					FA MNS MASS NOTIFICATION SYSTEM PANEL																																																																																						
					FA CARP CLEAN AGENT RELEASE PANEL																																																																																						
					FA PACP PREACTION RELEASE PANEL																																																																																						
					FA  CEILING-MOUNTED FIRE ALARM/MASS NOTIFICATION SPEAKER																																																																																						
					FA  WALL-MOUNTED FIRE ALARM/MASS NOTIFICATION SPEAKER																																																																																						
					FA  CEILING-MOUNTED FIRE ALARM SPEAKER/STROBE WITH CLEAR LENS AND WHITE FACEPLATE MARKED "FIRE"																																																																																						
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					FA WP EXTERIOR MASS NOTIFICATION LOUDSPEAKER																																																																																						
					FA P MANUAL FIRE ALARM PULL STATION																																																																																						
					FA  ADDRESSABLE HEAT DETECTOR																																																																																						
					FA  PHOTOELECTRIC SMOKE DETECTOR																																																																																						
					FA RS ADDRESSABLE PHOTOELECTRIC RETURN/SUPPLY DUCT DETECTOR (SUPPLIED BY FIRE ALARM CONTRACTOR; INSTALLED BY MECHANICAL CONTRACTOR)																																																																																						
					FA M ADDRESSABLE INPUT MONITOR MODULE																																																																																						
					FA C ADDRESSABLE OUTPUT CONTROL MODULE																																																																																						
					FA SPD SURGE PROTECTIVE DEVICE																																																																																						
					FA TS VALVE TAMPER SWITCH																																																																																						
					FA FS WATER FLOW SWITCH - VANE TYPE																																																																																						
					FA PS WATER FLOW SWITCH - PRESSURE TYPE																																																																																						
					FA H/L HIGH/LOW AIR PRESSURE SWITCH																																																																																						
					FA "WP" DESIGNATES WEATHER PROOF DEVICE AND BACKBOX																																																																																						
					FA  OMNI-DIRECTIONAL FIRE ALARM/MASS NOTIFICATION ANTENNA																																																																																						
E					DRAWING REFERENCE	A. FIRE ALARM AUDIBLE NOTIFICATION MUST BE PROVIDED FOR ALL AREAS AND ROOMS THROUGHOUT THE BUILDING. PROVIDE SPEAKERS AND SPEAKER/STROBES AS NECESSARY TO MEET INTELLIGIBILITY REQUIREMENTS OF NFPA 72, UFC 4-021-01, AND THE FOLLOWING CRITERIA. THE CRITERIA MUST BE MET WITH ALL DOORS AND WINDOWS CLOSED. WHERE THE CRITERIA CANNOT BE MET DURING SYSTEM ACCEPTANCE TESTING, THE CONTRACTOR MUST PROVIDE ADDITIONAL DEVICES AS NECESSARY TO MEET THE CRITERIA AT NO ADDITIONAL COST. B. ALL ROOMS MUST BE CONSIDERED NORMALLY OCCUPIED EXCEPT FOR STORAGE ROOMS, JANITORS CLOSET, ELECTRICAL ROOMS, COMMUNICATIONS EQUIPMENT ROOMS, AND OTHER SIMILAR UTILITY SPACES. IN NORMALLY OCCUPIED ROOMS OR AREAS, ENSURE A MINIMUM CIS VALUE OF 0.80 IS ACHIEVED. MEASUREMENTS MUST BE TAKEN AT HEAD LEVEL, APPROXIMATELY 1.5 METERS ABOVE THE FINISHED FLOOR IN THE MOST REMOTE PORTION OF THE ROOM. C. NORMALLY UNOCCUPIED AREAS ARE PERMITTED TO HAVE A CIS SCORE LESS THAN THE MINIMUM REQUIRED, PROVIDED THAT OCCUPANTS TRAVEL NO MORE THAN 15 METERS TO FIND A LOCATION WITH AT LEAST THE MINIMUM REQUIRED VALUE. D. WHERE OCCUPANTS ARE NORMALLY PRESENT, ENSURE A CIS SCORE GREATER THAN THE MINIMUM REQUIRED 0.80 CIS ACHIEVED. ROUNDING IS PERMITTED SUCH THAT A VALUE OF 0.75 MAY BE ROUNDED TO 0.8. E. IN NORMALLY OCCUPIED ROOMS WITH HARD WALL AND CEILING SURFACES SUCH AS METAL OR CONCRETE, OCCUPANTS MAY TRAVEL NO MORE THAN 9.75 METERS TO FIND A LOCATION WITH AT LEAST THE MINIMUM REQUIRED CIS VALUE. F. THE DISTANCE THE OCCUPANT MUST TRAVEL TO A LOCATION WITH THE MINIMUM REQUIRED CIS VALUE MUST BE ALONG THE CENTERLINE OF NATURAL PATH OF TRAVEL. MEASUREMENTS AROUND CORNERS OR OBSTRUCTIONS REQUIRE A MINIMUM OF 300 MM OF CLEARANCE. THE MEASURED DISTANCE MUST TERMINATE AT THE LOCATION WHERE THE MINIMUM REQUIRED CIS VALUE HAS BEEN ACHIEVED. G. EXTERIOR MASS NOTIFICATION LOUDSPEAKERS MUST HAVE A MINIMUM CIS SCORE OF 0.80 MEASURED AT 5 METERS FROM THE SPEAKER. H. STROBES ARE TO BE ACTIVATED IN CONJUNCTION WITH TRANSMISSION OF LIVE OR PRE-RECORDED MESSAGES. STROBES TO OPERATE FOR THE DURATION OF PRE-RECORDED MESSAGES AND NOT LESS THAN 15 SECONDS AFTER LIVE MESSAGES END. I. THE SOUND POWER OF THE AUDIBLE NOTIFICATION MUST BE 15 DBA ABOVE THE AMBIENT CONDITIONS OR 5 DBA ABOVE THE MAXIMUM SOUND LEVEL. SOUND POWER MUST NOT EXCEED 100 DBA AT THE MOST REMOTE LOCATION OF THE ROOM AREA. REFER TO MANUFACTURER'S DATASHEET FOR SOUND DISTRIBUTION DIAGRAMS. J. PROVIDE LOCAL OPERATING CONSOLES SO THAT OCCUPANTS NEED NOT TRAVEL GREATER THAN 200 FEET TO .																																																																																					
										US ARMY CORPS OF ENGINEERS JAPAN ENGINEER DISTRICT OKINAWA, JAPAN JACOBS JACOBS ENGINEERING GROUP, INC. 1717 MAIN STREET, SUITE 100, FORT WORTH, TEXAS 76102 ISO A1 FA001																																																																																	
														FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100 CAMP BUCKNER COMMUNICATIONS ANNEX OKINAWA, JAPAN PROJECT NUMBER: D3449395 FIRE ALARM / MASS NOTIFICATION GENERAL NOTES																																																																													
																		SHEET ID FA001																																																																									

F FD101-



A1 FIRE ALARM / MASS NOTIFICATION DEMOLITION PLAN
1 : 200

GENERAL SHEET NOTES

- A. REFER TO F-001 FOR GENERAL NOTES, SYMBOLS, AND ABBREVIATIONS.
- B. REFER TO FD101 FOR FIRE ALARM/MASS NOTIFICATION DEMOLITION PLAN.

XX

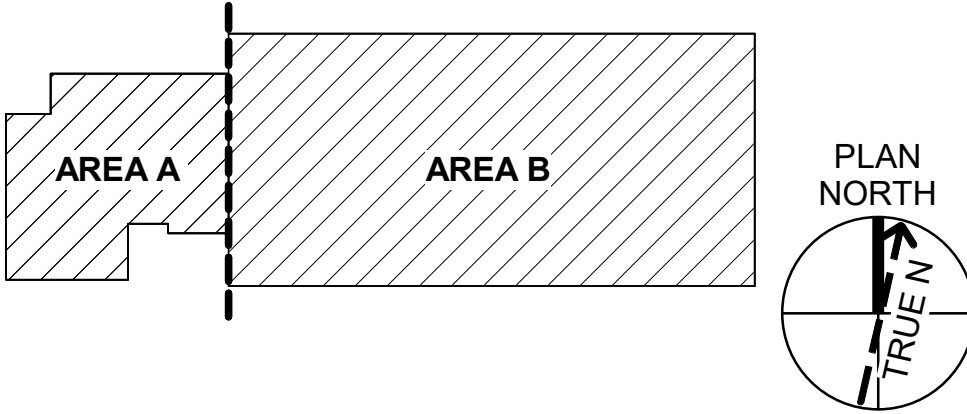
SHEET KEYNOTES

- | | |
|---|---|
| 1 | MNS PANEL TO BE DEMOLISHED. |
| 2 | EXISTING NOTIFICATION DEVICES TO BE DEMOLISHED. |
| 3 | ALL EXISTING INITATING DEVICES TO REMAIN. |

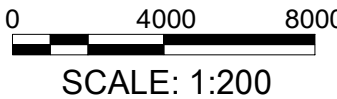
DEMOLITION LEGEND

- | | |
|--|------------------------------|
| | NOT IN RENOVATION SCOPE |
| | EXISTING EQUIPMENT TO REMIAN |
| | EQUIPMENT TO BE DEMOLISHED |

KEY PLAN



GRAPHIC SCALE(S)



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ALEX D. SHEFFERMAN
NAME
147009 04/28/2025
REGISTRATION # DATE
TEXAS REGISTERED ENGINEERING FIRM F-2966

DESIGNED BY: A.S. W.N. BY: E.H.	ISSUE DATE: 7/18/2025	SOLICITATION NO.:
CHECKED BY: B.B.	CONTRACT NO.:	W912H19D007
SUBMITTED BY: A. GUERRA	DRAWING CODE:	
SIZE:	FILE NAME:	FD101

ISO A1

US ARMY CORPS OF ENGINEERS
JAPAN ENGINEER DISTRICT
OKINAWA, JAPAN

Jacobs
JACOBS ENGINEERING SERVICES COMPANY
771 MAIN STREET, SUITE 100
IRVING, TEXAS 75039
USA

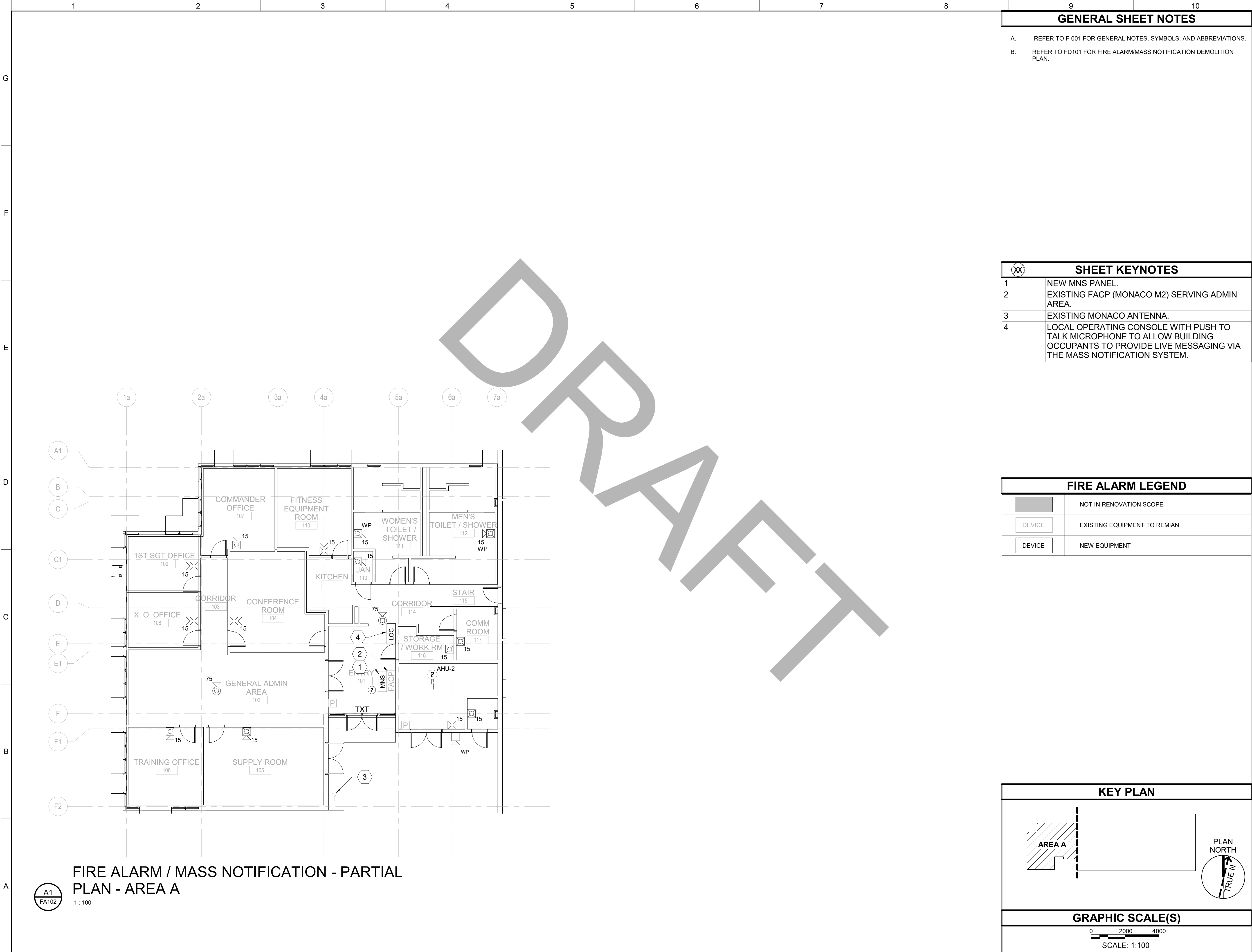
FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX
OKINAWA, JAPAN
PROJECT NUMBER: D3449305

FIRE ALARM / MASS NOTIFICATION DEMOLITION PLAN

SHEET ID

FD101

FINAL DESIGN



GENERAL SHEET NOTES

- A. REFER TO F-001 FOR GENERAL NOTES, SYMBOLS, AND ABBREVIATIONS.
- B. REFER TO FD101 FOR FIRE ALARM/MASS NOTIFICATION DEMOLITION PLAN.

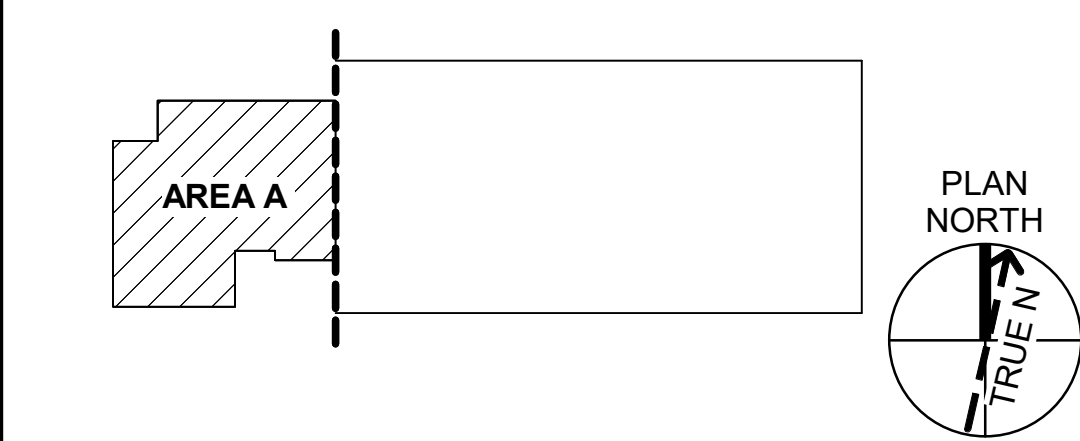
SHEET KEYNOTES

1	NEW MNS PANEL.
2	EXISTING FACP (MONACO M2) SERVING ADMIN AREA.
3	EXISTING MONACO ANTENNA.
4	LOCAL OPERATING CONSOLE WITH PUSH TO TALK MICROPHONE TO ALLOW BUILDING OCCUPANTS TO PROVIDE LIVE MESSAGING VIA THE MASS NOTIFICATION SYSTEM.

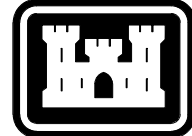
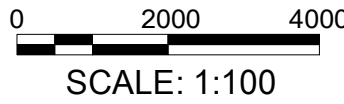
FIRE ALARM LEGEND

NOT IN RENOVATION SCOPE	
DEVICE	EXISTING EQUIPMENT TO REMIAN
DEVICE	NEW EQUIPMENT

KEY PLAN



GRAPHIC SCALE(S)



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ALEX D. SHEFFERMAN
NAME 04/28/2025
DATE
REGISTRATION #
TEXAS REGISTERED
ENGINEERING FIRM F-2966

ISSUE DATE: 7/18/2025	SOLICITATION NO.:
DESIGNED BY: S.B. WNW BY: E.H.	CONTRACT NO.:
CHECKED BY: B.B.	W912H19D007
SUBMITTED BY: A. GUERRA	DRAWING CODE
SIZE:	FILE NAME:
ISO A1	FA102

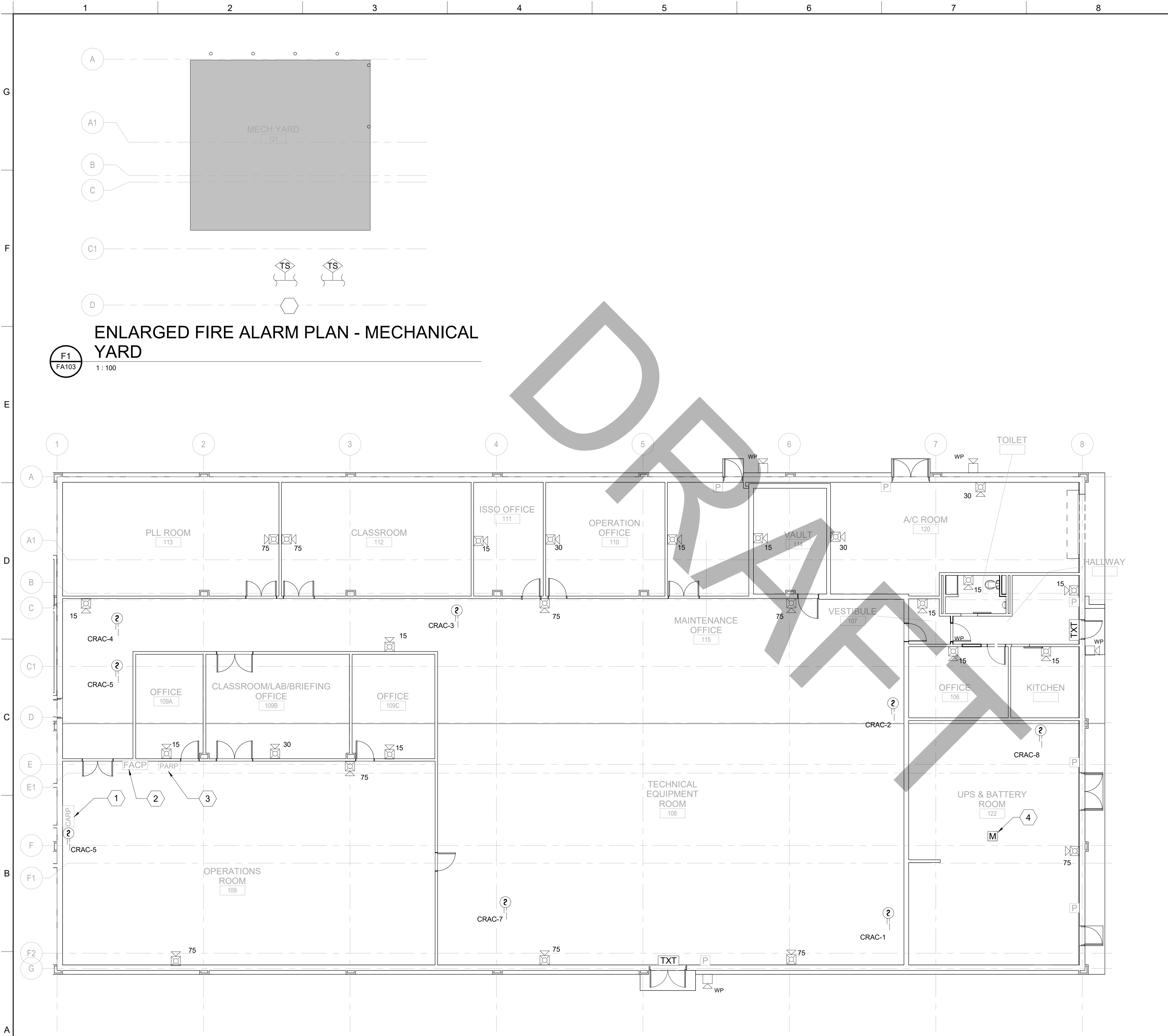
US ARMY CORPS OF ENGINEERS
JAPAN ENGINEER DISTRICT
OKINAWA, JAPAN
Jacobs
JACOBS ENGINEERING GROUP, INC.
777 MAIN STREET, SUITE 1000
USA

FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX
OKINAWA, JAPAN
PROJECT NUMBER: D3448305
FIRE ALARM / MASS NOTIFICATION PARTIAL FLOOR PLAN -
AREA A

SHEET ID

FA102

FINAL DESIGN



ENLARGED FIRE ALARM PLAN - MECHANICAL YARD

F1
FA103
1 : 100

FIRE ALARM / MASS NOTIFICATION - PARTIAL PLAN - AREA B

A1
FA103
1 : 100

GENERAL SHEET NOTES

- A. REFER TO F-001 FOR GENERAL NOTES, SYMBOLS, AND ABBREVIATIONS.
- B. REFER TO FD101 FOR FIRE ALARM/MASS NOTIFICATION DEMOLITION PLAN.



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NAME 04/28/2025
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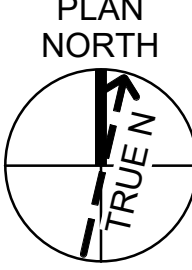
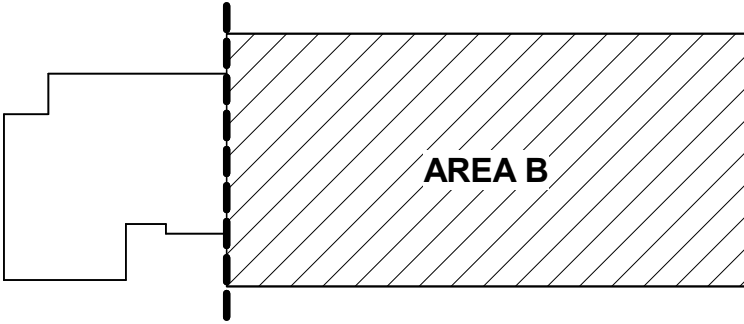
SHEET KEYNOTES

- 1 EXISTING FM200 RELEASE PANEL (AEGIS).
- 2 EXISTING FACP (FCI E3) SERVING TECHNICAL EQUIPMENT AREA.
- 3 EXISTING PREACTION RELEASE PANEL (VIKING VRF 400).
- 4 PROVIDE MONITOR MODULE FOR TROUBLE SIGNAL UPON HYDROGEN GAS DETECTION SYSTEM FAILURE. MODULE TO BE PROVIDED BY FIRE ALARM CONTRACTOR AND INSTALLED BY MECHANICAL CONTRACTOR.

FIRE ALARM LEGEND

	NOT IN RENOVATION SCOPE
DEVICE	EXISTING EQUIPMENT TO REMAIN
DEVICE	NEW EQUIPMENT

KEY PLAN



GRAPHIC SCALE(S)

0 2000 4000
SCALE: 1:100

ISSUE DATE:
7/18/2025
SOLICITATION NO.:
CONTRACT NO.:
W912H19D007
DRAWING CODE:
A. GUERRA
FILE NAME:
FA103
ISO A1

DESIGNED BY:
S.B. WNN BY:
E.H.
CHECKED BY:
B.B.
SUBMITTED BY:
A. GUERRA
SIZE:
ISO A1

US ARMY CORPS OF ENGINEERS
JAPAN ENGINEER DISTRICT
OKINAWA, JAPAN
PROJECT NUMBER: D3449305
FIRE ALARM / MASS NOTIFICATION PARTIAL FLOOR PLAN - AREA B

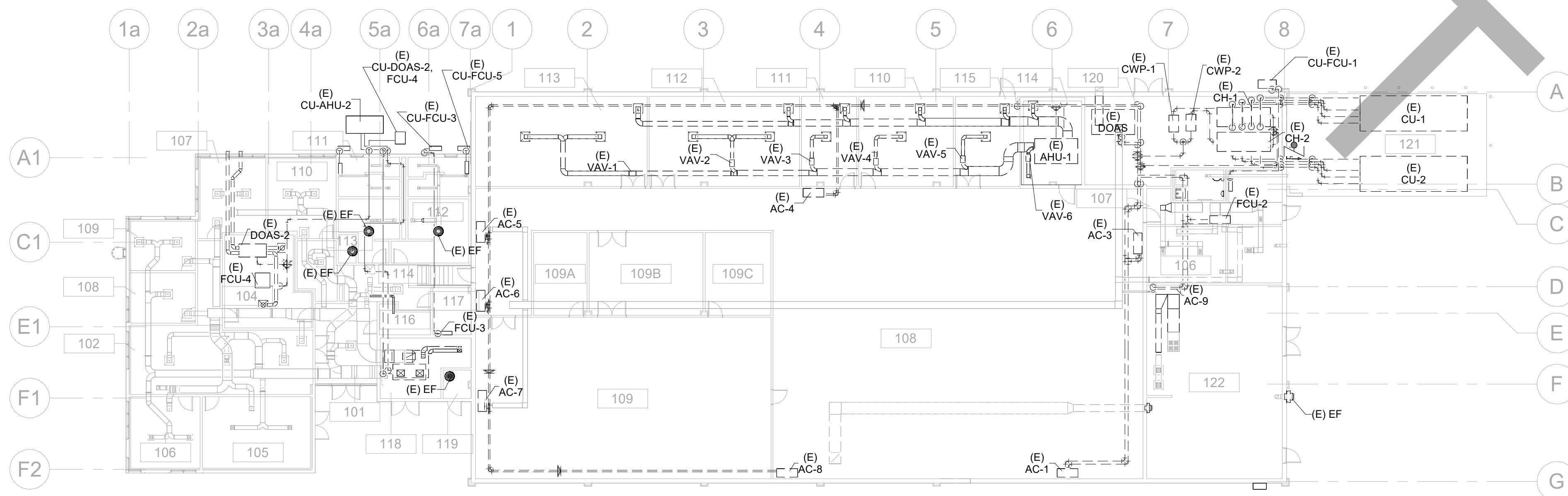
SHEET ID

FA103

FINAL DESIGN

DRAWING ABBREVIATIONS		LEGEND				HVAC DESIGN CRITERIA																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																
<table><tr><td>A</td><td>AIR CONDITIONER (EXISTING)</td><td>H</td><td>HEATING COIL</td></tr><tr><td>AC</td><td>AIR DEVICE</td><td>HP</td><td>HEAT PUMP</td></tr><tr><td>AD</td><td>ABOVE FINISHED FLOOR</td><td>HPS</td><td>HIGH PRESSURE STEAM</td></tr><tr><td>AF</td><td>ABOVE FINISHED GRADE</td><td>HVLS</td><td>HIGH VOLUME LOW SPEED</td></tr><tr><td>AFG</td><td>ABOVE FINISHED GRADE</td><td>HVU</td><td>HEAT AND VENT UNIT</td></tr><tr><td>AHU</td><td>AIR HANDLING UNIT</td><td>HWR</td><td>HOT WATER RETURN</td></tr><tr><td>AL</td><td>ACOUSTICAL LINING</td><td>HWS</td><td>HOT WATER SUPPLY</td></tr><tr><td>AS</td><td>AIR SEPARATOR</td><td>HX</td><td>HEAT EXCHANGER</td></tr><tr><td colspan="4"> </td></tr><tr><td>B</td><td>BRANCH SELECTOR BOX</td><td>I</td><td>INVERTER HEAT PUMP</td></tr><tr><td>BSB</td><td>BOTTOM OF DUCT</td><td>IRH</td><td>INFRARED RADIANT HEATER</td></tr><tr><td>BOD</td><td>BOTTOM OF PIPE</td><td></td><td></td></tr><tr><td>BOP</td><td>BOTTOM OF PIPE</td><td></td><td></td></tr><tr><td colspan="4"> </td></tr><tr><td>C</td><td>CONDENSATE</td><td>L</td><td>LEAKAGE AIR</td></tr><tr><td>CD</td><td>CONDENSER</td><td>LPS</td><td>LOW PRESSURE STEAM</td></tr><tr><td>CF</td><td>BYPASS CHEMICAL FEEDER</td><td>LV</td><td>LOUVER</td></tr><tr><td>CH</td><td>CHILLER</td><td>LEL</td><td>LOWER EXPLOSIVE LIMIT</td></tr><tr><td>CHR</td><td>CHILLED WATER RETURN</td><td>M</td><td></td></tr><tr><td>CHS</td><td>CHILLED WATER SUPPLY</td><td>MAU</td><td>MAKE-UP AIR UNIT</td></tr><tr><td>CO</td><td>CLEAN OUT</td><td>MD</td><td>MOTORIZED DAMPER</td></tr><tr><td>CR</td><td>CONDENSATE RETURN (STEAM)</td><td>MPS</td><td>MEDIUM PRESSURE STEAM</td></tr><tr><td>CRAC</td><td>COMPUTER ROOM AIR CONDITIONER</td><td>N</td><td></td></tr><tr><td>CU</td><td>CONDENSING UNIT</td><td>NTS</td><td>NOT TO SCALE</td></tr><tr><td>CWR</td><td>CONDENSER WATER RETURN</td><td>O</td><td></td></tr><tr><td>CWS</td><td>CONDENSER WATER SUPPLY</td><td>OA</td><td>OUTDOOR AIR</td></tr><tr><td colspan="4"> </td></tr><tr><td>D</td><td>DRY BULB</td><td>P</td><td>PUMP</td></tr><tr><td>DB</td><td>DEDICATED OUTDOOR AIR UNIT</td><td>PEL</td><td>PERMISSIBLE EXPOSURE LIMIT</td></tr><tr><td>DOAU</td><td>DEDICATED OUTDOOR AIR UNIT</td><td></td><td></td></tr><tr><td>DX</td><td>DIRECT EXPANSION</td><td>R</td><td></td></tr><tr><td>DP</td><td>DEW POINT</td><td>RA</td><td>RETURN AIR</td></tr><tr><td colspan="4"> </td></tr><tr><td>E</td><td>EXHAUST AIR</td><td>RH</td><td>RELATIVE HUMIDITY</td></tr><tr><td>EA</td><td>EXHAUST FAN</td><td>RM</td><td>RADIANT MANIFOLD</td></tr><tr><td>EF</td><td>ENERGY RECOVER PUMP</td><td>RTU</td><td>ROOF TOP UNIT</td></tr><tr><td>ERP</td><td>PUMP</td><td>S</td><td></td></tr><tr><td>ERR</td><td>ENERGY RECOVERY WATER RETURN</td><td>SA</td><td>SUPPLY AIR</td></tr><tr><td>ERS</td><td>ENERGY RECOVERY WATER SUPPLY</td><td>SCHP</td><td>SECONDARY CHILLED WATER PUMP</td></tr><tr><td>ERV</td><td>ENERGY RECOVERY VENTILATOR</td><td>SHWP</td><td>SECONDARY HOT WATER PUMP</td></tr><tr><td>ET</td><td>EXPANSION TANK</td><td>SM</td><td>SNOW MELT MANIFOLD</td></tr><tr><td colspan="4"> </td></tr><tr><td>F</td><td>FLEXIBLE CONNECTION</td><td>U</td><td>UNIT HEATER</td></tr><tr><td>FC</td><td>FAN COIL UNIT</td><td>UH</td><td></td></tr><tr><td>FCU</td><td>FIRE DAMPER</td><td>V</td><td>VARIABLE AIR VOLUME</td></tr><tr><td>FD</td><td>FAIL OPEN</td><td>VAV</td><td>VOLUME DAMPER</td></tr><tr><td>FO</td><td>FEET PER MINUTE</td><td>VD</td><td>VARIABLE</td></tr><tr><td>FP</td><td>FAN POWERED VAV</td><td>VRF</td><td>REFRIGERANT FLOW</td></tr><tr><td>FPVAV</td><td></td><td>VOC</td><td>VOLITILE ORGANIC COMPOUND</td></tr><tr><td colspan="4"> </td></tr><tr><td>G</td><td>GLYCOL FEEDER</td><td>VP</td><td>VACUUM PUMP</td></tr><tr><td>GF</td><td>GOVERNMENT FURNISH, INSTALLED</td><td>W</td><td></td></tr><tr><td>GFI</td><td>GOVERNMENT FURNISH, INSTALLED</td><td>WB</td><td>WET BULB</td></tr><tr><td colspan="4"> </td></tr><tr><td>GH</td><td>GRAVITY HOOD</td><td>Z</td><td></td></tr><tr><td>GR</td><td>GRAVITY RETURN</td><td>ZSB</td><td>ZONE SELECTOR BOX</td></tr><tr><td>GRV</td><td>GRAVITY ROOF VENTILATOR</td><td></td><td></td></tr><tr><td>GS</td><td>GLYCOL SUPPLY</td><td></td><td></td></tr></table>		A	AIR CONDITIONER (EXISTING)	H	HEATING COIL	AC	AIR DEVICE	HP	HEAT PUMP	AD	ABOVE FINISHED FLOOR	HPS	HIGH PRESSURE STEAM	AF	ABOVE FINISHED GRADE	HVLS	HIGH VOLUME LOW SPEED	AFG	ABOVE FINISHED GRADE	HVU	HEAT AND VENT UNIT	AHU	AIR HANDLING UNIT	HWR	HOT WATER RETURN	AL	ACOUSTICAL LINING	HWS	HOT WATER SUPPLY	AS	AIR SEPARATOR	HX	HEAT EXCHANGER					B	BRANCH SELECTOR BOX	I	INVERTER HEAT PUMP	BSB	BOTTOM OF DUCT	IRH	INFRARED RADIANT HEATER	BOD	BOTTOM OF PIPE			BOP	BOTTOM OF PIPE							C	CONDENSATE	L	LEAKAGE AIR	CD	CONDENSER	LPS	LOW PRESSURE STEAM	CF	BYPASS CHEMICAL FEEDER	LV	LOUVER	CH	CHILLER	LEL	LOWER EXPLOSIVE LIMIT	CHR	CHILLED WATER RETURN	M		CHS	CHILLED WATER SUPPLY	MAU	MAKE-UP AIR UNIT	CO	CLEAN OUT	MD	MOTORIZED DAMPER	CR	CONDENSATE RETURN (STEAM)	MPS	MEDIUM PRESSURE STEAM	CRAC	COMPUTER ROOM AIR CONDITIONER	N		CU	CONDENSING UNIT	NTS	NOT TO SCALE	CWR	CONDENSER WATER RETURN	O		CWS	CONDENSER WATER SUPPLY	OA	OUTDOOR AIR					D	DRY BULB	P	PUMP	DB	DEDICATED OUTDOOR AIR UNIT	PEL	PERMISSIBLE EXPOSURE LIMIT	DOAU	DEDICATED OUTDOOR AIR UNIT			DX	DIRECT EXPANSION	R		DP	DEW POINT	RA	RETURN AIR					E	EXHAUST AIR	RH	RELATIVE HUMIDITY	EA	EXHAUST FAN	RM	RADIANT MANIFOLD	EF	ENERGY RECOVER PUMP	RTU	ROOF TOP UNIT	ERP	PUMP	S		ERR	ENERGY RECOVERY WATER RETURN	SA	SUPPLY AIR	ERS	ENERGY RECOVERY WATER SUPPLY	SCHP	SECONDARY CHILLED WATER PUMP	ERV	ENERGY RECOVERY VENTILATOR	SHWP	SECONDARY HOT WATER PUMP	ET	EXPANSION TANK	SM	SNOW MELT MANIFOLD					F	FLEXIBLE CONNECTION	U	UNIT HEATER	FC	FAN COIL UNIT	UH		FCU	FIRE DAMPER	V	VARIABLE AIR VOLUME	FD	FAIL OPEN	VAV	VOLUME DAMPER	FO	FEET PER MINUTE	VD	VARIABLE	FP	FAN POWERED VAV	VRF	REFRIGERANT FLOW	FPVAV		VOC	VOLITILE ORGANIC COMPOUND					G	GLYCOL FEEDER	VP	VACUUM PUMP	GF	GOVERNMENT FURNISH, INSTALLED	W		GFI	GOVERNMENT FURNISH, INSTALLED	WB	WET BULB					GH	GRAVITY HOOD	Z		GR	GRAVITY RETURN	ZSB	ZONE SELECTOR BOX	GRV	GRAVITY ROOF VENTILATOR			GS	GLYCOL SUPPLY			<table><tr><td colspan="2">DUCTWORK SYMBOLS</td><td colspan="2">EQUIPMENT SYMBOLS</td></tr><tr><td></td><td>DUCT WITH ACOUSTICAL LINING MANUAL VOLUME DAMPER (SEE BELOW FOR OTHER TYPES)</td><td></td><td>NEW EQUIPMENT</td></tr><tr><td></td><td>DAMPER TYPE: FD - FIRE DAMPER SD - SMOKE DAMPER FSD - COMBINATION FIRE/SMOKE DAMPER BD - GRAVITY BACKDRAFT DAMPER CD - CONTROL DAMPER *NO MARK INDICATES MANUAL VOLUME DAMPER</td><td></td><td>EQUIPMENT CLEARANCE</td></tr><tr><td colspan="2"> </td><td colspan="2">CONTROLS SYMBOLS</td></tr><tr><td></td><td>DUCT ACCESS DOOR</td><td></td><td>ROOM THERMOSTAT</td></tr><tr><td></td><td>CEILING SUPPLY AIR DIFFUSER, ARROWS INDICATE PATTERN. 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CONTROL VALVE (PICV)	PROJECT: SATCOM OPS B100 HVAC REPAIR DESIGN CONDITIONS PER UFC 3-420-01, 3-410-01 WEATHER STATION - KADENA AFB, OKINAWA STATION #: 479310 ELEVATION: 48 METERS; LATITUDE: 26.35°N; LONGITUDE: 127.77°W ASHRAE CLIMATE ZONE 2A WINTER: 10.2°C DRY BULB (99%) SUMMER: DRY BULB CONDITIONS: 33.1°C DRY BULB (1%) DEHUMIDIFICATION: 29.9°C MEAN COINCIDENT DRY BULB 26.9°C MEAN COINCIDENT WET BULB 29.1°C DEW POINT INDOOR DESIGN CONDITIONS WINTER: 20°C DB (OFFICE/ADMIN AREAS) 22.2°C DB & 50%RH (OPERATIONS ROOM, COMM, TELEPHONE, NETWORK EQUIP) 20°C DB & 50%RH (UPS) SUMMER: 25.5°C DB/12.8°C WB(OFFICE/ADMIN AREAS) 22.2°C DB & 50%RH (OPERATIONS ROOM, COMM, TELEPHONE, NETWORK EQUIP) 20°C DB & 50%RH (UPS) CODE COMPLIANCE THE MECHANICAL AND PLUMBING EQUIPMENT IN THESE DRAWINGS ARE IN COMPLIANCE WITH 2024 INTERNATIONAL MECHANICAL CODE AS MODIFIED BY UFC 3-410-01. ENERGY CODE COMPLIANCE THE MECHANICAL AND PLUMBING EQUIPMENT IN THESE DRAWINGS ARE IN COMPLIANCE WITH ASHRAE 90.1-2019 AND UFC 1-200-02. THE HVAC LOADS WERE PERFORMED IN ACCORDANCE WITH ASHRAE FUNDAMENTALS. ENERGY CALCULATIONS WERE PERFORMED IN ACCORDANCE WITH ASHRAE 90.1-2019. GENERAL NOTES A. REFER TO GENERAL SHEETS FOR PHASING PLAN B. REFER TO SPECIFICATIONS FOR MATERIALS AND METHODS FOR CONSTRUCTION. C. COORDINATE LOCATION OF ALL PENETRATIONS WITH THE BUILDING STRUCTURE. D. COORDINATE ALL SLAB PENETRATIONS AND SLEEVES PRIOR TO EACH CONCRETE POUR. E. COORDINATE UNDERGROUND SITE PIPING AND TRENCH WITH SITE CIVIL INCLUDING FOUNDATIONS, SWALES, RETAINING WALLS, AND ANY OTHER POTENTIAL OBSTRUCTION. F. PERFORM WORK IN ACCORDANCE WITH THE LATEST EDITIONS, REVISIONS, AMENDMENTS OR SUPPLEMENTS OF APPLICABLE STATUTES, ORDINANCES, CODES, OR REGULATIONS OF FEDERAL, STATE AND LOCAL AUTHORITIES HAVING JURISDICTION IN EFFECT WHEN RFP WAS ISSUED. G. WHERE APPROVAL CODES HAVE BEEN ESTABLISHED BY OSHA, UNDERWRITERS LABORATORY, AMERICAN CODES, ANSI, ASME, ABAAG, ASHRAE, ASTM, ARI, NEC, NFPA, SMACNA, FOLLOW THESE STANDARDS WHETHER OR NOT INDICATED ON THE DRAWINGS. H. COORDINATE WORK WITH ARCHITECTURAL FEATURES AND COORDINATE WORK SUCH THAT INTERFERENCES BETWEEN PIPING, DUCTWORK, EQUIPMENT, PLUMBING WORK, ELECTRICAL WORK, AND BUILDING STRUCTURE WILL BE AVOIDED. I. PROVIDE THE ENTIRE SYSTEM AND ITS COMPONENT ITEMS OF EQUIPMENT IN OPERATING CONDITION WITHOUT OBJECTIONABLE VIBRATION OR NOISE IN ACCORDANCE WITH UFC-3-450-01 AND ASHRAE APPLICATIONS HANDBOOK. J. PROVIDE ALL WIRING AND ELECTRICAL CONTROLS IN ACCORDANCE WITH NFPA 70 (NEC). K. DO NOT SCALE THESE DRAWINGS FOR CONSTRUCTION PURPOSES. REFER TO STRUCTURAL DRAWINGS FOR FRAMING REQUIRED AROUND PIPES PENETRATING WALLS AND FLOORS. L. CONTRACTORS ARE FREE TO USE ANY PRODUCT WHICH MEETS THE GOVERNMENT'S MINIMUM REQUIREMENTS AND IS NOT IN CONFLICT WITH OTHER CONTRACT PROVISIONS. UNLESS INDICATED OTHERWISE ON THE PLANS OR SPECIFICATIONS (E.G. SOLE-SOURCE), ALL LISTED MANUFACTURERS ARE THE BASIS OF DESIGN. FOR THESE BASIS OF DESIGN PRODUCTS, OTHER MANUFACTURERS MAY BE SUBSTITUTED AS LONG AS THE STANDARD OF QUALITY AND CHARACTERISTICS OF THE SUBSTITUTED MANUFACTURER MEET OR EXCEED THE STANDARD OF QUALITY AND CHARACTERISTICS OF THE BASIS OF DESIGN. N. ALL DUCTWORK SIZES ARE FREE AIR DIMENSIONS. R. PROVIDE PLENUM RATED CABLE, DEVICES, PIPING, ETC. IN PLENUM SPACE. S. PROVIDE TEMPORARY 30% EFFICIENT, 50MM THICK FILTERS OVER RETURN AIR OPENINGS OF ANY AHU'S WHICH ARE PLACED IN SERVICE PRIOR TO COMPLETION OF CONSTRUCTION. REMOVE FILTERS FOR TEST & BALANCE. T. PROVIDE ACCESS PANELS FOR ALL DAMPER ACTUATORS AND MOTORS LOCATED INSIDE DUCT OR FAN HOUSING. U. COORDINATE LOCATION OF THERMOSTATS AND HUMIDITY SENSORS WITH FURNITURE, FIXTURES, AND EQUIPMENT PLANS.				<table><tr><td colspan="2">DRAWING/DETAIL REFERENCE KEY</td></tr><tr><td></td><td>REFER TO DRAWING/DETAIL NUMBER SHEET NUMBER ON WHICH DETAIL IS DRAWN</td></tr><tr><td>RE : 2 / M-201</td><td>M-201</td></tr><tr><td colspan="2">PIPING LABELS</td></tr><tr><td>CHS</td><td>CHILLED WATER SUPPLY</td></tr><tr><td>CHR</td><td>CHILLED WATER RETURN</td></tr><tr><td>CWS</td><td>CONDENSER WATER SUPPLY</td></tr><tr><td>CWR</td><td>CONDENSER WATER RETURN</td></tr><tr><td>HWS</td><td>HEATING WATER SUPPLY</td></tr><tr><td>HWR</td><td>HEATING WATER RETURN</td></tr><tr><td>CD</td><td>CONDENSATE DRAIN</td></tr><tr><td>D</td><td>DRAIN</td></tr><tr><td>STM</td><td>STEAM SUPPLY</td></tr><tr><td>STM</td><td>STEAM RETURN</td></tr><tr><td>MU</td><td>MAKE-UP WATER</td></tr><tr><td>RL</td><td>REFRIGERANT LIQUID</td></tr><tr><td>RS</td><td>REFRIGERANT SUCTION</td></tr><tr><td>PD</td><td>PUMP DISCHARGE</td></tr></table>				DRAWING/DETAIL REFERENCE KEY			REFER TO DRAWING/DETAIL NUMBER SHEET NUMBER ON WHICH DETAIL IS DRAWN	RE : 2 / M-201	M-201	PIPING LABELS		CHS	CHILLED WATER SUPPLY	CHR	CHILLED WATER RETURN	CWS	CONDENSER WATER SUPPLY	CWR	CONDENSER WATER RETURN	HWS	HEATING WATER SUPPLY	HWR	HEATING WATER RETURN	CD	CONDENSATE DRAIN	D	DRAIN	STM	STEAM SUPPLY	STM	STEAM RETURN	MU	MAKE-UP WATER	RL	REFRIGERANT LIQUID	RS	REFRIGERANT SUCTION	PD	PUMP DISCHARGE	<table><tr><td colspan="2">FOR REVIEW ONLY</td></tr><tr><td colspan="2">THESE DOCUMENTS ARE FOR CONCEPT DESIGN AND NOT INTENDED FOR CONSTRUCTION BIDDING OR PERMIT PURPOSES. 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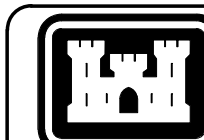
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G



A1 OVERALL DEMOLITION HVAC PLAN
1 : 200

GENERAL SHEET NOTES

- | | |
|---|--|
| A | ORIGINAL AIR FLOWS FOR EXISTING AIR DEVICES SHOWN. SYSTEM TO BE FULLY TESTED AND BALANCED PRIOR TO DEMOLITION. SUBMIT TAB REPORT TO GOVERNMENT. |
| B | REFER TO SHEET G010 AND G011 FOR PHASING PLAN. |
| C | EXISTING CEILING TO REMAIN, WITH PORTIONS REMOVED AS NECESSARY FOR ABOVE-CEILING CONSTRUCTION. REFER TO SHEETS AD122 AND AD123 FOR EXTENTS OF CEILING REMOVAL. |
| D | PHASE 1 TO BE FULLY TESTED AND BALANCED AND OPERATIONAL PRIOR TO BEGINNING PHASE 2. |
| E | PHASE 2 TO BE FULLY TESTED AND BALANCED AND OPERATIONAL PRIOR TO BEGINNING PHASE 3. |
| F | EXISTING CHW AND CRAC SYSTEMS TO REMAIN OPERATIONAL UNTIL NEW SYSTEM IS INSTALLED |
| G | ROUTE ALL CONDENSATE DRAIN LINES TO NEAREST FLOOR DRAIN. |
| H | AVOID ROUTING HYDRONIC PIPING OVER ELECTRICAL EQUIPMENT. |



**US Army Corps
of Engineers**
Japan District ®

FOR REVIEW
ONLY

THESE DOCUMENTS ARE FOR
CONCEPT DESIGN AND NOT
INTENDED FOR CONSTRUCTION
BIDDING OR PERMIT PURPOSES.
THEY WERE PREPARED BY, OR
UNDER SUPERVISION OF:

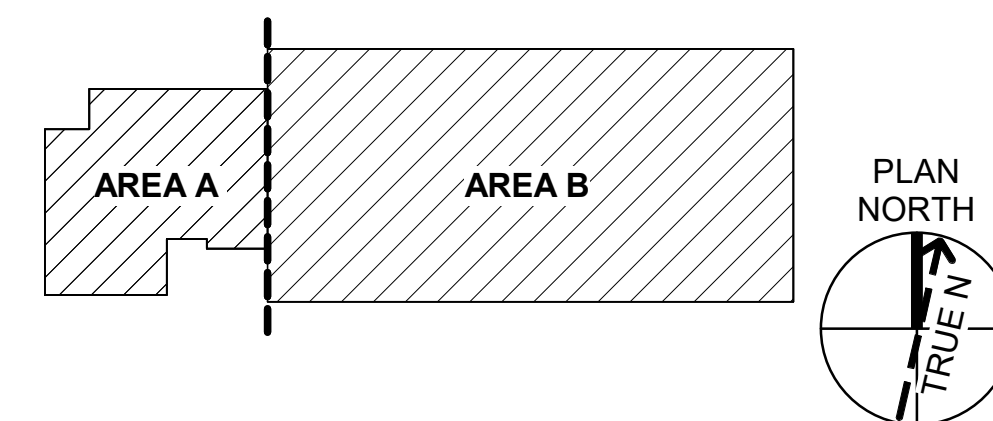
BRIAN C. ALLERHEILIGEN

NAME	
93876	09/04/2021
REGISTRATION #	DATE
TEXAS REGISTERED ENGINEERING FIRM F-2966	

[illegible]

US ARMY CORPS OF ENGINEERS JAPAN ENGINEER DISTRICT OKINAWA, JAPAN		DESIGNED BY: K. WILKINS		ISSUE DATE: 09/04/2025	
DRAWN BY: J. WILKINS		CHECKED BY: B. SCHNEIDER		SOLICITATION NO.: -	
SUBMITTED BY: J. WILKINS		CONTRACT NO.: -		DRAWING CODE: -	
FILE NAME: ISO A1		SIZE: MD101			

KEY PLAN



0 4000 8000
SCALE: 1:200

AIR HVAC WIDEBAND SATCOM OPS B100
 P BUCKNER COMMUNICATIONS ANNEX
 OKINAWA, JAPAN
 PROJECT NUMBER:D3449305

OVERALL DEMOLITION HVAC PLAN

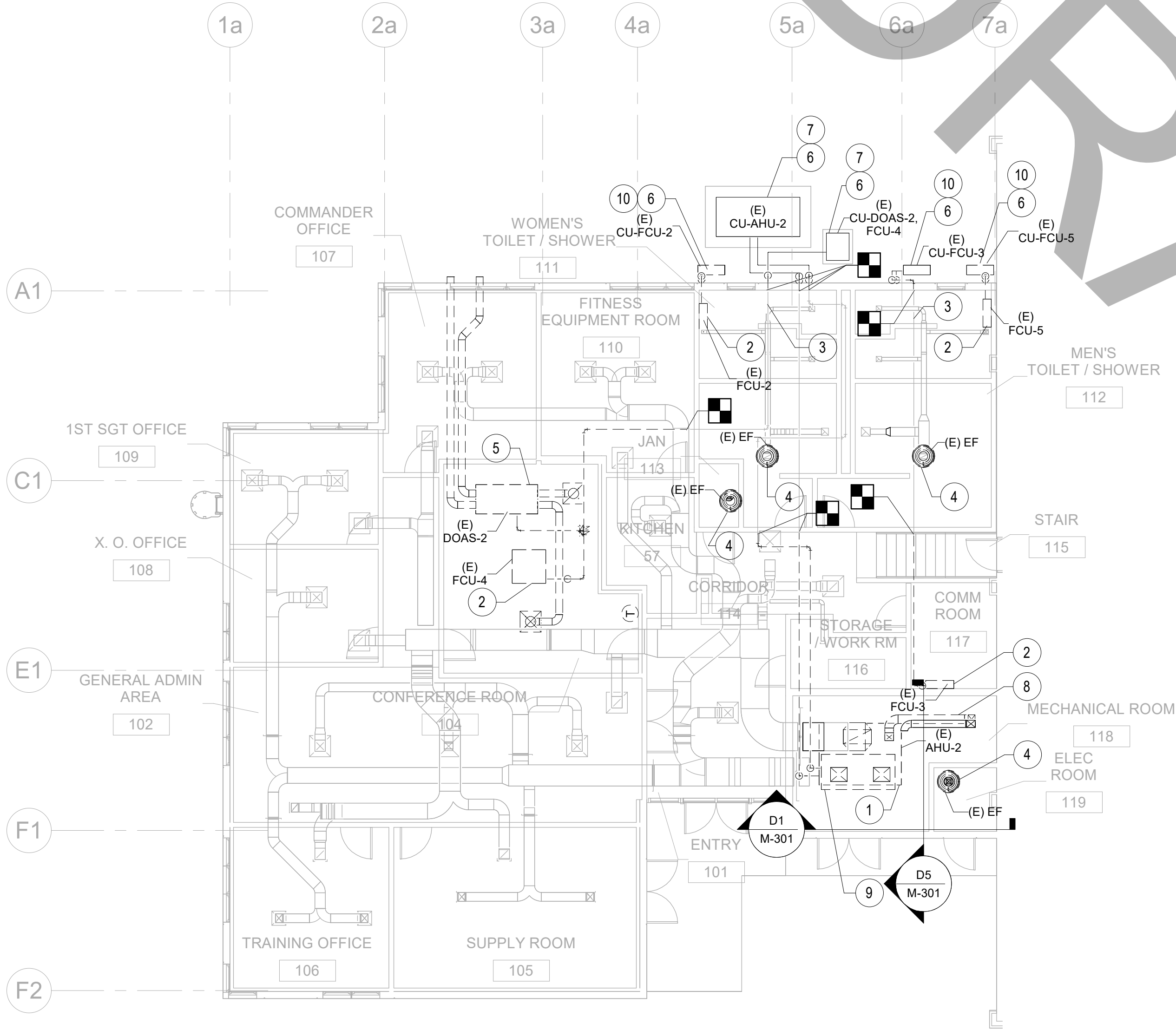
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1 2 3 4 5 6 7 8 9 10



A1 PARTIAL DEMOLITION HVAC - AREA A
1 : 100

GENERAL SHEET NOTES

- A ORIGINAL AIR FLOWS FOR EXISTING AIR DEVICES SHOWN. SYSTEM TO BE FULLY TESTED AND BALANCED PRIOR TO DEMOLITION. SUBMIT TAB REPORT TO GOVERNMENT.
- B REFER TO SHEET GI010 AND GI011 FOR PHASING PLAN.
- C EXISTING CEILING TO REMAIN, WITH PORTIONS REMOVED AS NECESSARY FOR ABOVE-CEILING CONSTRUCTION. REFER TO SHEETS AD122 AND AD123 FOR EXTENTS OF CEILING REMOVAL.
- D PHASE 1 TO BE FULLY TESTED AND BALANCED AND OPERATIONAL PRIOR TO BEGINNING PHASE 2.
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- G ROUTE ALL CONDENSATE DRAIN LINES TO NEAREST FLOOR DRAIN.
- H AVOID ROUTING HYDRONIC PIPING OVER ELECTRICAL EQUIPMENT.

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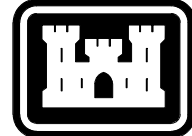
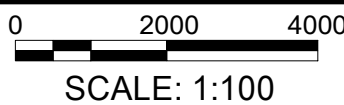
SHEET KEYNOTES

- 1 PHASE 3: REMOVE EXISTING AIR HANDLING UNIT, ASSOCIATED REFRIGERANT PIPING AND CONDENSING UNIT. EXISTING DUCTWORK TO REMAIN.
- 2 PHASE 3: REMOVE EXISTING SPLIT SYSTEM, ASSOCIATED REFRIGERANT PIPING, THERMOSTAT AND CONDENSING UNIT.
- 3 PHASE 3: EXISTING REFRIGERANT PIPING LOCATED ABOVE RESTROOMS TO BE ABANDONED IN PLACE.
- 4 PHASE 3: REMOVE EXISTING ROOF MOUNTED EXHAUST FAN. EXISTING EXHAUST DUCTWORK TO REMAIN.
- 5 PHASE 3: REMOVE EXISTING DOAS UNIT, ASSOCIATED REFRIGERANT PIPING, DUCTWORK AND CONDENSING UNIT. RE: SHEET AD-122 FOR EXTERIOR WALL PATCHING.
- 6 PHASE 3: REMOVE EXISTING CONDENSING UNIT AND ASSOCIATED REFRIGERANT PIPING. RE: ARCHITECTURE FOR PATCHING AND FOR REMOVAL OF PIPING THROUGH EXTERIOR WALL, RE: SHEET AD-122 FOR WALL PATCHING.
- 7 EXISTING CONCRETE PAD TO REMAIN.
- 8 REMOVE EXISTING DUCTWORK.
- 9 DISCONNECT EXISTING SUPPLY PLENUM AND ASSOCIATED DUCTWORK CONNECTED TO THE EXISTING AHU BEING REMOVED.
- 10 EXISTING WALL MOUNT TO REMAIN AND BE RE-USED.

KEY PLAN



GRAPHIC SCALE(S)



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REGISTRATION # DATE
TEXAS REGISTERED ENGINEERING FIRM F-2966

ISSUE DATE: 09/04/2025
SOLICITATION NO.:
CONTRACT NO.:
DRAWING CODE:
FILE NAME: MD102

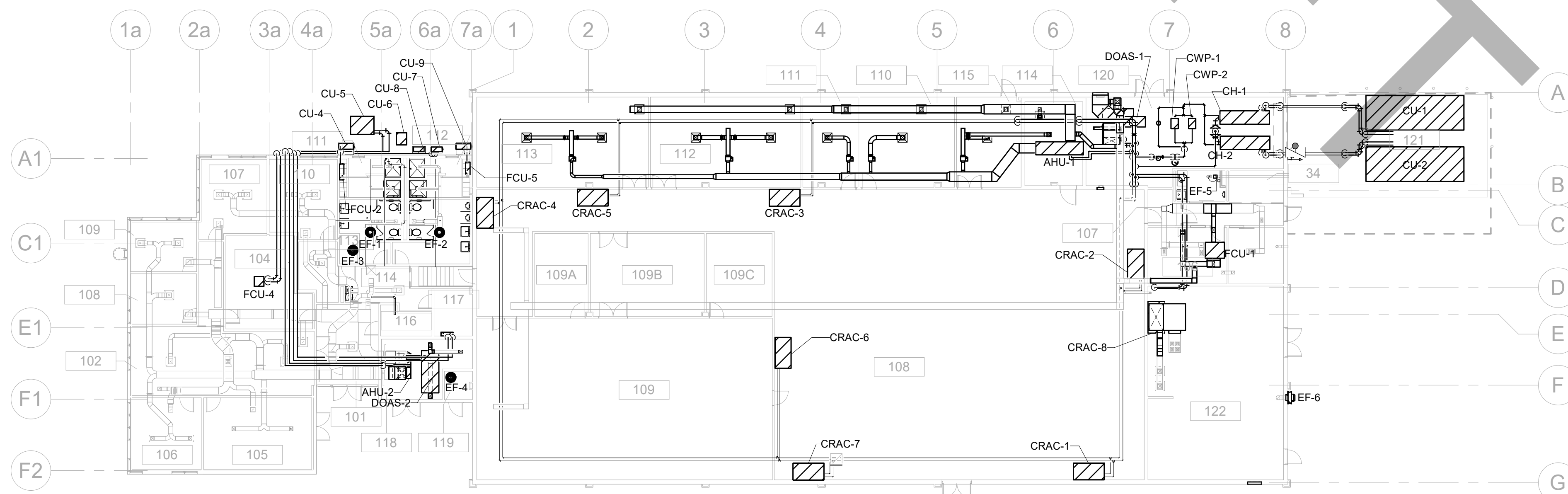
DESIGNED BY: R. WILKINS
CHECKED BY: C. HUFFINES
SUBMITTED BY: B. SCHNEIDER
SIZE: A
FILE NAME: MD102

Jacobs
JACOBS ENGINEERING GROUP, INC.
7700 WEST HIGHTWAY 141
HOUSTON, TEXAS 77057
USA

FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX
OKINAWA, JAPAN
PROJECT NUMBER: D3449305
PARTIAL DEMOLITION HVAC - AREA A

SHEET ID

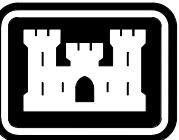
MD102



A1 OVERALL HVAC PLAN
1 : 200

GENERAL SHEET NOTES

- | | |
|---|--|
| A | ORIGINAL AIR FLOWS FOR EXISTING AIR DEVICES SHOWN. SYSTEM TO BE FULLY TESTED AND BALANCED PRIOR TO DEMOLITION. SUBMIT TAB REPORT TO GOVERNMENT. |
| B | REFER TO SHEET G1010 AND G1011 FOR PHASING PLAN. |
| C | EXISTING CEILING TO REMAIN, WITH PORTIONS REMOVED AS NECESSARY FOR ABOVE-CEILING CONSTRUCTION. REFER TO SHEETS AD122 AND AD123 FOR EXTENTS OF CEILING REMOVAL. |
| D | PHASE 1 TO BE FULLY TESTED AND BALANCED AND OPERATIONAL PRIOR TO BEGINNING PHASE 2. |
| E | PHASE 2 TO BE FULLY TESTED AND BALANCED AND OPERATIONAL PRIOR TO BEGINNING PHASE 3. |
| F | EXISTING CHW AND CRAC SYSTEMS TO REMAIN OPERATIONAL UNTIL NEW SYSTEM IS INSTALLED |
| G | ROUTE ALL CONDENSATE DRAIN LINES TO NEAREST FLOOR DRAIN. |
| H | AVOID ROUTING HYDRONIC PIPING OVER ELECTRICAL EQUIPMENT. |



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AN C. ALLERHEILIG

NAME 09/04

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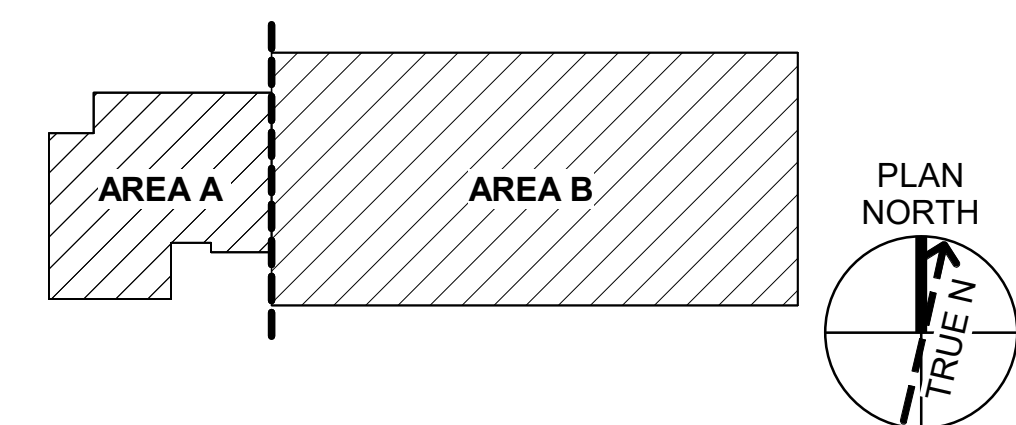
JACOBSON ENGINEER DISTRICT OKINAWA, JAPAN	DRAWN BY: K. WILKINS	SOLICITATION NO.: 09/04/2025
	CHECKED BY: C. HUFFINES	CONTRACT NO.: -
 JACOBS JACOBS GOVERNMENT SERVICES COMPANY 777 MAIN STREET, FORT WORTH, TX 76102 © 2005 Jacobs	SUBMITTED BY: A. GUERRA	DRAWING CODE: -
	FILE NAME: -	SIZE: 10.01

CAMP BUCKNER COMMUNICATIONS ANNEX
OKINAWA, JAPAN
PROJECT NUMBER:D3449305
OVERALL HVAC PLAN

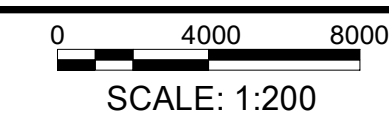
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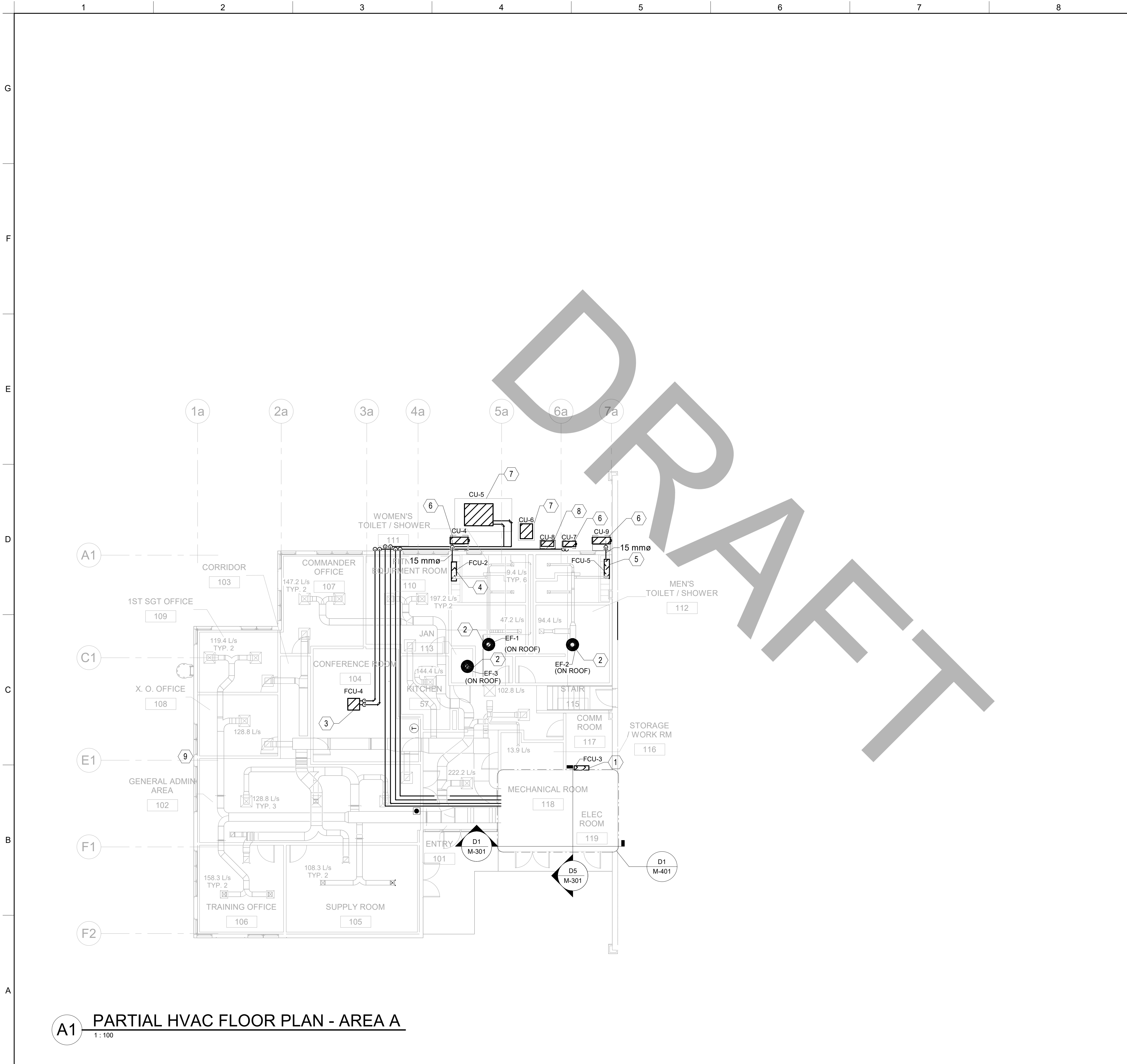
KEY PLAN



GRAPHIC SCALE(S)



M M-102-



GENERAL SHEET NOTES	
A	ORIGINAL AIR FLOWS FOR EXISTING AIR DEVICES SHOWN. SYSTEM TO BE FULLY TESTED AND BALANCED PRIOR TO DEMOLITION. SUBMIT TAB REPORT TO GOVERNMENT.
B	REFER TO SHEET GI010 AND GI011 FOR PHASING PLAN.
C	EXISTING CEILING TO REMAIN, WITH PORTIONS REMOVED AS NECESSARY FOR ABOVE-CEILING CONSTRUCTION. REFER TO SHEETS AD122 AND AD123 FOR EXTENTS OF CEILING REMOVAL.
D	PHASE 1 TO BE FULLY TESTED AND BALANCED AND OPERATIONAL PRIOR TO BEGINNING PHASE 2.
E	PHASE 2 TO BE FULLY TESTED AND BALANCED AND OPERATIONAL PRIOR TO BEGINNING PHASE 3.
F	EXISTING CHW AND CRAC SYSTEMS TO REMAIN OPERATIONAL UNTIL NEW SYSTEM IS INSTALLED
G	ROUTE ALL CONDENSATE DRAIN LINES TO NEAREST FLOOR DRAIN.
H	AVOID ROUTING HYDRONIC PIPING OVER ELECTRICAL EQUIPMENT.

SHEET KEYNOTES	
1	PHASE 3: PROVIDE NEW WALL MOUNTED MINI SPLIT UNIT, FCU-3, AND ASSOCIATED CONDENSING UNIT, CU-7.
2	PHASE 3: PROVIDE NEW ROOF MOUNTED EXHAUST FAN. CONNECT TO EXISTING EXHAUST DUCTWORK. REFER TO AD131 FOR ADDITIONAL INFORMATION.
3	PHASE 3: PROVIDE CEILING MOUNTED CASSETTE STYLE MINI SPLIT SYSTEM, FCU-4, AND ASSOCIATED CONDENSING UNIT, CU-8.
4	PHASE 3: PROVIDE NEW WALL MOUNTED MINI SPLIT UNIT, FCU-2 AND ASSOCIATED CONDENSING UNIT, CU-4. CONNECT TO EXISTING CONDENSATE DRAIN.
5	PHASE 3: PROVIDE NEW WALL MOUNTED MINI SPLIT UNIT, FCU-5 AND ASSOCIATED CONDENSING UNIT, CU-9. CONNECT TO EXISTING CONDENSATE DRAIN.
6	RE-USE EXISTING WALL MOUNT FOR NEW CONDENSING UNIT.
7	RE-USE EXISTING HOUSEKEEPING PAD FOR NEW CONDENSING UNIT.
8	PROVIDE NEW HOUSEKEEPING PAD FOR NEW CONDENSING UNIT. REFER TO STRUCTURAL FOR DETAIL.
9	EXISTING RADON FAN IS OPERATIONAL AND SHOULD REMAIN.

KEY PLAN

PLAN NORTH

GRAPHIC SCALE(S)

SCALE: 1:100

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BRIAN C. ALLERHEILIGEN
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ISSUE DATE: 09/04/2025	SOLICITATION NO.:	CONTRACT NO.:	DRAWING CODE:
DESIGNED BY: A. WILKINS	CHECKED BY: C. HUFFINES	SUBMITTED BY: B. SCHNEIDER	FILE NAME: A. GUERRA
SIZE: ISO A1			

US ARMY CORPS OF ENGINEERS
JAPAN ENGINEER DISTRICT
OKINAWA, JAPAN

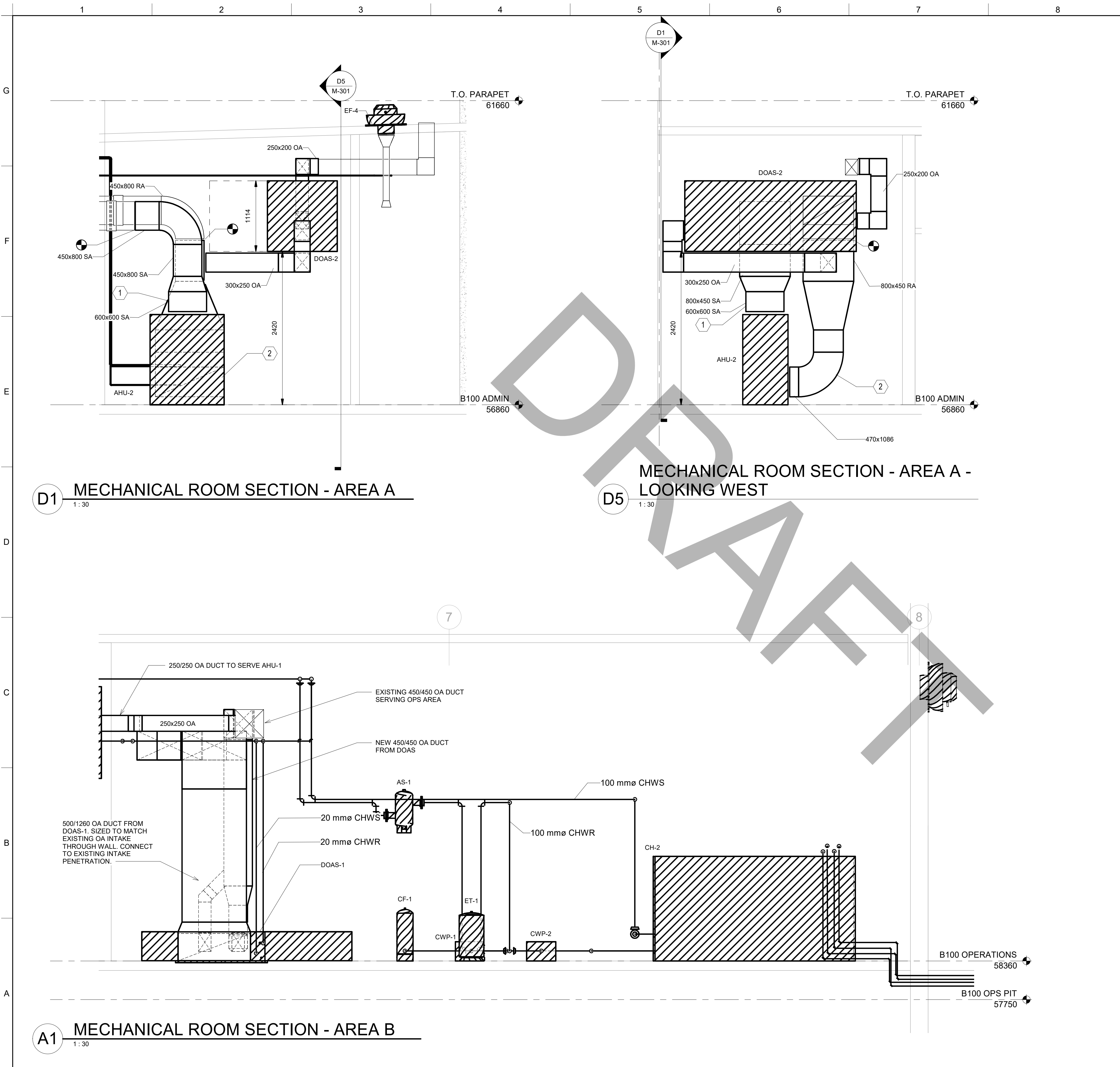
Jacobs
JACOBS ENGINEERING GROUP, INC.
171 MAIN STREET, SUITE 100
USA

FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX
OKINAWA, JAPAN
PROJECT NUMBER: D3448305

PARTIAL HVAC FLOOR PLAN - AREA A

SHEET ID

M-102



GENERAL SHEET NOTES

SHEET KEYNOTES

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NAME

09/04/2025

DATE

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TEXAS REGISTERED ENGINEERING FIRM F-2566

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DESCRIPTION

ISSUE DATE:

09/04/2025

SOLICITATION NO.:

CONTRACT NO.:

DRAWING CODE:

DESIGNED BY:

R. WILKINS

CHECKED BY:

C. HUFFINES

SUBMITTED BY:

B. SCHNEIDER

FILE NAME:

A. GUERRA

SIZE:

ISO A1

M-301

US ARMY CORPS OF ENGINEERS

JAPAN ENGINEER DISTRICT

OKINAWA, JAPAN

JACOBS

JACOBS ENGINEERING GROUP, INC.

77 MAWA STREET, SUITE 100, WYOMING, USA

FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100

CAMP BUCKNER COMMUNICATIONS ANNEX

OKINAWA, JAPAN

PROJECT NUMBER: D3449305

HVAC BUILDING SECTION

SHEET ID

M-301

GRAPHIC SCALE(S)

KEY PLAN

AREA A

AREA B

PLAN NORTH

TRUE N

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TEXAS REGISTERED
ENGINEERING FIRM F-2966

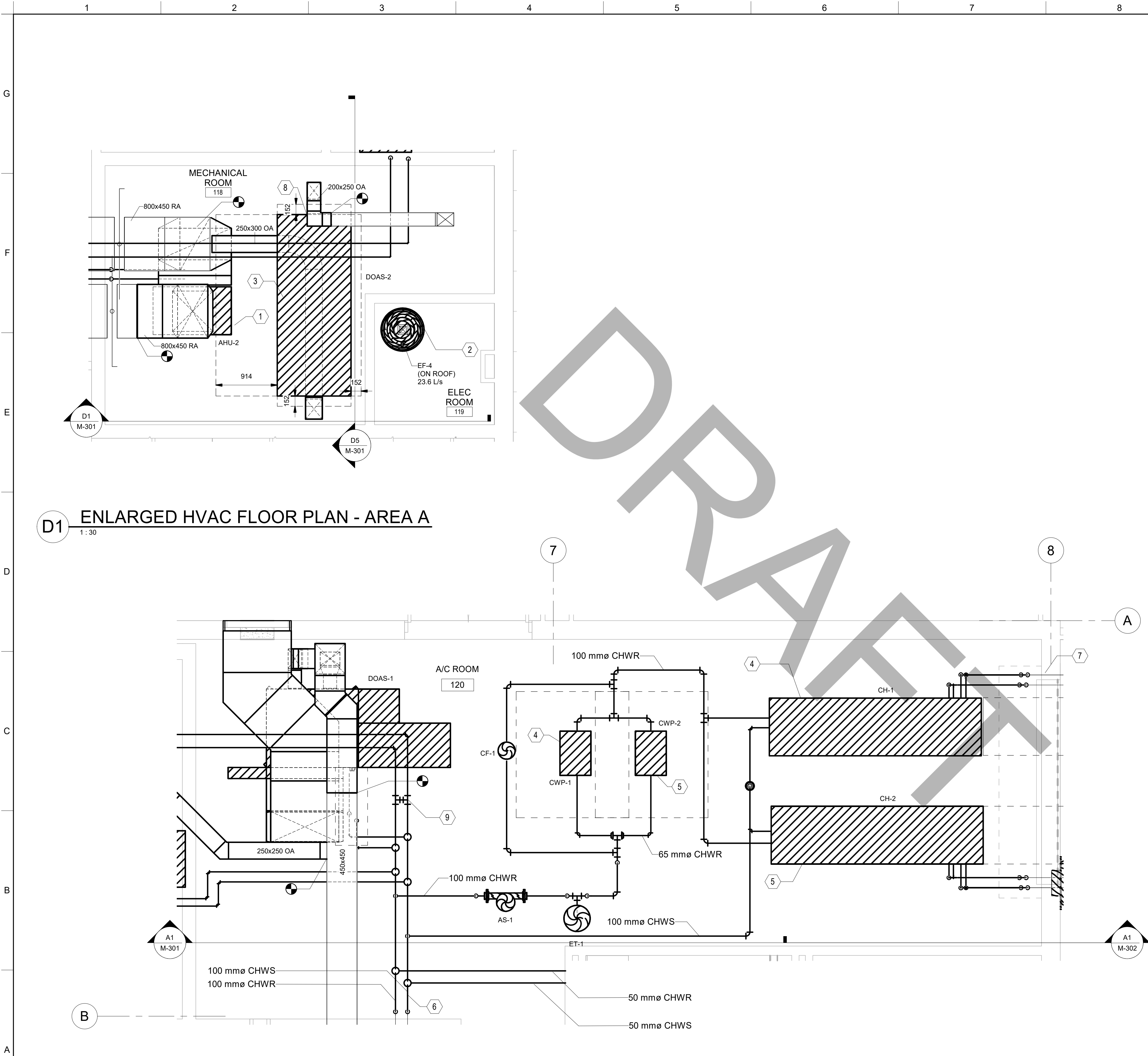
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US ARMY CORPS OF ENGINEERS JAPAN ENGINEER DISTRICT OKINAWA, JAPAN	DESIGNED BY: K. WILKINS C. HARRIS C. HARRIS	ISSUE DATE: 09/04/2025
	CHECKED BY: B. SCHNEIDER	CONTRACT NO.: .
 JACOBS	SUBMITTED BY: A. GUERRA	DRAWING CODE: .
	SIZE: ISO A1	FILE NAME: M-302

Diagram illustrating the area to be measured (AREA B). The area is bounded by a stepped line on the left and a hatched area labeled "AREA B".

0 500 1000 2000
SCALE: 1:50

M-302



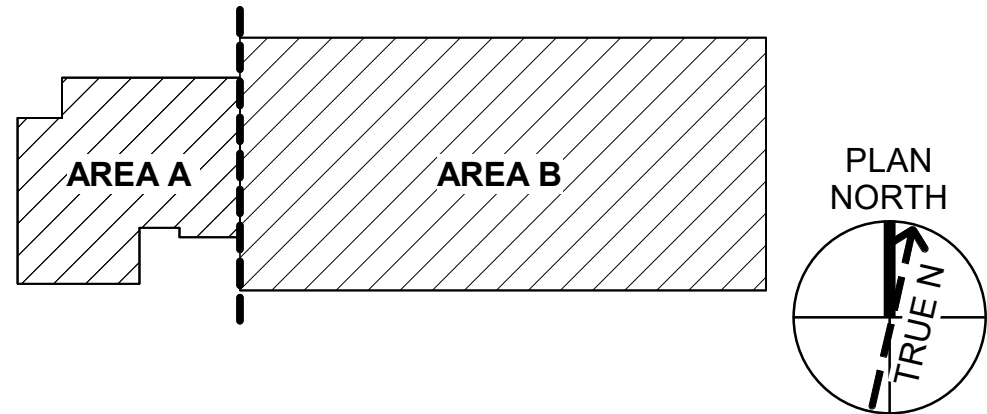
D1 ENLARGED HVAC FLOOR PLAN - AREA A
1:30

A1 ENLARGED HVAC FLOOR PLAN - AREA B
1:30

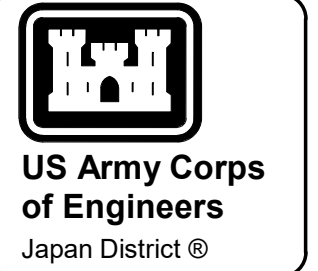
GENERAL SHEET NOTES

SHEET KEYNOTES	
1	PHASE 3: PROVIDE NEW DX AIR HANDLING UNIT, AHU-2, AND ASSOCIATED CONDENSING UNIT, CU-5. CONNECT TO EXISTING DUCTWORK.
2	PHASE 3: PROVIDE NEW ROOF MOUNTED EXHAUST FAN. CONNECT TO EXISTING EXHAUST DUCTWORK. REFER TO AD131 FOR ADDITIONAL INFORMATION.
3	PHASE 3: PROVIDE NEW DOAS UNIT, DOAS-2, AND ASSOCIATED CONDENSING UNIT, CU-6. SUSPEND UNIT FROM STRUCTURE ABOVE. DUCT DIRECTLY INTO THE RETURN DUCT OF AHU-2.
4	PHASE 2: PROVIDE NEW CHILLER, CH-1, ASSOCIATED PUMP, CWP-1, CONDENSING UNIT, CU-1 AND NEW REFRIGERANT PIPING.
5	PHASE 1: PROVIDE NEW CHILLER, CH-2, ASSOCIATED PUMP, CWP-2, CONDENSING UNIT, CU-2 AND NEW REFRIGERANT PIPING.
6	PHASE 1: PROVIDE NEW CHILLED WATER PIPING LOOP.
7	REFRIGERANT LINES ROUTED IN EXISTING TRENCH OUT TO CONDENSING UNITS.
8	BLANK OFF UNUSED PORTION OF UNIT OPENING AND CONNECT TO 200MM X 250MM INLET.
9	CHW BYPASS.

KEY PLAN



GRAPHIC SCALE(S)



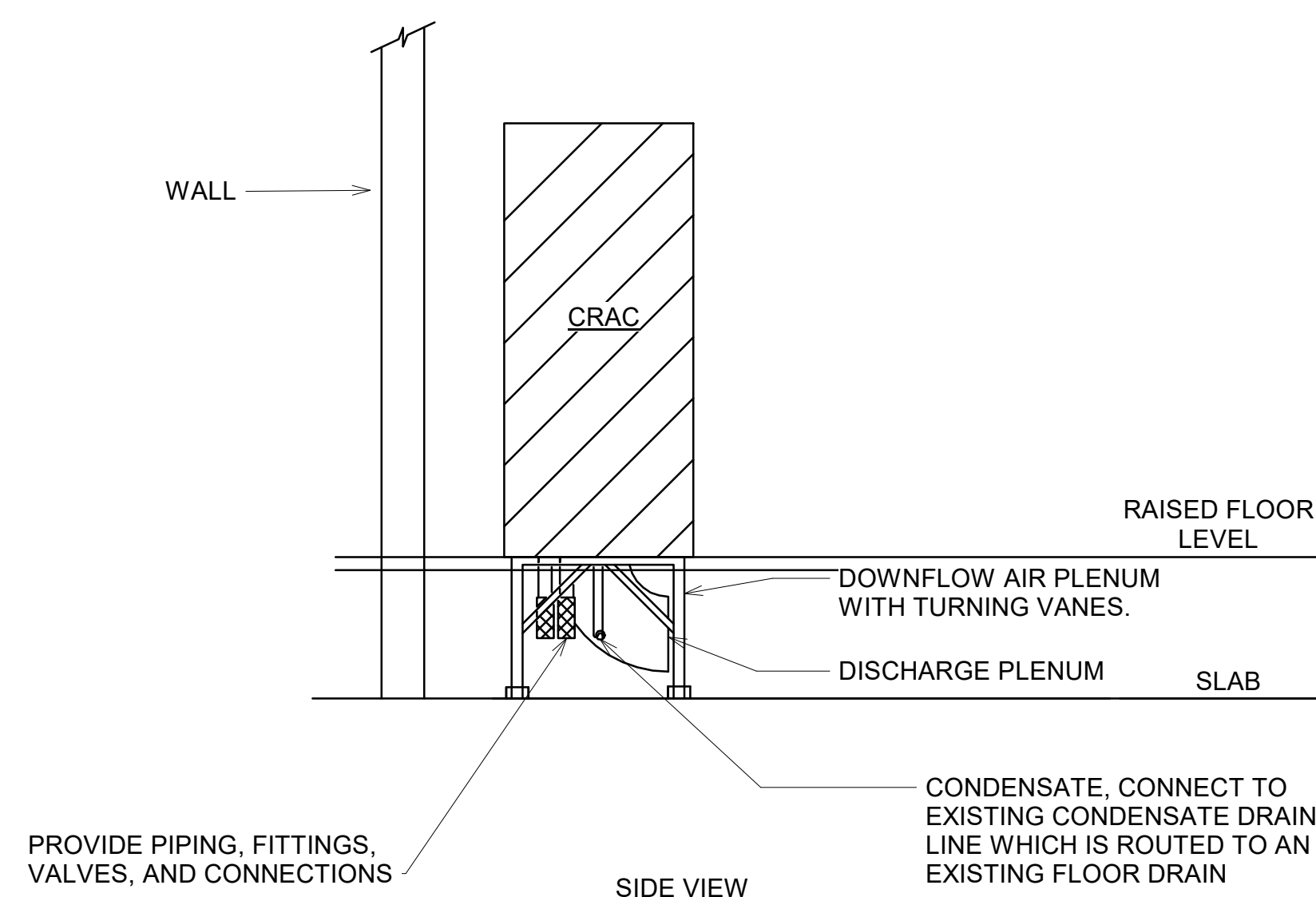
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BRIAN C. ALLERHEILIGEN
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SUBMITTED BY: B. SCHNEIDER	DRAWING CODE:	
FILE NAME: A. GUERRA	SIZE:	ISO A1

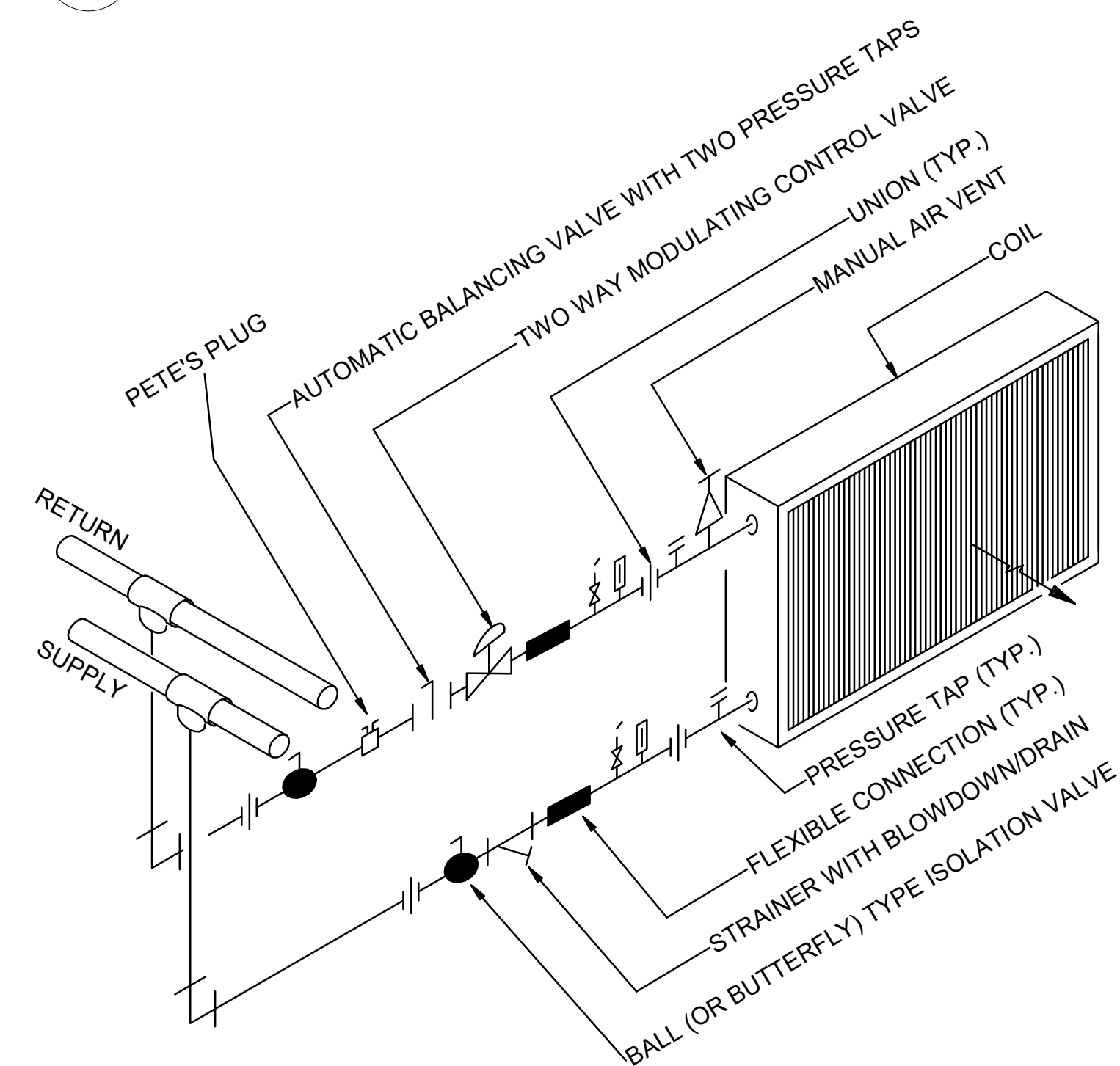
US ARMY CORPS OF ENGINEERS
JAPAN ENGINEER DISTRICT
OKINAWA, JAPAN
PROJECT NUMBER: D3449305
HVAC ENLARGED PLANS

SHEET ID

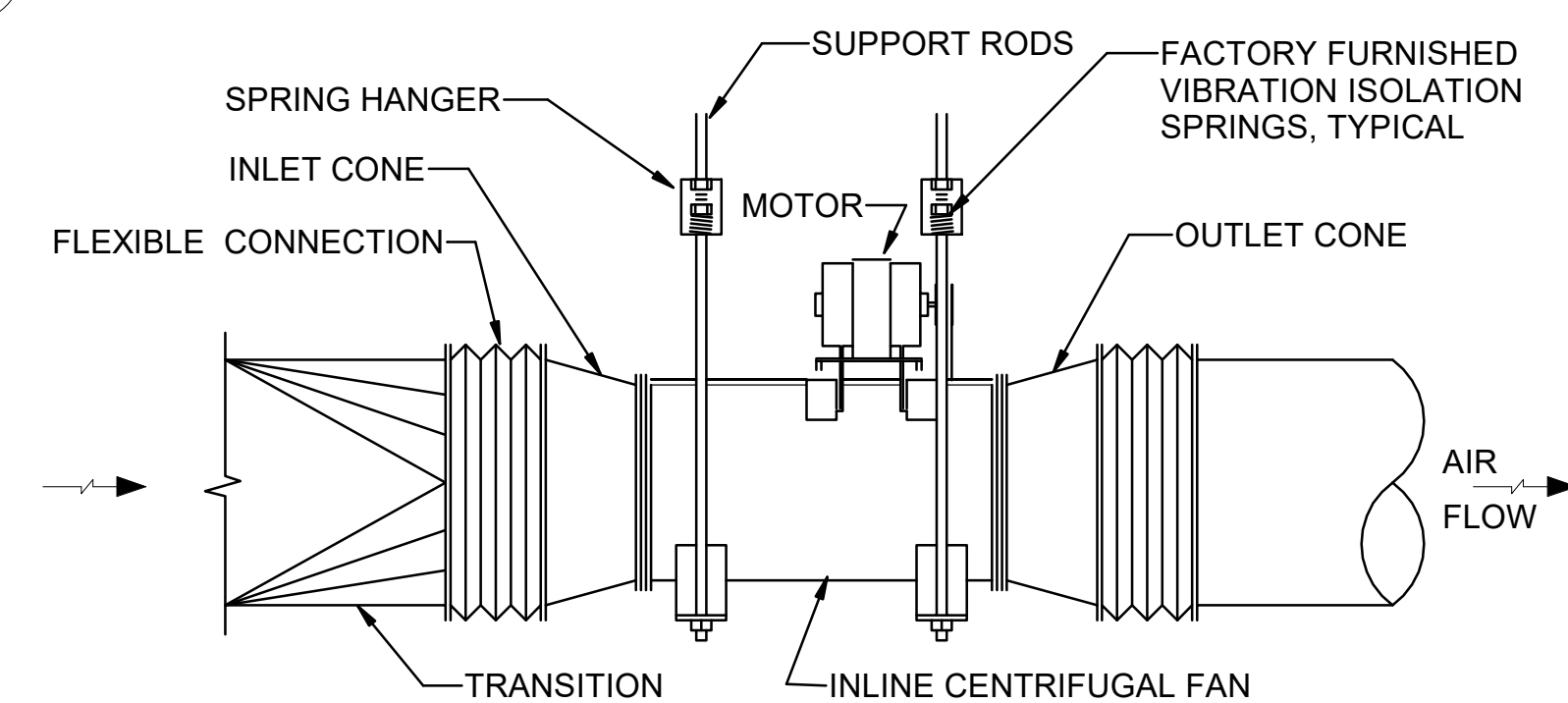
M-401



E1 TYPICAL CRAC DETAIL (DOWNFLOW)
NTS

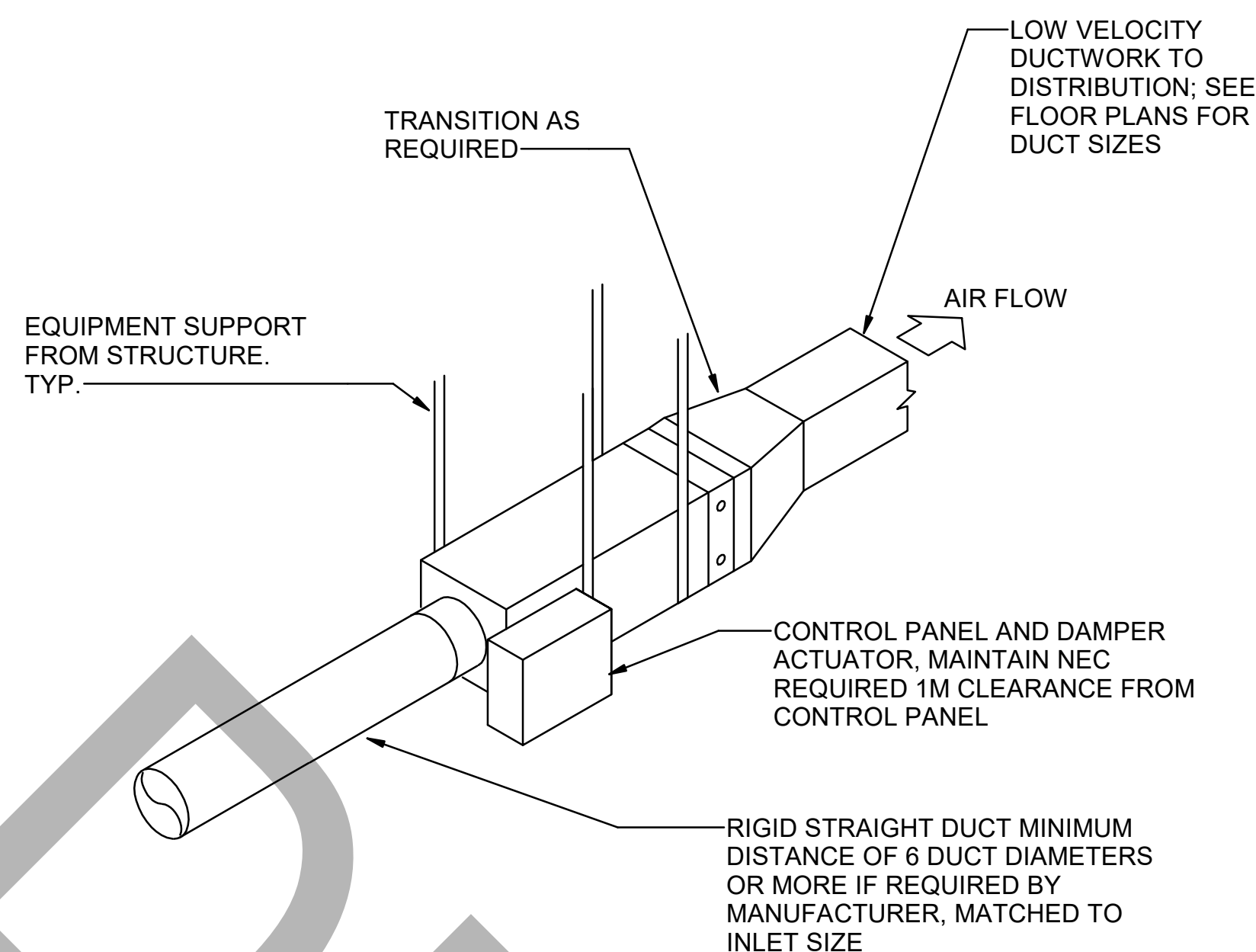


WATER COIL PIPING (TWO-WAY VALVE SINGLE COIL)



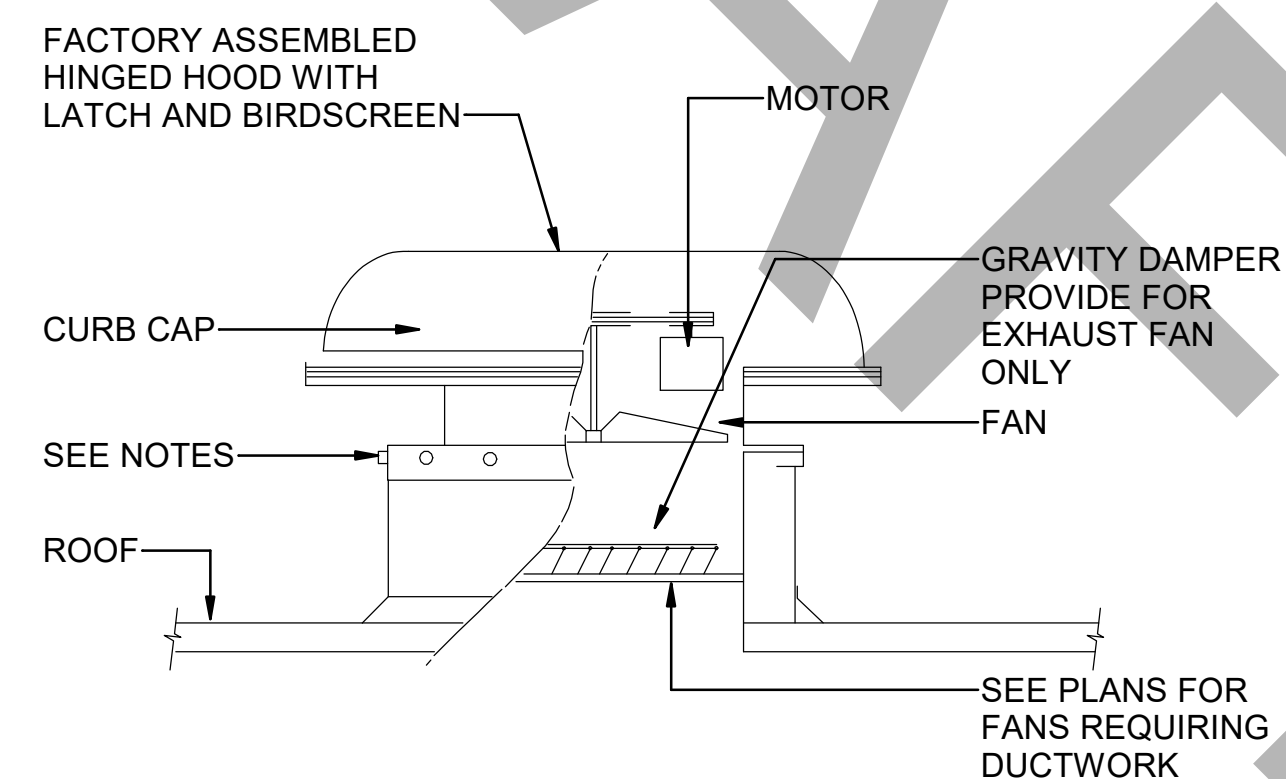
NOTE : PROVIDE INLET BELL FOR FANS WITHOUT INLET DUCT

A1 INLINE CENTRIFUGAL FAN DETAIL



NOTES:
1. DO NOT SUPPORT UNIT WITH DUCTWORK (SUPPORT SEPARATELY).

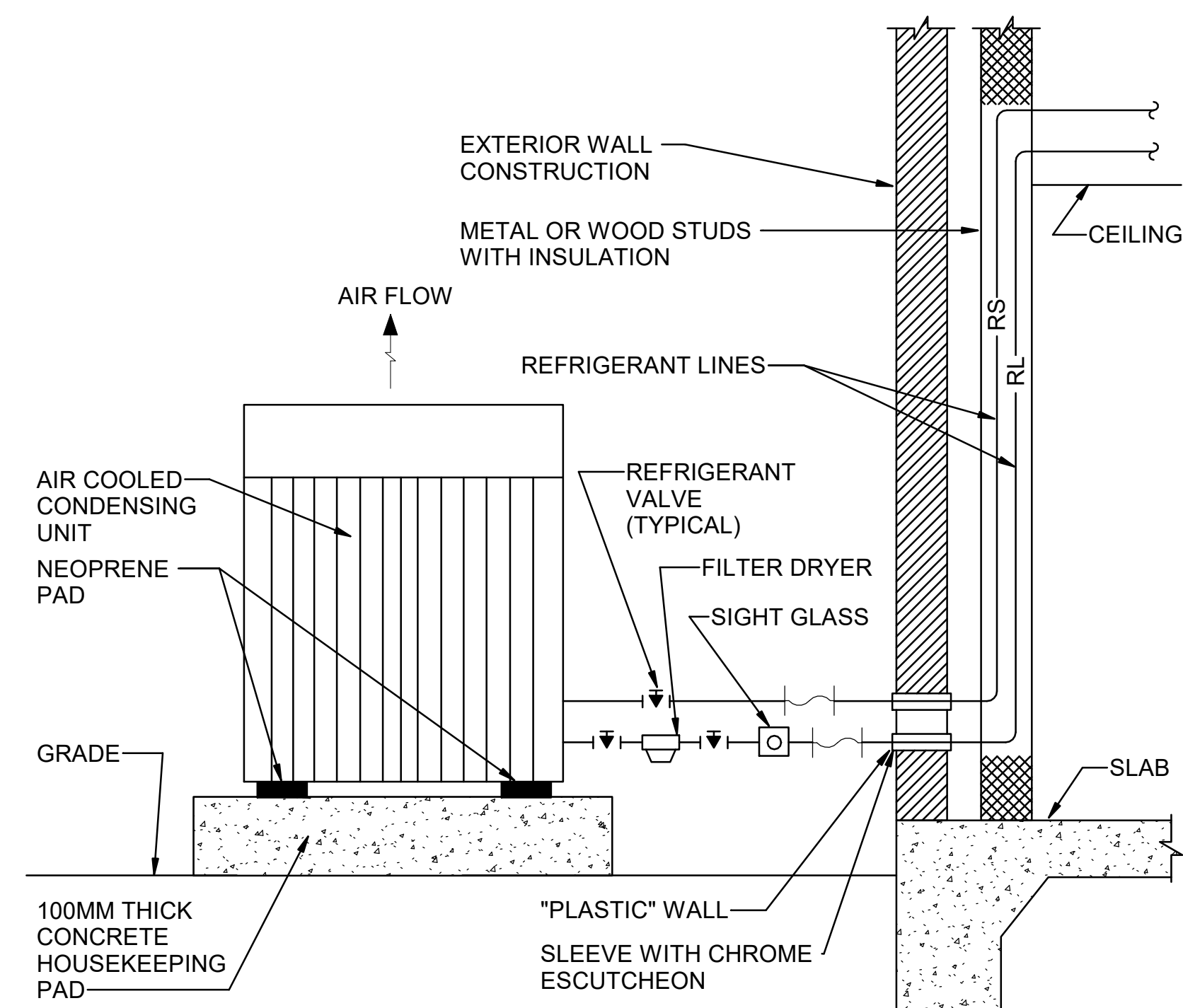
D4 TYPICAL SINGLE DUCT TERMINAL UNIT



NOTES:

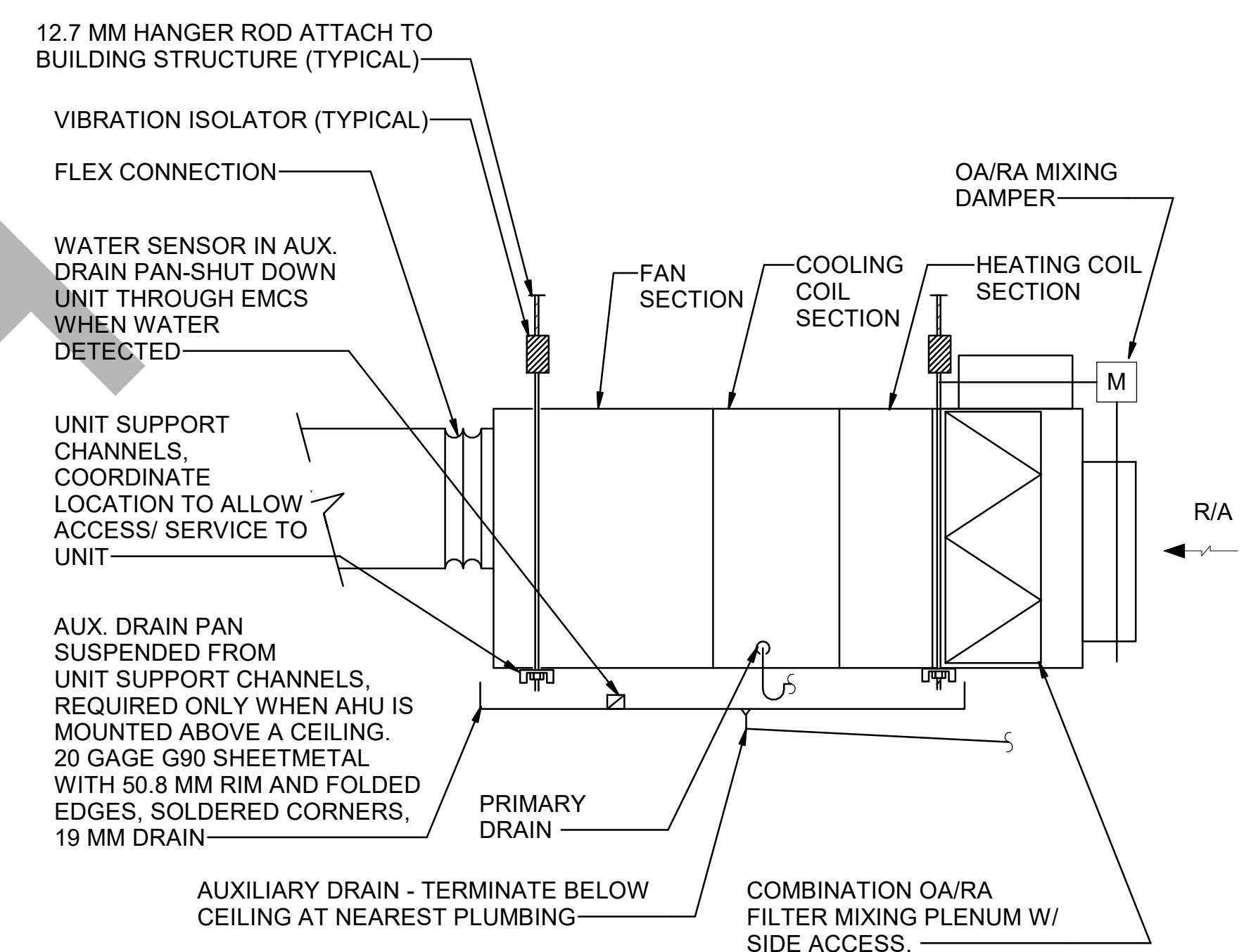
1. WHEN PLATE IS PROVIDED AROUND TOP OF CURB:
 - A. SECURE HOOD TO PLATE WITH 6.35 MM X 38.1 MM LG CADMIUM PLATED LAG BOLTS NOT OVER 304.8 MM OC.
 - B. EXTEND DUCT TO FULL HEIGHT OF CURB AND FASTEN HEM EDGE OF DUCT TO PLATE WITH SCREWS NOT OVER 152.4 MM OC.
2. WHEN PREFABRICATED METAL CURB IS USED, SECURE HOOD AND DUCTWORK WITH SHEETMETAL SCREWS AS REQUIRED FOR TIGHT JOINTS AND RIGID INSTALLATION.
3. PROVIDE PREFINISHED ALUMINUM BASE FLASHING AT EXISTING ROOF EQUIPMENT CURB FOR WATERTIGHT EXHAUST FAN INSTALLATION.

B4 ROOF EXHAUST FAN

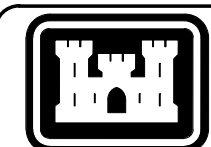


NOTE: REFRIGERANT LINE ACCESSORIES SHOWN ARE MINIMUM REQUIRED. PROVIDE ADDITIONAL ACCESSORIES AS REQUIRED BY MANUFACTURER.

D8 AIR COOLED CONDENSING UNIT



A8 FCU MOUNTING DETAIL



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93876 09/04/2025

REGISTRATION # DATE

TEXAS REGISTERED
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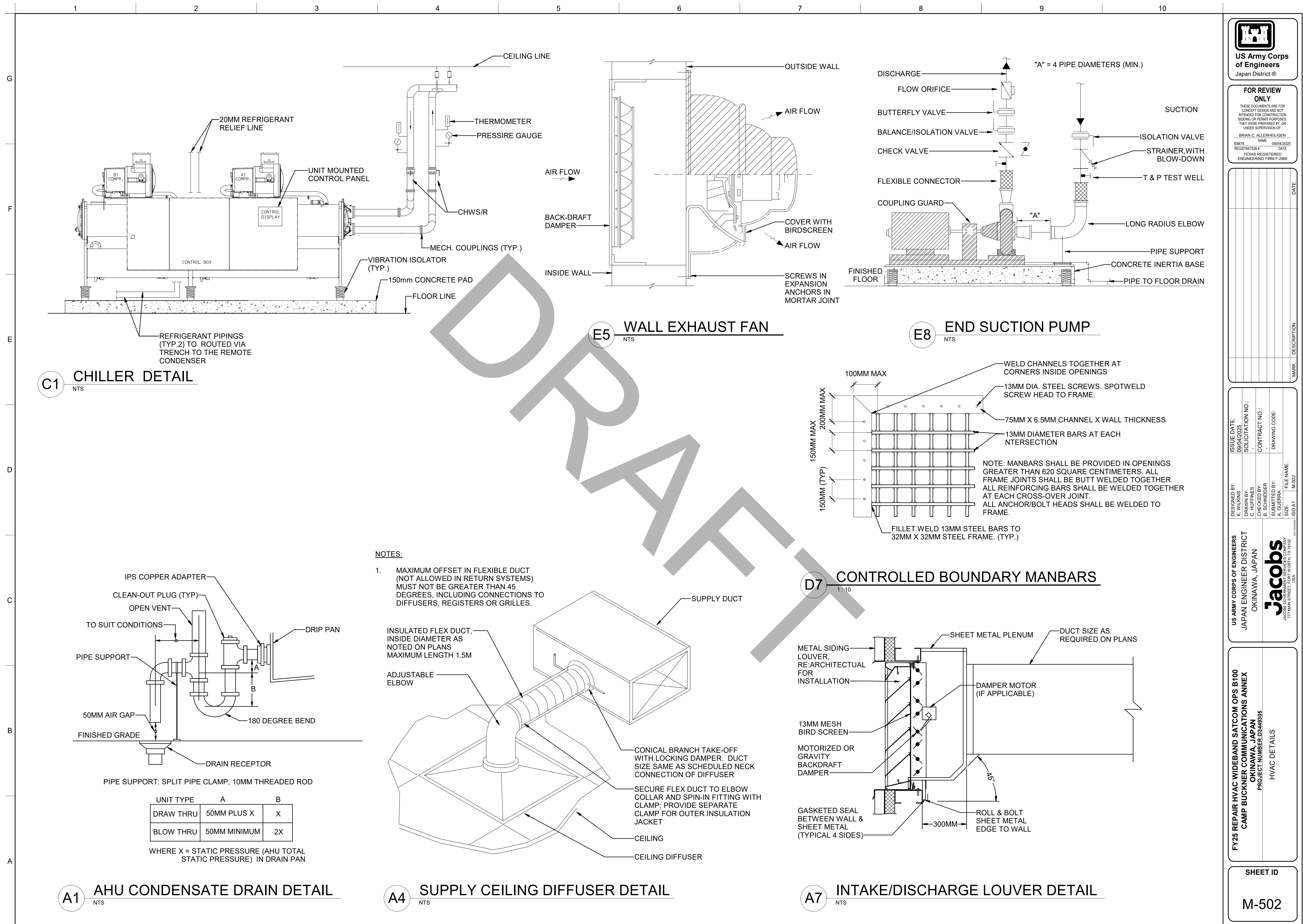
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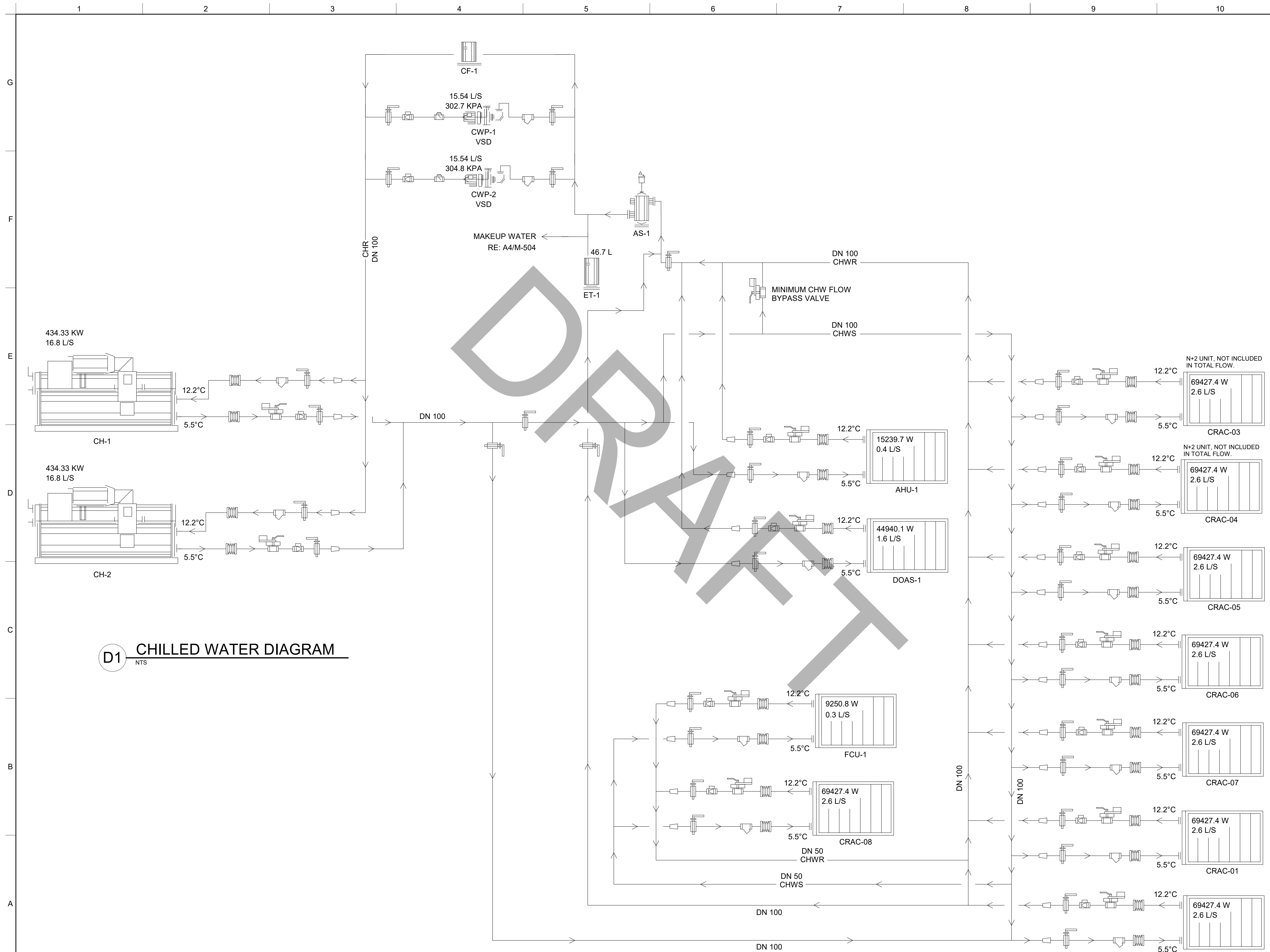
Jacobs JACOBS GOVERNMENT SERVICES COMPANY 777 MAIN STREET • FORT WORTH, TEXAS 76102-3899 USA	OKINAWA, JAPAN	
	US ARMY CORPS OF ENGINEERS JAPAN ENGINEER DISTRICT	
DESIGNED BY: DRAWN BY: C. HUFFINES CHECKED BY: B. SCHNEIDER APPROVED BY: A. GUERRA	ISSUE DATE: 09/01/2025 SOLICITATION NO.: CONTRACT NO.: DRAWING CODE: FILE NAME: ISO A1	M-501

FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX
OKINAWA, JAPAN
PROJECT NUMBER:D3449305
HVAC DETAILS

SHEET ID

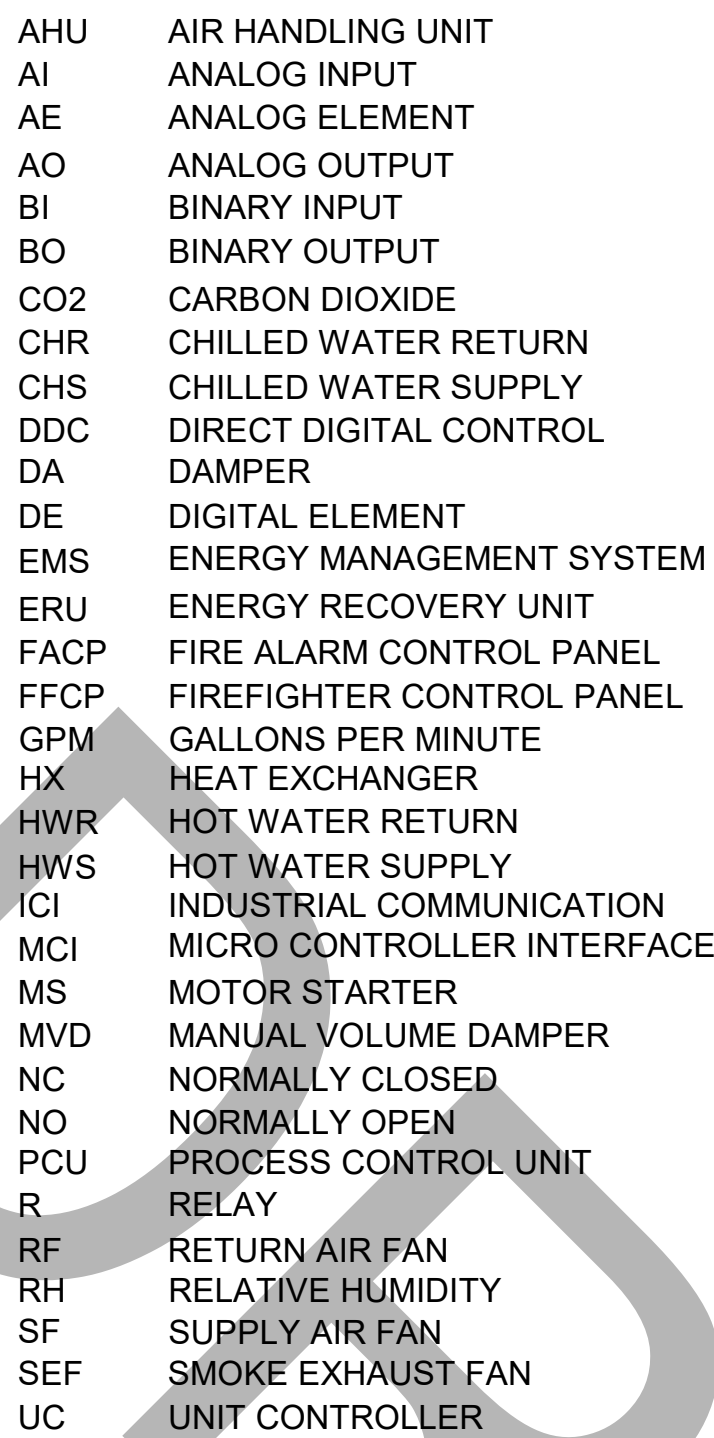
M-501





(ALL SYMBOLS SHOWN ARE NOT NECESSARILY USED ON THE DRAWINGS)

ABBREVIATIONS



CONTROL DEFINITIONS:

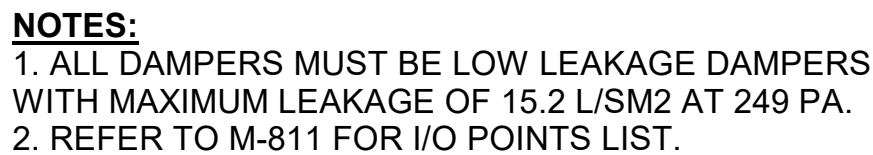
NORMAL POSITION: INDICATES DEVICE POSITION WHEN THE SYSTEM IS DISABLED OR NOT ENERGIZED

FAIL POSITION: INDICATES DEVICE POSITION WHEN THE CONTROL SIGNAL IS LOST OR UPON A LOSS OF POWER.

CONTROLS GENERAL NOTES:

1. GENERAL NOTES MUST BE APPLIED FOR ALL HVAC CONTROLS SHEETS.
2. AFFIX PERMANENT TAGS/LABELS TO ALL DEVICES AS SPECIFIED.
3. CONTRACTOR MUST LABEL ALL DDC I/O SIGNAL LINES: 4-20 mA, VDC, OR SNVT.
4. SHOW A UNIQUE IDENTIFIER FOR EACH DEVICE. WHERE MULTIPLE IDENTICAL DEVICES ARE SHOWN (FOR EXAMPLE: DDC CONTROLLER, OA SENSOR, OR EP TRANSDUCER) EACH MUST BE SEQUENTIALLY NUMBERED. WHERE SEPERATE DDC CONTROLLER BUBBLES ARE USED THE SAME IDENTIFIER AND NUMBER, DEVICE AND SIGNAL IDENTIFIERS MUST BE CONSISTENT BETWEEN DRAWINGS.
5. UNIQUELY NUMBER ALL ANSI 709.1 DEVICES. THIS NUMBERING MUST BE CONSISTENT BETWEEN ALL DRAWINGS.





DEDICATED OUTSIDE AIR SYSTEM (DOAS-2):
THE DEDICATED OUTSIDE AIR UNIT SHALL BE A CONSTANT VOLUME SYSTEM TO PROVIDE BUILDING VENTILATION FOR THE ADMINISTRATION AREAS. IT SHALL BE A CONSTANT VOLUME TYPE WITH A SUPPLY FAN PKG, DX COIL, ELECTRIC REHEAT. IT SHALL DEHUMIDIFY THE OUTDOOR AIR DOWN TO 12.8°C DRY BULB THEN REHEAT IT UP TO 15.6°C DRY BULB. VENTILATION AIR SHALL BE DELIVERED TO AHU-2.

THE DOAS SHALL RUN AS DEFINED BY THE BUILDING AUTOMATION SYSTEM (BAS) OPERATOR FOR OCCUPIED OR UNOCCUPIED HOURS CONSISTENT WITH THE BUILDING'S OPERATING SCHEDULE. THE SUPPLY FAN SHALL RUN ANYTIME THE UNIT IS COMMANDED TO RUN, UNLESS SHUTDOWN ON SAFETIES. TO PREVENT SHORT CYCLING, THE FANS SHALL HAVE A USER DEFINABLE MINIMUM RUNTIME OF 30 MINUTES (ADJUSTABLE). THE BAS SHALL ALLOW DOAS TO BE PROGRAMMED FOR START AND STOP BASED ON A USER DEFINABLE SCHEDULE UNIQUE TO EACH DAY OF THE WEEK (SUNDAY THROUGH SATURDAY). UPON RECEIVING A START COMMAND, THE MOTORIZED CONTROL DAMPERS AT THE INTAKE AIR OPENING SHALL OPEN AND THE STATUS SHALL BE CONFIRMED. IF DAMPER STATUS IS NOT CONFIRMED OPEN AFTER 1 MINUTE (ADJUSTABLE) AN ALARM SHALL BE GENERATED AT THE BAS.

THE DX COIL SHALL OPERATE UNDER MANUFACTURERED SUPPLIED SAFETIES AND CONTROLS TO MAINTAIN THE FOLLOWING CONDITIONS. IN COOLING/DEHUMIDIFICATION MODE DX COOLING COIL SHALL DEHUMIDIFY SUPPLY AIR DOWN TO 12.8°C DB (ADJUSTABLE) THEN USING ELECTRIC REHEAT, REHEAT THE SUPPLY AIR UP TO 15.6°C (ADJUSTABLE). IF ENTERING AIR TEMPERATURE DROPS BELOW 11.8°C (ADJUSTABLE) FOR MORE THAN 15 MINUTES (ADJUSTABLE) THE DX COOLING SECTION SHALL DENERGIZE.

WHEN OUTSIDE AIR TEMPERATURE IS BELOW 11.7°C (ADJUSTABLE) FOR MORE THAN 15 MINUTES (ADJUSTABLE) ELECTRIC HEATING SECTION SHALL
MODULATE UNDER MANUFACTURER SUPPLIED CONTROL TO MAINTAIN SUPPLY AIR TEMPERATURE OF 15.6°C +/- 1.1°C (ADJUSTABLE; T-03). AN ALARM
SHALL BE PROVIDED FOR SUPPLY AIR TEMPERATURE 2.8°C (ADJUSTABLE) GREATER THAN THE COIL DISCHARGE SETPOINT.

THE BAS SHALL MONITOR THE OUTSIDE AIRFLOW THROUGH THE OUTSIDE AIRFLOW STATION.

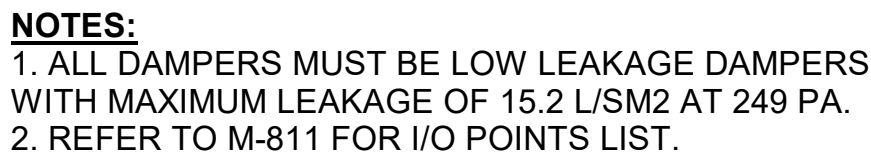
THE BAS SHALL MONITOR THE DIFFERENTIAL PRESSURE ACROSS THE FILTER BANK. ALARMS SHALL BE PROVIDED FOR THE FOLLOWING:
 FILTER CHANGE REQUIRED: FILTER DIFFERENTIAL PRESSURE EXCEEDS A USER DEFINABLE LIMIT (ADJUSTABLE).

THE UNIT SHALL SHUT DOWN AND GENERATE AN ALARM AT THE BAS IF ONE OF THE FOLLOWING CONDITIONS OCCURS:

1. SUPPLY FAN: SUPPLY FAN FAILURE ALARM PER THE FOLLOWING:
 - a. SUPPLY FAN FAILURE: COMMANDED ON, BUT STATUS IS OFF.
 - b. SUPPLY FAN IN HAND: COMMANDED OFF, BUT STATUS IS ON.
3. AN EMERGENCY SHUTDOWN SIGNAL FROM A CENTRAL SWITCH (PER UFC 4-010-01) WHEN A CHEMICAL OR BIOLOGICAL THREAT IS IMMINENT.
4. COMMANDED OFF BY BAS.

UNIT SHUTDOWN SHALL ALIGN THE UNIT COMPONENTS TO THE FOLLOWING: SUPPLY AND CORRESPONDING ADMIN BUILDING EXHAUST FANS OFF AND OUTSIDE AIR DAMPER CLOSED. UPON SHUTDOWN ON ONE OF THE ABOVE FAULT CONDITIONS, THE BAS SHALL GENERATE AN ALARM AT THE USER INTERFACE.

B1



DEDICATED OUTSIDE AIR SYSTEM (DOAS-1):
THE DEDICATED OUTSIDE AIR UNIT SHALL BE A CONSTANT VOLUME SYSTEM TO PROVIDE BUILDING VENTILATION FOR THE OPERATIONS AREAS. IT SHALL BE A CONSTANT VOLUME TYPE WITH A SUPPLY FAN PKG, CHILLED WATER COIL, AND ELECTRIC REHEAT. IT SHALL DEHUMIDIFY THE OUTDOOR AIR DOWN TO 12.8°C WET BULB THEN REHEAT IT UP TO 15.6°C DRY BULB. VENTILATION AIR SHALL BE DELIVERED TO EACH AIR HANDLER IN THE OPERATIONS AREA..

THE DOAS SHALL RUN AS DEFINED BY THE BUILDING AUTOMATION SYSTEM (BAS) OPERATOR FOR OCCUPIED OR UNOCCUPIED HOURS CONSISTENT WITH THE BUILDING'S OPERATING SCHEDULE. THE SUPPLY FAN SHALL RUN ANYTIME THE UNIT IS COMMANDED TO RUN, UNLESS SHUTDOWN ON SAFETIES. TO PREVENT SHORT CYCLING, THE FANS SHALL HAVE A USER DEFINABLE MINIMUM RUNTIME OF 30 MINUTES (ADJUSTABLE). THE BAS SHALL ALLOW DOAS TO BE PROGRAMMED FOR START AND STOP BASED ON A USER DEFINABLE SCHEDULE UNIQUE TO EACH DAY OF THE WEEK (SUNDAY THROUGH SATURDAY). UPON RECEIVING A START COMMAND, THE MOTORIZED CONTROL DAMPERS AT THE INTAKE AIR OPENING SHALL OPEN AND THE STATUS SHALL BE CONFIRMED. IF DAMPER STATUS IS NOT CONFIRMED OPEN AFTER 1 MINUTE (ADJUSTABLE) AN ALARM SHALL BE GENERATED AT THE BAS.

THE CHILLED WATER COIL MUST OPERATE UNDER MANUFACTURER SUPPLIED SAFETIES AND CONTROLS TO MAINTAIN THE FOLLOWING CONDITIONS. THE CHILLED WATER COOLING COIL MUST DEHUMIDIFY SUPPLY AIR DOWN TO 12.8°C DRY BULB (ADJUSTABLE) BY MODULATING THE CHILLED WATER CONTROL VALVE. USING ELECTRIC REHEAT, THE REHEAT COIL SHALL MODULATE TO REHEAT THE AIR TO 15.6°C DRY BULB (ADJUSTABLE). IF ENTERING AIR TEMPERATURE DROPS BELOW 11.7°C (ADJUSTABLE) FOR MORE THAN 15 MINUTES (ADJUSTABLE) THE CHILLED WATER CONTROL VALVE COOLING CHILL CLOSE. AN ALARM MUST BE PROVIDED FOR HIGH SUPPLY AIR IF THE SUPPLY AIR TEMPERATURE IS 2.8°C (ADJUSTABLE) GREATER THAN THE CHILL DISCHARGE SETPOINT.

WHEN OUTSIDE AIR TEMPERATURE IS BELOW 13.3°C (ADJUSTABLE) FOR MORE THAN 15 MINUTES (ADJUSTABLE) ELECTRIC HEATING SECTION SHALL
MODULE UNDER MANUFACTURER SUPPLIED CONTROL TO MAINTAIN SUPPLY AIR TEMPERATURE OF 15.6°C +/- 1.1°C (ADJUSTABLE; T-03). AN ALARM
SHALL BE PROVIDED FOR SUPPLY AIR TEMPERATURE 2.8°C (ADJUSTABLE) GREATER THAN THE COIL DISCHARGE SETPOINT.

THE BAS SHALL MONITOR THE OUTSIDE AIRFLOW THROUGH THE OUTSIDE AIRFLOW STATION.

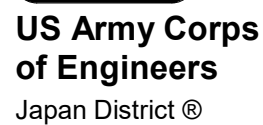
THE BAS SHALL MONITOR THE DIFFERENTIAL PRESSURE ACROSS THE FILTER BANK. ALARMS SHALL BE PROVIDED FOR THE FOLLOWING:
 FILTER CHANGE REQUIRED: FILTER DIFFERENTIAL PRESSURE EXCEEDS A USER DEFINABLE LIMIT (ADJUSTABLE).

THE UNIT SHALL SHUT DOWN AND GENERATE AN ALARM AT THE BAS IF ONE OF THE FOLLOWING CONDITIONS OCCURS:

1. SUPPLY AIR SMOKE DETECTION: SA DUCT DETECTORS SHALL BE HARD WIRED TO SHUT DOWN THE UNIT SUPPLY FAN UPON DETECTION OF SMOKE. THE SA SMOKE DETECTOR SHALL ALSO ADDRESS THE FIRE ALARM CONTROL PANEL, RE: FIRE PROTECTION.
2. SUPPLY FAN: SUPPLY FAN FAILURE ALARM PER THE FOLLOWING:
 - a. SUPPLY FAN FAILURE: COMMANDED ON, BUT STATUS IS OFF.
 - b. SUPPLY FAN IN HAND: COMMANDED OFF, BUT STATUS IS ON.
3. AN EMERGENCY SHUTDOWN SIGNAL FROM A CENTRAL SWITCH (PER UFC 4-010-01) WHEN A CHEMICAL OR BIOLOGICAL THREAT IS IMMINENT.
4. COMMANDED OFF BY BAS.

UNIT SHUTDOWN SHALL ALIGN THE UNIT COMPONENTS TO THE FOLLOWING: SUPPLY AND CORRESPONDING OPERATIONS BUILDING EXHAUST FANS OFF AND OUTSIDE AIR DAMPER CLOSED. UPON SHUTDOWN ON ONE OF THE ABOVE FAULT CONDITIONS, THE BAS SHALL GENERATE AN ALARM AT THE USER INTERFACE.

B6



THESE DOCUMENTS ARE FOR
CONCEPT DESIGN AND NOT
INTENDED FOR CONSTRUCTION
BIDDING OR PERMIT PURPOSES.
THEY WERE PREPARED BY, OR
UNDER SUPERVISION OF:

BRIAN C. ALLERHEILIGEN
NAME 09/04/2025
REGISTRATION # DATE
TEXAS REGISTERED
ENGINEERING FIRM F-2966

[illegible]

		4010000000 1000000000	
US ARMY CORPS OF ENGINEERS JAPAN ENGINEER DISTRICT OKINAWA, JAPAN		DRAWN BY: K. WILKINS C. HUFFINES CHECKED BY: S. COOPER SUBMITTED BY: A. GUERRA	
SOLICITATION NO.: 09/04/2029		CONTRACT NO.: _____	
DRAWING CODE: _____		FILE NAME: M-303	
SIZE: A0/A1		177 MAIN STREET, FORT WORTH, TX 76102 USA	

FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX
OKINAWA, JAPAN
PROJECT NUMBER:D3449305
HVAC CONTROLS

M-803

NOTES:

- | OCCUPANCY SCHEDULE | | | | | |
|--------------------|------------------------|---|---|----------------------------------|---|
| SYSTEM | UNOCCUPIED | SYSTEM
DEFAULT
OCCUPIED
SCHEDULE | SUPERVISORY
MONITORING &
CONTROL OCCUPIED
SCHEDULE | BUILDING
WARM-UP/CO
OLDOWN | # OCC
SENSORS TO
PUT IN OCC
MODE |
| AHU-1 | UNLESS OTHERWISE NOTED | M-F 0700-1900 | M-F 0800-1800 | NOTE: 1 | 1 |
| AHU-2 | UNLESS OTHERWISE NOTED | M-F 0700-1900 | M-F 0800-1800 | NOTE: 1 | N/A |
| DOAS-1 | UNLESS OTHERWISE NOTED | M-F 0700-1900 | M-F 0800-1800 | NOTE: 1 | 1 |
| DOAS-2 | UNLESS OTHERWISE NOTED | M-F 0700-1900 | M-F 0800-1800 | NOTE: 1 | N/A |
| CRAC | UNLESS OTHERWISE NOTED | 24/7 | 24/7 | N/A | N/A |
| FCU | UNLESS OTHERWISE NOTED | 24/7 | 24/7 | N/A | N/A |
| AHU-3 | UNLESS OTHERWISE NOTED | M-F 0700-1900 | M-F 0800-1800 | NOTE: 1 | N/A |

NOTES:

- | Descriptor | System | Location | Design Flow Rate | Design Pressure Drop | Valve Size | Function | Type | Notes |
|--------------|---------------|-----------------|------------------|----------------------|------------|---------------------|-------------------------------|-------|
| CHW Min Flow | Chilled Water | Mechanical Room | | 10 | 50 | Minimum Flow Bypass | 2-Way, Linear Characteristics | 1 |
| CRACS | Chilled Water | Refer to Plans | 16.8 LPS | 10 | 50 | Temperature Control | 2-Way, Equal Percentage | 2 |
| DOAS-1 CC | Chilled Water | Mechanical Room | 1.6 LPS | 10 | 20 | Temperature Control | 2-Way, Equal Percentage | |
| AHU-1 CC | Chilled Water | Mechanical Room | 0.44 LPS | 10 | 25 | Temperature Control | 2-Way, Equal Percentage | |
| AHU-3 CC | Chilled Water | Mechanical Room | 0.33 LPS | 10 | 20 | Temperature Control | 2-Way, Equal Percentage | |

NOTES

- ## BACKCHECK SUBMITTAL DESIGN

DEDICATED OUTDOOR AIR SYSTEM (DOAS-1)

NODE: <DDC##>
 NODE LOCATION: <____>
 NODE ADDRESS DOMAIN = <____>, SUBNET = <____>, NODE = <____>
 NODE ID: <____>

FUNCTION	POINT ID	DESCRIPTION	DDC HARDWARE ID	SETTING (WITH UNITS)	RANGE (WITH UNITS)	IO TYPE	HOA REQ'D	CONFIG. TYPE	M&C VIEW AND OVRD	LDP VIEW AND OVRD	TREND REQ'D	ALARMING			PRIMARY POINT INFORMATION		NETWORK DATA EXCHANGE		OVERRIDES		ALARM INFORMATION		CONFIGURATION INFORMATION
												ALARM CONDITION (SEE NOTES)	ALARM PRIORITY	M&C ROUTING	OBJECT TYPE AND INSTANCE NUMBER (AND PROPERTY ID)	NIAGARA ID	DATA FROM	DATA TO	OBJECT TYPE & INSTANCE NUMBER OR COMMANDABLE	ALARM TYPE (PROVIDE DDC ID FOR REMOTE IN)	NOTIFICATION CLASS OBJECT INSTANCE NUMBER		
PROOFS & SAFETIES	SF-S	SF VFD STATUS	<___>	~	ON/OFF	BI	~	~	V	V	Y	MISMATCH COMMAND VS STATUS	INFO	<___>									
	SF-FAULT FLT-P	SF FAULT OUTSIDE AIR FILTER STATUS	<___>	~	ALM/NORMAL <0-249> PA.	BI AI	~ ~	~ H	V V	V V	Y Y	SF VFD FAULT DIRTY FILTER	INFO INFO	<___> <___>									
START/STOP	SF-SS	SF START/STOP	<___>	~	ON/OFF	BO	~	~	VO	V	Y	~	~	~									
FAN CONTROL	SF-SPD-C	SF SPEED	<___>	~	0-100%	AI	~	~	V	V	Y	RE: B6/M-803	INFO	<___>									
	SA-P	SA PRESSURE SENSOR	<___>	~	RE: B6/M-803	AI	~	~	V	V	Y	RE: B6/M-803	INFO	<___>									
FLOW MEASURING STATION	OA-FMS	OA FLOW MEASURING STATION	<___>	~	0 L/S - 9,439 L/S	AI	~	~	V	V	Y	RE: B6/M-803	INFO	<___>									
TEMPERATURE CONTRL	SA-T	SA TEMPERATURE SENSOR	<___>	~	4-35°C	AI	~	~	V	V	Y	RE: B6/M-803	INFO	<___>									
	OA-T	OA TEMPERATURE SENSOR	<___>	~	-18-49°C	AI	~	~	V	V	Y	RE: B6/M-803	INFO	<___>									
	CC-T	CC TEMPERATURE SENSOR	<___>	~	4-35°C	AI	~	~	V	V	Y	RE: B6/M-803	INFO	<___>									
	CC-V-C	CC VALVE CONTROL	<___>	~	0-100%	AO	~	~	V	V	Y	MISMATCH COMMAND VS POSITION	INFO	<___>									
	ELEC-ENA	ELECTRIC HEATING COIL CONTROL	<___>	~	0-100%	AO	~	~	V	V	Y	MISMATCH COMMAND VS POSITION	INFO	<___>									
OTHER POINTS	OA-D-2P	OUTSIDE AIR DAMPER COMMAND	<___>	~	OPEN/CLOSE	AO	~	~	V	~	~	MISMATCH COMMAND VS POSITION	~	~									

NOTES:

1. THE CONTRACTOR MUST COMPLETE THE POINTS SCHEDULE AS INDICATED IN THE SPECIFICATION AND IN ACCORDANCE WITH THE POINTS SCHEDULE INSTRUCTIONS DRAWING.
2. UNIT MANUFACTURERS PROOFS AND SAFETIES: THE CONTRACTOR MUST SHOW EACH PROOF AND SAFETY AS A SEPARATE ROW.
3. UNIT STATUS: SERVES AS A MONITORED POINT AT THE M&C SOFTWARE (FRONT-END) AND AS A HEATING/COOLING REQUEST TO THE BOILER, HEAT EXCHANGER, AND/OR CHILLER SERVING THIS SYSTEM (IF APPLICABLE)

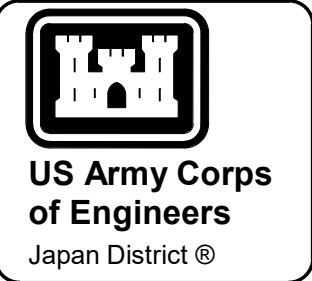
DEDICATED OUTDOOR AIR SYSTEM (DOAS-2)

NODE: <DDC##>
 NODE LOCATION: <____>
 NODE ADDRESS DOMAIN = <____>, SUBNET = <____>, NODE = <____>
 NODE ID: <____>

FUNCTION	POINT ID	DESCRIPTION	DDC HARDWARE ID	SETTING (WITH UNITS)	RANGE (WITH UNITS)	IO TYPE	HOA REQ'D	CONFIG. TYPE	M&C VIEW AND OVRD	LDP VIEW AND OVRD	TREND REQ'D	ALARMING			PRIMARY POINT INFORMATION		NETWORK DATA EXCHANGE		OVERRIDES	ALARM INFORMATION		CONFIGURATION INFORMATION
												ALARM CONDITION (SEE NOTES)	ALARM PRIORITY	M&C ROUTING	OBJECT TYPE AND INSTANCE NUMBER (AND PROPERTY ID)	NIAGARA ID	DATA FROM	DATA TO	OBJECT TYPE & INSTANCE NUMBER OR COMMANDABLE	ALARM TYPE (PROVIDE DDC ID FOR REMOTE IN)	NOTIFICATION CLASS OBJECT INSTANCE NUMBER	
PROOFS & SAFETIES	SF-S	SF STATUS	<___>	~	ON/OFF	BI	~	~	V	V	Y	MISMATCH COMMAND VS STATUS	INFO	<___>								
	CU-S	CONDENSING UNIT STATUS	<___>	~	ON/OFF	BI	~	~	V	V	Y	MISMATCH COMMAND VS STATUS	INFO	<___>								
	SF-FAULT	SF FAULT	<___>	~	ALM/NORMAL	BI	~	~	V	V	Y	SF VFD FAULT	INFO	<___>								
	FLT-P	OUTSIDE AIR FILTER STATUS	<___>	~	<0-249> PA.	AI	~	H	V	V	Y	DIRTY FILTER	INFO	<___>								
START/STOP	SF-SS	SF START/STOP	<___>	~	ON/OFF	BO	~	~	VO	V	Y	~	~	~								
	CU-SS	CONDENSING UNIT START/STOP	<___>	~	ON/OFF	BO	~	~	VO	V	Y	~	~	~								
FAN CONTROL	SA-P	SA PRESSURE SENSOR	<___>	~	RE: B1/M-802	AI	~	~	V	V	Y	RE: B1/M-802	INFO	<___>								
FLOW MEASURING STATION	OA-FMS	OA FLOW MEASURING STATION	<___>	~	0 L/S - 9,439 L/S	AI	~	~	V	V	Y	RE: B1/M-803	INFO	<___>								
TEMPERATURE CONTORL	SA-T	SA TEMPERATURE SENSOR	<___>	~	4-35°C	AI	~	~	V	V	Y	RE: B1/M-803	INFO	<___>								
	OA-T	OA TEMPERATURE SENSOR	<___>	~	-18-49°C	AI	~	~	V	V	Y	RE: B1/M-803	INFO	<___>								
	CC-T	CC TEMPERATURE SENSOR	<___>	~	4-35°C	AI	~	~	V	V	Y	RE: B1/M-803	INFO	<___>								
	ELEC-ENA	ELECTRIC HEATING COIL CONTROL	<___>	~	0-100%	AO	~	~	V	V	Y	MISMATCH COMMAND VS POSITION	INFO	<___>								
OTHER POINTS	OA-D-2P	OUTSIDE AIR DAMPER COMMAND	<___>	~	OPEN/CLOSE	AO	~	~	V	~	~	MISMATCH COMMAND VS POSITION	~	~								

NOTES:

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A rectangular stamp with a double-line border. At the top, the words "FOR REVIEW" and "ONLY" are printed in large, bold, sans-serif capital letters. Below this, a paragraph of text in a smaller, standard sans-serif font reads: "THESE DOCUMENTS ARE FOR CONCEPT DESIGN AND NOT INTENDED FOR CONSTRUCTION BIDDING OR PERMIT PURPOSES. THEY WERE PREPARED BY, OR UNDER SUPERVISION OF:". This is followed by the name "BRIAN C. ALLERHEILIGEN" in bold, sans-serif capital letters. A horizontal line separates the name from the registration information. Below the line, the word "NAME" is printed in a small font. To the left of "NAME" is the number "93876", and to the right is the date "09/04/2025". Below "NAME" is the text "REGISTRATION #", and to the right is the word "DATE". At the bottom of the stamp, the text "TEXAS REGISTERED ENGINEERING FIRM F-2966" is printed in bold, sans-serif capital letters.

[illegible]

US ARMY CORPS OF ENGINEERS JAPAN ENGINEER DISTRICT OKINAWA, JAPAN		Jacobs JACOBS GOVERNMENT SERVICES COMPANY 1707 MAIN STREET, 14TH FLOOR ARLINGTON, VA 22202-4302 USA	
DESIGNED BY:	ISSUE DATE:	CONTRACT NO.:	DRAWING CODE:
DRAWN BY:	09/04/2025	-	-
CHECKED BY:	SOLICITATION NO.:	-	-
A. GUERRAS:			
B. SCHNEIDER:			
C. HUFFINES:			
D. GUERRAS:			
FILE NAME:	SIZE:	SCALE:	M-S-11
		ISO-A1	

FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX
OKINAWA, JAPAN
PROJECT NUMBER:D3449305

SHEET ID

M-811

FCU SYSTEMS

NODE: <DDC##>
 NODE LOCATION: <____>
 NODE ADDRESS DOMAIN = <____>, SUBNET = <____>, NODE = <____>
 NODE ID: <____>

FUNCTION	POINT ID	DESCRIPTION	DDC HARDWARE ID	SETTING (WITH UNITS)	RANGE (WITH UNITS)	IO TYPE	HOA REQ'D	CONFIG. TYPE	M&C VIEW AND OVRD	LDP VIEW AND OVRD	TREND REQ'D	ALARMING			PRIMARY POINT INFORMATION		NETWORK DATA EXCHANGE		OVERRIDES	ALARM INFORMATION		CONFIGURATION INFORMATION
												ALARM CONDITION (SEE NOTES)	ALARM PRIORITY	M&C ROUTING	OBJECT TYPE AND INSTANCE NUMBER (AND PROPERTY ID)	NIAGARA ID	DATA FROM	DATA TO	OBJECT TYPE & INSTANCE NUMBER OR COMMANDABLE	ALARM TYPE (PROVIDE DDC ID FOR REMOTE IN)	NOTIFICATION CLASS OBJECT INSTANCE NUMBER	
PROOF AND SAFETIES	MS-S	MINI-SPLIT STATUS	<___>	~	ON/OFF	BI	~	~	V	V	Y	MISMATCH COMMAND VS STATUS	INFO	<___>								
START STOP	MS-SS	MINI-SPLIT START/STOP	<___>	~	ON/OFF	BO	~	~	VO	V	Y	~	~	~								
TEMPERATURE CONTORL	ZN-T	ZONE TEMPERATURE SENSOR	<___>	~	4-35°C	AI	~	~	V	V	Y	~	INFO	<___>								

NOTES:

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2. UNIT MANUFACTURERS PROOFS AND SAFETIES: THE CONTRACTOR MUST SHOW EACH PROOF AND SAFETY AS A SEPARATE ROW.
3. UNIT STATUS: SERVES AS A MONITORED POINT AT THE M&C SOFTWARE (FRONT-END) AND AS A HEATING/COOLING REQUEST TO THE BOILER, HEAT EXCHANGER, AND/OR CHILLER SERVING THIS SYSTEM (IF APPLICABLE)

CRAC UNITS

NODE: <DDC##>
 NODE LOCATION: <____>
 NODE ADDRESS DOMAIN = <____>, SUBNET = <____>, NODE = <____>
 NODE ID: <____>

FUNCTION	POINT ID	DESCRIPTION	DDC HARDWARE ID	SETTING (WITH UNITS)	RANGE (WITH UNITS)	IO TYPE	HOA REQ'D	CONFIG. TYPE	M&C VIEW AND OVRD	LDP VIEW AND OVRD	TREND REQ'D	ALARMING			PRIMARY POINT INFORMATION		NETWORK DATA EXCHANGE		OVERRIDES	ALARM INFORMATION		CONFIGURATION INFORMATION
												ALARM CONDITION (SEE NOTES)	ALARM PRIORITY	M&C ROUTING	OBJECT TYPE AND INSTANCE NUMBER (AND PROPERTY ID)	NIAGARA ID	DATA FROM	DATA TO	OBJECT TYPE & INSTANCE NUMBER OR COMMANDABLE	ALARM TYPE (PROVIDE DDC ID FOR REMOTE IN)	NOTIFICATION CLASS OBJECT INSTANCE NUMBER	
START STOP	SF-SS	START/STOP	<__>	~	ON/OFF	BO	~	~	VO	V	Y	~	~									
	SF-S	STATUS	<__>	~	ON/OFF	BI	~	~	VO	V	Y	MISMATCH COMMAND VS STATUS	~	~								
TEMPERATURE CONTRL	CLG-V-C	COOLING VALVE CONTROL	<__>	~	0-100%	AO	~	~	V	V	Y	MISMATCH COMMAND VS POSITION	INFO	<__>								
	CLG-V-P	COOLING VALVE POSITION	<__>	~	0-100%	AI	~	~	V	V	Y	MISMATCH COMMAND VS POSITION	INFO	<__>								
	ZN-T	ZONETEMPERATURE SENSOR	<__>	~	4-35°C	AI	~	~	V	V	Y	~	INFO	<__>								

NOTES:

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MECH EF

NODE: <DDC##>
 NODE LOCATION: <____>
 NODE ADDRESS DOMAIN = <____>, SUBNET = <____>, NODE = <____>
 NODE ID: <____>

FUNCTION	POINT ID	DESCRIPTION	DDC HARDWARE ID	SETTING (WITH UNITS)	RANGE (WITH UNITS)	IO TYPE	HOA REQ'D	CONFIG. TYPE	M&C VIEW AND OVRD	LDP VIEW AND OVRD	TREND REQ'D	ALARMING			PRIMARY POINT INFORMATION		NETWORK DATA EXCHANGE		OVERRIDES	ALARM INFORMATION		CONFIGURATION INFORMATION
												ALARM CONDITION (SEE NOTES)	ALARM PRIORITY	M&C ROUTING	OBJECT TYPE AND INSTANCE NUMBER (AND PROPERTY ID)	NIAGARA ID	DATA FROM	DATA TO		OBJECT TYPE & INSTANCE NUMBER OR COMMANDABLE	ALARM TYPE (PROVIDE DDC ID FOR REMOTE IN)	
START STOP	EF-SS	START/STOP	< ___ >	~	ON/OFF	BO	~	~	VO	V	Y	~	~	~								
	EF-S	STATUS	< ___ >	~	ON/OFF	BI	~	~	VO	V	Y	MISMATCH COMMAND VS STATUS	~	~								
OTHER POINTS	ZN-HY	HYDROGEN SENSOR (WHERE APPLICABLE)	< ___ >	~	0-2%	AI	~	~	V	~	~	0.5% HYDROGEN	~	~								
	EA-D-2P	EXHAUST AIR DAMPER COMMAND	< ___ >	~	OPEN/CLOSE	AO	~	~	V	~	~	MISMATCH COMMAND VS POSITION	~	~								

NOTES:

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**US Army Corps
of Engineers**
Japan District ®

FOR REVIEW
ONLY

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BIDDING OR PERMIT PURPOSES.
THEY WERE PREPARED BY, OR
UNDER SUPERVISION OF:

BRIAN C. ALLERHEILIGEN
NAME

3876 09/04/2021
REGISTRATION # DATE

TEXAS REGISTERED
ENGINEERING FIRM F-2966

[illegible]

Jacobs JACOBS GOVERNMENT SERVICES COMPANY 177 MAIN STREET, SUITE 100 WESTPORT, MA 01886 REG. U.S. PAT. & TM. OFF.		OKINAWA, JAPAN	
U.S. ARMY CORPS OF ENGINEERS JAPAN ENGINEER DISTRICT		DESIGNED BY: DRAWING NO.: C. HUFFINES CHECKED BY: B. SCHNEIDER APPROVED BY: A. MUESTER	
ISSUE DATE: 09/04/2025		SOLICITATION NO.: CONTRACT NO.: DRAWING CODE:	
FILE NAME: M-814		ISO A1	

Jacobs
JACOBS GOVERNMENT SERVICES COMPANY
777 MAIN STREET, FORT WORTH, TX 76102
USA

PAC WIDEBAND SATCOM OPS B100
 KNER COMMUNICATIONS ANNEX
 OKINAWA, JAPAN
 PROJECT NUMBER:D3449305

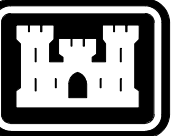
SHEET ID

M-814

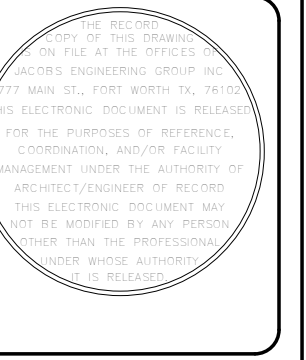
PLUMBING FIXTURE SCHEDULE

MARK	ROUGH-IN SIZES				DESCRPTION	NOTES
	W	V	CW	HW		
L-1	40 mm	50mm	15mm	15mm	LAVATORY, VITREOUS CHINA, UNDERMOUNT, 14" X 17" BOWL, ABA COMPLIANT.	
WC-1	100 mm	50mm	15mm		WATER CLOSET, WALL-MOUNT, MANUAL FLUSH VALVE, 4.8 LPF.	

GENERAL SHEET NOTES



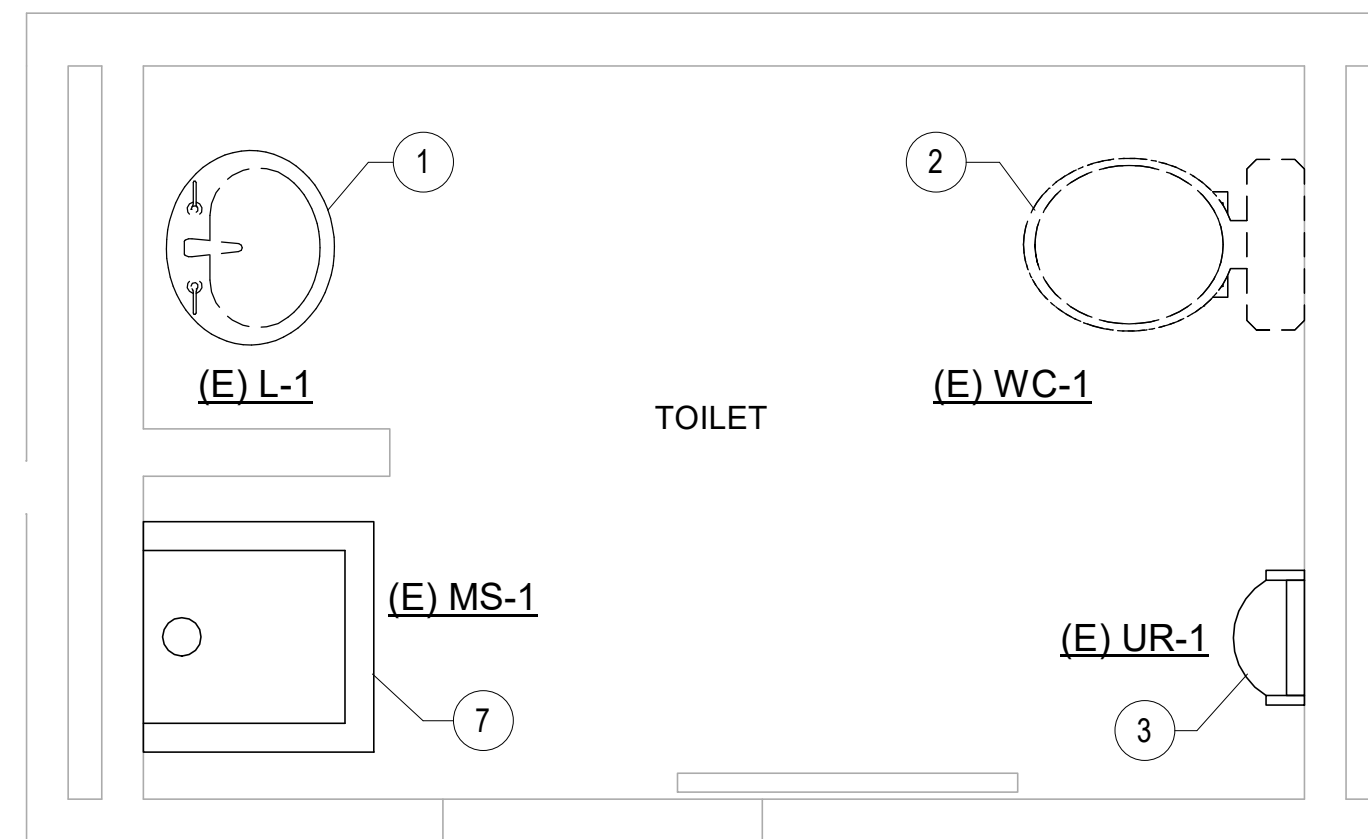
**US Army Corps
of Engineers**
Japan District ®



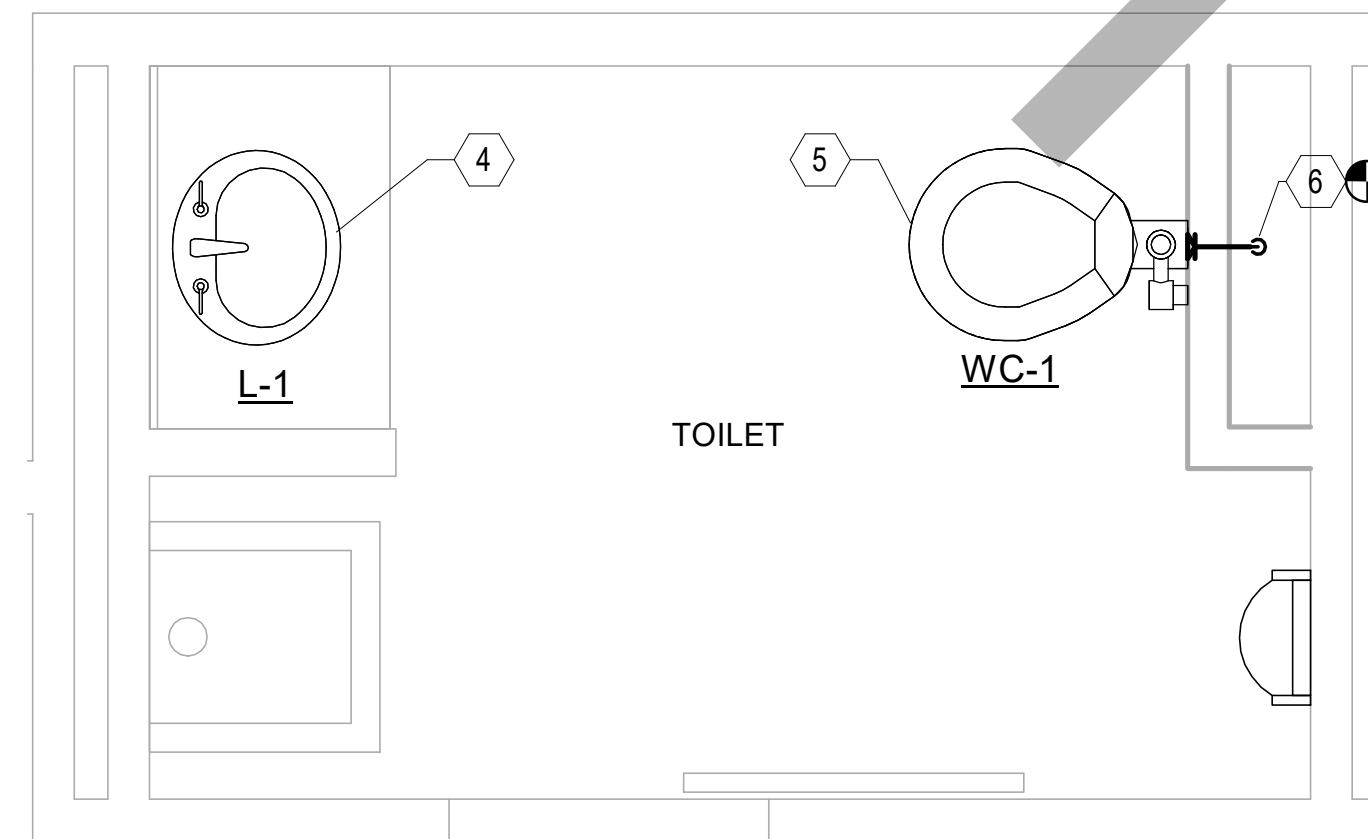
SHEET KEYNOTES

1	REMOVE EXISTING LAVATORY. EXISTING PIPING TO REMAIN TO SERVE NEW.
2	REMOVE EXISTING WATER CLOSET. EXISTING PIPING TO REMAIN TO SERVE NEW.
3	EXISTING URINAL TO BE REMOVED, SALVAGED, AND REINSTALLED.
4	PROVIDE NEW LAVATORY, L-1. CONNECT TO EXISTING COLD WATER, HOT WATER, WASTE AND VENT.
5	PROVIDE NEW WATER CLOSET, WC-1. CONNECT TO EXISTING COLD WATER, WASTE AND VENT.
6	CONNECT TO EXISTING SANITARY PIPING.
7	EXISTING MOP SINK TO REMAIN.

JAPAN ENGINEER DISTRICT OKINAWA, JAPAN Jacobs JACOBS ENGINEERING SERVICES COMPANY 777 MAIN STREET, FORT WORTH, TX 76102 USA	09/04/2025
	K. WILKINS
	CH. HAINES
	CONTRACT NO.:
	CONTRACT NO.:
CHECKED BY:	DRAWING CODE:
B. SCHNEIDER	A. GUERRA
SUBMITTED BY:	FILE NAME:
ISO A1	P-101

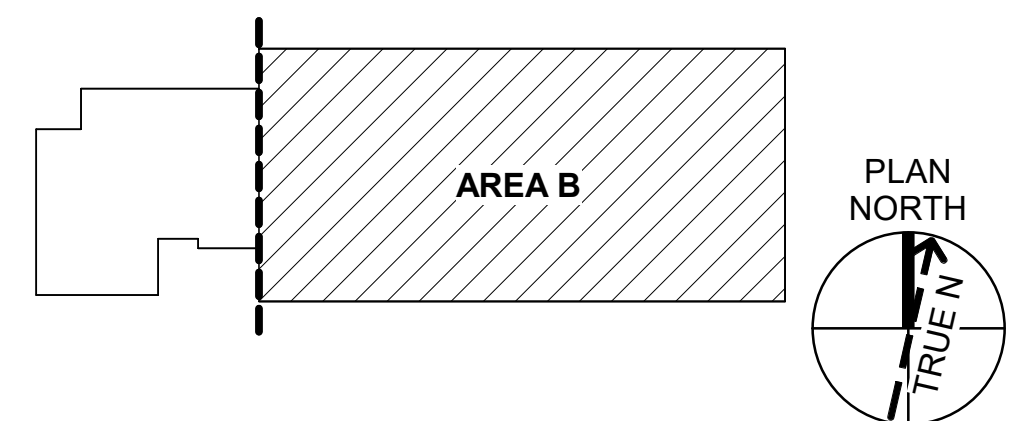


ENLARGED PLUMBING DEMOLITION PLAN - AREA B

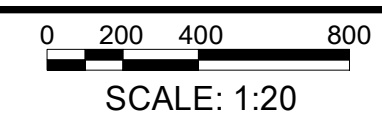


ENLARGED PLUMBING FLOOR PLAN - AREA B

KEY PLAN



GRAPHIC SCALE(S)



CAMP BUCKNER COMMUNICATIONS ANNEX
OKINAWA, JAPAN
PROJECT NUMBER:D3449305

PARTIAL PLUMBING FLOOR PLAN - AREA B

HEET ID

P-101

KEYNOTES	
101	PHASE 1: DISCONNECT POWER TO MECHANICAL EQUIPMENT TO BE DEMOLISHED. REMOVE DISCONNECT SWITCH, CABLE AND CONDUIT BACK TO SOURCE. TURN CIRCUIT BREAKER OFF AND UPDATE THE PANEL DIRECTORY AS SPARE IF ALL LOADS HAVE BEEN REMOVED FROM THE CIRCUIT.
102	PHASE 2: DISCONNECT POWER TO MECHANICAL EQUIPMENT TO BE DEMOLISHED. REMOVE DISCONNECT SWITCH, CABLE AND CONDUIT BACK TO SOURCE. TURN CIRCUIT BREAKER OFF AND UPDATE THE PANEL DIRECTORY AS SPARE IF ALL LOADS HAVE BEEN REMOVED FROM THE CIRCUIT.
103	PHASE 3: DISCONNECT POWER TO MECHANICAL EQUIPMENT TO BE DEMOLISHED. REMOVE DISCONNECT SWITCH, CABLE AND CONDUIT BACK TO SOURCE. TURN CIRCUIT BREAKER OFF AND UPDATE THE PANEL DIRECTORY AS SPARE IF ALL LOADS HAVE BEEN REMOVED FROM THE CIRCUIT.
119	PRIOR TO DEMOLITION OF RF FILTER INSTALL 4-38MM, 1-14MM G, G54C FROM PANEL (E) NO.8 TO TPP PANEL ASSOCIATED WITH TO BE DEMOLISHED RF FILTER. TYP.
120	REMOVE RF FILTER AND CABLE AND CONDUIT BEING FED TO AND FROM RF FILTER. TYP.

GENERAL NOTES - POWER - DEMO	
1	REFER TO E-001 FOR GENERAL NOTES AND LEGENDS.
2	COORDINATE ALL OUTAGES WITH THE CONTRACTING OFFICER IN WRITING 21 DAYS PRIOR TO OUTAGE. OUTAGES MAY NOT LAST LONGER THAN 6 HOURS. PROVIDE THE CONTRACTING OFFICER WITH DETAILED TEMPORARY POWER PLANS IF NECESSARY.
3	PERFORM AS MUCH WORK AS FEASIBLE WITH LOCKED OUT, UNTERMINATED CONDUCTORS TO NEW POWER FEEDS BEFORE DE-ENERGIZING EXISTING EQUIPMENT TO ENSURE QUICK TRANSITIONS FROM OLD TO NEW EQUIPMENT.
4	WHEN OLD WIRING AND CONDUIT INFRASTRUCTURE IS DISCONNECTED, DEMOLISH BACK TO THE SOURCE AND RECONDITION THE PATHWAY TO PROVIDE LIKE-NEW CONDITION WITH SEALED PENETRATIONS AS APPROPRIATE AND SPECIFIED. PROVIDE A FINISHED LOOK WHEREVER FEASIBLE. COORDINATE WITH ARCHITECTURAL TRADES.



**US Army Corps
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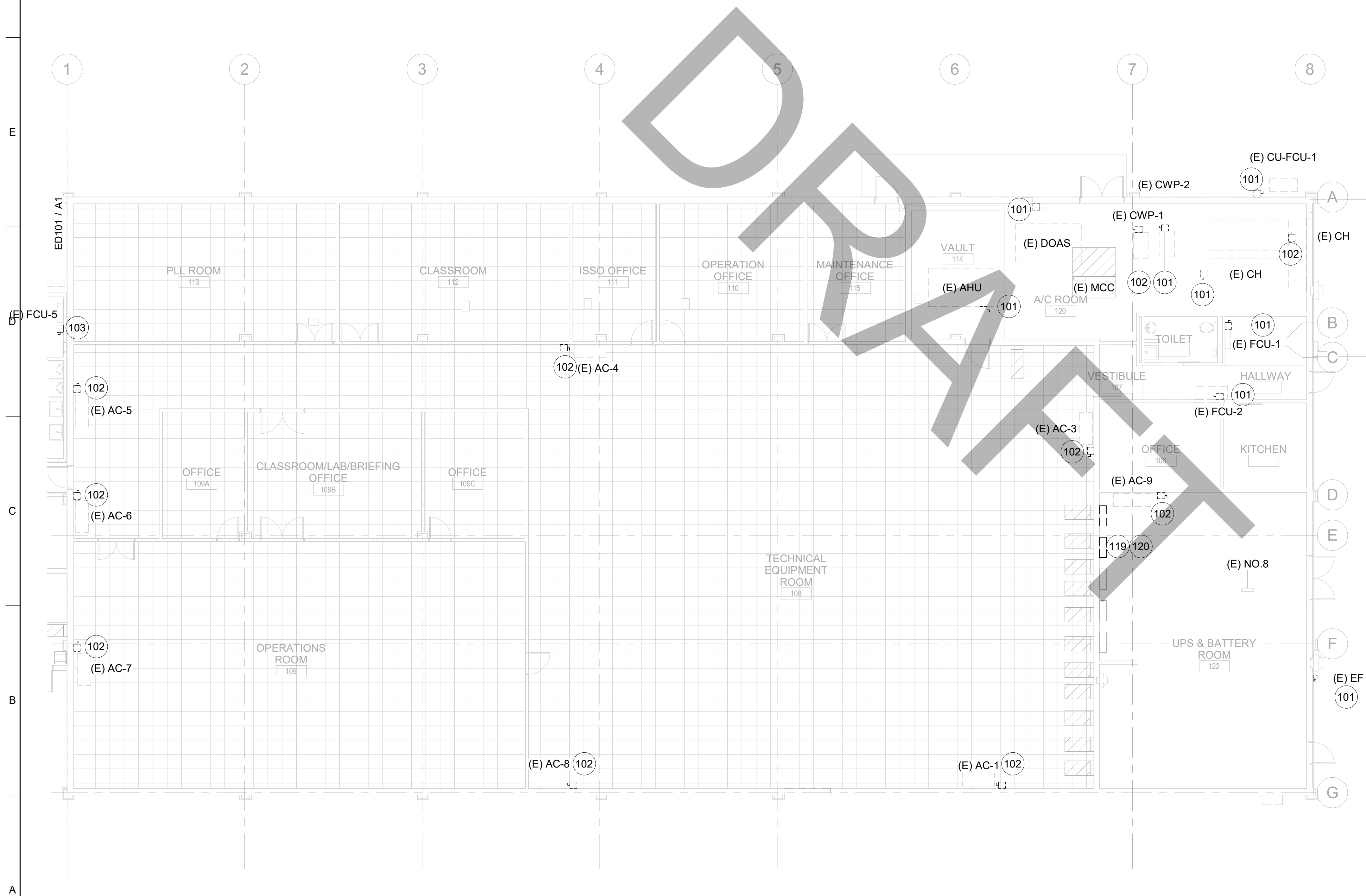
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Jacobs JACOBS GOVERNMENT SERVICES COMPANY 177 MAIN STREET, SUITE 100 #B-2000 FARMINGTON, CT 06030 USA	DESIGNED BY:		ISSUE DATE:	
	CZARANSKY /		09/04/2023	
	C ZABRANSKY /		SOLICITATION NO.:	
	C. ZABRANSKY /		CONTRACT NO.:	
	C. KOHRMANN /		DRAWING CODE:	
	C. KOHRMANN /			
		SUBMITTED BY:	A. GUERRA	
		SIZE:	FILE NAME:	
		ISO A1	ED102	

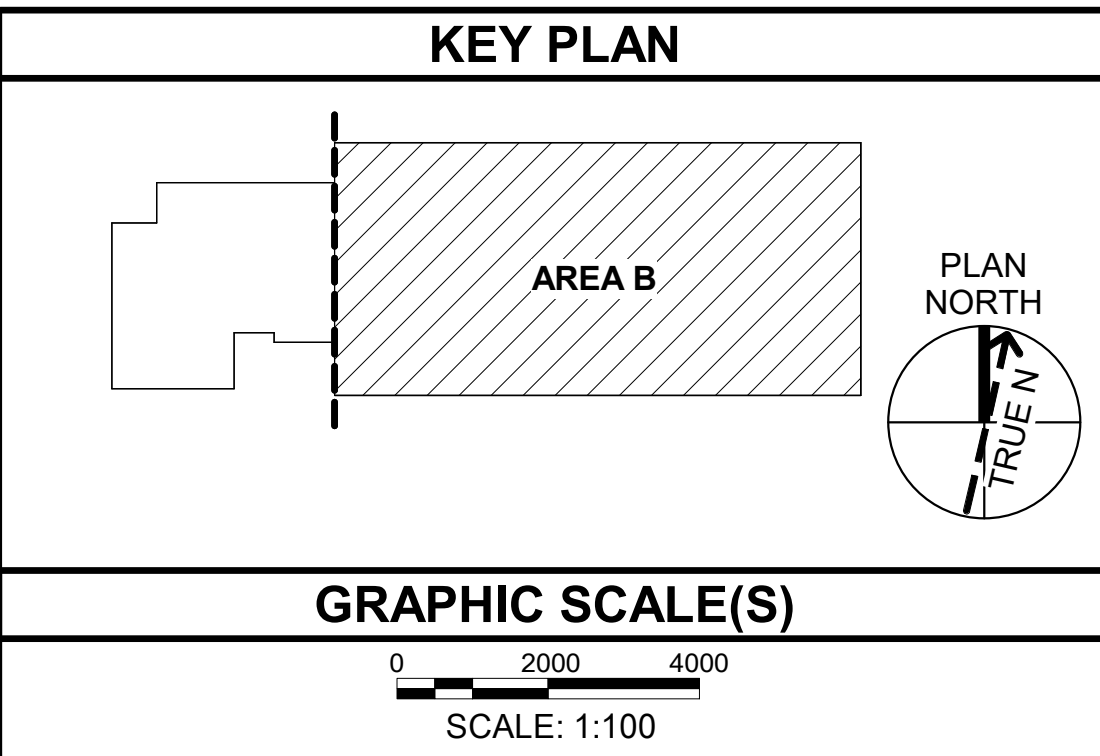
**FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX
OKINAWA, JAPAN
PROJECT NUMBER: D3448305**

PARTIAL DEMOLITION POWER PLAN - AREA B

SHEET ID
ED102



A1 PARTIAL POWER PLAN - AREA B - DEMO
1 : 100





A	REFER TO SHEET E-001 FOR GENERAL NOTES AND SYMBOL LEGEND.
B	REMOVE LIGHT FIXTURES, WIRE, AND THE ASSOCIATED CONDUITS BACK TO NEAREST JUNCTION BOX. VERIFY EXISTING CONDITIONS IN FIELD.



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THEY WERE PREPARED BY, OR
UNDER SUPERVISION OF:

CHRIS A. ZABRANSKY

NAME	
497	09/04/2025
ISTRATION #	DATE

TEXAS REGISTERED
ENGINEERING FIRM F-2966

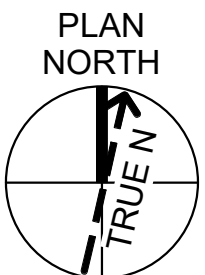
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118	DEMO ASSOCIATED LOCAL LIGHTING CONTROLS, PULL WIRE BACK TO NEAREST JUNCTION BOX.
-----	--

JACOBSON ENGINEERING DISTRICT OKINAWA, JAPAN	 JACOBSON ENGINEERING DISTRICT COMPANY 177 BANK STREET, SUITE 1000 NEW YORK, NY 10038 USA	DRAWN BY: C. ZABRANSKY	SOLICITATION NO.:
		CHECKED BY: C. KOHRMANN	CONTRACT NO.:
		SUBMITTED BY: A. GUERRA	DRAWING CODE:
		FILE NAME: SIZE	ISO A1 ED103

SHEET ID

ED103



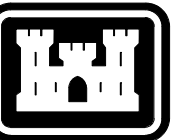
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SCALE: 1:100

KEYNOTES

118	DEMO ASSOCIATED LOCAL LIGHTING CONTROLS, PULL WIRE BACK TO NEAREST JUNCTION BOX.
-----	--

GENERAL NOTES - LIGHTING - DEMO

A	REFER TO SHEET E-001 FOR GENERAL NOTES AND SYMBOL LEGEND.
B	REMOVE LIGHT FIXTURES, WIRE, AND THE ASSOCIATED CONDUITS BACK TO NEAREST JUNCTION BOX. VERIFY EXISTING CONDITIONS IN FIELD.



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CHRIS A. ZABRANSKY
NAME
2497 09/04/2025
REGISTRATION # DATE
TEXAS REGISTERED
ENGINEERING FIRM F-2966

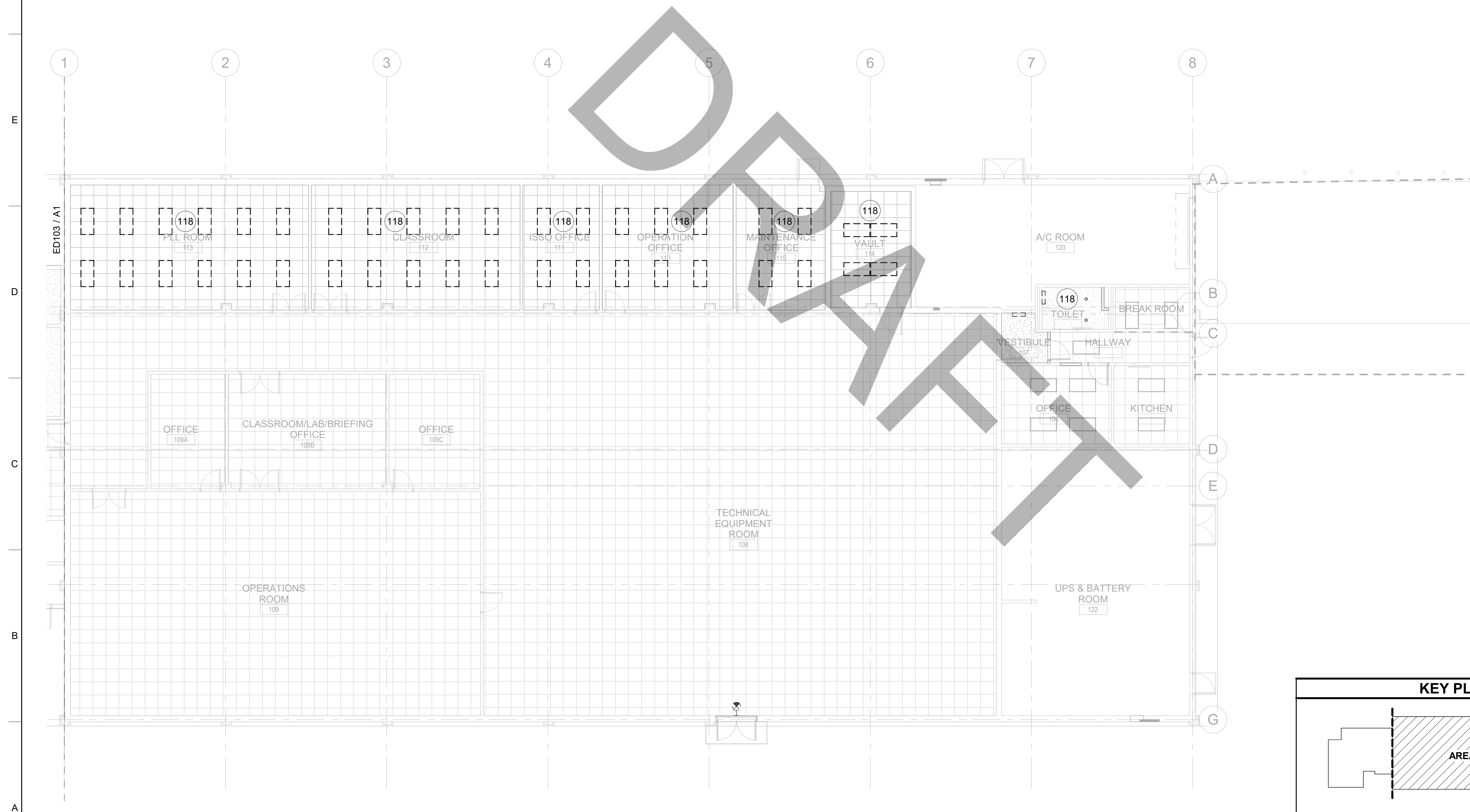
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Jacobs JACOBS GOVERNMENT SERVICES COMPANY 77 BAY STREET, SUITE 1000, NEW YORK, NY 10013 NEW YORK, NY 10013-1000, USA		(OFFICE ADDRESS)	
OKINAWA, JAPAN	SIZE: A1	FILE NAME: E0104	ISO:
DRAWING CODE:	CHECKED BY: A. GUERRA	DRAWN BY: C. KORMANN	CONTRACT NO.:
SOLICITATION NO.:	DRAWN BY: C. ZABRANSKY	DESIGNED BY: C. ZABRANSKY	ISSUE DATE: 04/07/2025

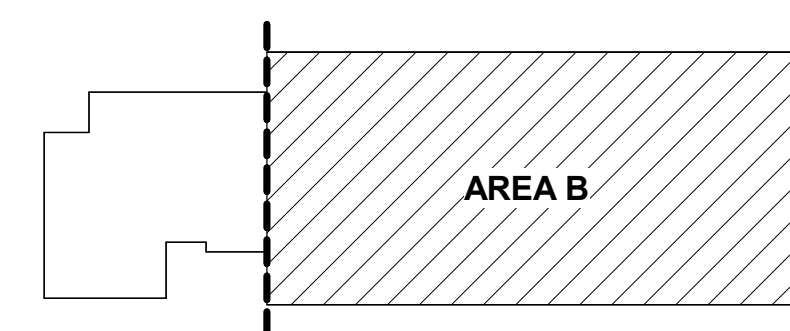
FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX
OKINAWA, JAPAN
PROJECT NUMBER:D3449305

SHEET ID

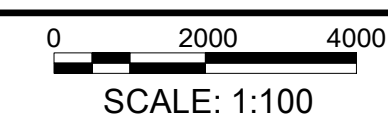
ED104



KEY PLAN



GRAPHIC SCALE(S)



A1

PARTIAL LIGHTING PLAN - AREA B - DEMO

1 : 100



KEY PLAN

AREA A

PLAN NORTH

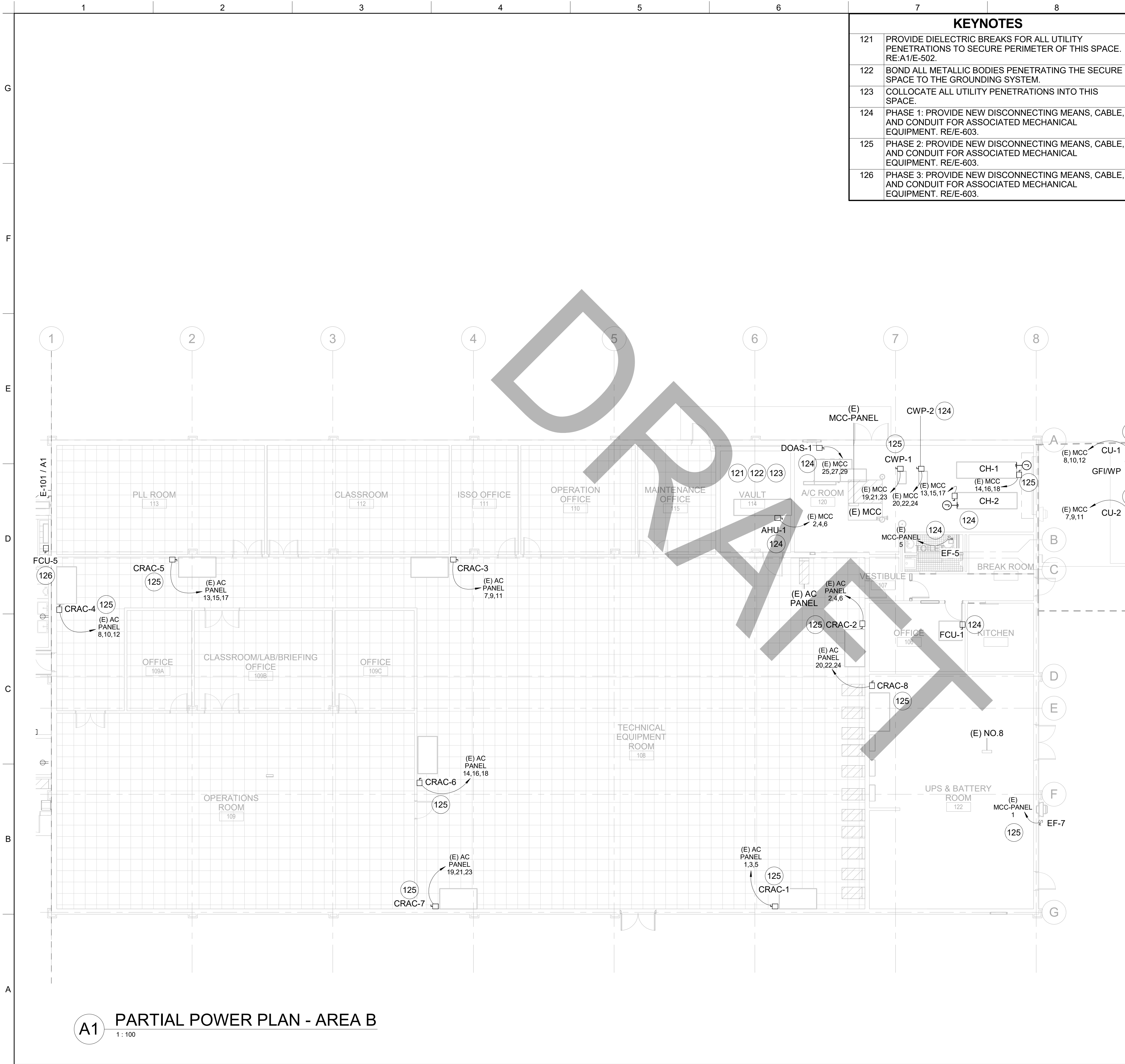
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SCALE: 1:100

SHEET ID
E-101

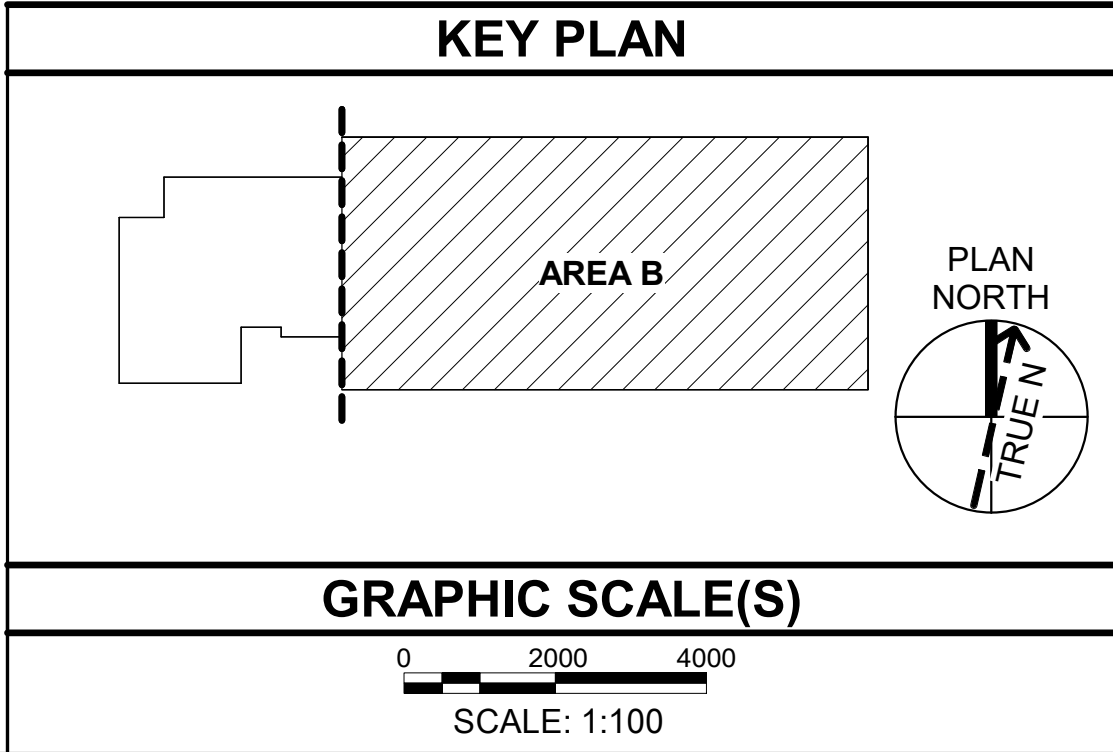
E E-102-



A1 PARTIAL POWER PLAN - AREA B
1 : 100

KEYNOTES	
121	PROVIDE DIELECTRIC BREAKS FOR ALL UTILITY PENETRATIONS TO SECURE PERIMETER OF THIS SPACE. RE:A1/E-502.
122	BOND ALL METALLIC BODIES PENETRATING THE SECURE SPACE TO THE GROUNDING SYSTEM.
123	COLLOCATE ALL UTILITY PENETRATIONS INTO THIS SPACE.
124	PHASE 1: PROVIDE NEW DISCONNECTING MEANS, CABLE, AND CONDUIT FOR ASSOCIATED MECHANICAL EQUIPMENT. RE/E-603.
125	PHASE 2: PROVIDE NEW DISCONNECTING MEANS, CABLE, AND CONDUIT FOR ASSOCIATED MECHANICAL EQUIPMENT. RE/E-603.
126	PHASE 3: PROVIDE NEW DISCONNECTING MEANS, CABLE, AND CONDUIT FOR ASSOCIATED MECHANICAL EQUIPMENT. RE/E-603.

GENERAL NOTES - POWER	
A	REFER TO SHEET E-001 FOR GENERAL NOTES AND SYMBOL LEGEND.
B	ALL EQUIPMENT LOCATED ON THE EXTERIOR OF THE BUILDING MUST BE NEMA 4X STAINLESS STEEL UON.
C	COORDINATE POWER REQUIREMENTS FOR ALL EQUIPMENT WITH OTHER TRADES.
D	BOND NEW ROOFTOP MECHANICAL EQUIPMENT TO EXISTING LIGHTING PROTECTION SYSTEM.
E	PHASE 1 TO BE FULLY OPERATIONAL PRIOR TO BEGINNING PHASE 2.
F	PHASE 2 TO BE FULLY OPERATIONAL PRIOR TO BEGINNING PHASE 3.
G	REFER TO SHEET G1010 AND G1011 FOR PHASING PLAN.
H	PERFORM AS MUCH WORK AS FEASIBLE WITH LOCKED OUT, UNTERMINATED CONDUCTORS TO NEW POWER FEEDS BEFORE DE-ENERGIZING EXISTING EQUIPMENT TO ENSURE QUICK TRANSITIONS FROM OLD TO NEW EQUIPMENT.





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Japan District @

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CHRIS A. ZABRANSKY
NAME
152497
REGISTRATION #
090402025
DATE
TEXAS REGISTERED ENGINEERING FIRM F-2596

ISSUE DATE: 09/04/2025	SOLICITATION NO.:
DESIGNED BY: C. ZABRANSKY	CONTRACT NO.:
CHECKED BY: C. ZABRANSKY	DRAWING CODE:
SUBMITTED BY: A. GUERRA	FILE NAME: E-102
SIZE:	ISO A1

Jacobs
JACOBS ENGINEERING GROUP, INC.
7700 W. HUNTERS TRAIL, SUITE 100
DALLAS, TEXAS 75247-1411
USA

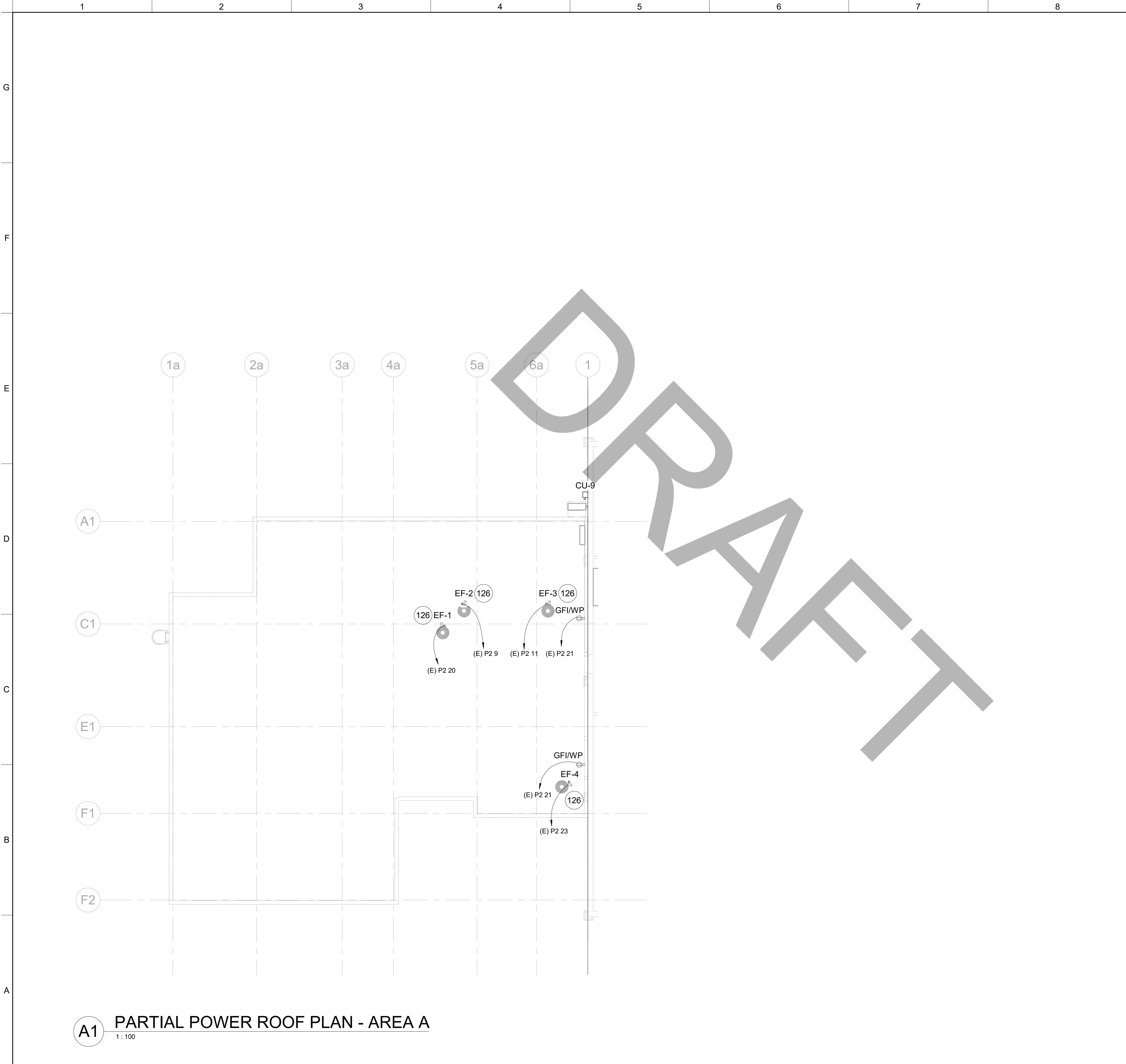
US ARMY CORPS OF ENGINEERS
JAPAN ENGINEER DISTRICT
OKINAWA, JAPAN

FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX
OKINAWA, JAPAN
PROJECT NUMBER: D3448305

PARTIAL POWER PLAN - AREA B

SHEET ID
E-102

E E-103-



GENERAL NOTES - POWER

A

REFER TO SHEET E-001 FOR GENERAL NOTES AND SYMBOL LEGEND.

B

ALL EQUIPMENT LOCATED ON THE EXTERIOR OF THE BUILDING MUST BE NEMA 4X STAINLESS STEEL UON.

C

COORDINATE POWER REQUIREMENTS FOR ALL EQUIPMENT WITH OTHER TRADES.

D

BOND NEW ROOFTOP MECHANICAL EQUIPMENT TO EXISTING LIGHTING PROTECTION SYSTEM.

E

PHASE 1 TO BE FULLY OPERATIONAL PRIOR TO BEGINNING PHASE 2.

F

PHASE 2 TO BE FULLY OPERATIONAL PRIOR TO BEGINNING PHASE 3.

G

REFER TO SHEET GI010 AND GI011 FOR PHASING PLAN.

H

PERFORM AS MUCH WORK AS FEASIBLE WITH LOCKED OUT, UNTERMINATED CONDUCTORS TO NEW POWER FEEDS BEFORE DE-ENERGIZING EXISTING EQUIPMENT TO ENSURE QUICK TRANSITIONS FROM OLD TO NEW EQUIPMENT.

KEYNOTES

126

PHASE 3: PROVIDE NEW DISCONNECTING MEANS, CABLE, AND CONDUIT FOR ASSOCIATED MECHANICAL EQUIPMENT. RE/E-603.

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CHRIS A. ZABRANSKY

NAME

09/04/2025

DATE

152497

REGISTRATION #

TEXAS REGISTERED ENGINEERING FIRM F-2966

MARK

DESCRIPTION

DATE

ISSUE DATE:
09/04/2025

SOLICITATION NO.:

CONTRACT NO.:

DRAWING CODE:

DESIGNED BY:
C. ZABRANSKY

CHECKED BY:
C. ZABRANSKY

SUBMITTED BY:
A. GUERRA

FILE NAME:
E-103

ISO A1

SIZE:

US ARMY CORPS OF ENGINEERS

JAPAN ENGINEER DISTRICT

OKINAWA, JAPAN

Jacobs

JACOBS ENGINEERING GROUP, INC.

7700 WEST 11TH AVENUE, SUITE 100

DENVER, CO 80202

USA

FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100

CAMP BUCKNER COMMUNICATIONS ANNEX

OKINAWA, JAPAN

PROJECT NUMBER: D3448305

PARTIAL POWER ROOF PLAN - AREA A

SHEET ID

E-103

KEY PLAN

AREA A

PLAN NORTH

TRUE N

GRAPHIC SCALE(S)

0

2000

4000

SCALE: 1:100

FILE PATH: C:\Users\zabranca\Documents\D3448305-E-3D_chris.zabransky.rvt

PLOT DATE: 8/29/2025 9:17:10 AM

PLOTTED BY:

PLOT SCALE: 1 : 100

BACKCHECK SUBMITTAL DESIGN

GENERAL NOTES - LIGHTING

- | | |
|---|--|
| A | REFER TO SHEET E-001 FOR GENERAL NOTES AND SYMBOL LEGEND. |
| B | CIRCUIT NEW LIGHT FIXTURES USING NEW WIRES FROM THE FIXTURES NEAREST JUNCTION BOX. ENSURE CONTINUITY OF THE EXISTING CIRCUIT AND THE EXISTING LIGHTS TO REMAIN STILL WORK PROPERLY, VERIFY EXISTING CONDITIONS IN FIELD. |



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CHRIS A. ZABRANSKY
NAME
2497 09/04/2025
REGISTRATION # DATE
TEXAS REGISTERED
ENGINEERING FIRM F-2966

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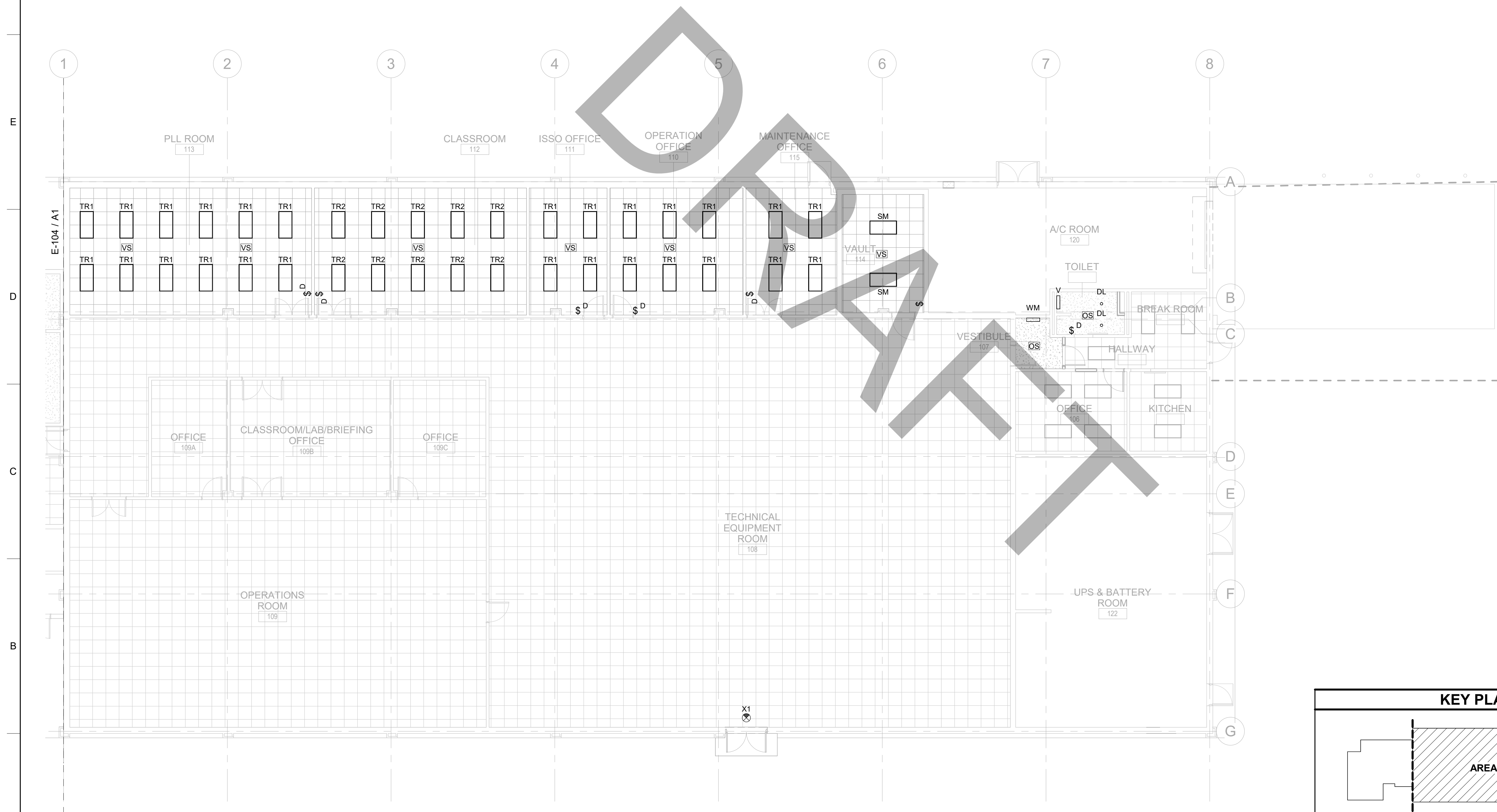
Jacobs JACOBS ENGINEERING GROUP, INC. 777 MAIN STREET, FORT WORTH, TX 76102 USA	JACOBS ENGINEERING GROUP, INC. 777 MAIN STREET, FORT WORTH, TX 76102 USA	
	US ARMY CORPS OF ENGINEERS ENGINEER DISTRICT OKINAWA, JAPAN	DRAWN BY: ZABRANSKY CHECKED BY: ZABRANSKY SUBMITTED BY: C. KOHRMANN DATE: 09/04/2025 CONTRACT NO.: SOLICITATION NO.:

OKINAWA, JAPAN
PROJECT NUMBER:D3449305

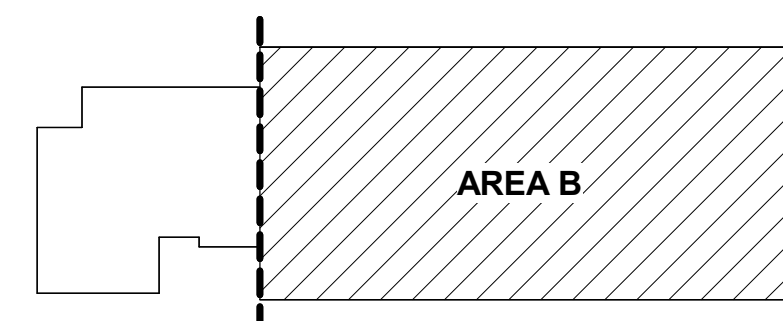
PARTIAL LIGHTING PLAN - AREA B

SHEET ID

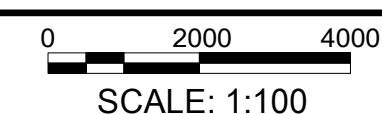
E-105



KEY PLAN



GRAPHIC SCALE(S)



SCALE: 1:100

A1

PARTIAL LIGHTING PLAN - AREA B

1 : 100

KEYNOTES

127	BOND FENCE TO GROUND RING USING 14MM2 CU.
-----	---

GENERAL NOTES - GROUNDING

- | | |
|---|--|
| A | THE DESIGN IS INTENDED TO SHOW THE BASIC GROUNDING ELECTRODE SYSTEM. VERIFY THE DIMENSIONS AND SITE CONDITIONS. |
| B | BOND ALL METALLIC UNDERGROUND PIPING ENTERING THE STRUCTURE TO THE GROUND RING AT ITS NEAREST POINT. |
| C | ALL BURIED CONNECTIONS MUST BE EXOTHERMIC WELDS. |
| D | GROUND METALLIC PIPES AT TWO LOCATIONS USING AN APPROVED PIPE CLAMP WITHIN 1.5M OF SERVICE ENTRANCE. |
| E | THOROUGHLY CLEAN AND COAT ALL CONTACT SURFACES AND APPLY TO CONDUCTOR AND LUG UL-LISTED COPPER BASED CONDUCTIVE GREASE RATED FOR GROUNDING AND PROTECTION OF COPPER AND STEEL SURFACE. |
| F | PROVED MAGNETIC TAPE OVER ALL BURIED GROUND CONDUCTORS. |



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CHRIS A. ZABRANSKY

NAME 09/04/2025

ISTRATION # DATE

TEXAS REGISTERED
ENGINEERING FIRM F-2966

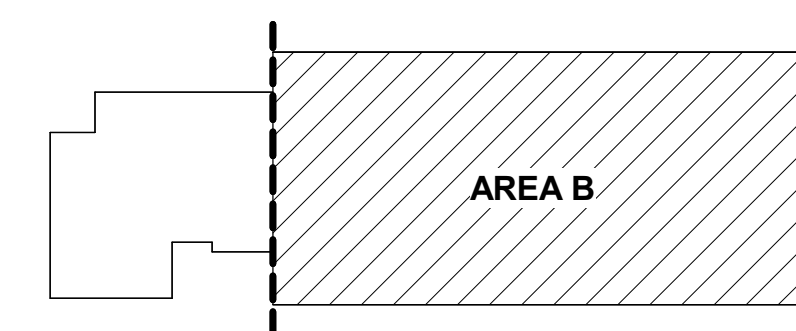
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Jacobs JACOBS ENGINEERING GROUP, INC. 777 MAIN STREET FORT WORTH, TX 76102 USA	JPE (Standard)
	US ARMY CORPS OF ENGINEERS JAPAN ENGINEERING DISTRICT OKINAWA, JAPAN

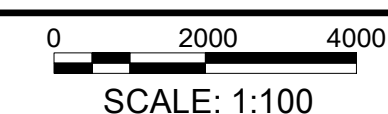
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FY25 REPAIR HVAC WIDESAND SAILCOW OPS BTU0
CAMP BUCKNER COMMUNICATIONS ANNEX
OKINAWA, JAPAN
PROJECT NUMBER:D3448305
PARTIAL GROUNDING PLAN - AREA B

SHEET ID
E-106



GRAPHIC SCALE(S)



SCALE: 1:100

PARTIAL GROUNDING PLAN - AREA B

1 : 100

E E-502-

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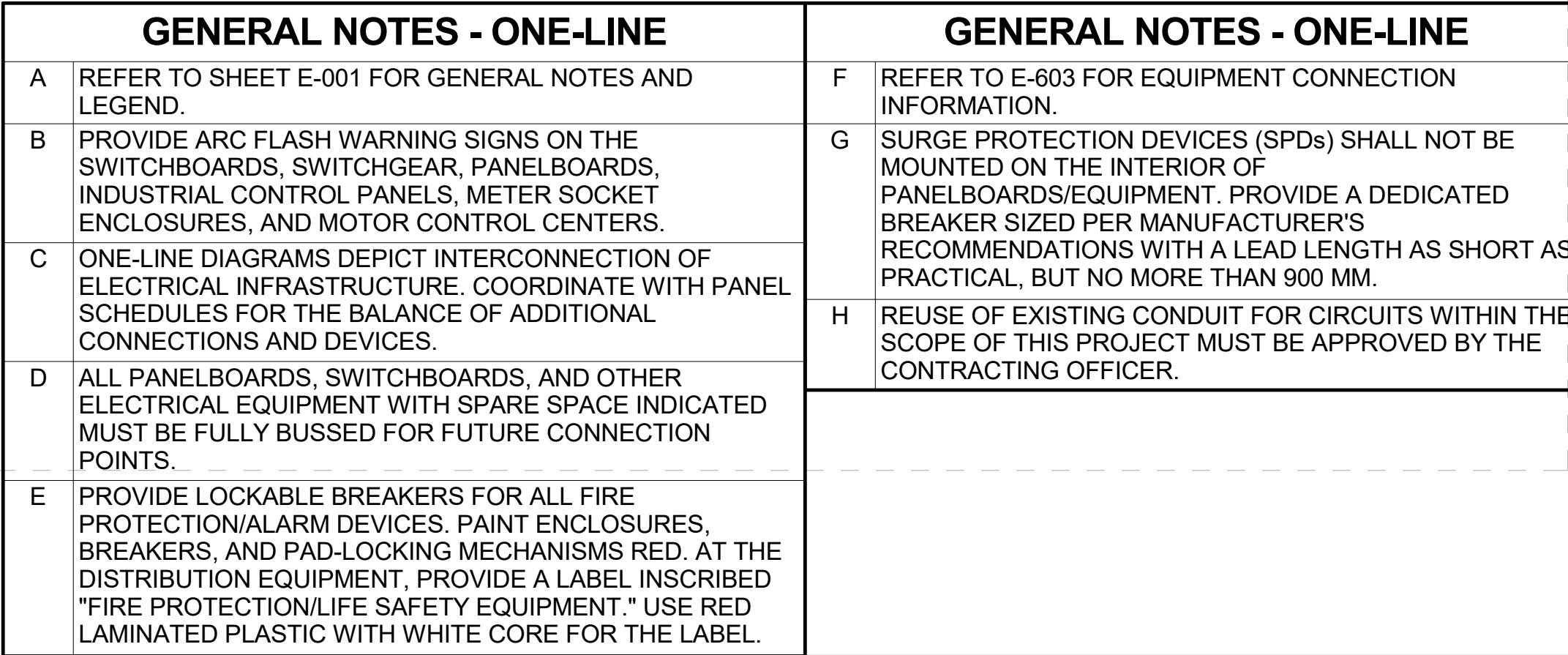
CHRIS A. ZABRANSKY	
NAME	
2497	09/04/2025
REGISTRATION #	DATE
TEXAS REGISTERED	
ENGINEERING FIRM F-2966	

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Jacobs JACOBS GOVERNMENT SERVICES COMPANY 777 MAIN STREET, FORT WORTH, TX 76102 USA	SIZE: ISO A	FILE NAME: E-602
	SUBMITTED BY: A. GUERRA	
JAPAN ENGINEER DISTRICT OKINAWA, JAPAN	CHECKED BY: C. KOHRMANN	DRAWING CODE:
	DRAWN BY: C. ZABRANSKY	CONTRACT NO.:
	SOLICITATION NO.:	09/04/2025
	C. ZABRANSKY	09/04/2025

OKINAWA, JAPAN
PROJECT NUMBER:D3449305
ONE-LINE DIAGRAM

E-602



LOCATION: A/C ROOM 120
SUPPLIED FROM:
MOUNTING: Recessed
ENCLOSURE: Type 1

VOLTAGE: 480/277 Wye
PHASES: 3
WIRES: 4
FREQUENCY: 60 HZ

A.I.C. RATING: 65000
MAINS TYPE: MCB
MAINS RATING: 1200 A
MCB RATING: 1200 A

LOCATION: TECHNICAL EQUIPMENT ROOM 108
SUPPLIED FROM: (E) MCC
MOUNTING: Surface
ENCLOSURE: Type 1

VOLTAGE: 480/277 Wye
PHASES: 3
WIRES: 4
FREQUENCY: 60 HZ

A.I.C. RATING: 25000
MAINS TYPE: MCB
MAINS RATING: 400 A

NOTES:

CKT	DESCRIPTION	TRIP	L	P	A	B	C	P	L	TRIP	DESCRIPTION	CKT
1					22701 VA	2084 VA						2
3	(E) AC PANEL	100 A	3			22701 VA	2084 VA		3	15 A	AHU-1	4
5							22701 VA	2084 VA				6
7					14765 VA	14765 VA						8
9	CU-2	100 A	3			14765 VA	14765 VA		3	100 A	CU-1	10
11							14765 VA	14765 VA				12
13	CH-2	400 A	3		64582 VA	64582 VA			3	400 A	CH-1	14
15						64582 VA	64582 VA					16
17							64582 VA	64582 VA				18
19	CWP-1	30 A	3		5820 VA	5820 VA			3	30 A	CWP-2	20
21						5820 VA	5820 VA					22
23							5820 VA	5820 VA				24
25	DOAS-1	30 A	3		5625 VA	50 VA			3	15 A	(E) AIR COMP	26
27						5625 VA	50 VA					28
29							5625 VA	50 VA				30
31						10000 VA						32
33							10000 VA		3	50 A	(E) DTTX	34
35								10000 VA				36
37					500 VA	30000 VA						38
39	(E) MOV	15 A	3			500 VA	30000 VA		3	150 A	(E) CAP BANK	40
41							500 VA	30000 VA				42
TOTAL LOAD:					241294 VA	241294 VA	241294 VA					
TOTAL AMPS:					871 A	871 A	871 A					

LEGEND:

* = GFI BREAKER, ** = AFCI BREAKER, S = SHUNT TRIP BREAKER, F = PAINTED RED, L = LOCKABLE IN OPEN POSITION REQUIRED

LOAD CLASSIFICATION	CONNECTED LOAD	DEMAND FACTOR	ESTIMATED DEMAND	PANEL TOTALS
Motor	602232 VA	85.00%	511897 VA	
Spare	121650 VA	100.00%	121650 VA	
				TOTAL CONNECTED: 723882 VA
				TOTAL DEMAND: 633547 VA
				TOTAL CONNECTED: 871 A
				TOTAL DEMAND: 762 A

NOTES:

LOCATION: A/C ROOM 120
SUPPLIED FROM:
MOUNTING: Recessed
ENCLOSURE: Type 1

VOLTAGE: 120/208 Wye
PHASES: 3
WIRES: 4
FREQUENCY: 60 HZ

A.I.C. RATING: 10000
MAINS TYPE: MLO
MAINS RATING: 100 A

NOTES:

CKT	DESCRIPTION	TRIP	L	P	A	B	C	P	L	TRIP	DESCRIPTION	CKT
1	EF-7	15 A	1		864 VA	500 VA						2
3	EXT REC EAST	20 A	1			180 VA	500 VA		1	20 A	CH-1 CONTROLS	4
5	EF-5	20 A	1					19 VA		20 A	CH-2 CONTROLS	6
7												8
9												10
11												12
13												14
15												16
17												18
19												20
21												22
23												24
25												26
27												28
29												30
31												32
33												34
35												36
37												38
39												40
41												42
TOTAL LOAD:		1364 VA				680 VA		19 VA				
TOTAL AMPS:		12 A				7 A		0 A				

LEGEND:

* = GFI BREAKER, ** = AFCI BREAKER, S = SHUNT TRIP BREAKER, F = PAINTED RED, L = LOCKABLE IN OPEN POSITION REQUIRED

LOAD CLASSIFICATION	CONNECTED LOAD	DEMAND FACTOR	ESTIMATED DEMAND	PANEL TOTALS
Motor	883 VA	85.00%	750 VA	
REC	180 VA	100.00%	180 VA	TOTAL CONNECTED: 2063 VA
OTHERS	1000 VA	100.00%	1000 VA	TOTAL DEMAND: 1930 VA
				TOTAL CONNECTED: 6 A
				TOTAL DEMAND: 5 A

NOTES:

CKT	DESCRIPTION	TRIP	L	P	A		B		C		P	L	TRIP	DESCRIPTION	CKT
1					2838 VA	2838 VA									2
3	CRAC-1	20 A	3				2838 VA	2838 VA				3	20 A	CRAC-2	4
5									2838 VA	2838 VA					6
7					2838 VA	2838 VA									8
9	CRAC-3	20 A	3				2838 VA	2838 VA				3	20 A	CRAC-4	10
11									2838 VA	2838 VA					12
13					2838 VA	2838 VA									14
15	CRAC-5	20 A	3				2838 VA	2838 VA				3	20 A	CRAC-6	16
17									2838 VA	2838 VA					18
19					2838 VA	2838 VA									20
21	CRAC-7	20 A	3				2838 VA	2838 VA				3	20 A	CRAC-8	22
23									2838 VA	2838 VA					24
25															26
27															28
29															30
31															32
33															34
35															36
37															38
39															40
41															42
TOTAL LOAD:					22701 VA		22701 VA		22701 VA						
TOTAL AMPS:					82 A		82 A		82 A						

LEGEND:

* = GFI BREAKER, ** = AFCI BREAKER, S = SHUNT TRIP BREAKER, F = PAINTED RED, L = LOCKABLE IN OPEN POSITION REQUIRED

LOAD CLASSIFICATION	CONNECTED LOAD	DEMAND FACTOR	ESTIMATED DEMAND	PANEL TOTALS
Motor	68104 VA	85.00%	57888 VA	
				TOTAL CONNECTED: 68104 VA
				TOTAL DEMAND: 57888 VA
				TOTAL CONNECTED: 82 A
				TOTAL DEMAND: 70 A

NOTES:

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UNDER SUPERVISION OF:
RIS A. ZABRANSKY
NAME **09/04/2025**
DATE
EXAMINATION #
EXAMINER
EXAMINER'S FIRM F-2966

[illegible]

JAPAN ENGINEER DISTRICT OKINAWA, JAPAN		09/04/2025	
DRAWN BY: C. ZAGRANSKY		SOLICITATION NO.:	
CHECKED BY: C. KOHRMANN		CONTRACT NO.:	
SUBMITTED BY: A. GUERRA		DRAWING CODE:	
SIZE:	FILE NAME:	ISO 1	E-B04

CAMP BUCKNER COMMUNICATIONS ANNEX
OKINAWA, JAPAN
PROJECT NUMBER: D3448305
PANEL SCHEDULES

SHEET ID

E-604

G

F

E

D

C

B

A

BRANCH PANEL: (E) P4

LOCATION: ELEC...

SUPPLIED FROM:

MOUNTING: Surface

ENCLOSURE: Type 1

VOLTAGE: 480/277 Wye

PHASES: 3

WIRES: 4

FREQUENCY: 60 HZ

A.I.C. RATING: 14000

MAINS TYPE: MCB

MAINS RATING: 225 A

MCB RATING: 225 A

NOTES:

CKT	DESCRIPTION	TRIP	L	P	A	B	C	P	L	TRIP	DESCRIPTION	CKT
1					14325 VA	1109 VA						2
3	(E) T-P2	60 A		3		12305 VA	1109 VA			30 A	AHU-2	4
5							11185 VA	1109 VA				6
7					1432 VA	5000 VA						8
9	DOAS-2	30 A		3		1432 VA	5000 VA			20 A	(E) SPARE	10
11							1432 VA	5000 VA				12
13					1197 VA	4212 VA						14
15	CU-6	20 A		3		1197 VA	4212 VA			30 A	CU-5	16
17							1197 VA	4212 VA				18
19												20
21												22
23												24
25												26
27												28
29												30
31												32
33												34
35												36
37												38
39												40
41												42
TOTAL LOAD:					27275 VA	25255 VA	24135 VA					
TOTAL AMPS:					99 A	92 A	87 A					

LEGEND:

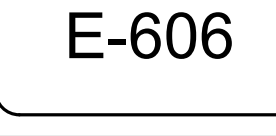
* = GFI BREAKER, ** = AFCI BREAKER, S = SHUNT TRIP BREAKER, F = PAINTED RED, L = LOCKABLE IN OPEN POSITION REQUIRED

LOAD CLASSIFICATION	CONNECTED LOAD	DEMAND FACTOR	ESTIMATED DEMAND	PANEL TOTALS
Motor	23849 VA	85.00%	20272 VA	TOTAL CONNECTED: 76664 VA
Spare	52815 VA	100.00%	52815 VA	TOTAL DEMAND: 73087 VA
				TOTAL CONNECTED: 92 A
				TOTAL DEMAND: 88 A

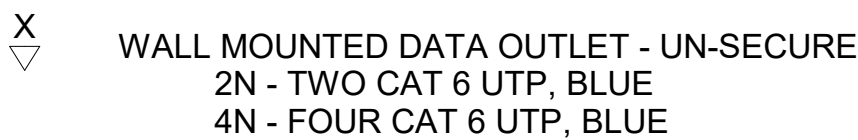
NOTES:

BRANCH PANEL: (E) P2-1												
LOCATION: MECHANICAL ROOM 118					VOLTAGE: 120/208 Wye					A.I.C. RATING:		
SUPPLIED FROM: T-P2					PHASES: 3					MAINS TYPE:		
MOUNTING: Surface					WIRES: 4					MAINS RATING: 225 A		
ENCLOSURE: Type 1					FREQUENCY: 60 HZ					MCB RATING: 175 A		
NOTES:												
CKT	DESCRIPTION	TRIP	L	P	A	B	C	P	L	TRIP	DESCRIPTION	CKT
1					7770 VA							2
3	(E) PANEL D	20 A		3		7580 VA						4
5							7380 VA					6
7					555 VA	2750 VA				2	20 A (E) EWH	8
9	(E) SEWAGE PUMP UNIT	20 A		3		555 VA	2750 VA					10
11							555 VA	2500 VA		2	20 A (E) DRYER	12
13						2500 VA						14
15							750 VA			2	20 A (E) SPARE	16
17								750 VA				18
19						750 VA				1	20 A (E) SPARE	20
21							670 VA			1	20 A (E) D.H.W. CIRC PUMP	22
23												24
25												26
27												28
29												30
31												32
33												34
35												36
37												38
39												40
41												42
TOTAL LOAD:					14325 VA	12305 VA	11185 VA					
TOTAL AMPS:					121 A	104 A	93 A					
LEGEND:												
* = GFI BREAKER, ** = AFCI BREAKER, S = SHUNT TRIP BREAKER, F = PAINTED RED, L = LOCKABLE IN OPEN POSITION REQUIRED												
LOAD CLASSIFICATION					CONNECTED LOAD	DEMAND FACTOR	ESTIMATED DEMAND	PANEL TOTALS				
Spare					37815 VA	100.00%	37815 VA	TOTAL CONNECTED: 37815 VA				
								TOTAL DEMAND: 37815 VA				
								TOTAL CONNECTED: 105 A				
								TOTAL DEMAND: 105 A				
NOTES:												

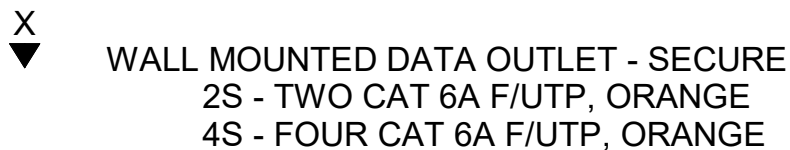
BRANCH PANEL: (E) P2														
LOCATION: MECHANICAL ROOM 118					VOLTAGE: 120/208 Wye					A.I.C. RATING: 10000				
SUPPLIED FROM:					PHASES: 3					MAINS TYPE: MCB				
MOUNTING: Surface					WIRES: 4					MAINS RATING: 225 A				
ENCLOSURE: Type 1					FREQUENCY: 60 HZ									
NOTES:														
CKT	DESCRIPTION	TRIP	L	P	A	B	C	P	L	TRIP	DESCRIPTION	CKT		
1	(E) LIGHTS	20 A		1	980 VA	630 VA				1	20 A (E) LIGHTS	2		
3	(E) LIGHTS	20 A		1		870 VA	820 VA			1	20 A (E) LIGHTS	4		
5	(E) LIGHTS	20 A		1			840 VA	890 VA		1	20 A (E) LIGHTS	6		
7	(E) LIGHTS	20 A		1	445 VA	760 VA				1	20 A (E) LIGHTS	8		
9	EF-2	15 A		1		336 VA	900 VA			1	20 A (E) RECEPTACLES	10		
11	EF-3	15 A		1			154 VA	510 VA		1	20 A (E) EWC	12		
13	(E) MICROWAVE OVEN	20 A		1	850 VA	1130 VA				1	20 A (E) WASHER	14		
15	(E) RECEPTACLES	20 A		1		1000 VA	1000 VA			1	20 A (E) PROJECTOR	16		
17	(E) REFRIGERATOR	20 A		1			670 VA	720 VA		1	20 A (E) RECEPTACLES	18		
19	(E) DISPOSER	20 A		1	750 VA	9 VA				1	15 A EF-1	20		
21	ROOFTOP REC	20 A		1		360 VA	750 VA			1	20 A (E) EXHAUST FAN	22		
23	EF-4	15 A		1			154 VA	1184 VA		2	30 A CU-4	24		
25	EXTERIOR REC NORTH	20 A		1	180 VA	1184 VA						26		
27	F MNS	20 A		1		1200 VA	641 VA			2	15 A CU-7	28		
29								641 VA				30		
31					761 VA							32		
33	CU-8	15 A		2		761 VA						34		
35								1184 VA				36		
37	CU-9	20 A		2	1184 VA							38		
39												40		
41												42		
TOTAL LOAD:					8862 VA	8638 VA	6946 VA							
TOTAL AMPS:					76 A	74 A	58 A							
LEGEND:														
* = GFI BREAKER, ** = AFCI BREAKER, S = SHUNT TRIP BREAKER, F = PAINTED RED, L = LOCKABLE IN OPEN POSITION REQUIRED														
LOAD CLASSIFICATION					CONNECTED LOAD	DEMAND FACTOR	ESTIMATED DEMAND	PANEL TOTALS						
Motor					8190 VA	85.00%	6962 VA	TOTAL CONNECTED: 24445 VA						
Spare					14515 VA	100.00%	14515 VA	TOTAL DEMAND: 23217 VA						
REC					540 VA	100.00%	540 VA	TOTAL CONNECTED: 68 A						
OTHERS					1200 VA	100.00%	1200 VA	TOTAL DEMAND: 64 A						
NOTES: 1. PROVIDE RED LOCK-ON BREAKER FOR FA EQUIPMENT.														



A
B
C
D
E
F
G



SEE SHEET T-502 FOR FACE PLATE DETAILS
SEE GENERAL NOTE P/T-001 FOR MOUNTING HEIGHTS
CABLE TO MATCH OUTLET JACK COLORING



SEE SHEET T-502 FOR FACE PLATE DETAILS
SEE GENERAL NOTE P/T-001 FOR MOUNTING HEIGHTS
CABLE TO MATCH OUTLET JACK COLORING



WAP - TWO UNSECURE CAT6 UTP FOR ACCESS
POINT
CAM - ONE CAT 6 UTP FOR IP CAMERA

SEE SHEET T-502 FOR FACE PLATE DETAILS
SEE GENERAL NOTE P/T-001 FOR MOUNTING HEIGHTS
CABLE TO MATCH OUTLET JACK COLORING



FB2 FLOOR BOX OUTLET DATA WITH TWO (2)
CAT 6. ONE INCH CONDUIT FROM NEAREST
BLACK CABLE TRAY OR AS INDICATED ON
PLANS.

ACS
ACPB
AFC
AFF
AFG
ANN
ARCH
AV

BAS	BUILDING AUTOMATION SYSTEM
BATT	BATTERY
BB	BACKBONE
BBFPP	BACKBONE FIBER PATCH PANEL
BBV	BACKBONE VOICE
BEPB	BUILDING ENTRANCE PROTECTION BLOCKS
BLDG	BUILDING
BMS	BALANCED MAGNETIC SWITCH

C	CONDUIT
CAB	CABINET
CAT	CATEGORY
CCTV	CLOSED CIRCUIT TELEVISION
CLG	CEILING
CMH	COMMUNICATIONS MANHOLE
COB	CENTER OF BOX
COMM	COMMUNICATIONS
CONST	CONSTRUCTION
CONT	CONTINUATION

DIST	DISTRIBUTION
DCPB	DC POWER BLOCK
DWG	DRAWING

(E)	EXISTING
EF	ENTRANCE FACILITY
EL	ELEVATION
ELEC	ELECTRICAL
ELEV	ELEVATOR
EMER	EMERGENCY
ESM	ENERGY STORAGE M
ESS	ENERGY STORAGE S
ESU	ENERGY STORAGE U
EQPT	EQUIPMENT
EXIST	EXISTING

FA	FIRE ALARM
FACP	FIRE ALARM CONTROL PANEL
FIXT	FIXTURE
FLR	FLOOR
FO	FIBER OPTIC
FTP	FIELD TERM PLUG

GSF GRIDSTAR FLOW

HCPP	HORIZONTAL COPPER PATCH PANEL
HFPP	HORIZONTAL FIBER PATCH PANEL
HMI	HUMAN MACHINE INTERFACE
HOV	HORIZONTAL VOICE

IDS INTRUSION DETECTION SYSTEM

JB JUNCTION BOX

MFR	MANUFACTURER
MH	MOUNTING HEIGHT
MMF	MULTI MODE FIBER
MTD	MOUNTED

NIC NOT IN CONTRACT
NTS NOT TO SCALE

OH	OVERHEAD
OSP	OUTSIDE PLANT

PA	PUBLIC ADDRESS
PB	PULL BOX
PBB	PRIMARY BONDING BUSBAR
PBX	PRIVATE BRANCH EXCHANGE
PC	PERSONAL COMPUTER
PCS	POWER CONVERSION SYSTEM (INVERTER)
PLAS	PLASTIC
PP	PATCH PANEL
PR	PAIR
PWR	POWER

QTY QUANTITY

RBB	RACK BONDING BUSBAR
RBC	RACK BONDING CONDUCTOR
RECP	RECEPTACLE
REQD	REQUIRED
RGS	RIGID GALVANIZED STEEL
RM	ROOM
RSC	RIGID STEEL CONDUIT

SBB	SECONDARY BONDING BUSBAR
SC	SITE CONTROLLER
SCC	SITE CONTROLLER CABINET
SCRT	SECURITY
SECT	SECTION
SPEC	SPECIFICATION
SMF	SINGLE MODE FIBER
SP	SERVICE PROVIDER
SURF	SURFACE
SW	SWITCH
SYM	SYMMETRICAL
SYS	SYSTEMS

TBB	TELECOMMUNICATIONS BONDING BACKBONE
TEL	TELEPHONE
TELECOM	TELECOMMUNICATIONS
TER	TELECOMMUNICATIONS EQUIPMENT ROOM
TR	TELECOMMUNICATIONS ROOM
TYP	TYPICAL

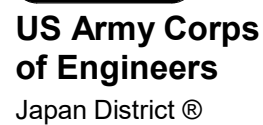
UG	UNDERGROUND
UON	UNLESS OTHERWISE NOTED
UPS	UNINTERRUPTIBLE POWER SUPPLY

WP WEATHERPROOF

XFR TRANSFER
XP EXPLOSION PROOF

NOTE: TELECOMMUNICATIONS PLANS AND SPECIFICATIONS ARE FOR INFORMATION ONLY. THERE IS NO NEW TELECOMMUNICATIONS WORK ASSOCIATED WITH THE MECHANICAL RENOVATIONS. TELECOMMUNICATIONS PLANS AND SPECIFICATIONS SERVE AS A GUIDEPOINT AND REFERENCE OF EXISTING CONDITIONS SHOULD ANY DAMAGE OCCUR TO NETWORK OPERATIONS DURING CONSTRUCTION. THE TELECOMMUNICATIONS PLANS CAN BE USED AS A GUIDE FOR REMEDIAL ACTION IF / AS REQUIRED. UNINTERRUPTED NETWORK OPERATIONS MUST BE MAINTAINED DURING CONSTRUCTION. IT IS THE CONTRACTORS RESPONSIBILITY TO REPAIR ANY DAMAGE MADE AS A RESULT OF CONSTRUCTION.

A	REFER TO G-1002 AND G-1003 FOR FULL LIST OF INTERDISCIPLINARY ABBREVIATIONS.
B	COORDINATE WITH THE COMMUNICATIONS POINT OF CONTACT FOR CHANGES TO EXISTING COMMUNICATION CABLING.
C	PROVIDE U.L. CLASSIFIED FIRE RATED ASSEMBLY AT PENETRATIONS OF FIRE RATED WALLS. COORDINATE WITH ARCHITECTURAL DRAWINGS/DETAILS/SPECIFICATIONS.
D	PROVIDE INFORMATION TECHNOLOGY WIRING (I.E. LAN, PHONE, ETC.) UP TO AND INCLUDING THE FACE PLATE FOR FREESTANDING AND/OR SYSTEMS FURNITURE DESK TOPS.
E	DIMENSIONS SHOWN ON THESE DRAWINGS ARE INTENDED TO PROVIDE GENERAL LOCATIONS AS REQUESTED BY THE OWNER. THE CONTRACTOR IS RESPONSIBLE FOR CONFIRMING DIMENSIONS PRIOR TO ROUGH-IN AND IMMEDIATELY REPORT CONFLICTS WITH THE OUTLET PLACEMENT TO THE GENERAL CONTRACTOR
F	PLASTIC TIE WRAPS ARE NOT PERMITTED ON THIS INSTALLATION. CABLE ROUGH-IN AND DRESS-OUT MUST BE DONE WITH VELCRO ONLY. CABLES BUNDLED WITH PLASTIC TIE WRAPS MUST BE IMMEDIATELY REPLACED AT THE CONTRACTOR'S EXPENSE.
G	DO NOT INSTALL CABLES IN CONDUIT SLEEVE, STUB UP, OR WALL CAP WITHOUT A PROTECTIVE BUSHING. CABLES PULLED INTO UNPROTECTED CONDUITS MUST BE REPLACED AT THE CONTRACTOR'S EXPENSE. DO NOT INSTALL CABLING INTO CONDUIT THAT HAS NOT BEEN CONFIRMED TO BE BLOWN CLEAR. COORDINATE WITH ELECTRICAL PRIOR TO ROUGH-IN OF CABLING.
H	CABLING FOUND PAINTED DURING THE CONSTRUCTION PROCESS MUST BE REPLACED AT THE CONTRACTOR'S EXPENSE. DIV 27 CONTRACTOR WILL COORDINATE WITH THE GENERAL CONTRACTOR AND PAINTER TO AVOID CONFLICTS.
I	IT IS THE RESPONSIBILITY OF THE CONTRACTORS TO REFERENCE BOTH THE PROJECT DRAWINGS AND THE PROJECT SPECIFICATIONS. TOGETHER THEY FORM THE COMPLETE CONTRACT DOCUMENTS. PLEASE REFER TO THE PROJECT SPECIFICATIONS FOR MATERIALS, EQUIPMENT AND COMPONENTS NOT INDICATED ON THE DRAWING SET.
J	CABLING INSTALLED IN AREAS WITH EXPOSED CEILINGS MUST BE INSTALLED IN A PROPERLY SIZED CONDUIT PATHWAY. CONDUIT PATHWAYS MUST BE PROVIDED BY THE CONTRACTOR. THE CONDUIT MAY BE PAINTED TO MATCH CEILING COLOR AS DIRECTED BY THE CONTRACT DOCUMENTS.
K	CABLE, PATHWAYS, TERMINATIONS AND EQUIPMENT MUST MAINTAIN MINIMUM SEPARATION DISTANCES BETWEEN SECURE AND UN-SECURE DATA . REFERENCE A1/T-501
L	HORIZONTAL COPPER CABLING MUST BE ROUTED IN BASKET TYPE TRAYS OVERHEAD. CABLES ROUTING BETWEEN THE DATA OUTLET AND THE CABLE TRAY MUST BE INSTALLED WITHIN METALLIC CONDUITS. SECURE PATHWAYS ON STC WALLS MUST BE SURFACE MOUNTED AND VISIBLE FOR INSPECTION.
M	SYMBOLS, ABBREVIATIONS AND GENERAL NOTES SHOWN ARE NOT NECESSARILY USED ON THIS PROJECT.
N	SEE RISER DIAGRAM FOR CABLE SIZES, CONDUIT SIZES, AND DEVICES NOT SHOWN ON PLANS.
O	SECURITY CONDUITS MUST BE EMT UTILIZING COMPRESSION TYPE FITTINGS AND INCLUDE 200 LBS TENSILE STRENGTH PULL TAPES.
P	MOUNTING HEIGHTS ABOVE FINISHED FLOOR, TO CENTERLINE OF DEVICE, ARE AS FOLLOWS, UNLESS OTHERWISE NOTED:
	DATA OUTLETS-----457MM
	TELEPHONE OUTLETS-----1060MM
	COUNTER TOP OUTLETS (CT)-----200MM ABOVE COUNTER
	WIRELESS ACCESS POINT-----300MM ABOVE FINISHED CEILING
	CEILING SPEAKERS-----FLUSH WITH FINISHED CEILING
	TELEVISION OUTLET-----1800MM
	CARD READER BACKBOX-----1060MM



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RENE RUBIO

NAME	09/04/2021
REGISTRATION #	DATE
TEXAS REGISTERED ENGINEERING FIRM F-2966	

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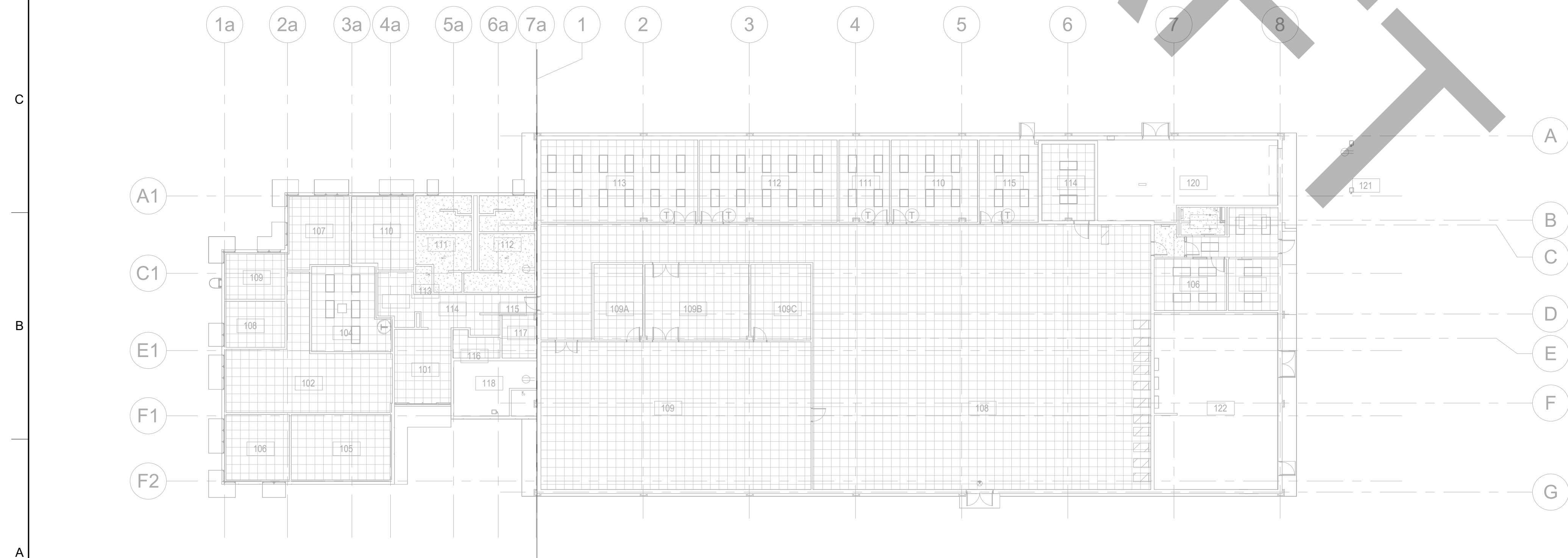
U.S. ARMY CORPS OF ENGINEERS JAPAN ENGINEER DISTRICT OKINAWA, JAPAN Jacobs JACOBS ENGINEERING GROUP, INC. 7000 WESTPORT AVENUE, SUITE 200 WESTPORT, TX 75086 TEL: 972.346.7000 FAX: 972.346.7001 WWW.JACOBS.COM	DESIGNED BY: R. RUBIO	ISSUE DATE: 09/04/2025
	DRAWN BY: S. LEFFLER	SOLICITATION NO.: -
	CHECKED BY: S. LEFFLER	CONTRACT NO.: -
	SUBMITTED BY: S. LEFFLER	DRAWING CODE: -
	QUANTITY: 1	FILE NAME: T-001
SIZE: ISO A1		

FY25 REPAIR HVAC WIDEBAND SATCOM OPS B100
CAMP BUCKNER COMMUNICATIONS ANNEX
OKINAWA, JAPAN
PROJECT NUMBER:D3449305
GENERAL NOTES & LEGEND

SHEET ID

T-001

T T-001- T



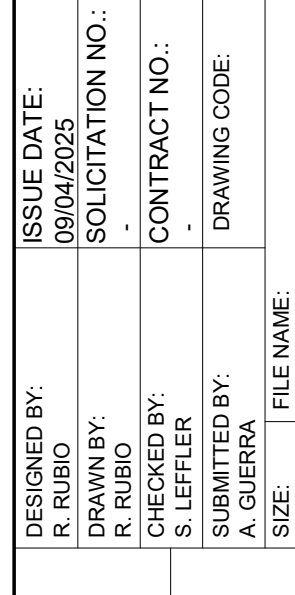


A	MAINTAIN SEPARATION BETWEEN SECURE AND UNSECURE CABLING IN ACCORDINACE WITH A1/T-501.
B	TELECOMMUNICATIONS CONDUITS MUST BE BONDED TO THE PBB/SBB, BUILDING STEEL OR TO THE NEAREST CABLE TRAY.
C	REFER TO SHEETS GI010 AND GI011 FOR OVERALL PHASING INFORMATION.
D	CONTRACTOR MUST VERIFY FIELD CONDITIONS PRIOR TO START OF WORK TO INCLUDE BUT NOT LIMITED TO ROUTING OF OVERHEAD AND UNDERFLOOR CABLE TRAYS, AVAILABILITY OF PATCH PANEL AND RACK SPACE LOCATON OF SLEEVES AND PENETRATIONS, ROUTING OF FIBER AND COPPER FEEDERS.

002	EXISTING NETWORK TWO-POST EQUIPMENT RACK IN COMM ROOM 117 WITH CAT6 COPPER PATCH PANEL. THIS RACK IS NOT PART OF THE DEMOLITION SCOPE. ROOM 117 IS SUPPORTED FROM ISSO RM 111 VIA FIBER OPTIC CABLE ROUTED OVERHEAD. CARE MUST BE TAKEN TO ENSURE THAT DAMAGE DOES NOT OCCUR AND PERSONEL MUST BE PREPARED TO REMEDY DISRUPTION SHOULD IT RESULT FROM RENOVATIONS. NETWORK OPERATIONS MUST REMAIN IN CONTINUOUS SERVICE. THIS ROOM MAY BE USED FOR ADDITIONAL CONNECTIONS TO HVAC MONITORING EQUIPMENT AS REQUIRED.
007	CAUTION MUST BE EXERCISED SO THAT DAMAGE DOES NOT OCCUR TO AUDIO-VISUAL SYSTEMS. BE PREPARED TO REMEDY DISRUPTION SHOULD IT RESULT FROM RENOVATIONS. AV SYSTEMS MUST REMAIN FUNCTIONAL AND IN CONTINUOUS OPERATION.
008	CAUTION MUST BE EXERCISED SO THAT DAMAGE DOES NOT OCCUR TO CABLE TELEVISION (CATV) SYSTEMS. BE PREPARED TO REMEDY DISRUPTION SHOULD IT RESULT FROM RENOVATIONS. CATV SYSTEMS MUST REMAIN FUNCTIONAL AND IN CONTINUOUS OPERATION.
010	NETWORK CABLES ARE DISTRIBUTED THROUGH THE BUILDING SUPPORTING VARIOUS VOICE, DATA, AND OTHER SIGNALS. THIS NETWORK OF CABLES COMPRISES THE LOCAL AREA NETWORK (LAN). CAUTION MUST BE EXERCISED SO THAT DAMAGE DOES NOT OCCUR TO THE LAN CABLING. BE PREPARED TO REMEDY DISRUPTION SHOULD IT RESULT FROM RENOVATIONS. THE BUILDING LAN MUST REMAIN FUNCTIONAL AND IN CONTINUOUS OPERATION.

A diagram showing a shaded area labeled "AREA A" on the left side of a vertical dashed line. The shaded area is a complex polygon with a stepped profile. The area to the right of the dashed line is unshaded.

0 2000 4000
SCALE: 1:100



US ARMY CORPS OF ENGINEERS
JAPAN ENGINEER DISTRICT
OKINAWA, JAPAN

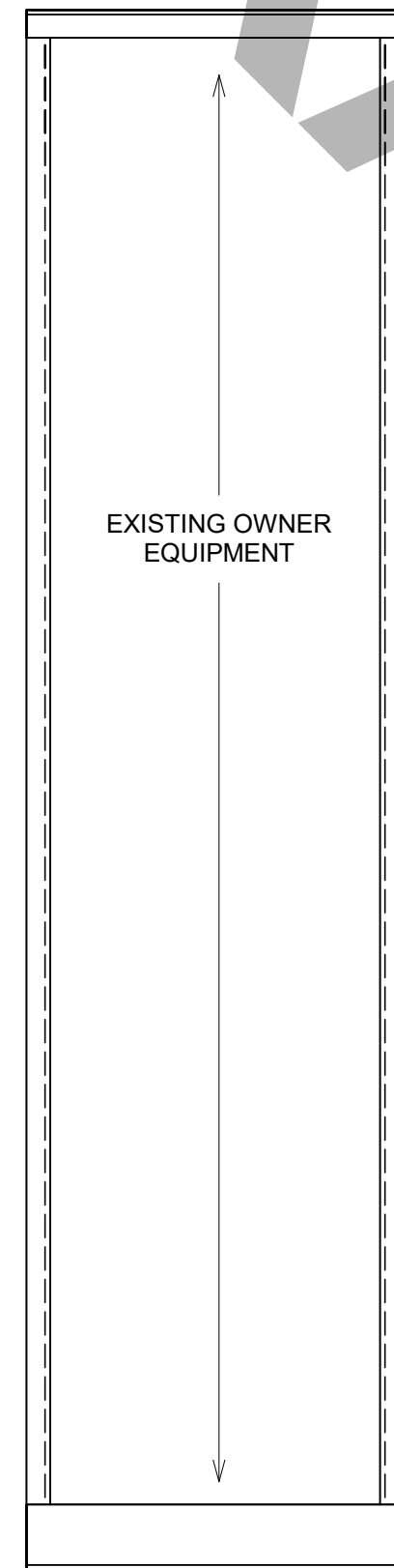
Jacobs
JACOBS GOVERNMENT SERVICES COMPANY
7721 MAIN STREET, FORT WORTH, TX 76102

FY25 REPAIR HVAC WIDEBAND SATCOM OPS B7100
CAMP BUCKNER COMMUNICATIONS ANNEX
OKINAWA, JAPAN
PROJECT NUMBER:D3449305

TELECOM - PARTIAL FLOOR PLAN - AREA A

T-102

A
B
C
D
E
F
G

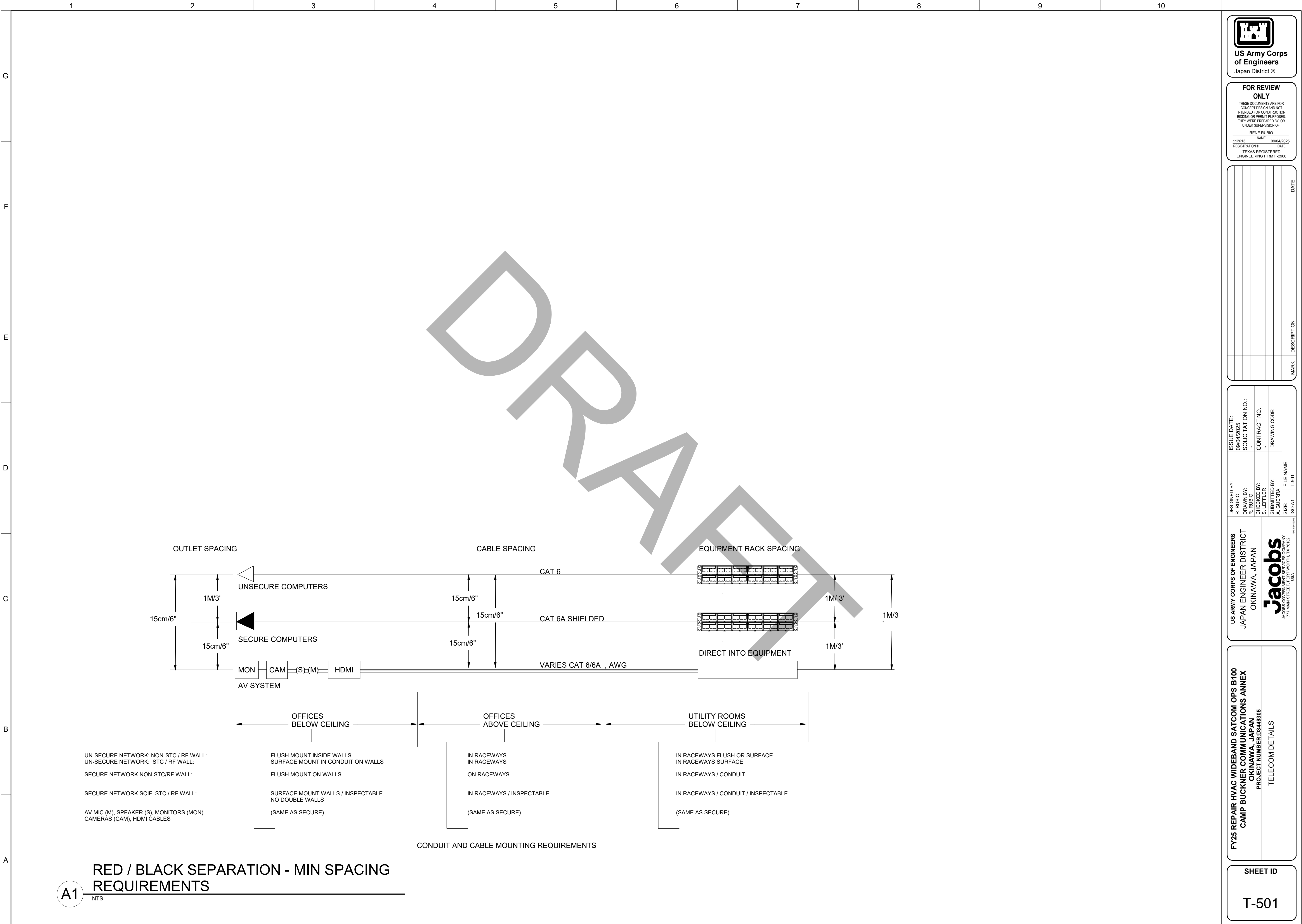


A4 EQUIPMENT RACK COMM RM 117
NTS



T T-201- T

T T-501- T

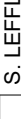


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RENE RUBIO	
2613	09/04/2025
NAME	DATE
REGISTRATION #	DATE
TEXAS REGISTERED	
ENGINEERING FIRM F-2966	

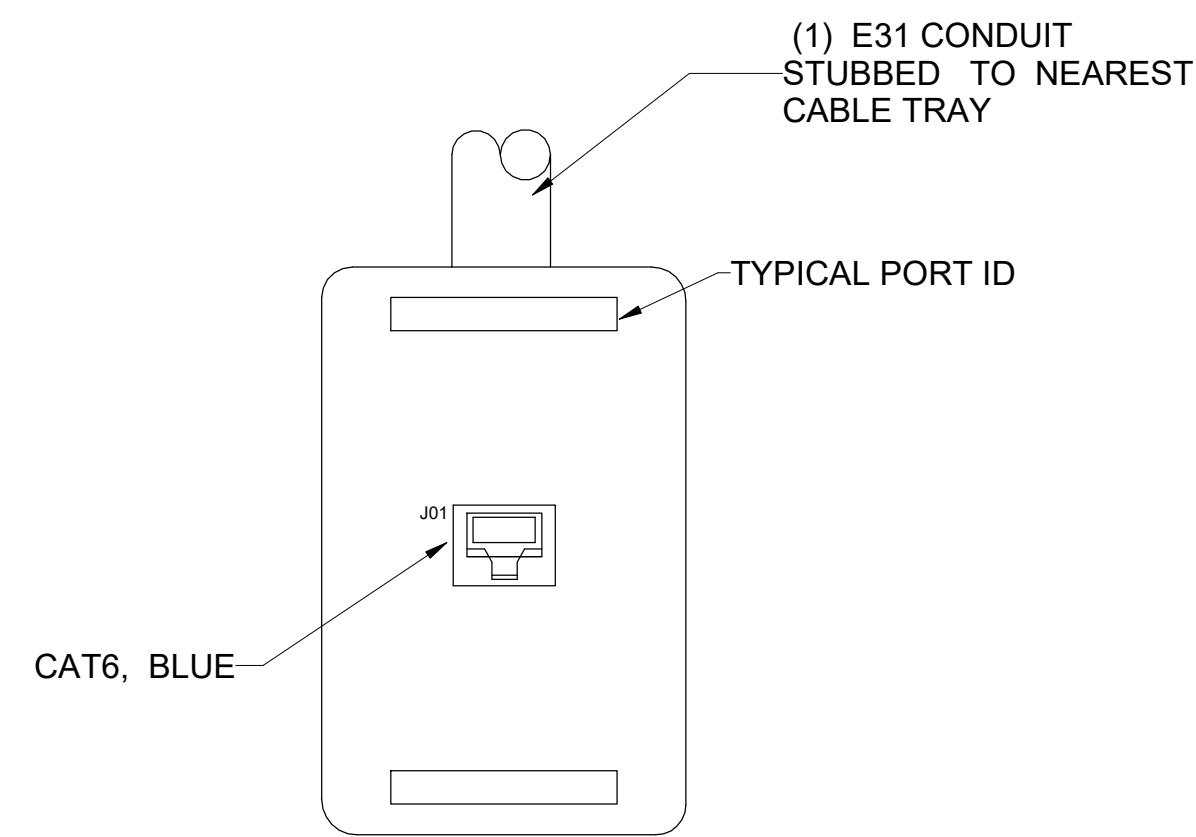
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JAPAN ENGINEER DISTRICT OKINAWA, JAPAN  JACOBS ENGINEERING GROUP, INC. 777 MAIN STREET, FORT WORTH, TX 76102 USA	R. RUBIO		09/04/2025
	R. RUBIO		SOLICITATION NO.:
	R. RUBIO		-
	CHECKED BY:		CONTRACT NO.:
	S. LEFFLER	-	DRAWING CODE:
	SUBMITTED BY:	A. GUERRA	FILE NAME:
			T-502
		ISO A1	

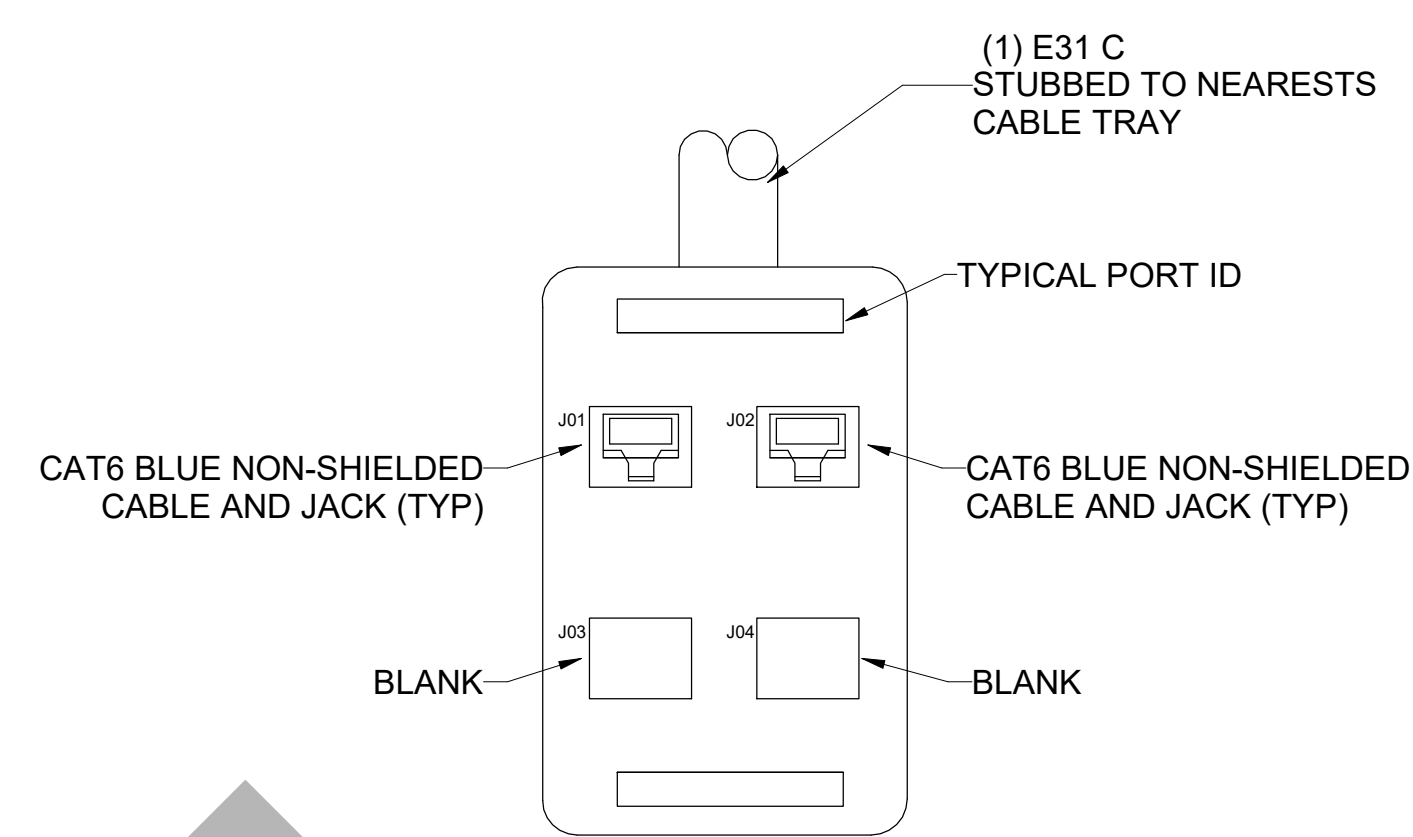
CAMP BUCKNER COMMUNICATIONS ANNEX OKINAWA, JAPAN PROJECT NUMBER D3448305	TELECOM DETAILS
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SHEET ID

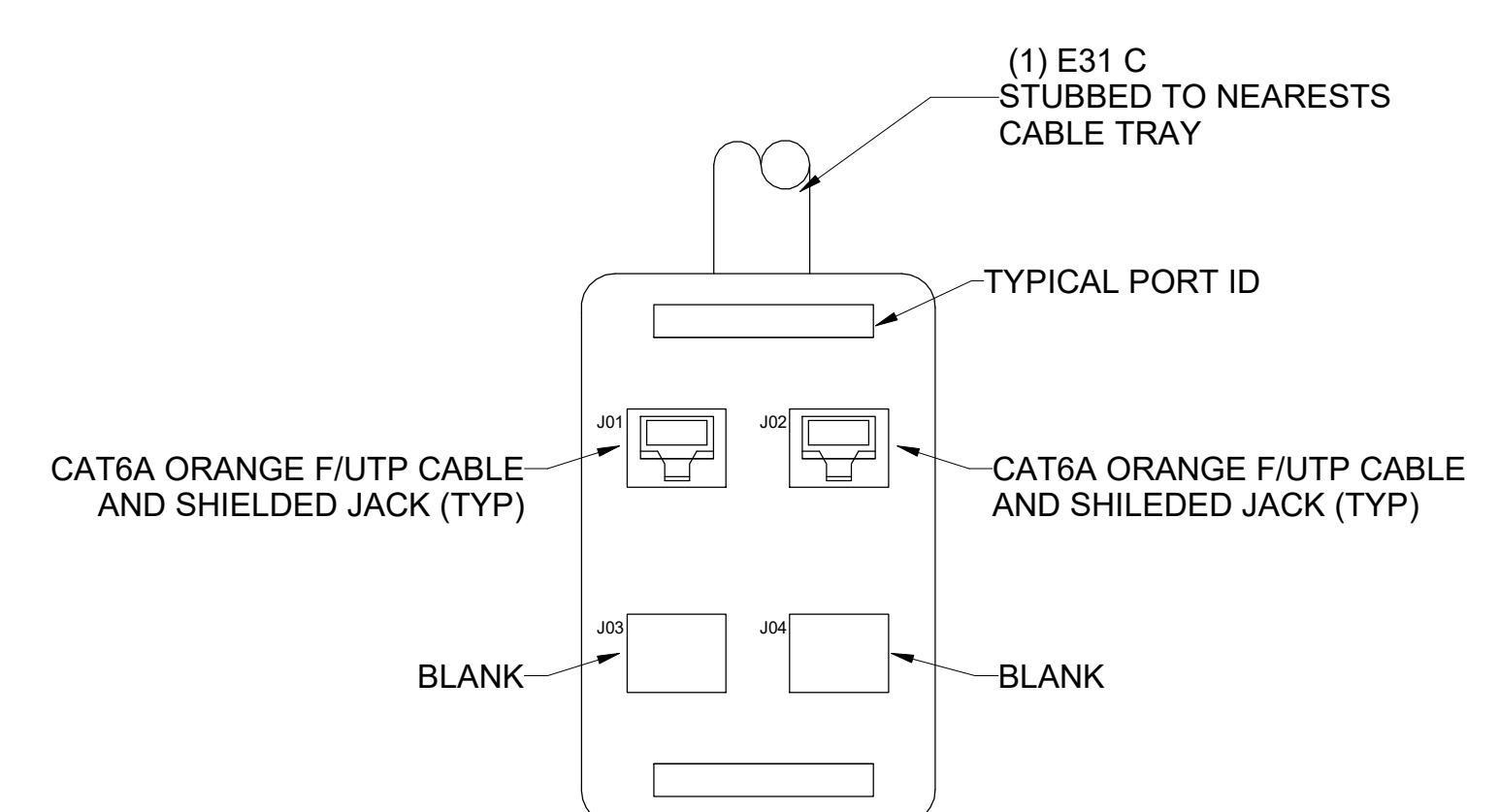
T-502



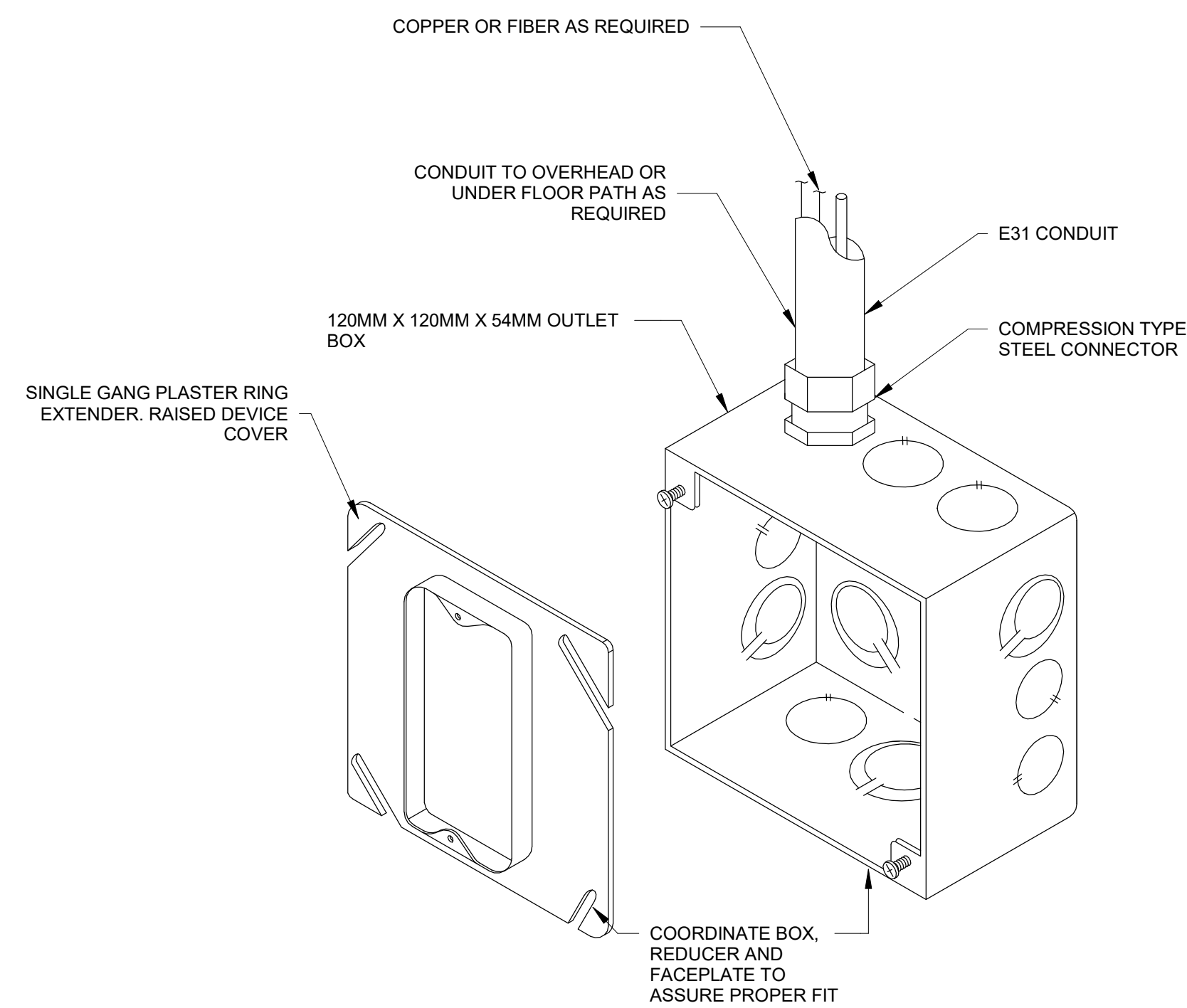
**TYPICAL 1-CAT6 "1N" DATA FACE PLATE
(UNSECURED)**



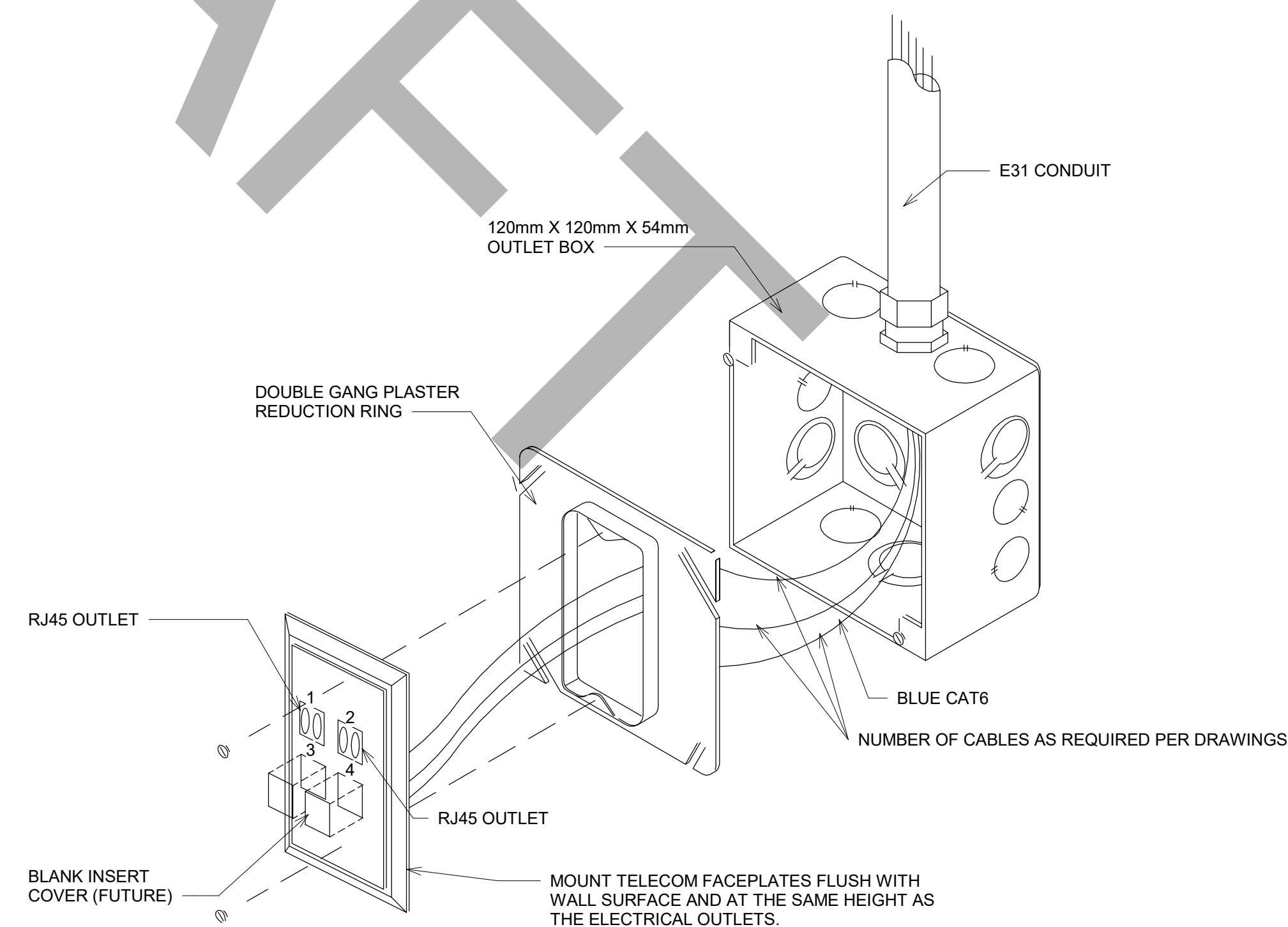
E4 TYPICAL 2-CAT6 "2N", DATA FACE PLATE (UNSECURE)



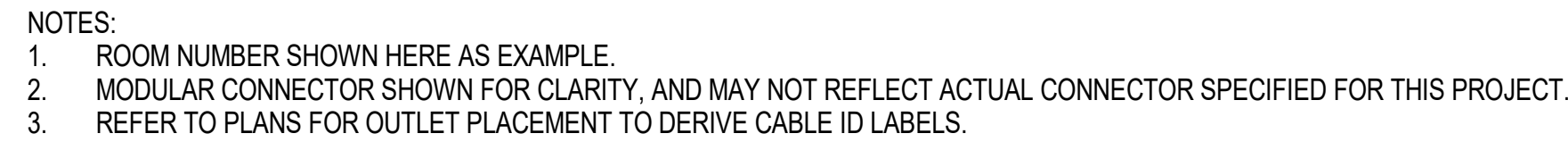
E8 TYPICAL 2-CAT6 "2S", DATA FACE PLATE (SECURE)



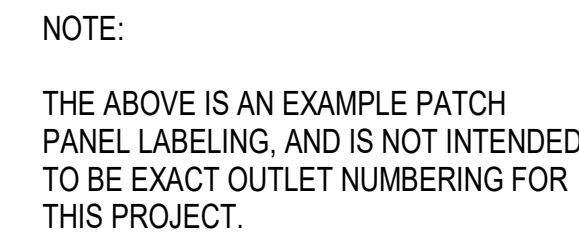
A1 WALL-MOUNTED OUTLET BOX DETAIL



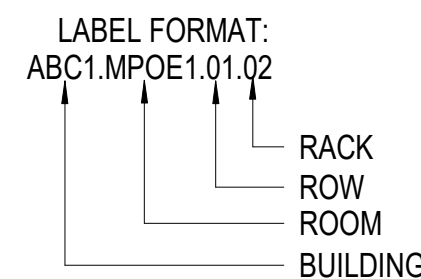
A5 TYPICAL DATA OUTLET DETAIL



NTS



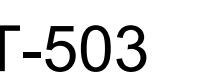
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A1

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NTS



1	2	3	4	5	6	7	8	9	10
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G

F

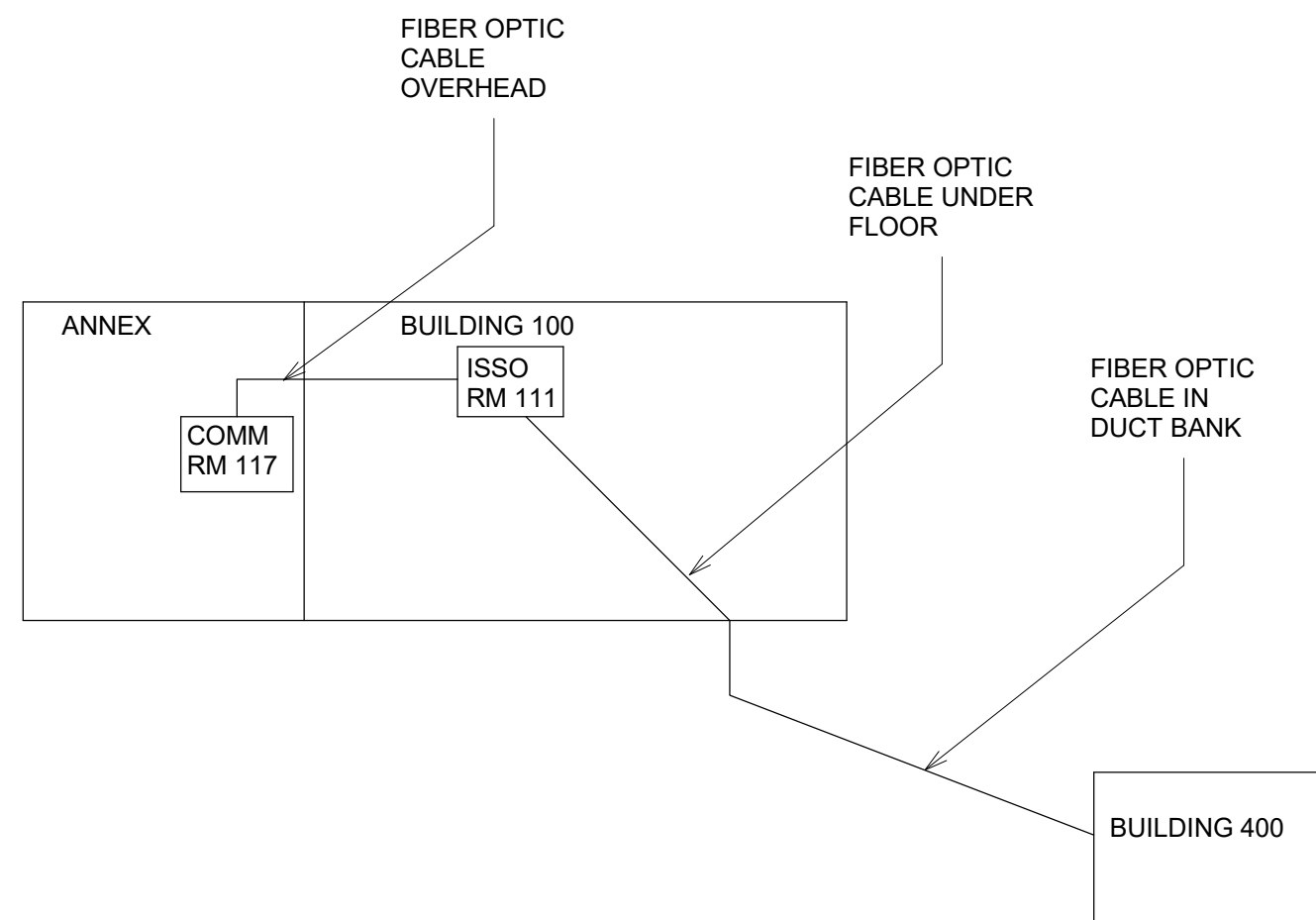
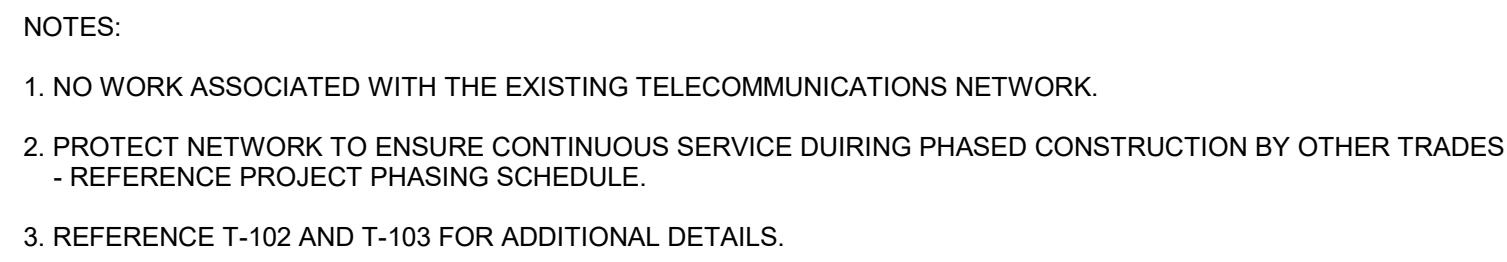
E

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A



A1 FIBER RISER DIAGRAM (E) - PHASING



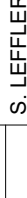
**US Army Corps
of Engineers**
Japan District ®

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UNDER SUPERVISION OF:

RENE RUBIO	
NAME	
2613	09/04/2025
REGISTRATION #	DATE
TEXAS REGISTERED	
ENGINEERING FIRM F-2966	

[illegible]

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CHECKED BY: S. LEFFLER		SOLICITATION NO.:		09/10/2025	

FY25 KEPAIK HVAC WIDEBAND SATCOM OPS B100
 CAMP BUCKNER COMMUNICATIONS ANNEX
 OKINAWA, JAPAN
 PROJECT NUMBER:D3449305
 TELECOM RISER DIAGRAM

SHEET ID

T-601

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